



Evaluating the Contribution of Demand Response to the Electrical Power System: Generation Adequacy and Market Implications

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Abstract

Interest in demand response has been increasing in recent years and a large number of initiatives are being developed and deployed. This interest stems from the ability of demand response to change the energy usage of customers from normal consumption patterns in response to events on the power system or to the varying price of electricity over time. As a consequence of changing energy usage, demand response can contribute to generation adequacy. Demand response can also provide ancillary services and by doing so it is capable of assisting with the integration of variable renewable energy, such as wind and solar photovoltaic energy.

Despite these potential benefits there is still considerable uncertainty surrounding demand response from the power system point of view and the potential benefit for those engaged in demand response programs. Despite this uncertainty, it is apparent from the literature that there is considerable potential for demand response to provide significant value to the power system. Thus, if demand response proves to be the neat and sophisticated solution that many claim it to be, failure to exploit the resource is clearly suboptimal. On the other hand, if the potential value of demand response is over estimated, considerable resources could be invested in order to exploit a resource which ultimately cannot compete effectively. There is evidently a pressing need to quantify the potential system and market value of demand response.

This thesis aims to provide an understanding of the value of demand response to the electrical power system, specifically in relation to generation adequacy, and the impact of demand response on electricity markets. The key challenges and barriers to demand response deployment and evaluation are first identified, while the main opportunities for demand response are also considered. Subsequently, a number of methodologies for evaluating demand response, with particular emphasis on the contribution to generation adequacy, are utilized.

Firstly, a preliminary methodology for estimating the contribution of different demand response resources to generation adequacy utilizing availability profiles and applying a response duration constraint is employed. This methodology is then extended to the analysis of a specific load-shifting demand response resource that provides multiple services but also inherently contributing to generation adequacy. A methodology that couples a load model with a power system operations model, permitting co-optimization of the supply-side and the demand-side, in conjunction with Monte Carlo simulations, is applied to determine the generation adequacy contribution of this specific load-shifting resource.

The results highlight that the contribution of demand response resources to generation adequacy can be quite low in comparison with conventional thermal generation. It is found that the contribution of the resources to adequacy are influenced by the operation of the resource and by the end-user constraints. Furthermore, it is found that the contribution to adequacy of demand response is sensitive to the energy limitation of the resource and to the underlying system characteristics. It is also evident that there is a distinct trade-off between the contribution of a demand response resource to generation adequacy and its ability to provide reserve services. This trade-off indicates that perhaps there is an optimal portfolio for demand response participation and indicates the importance of evaluating demand response within a market environment.

Consequently, an electricity system with energy and reserve markets, as well as a Capacity Remuneration Mechanism, in the form of a capacity market, is modeled. A Mixed Complementarity Problem is formulated to represent the decisions of generators and a demand response aggregator participating in multiple markets, permitting investment and exit of generating capacity. An approach to approximate the ability of the demand response resource to contribute to generation adequacy, thereby permitting participation in a capacity market, is presented.

Analysis reveals the contribution of demand response to electricity markets in terms of operating cost savings and the ability of demand response to reduce peak electricity prices. It is also found that considerable cost savings are attainable by the demand response aggregator through participation in the energy market. However, significant further cost savings are not forthcoming through participation in the reserve market or in the capacity market.

Overall it is found that demand response resources can have a positive contribution to the electrical power system. However, this impact is sensitive to the characteristics of the underlying power system; the physical constraints and operation of the resource; the necessity to ensure end-user requirements are satisfied as well as the interaction of the demand response resource with the power system as a whole.

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Statement of Original Authorship

I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree stated on the Title Page, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.

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Nomenclature

Indices

<i>arch</i>	Building archetype
<i>capacity</i>	Relates to capacity market
<i>elec</i>	Electrical output
<i>energy</i>	Relates to energy market
<i>g</i>	Conventional generators
<i>h</i>	Hydro generators
<i>heat</i>	Heat output
<i>i</i>	Firm
<i>j</i>	Technology
<i>reserve</i>	Relates to reserve market
<i>space</i>	Refers to space heating devices
<i>t</i>	Time in hours
<i>water</i>	Refers to water heating devices

Parameters

η	Efficiency
μ_{gen}	Mean time between generator outages
CAP	Initial installed generating capacity
$COST$	Marginal cost of operating generator
DEM	Non-interruptible system demand
DEM_{BASE}	System demand minus heating requirements

$DMAX$	Maximum capacity of demand response resource
DR_t	Demand response operation at hour t
$DREF$	Reference demand
E	Elasticity of demand
E_{MAX}	Installed energy capacity of DR resource
FOR_g	Forced outage rate for generator g
GEN	Total generator installed capacity
$ICOST$	Investment cost
MC	Marginal operating cost
$MCOST$	Maintenance cost
$MTTR_g$	Mean time to repair for generator g
$OUTAGES_g^{avg}$	Average number of generator g outages per year
P^{MAX}	Installed capacity of generator
$PEAK_{AVAIL}$	Peak availability of DR
$RATING$	Power rating of DR resource
$RESERVE_{REQ}$	Reserve requirement
$TARGET$	Capacity market target
$TRIGGER$	Peak load level at which DR is dispatched is triggered
$VOIR$	Value of Insufficient Reserve
$VOLL$	Value of Lost Load
$WIND$	Wind generation

Variables

ΔL	The amount of extra load that added to each hour of the year
κ	Capacity price
λ	System marginal price
μ	Reserve price
Π	Profit
$\tilde{R}$	Random number
cap_{bid}	Capacity market bid

dem_{lost}	Load shed
dr_{down}	Load-shifting downwards
dr_{up}	Load-shifting upwards
$dres$	Downward Reserve
e	Energy Storage
eh	Time of next outage event
$event_{num}$	Outage event number
$exit$	Generation capacity exit decision variable
gen	Generator power output
$invest$	Generation capacity investment decision variable
$LOLE_{DR}$	Loss of load expectation with DR
$LOLE_{ref}$	Loss of load expectation without DR
$LOLP_{DR}^t$	Loss of load probability with DR
$LOLP_{ref}^t$	Loss of load probability without DR
$losses$	losses from the DR resource energy component
p	Power
por_{total}^t	POR requirement at time t
por_{unmet}^t	POR unmet at time t
$res_{max}^{t,down}$	Max Downward Reserve available from DR
$reserve_{dr}$	Reserve provision from the demand response resource
$reserve_{gen}$	Reserve provision from a generating unit
U	Binary variable for availability of generator
Sets	
H	Set containing the first hour of each day

List of Acronyms

AS	Ancillary services
AVA	Avista Corporation
BEM	Building Energy Model
COPT	Capacity Outage Probability Table
CPP	Critical Peak Pricing
CRM	Capacity Remuneration Mechanism
CSP	Concentrating Solar Power
CV	Capacity Value
DR	Demand response
DRH	Direct resistive heating
ECC	Equivalent Conventional Capacity
EFC	Equivalent Firm Capacity
ELCC	Effective Load Carrying Capability
EU	European Union
FOR	Forced Outage Rate
GAMS	Generic Algebraic Modeling System
I-SEM	Integrated Single Electricity Market
KKT	Karush Kuhn Tucker
kW	Kilowatt
kWh	Kilowatt-Hour
LBNL	Lawrence Berkely National Laboratory
LDWP	Los Angeles Department of Water and Power
LOEE	Loss of Energy Expectation
LOLE	Loss of Load Expectation
LOLH	Loss of Load Hours

LOLM	Loss of Load Magnitude
LOLP	Loss of Load Probability
MC	Monte Carlo
MCP	Mixed Complementarity Problem
MTTR	Mean Time to Repair
MW	Megawatt
NEVP	Nevada Power
NREL	National Renewable Energy Laboratory
POR	Primary Operating Reserve
PSE	Puget Sound Energy
RTP	Real time pricing
SEM	Single Electricity Market
SMP	System marginal price
TUS	Time-use survey
UCED	Unit Commitment and Economic Dispatch
UK	United Kingdom
VOIR	Value of Insufficient Reserve
VOLL	Value of Lost Load

Publications Arising

The following papers and reports have been completed based on the research presented in this thesis:

Conference Papers

1. S Nolan, M O'Malley. "Integration cost of demand response", *31st USAEE/IEEE North American Conference*, Anchorage, Alaska, USA, 2013
2. S Nolan, M O'Malley, M Hummon, S Kiliccote, O Ma, "A methodology for estimating the capacity value of demand response". *IEEE PES General Meeting*, Washington DC, USA, 2014. doi: 10.1109/PESGM.2014.6939174
3. S Nolan, H Wajahat Qazi, D Burke, D Flynn, M O'Malley, J Kiviluoma, M Hummon, M Milligan, B Kirby. "Synergies between Wind and Solar Generation and Demand Response: An IEA Task 25 Collaboration", *13th Wind Integration Workshop*, Berlin, Germany, 2014.

Journal Papers

1. S Nolan, M O'Malley. "Challenges and barriers to demand response deployment and evaluation", *Applied Energy*, Volume 152, 15 August 2015, Pages 1-10, ISSN 0306-2619, <http://dx.doi.org/10.1016/j.apenergy.2015.04.083>.
2. S Nolan, O Neu, M O'Malley. "Capacity Value Estimation of a Load-Shifting Resource using a Coupled Building and Power System Model", *Applied Energy*, *In Review*

3. S Nolan, M Devine, M Lynch, M O'Malley. "Impact of Demand Response Participation in Energy, Reserve and Capacity Markets ", *The Energy Journal*, *In Review*

Technical Reports

1. D Burke, S Nolan, O Neu, WH Qazi, D Flynn, D Finn, M O'Malley. (2014), Initial Cost Benefit Analysis Comparison for Domestic Smart Electric Thermal Storage (SETS) vs Direct Resistive Heating (DRH) on the Irish All-Island Power System, Electricity Research Centre & GlenDimplex collaboration.

Introduction

There are three pillars of energy policy: security of supply, environmental sustainability and economic competitiveness. Security of supply relates to the availability of indigenous sources of natural resources for energy production and the need to import resources in the absence of sufficient indigenous sources. Recent decades have seen a diversification of generation technology, with the integration of greater levels of variable renewable generation into the power system in a bid to enhance security of supply and reduce dependence on imports. World-wide the share of renewables is rapidly increasing and, fortunately, this is also contributing to the second pillar of energy policy - environmental sustainability.

It is now irrefutable that anthropogenic emissions of carbon dioxide, and other greenhouse gases, are contributing to climate change. In an effort to halt the inevitable global temperature increases and sea level rises, in conjunction with the associated detrimental impact on vulnerable species, many nations have adopted energy policies under the remit of environmental sustainability. This is in an effort to reduce carbon emissions and increase the penetration of renewables and shift away from carbon intensive fossil fuel-based generating technologies.

It is vitally important that generating technologies are competitive and sustainable to ensure low electricity prices and overall competitiveness of economies, the third energy policy pillar. There are always tensions between the three pillars of energy policy; indigenous resources may not be environmentally sustainable and while importing fuels may, in some instances, be economically competitive, the regions from which those fuels are sourced may not be politically stable, thereby endangering security of supply.

In conjunction with the advent of increased penetration of variable renewable generation and the adoption of energy policies, there has been significant developments and cost reductions in the area of telecommunications and control systems which has facilitated greater controllability of demand-side resources. As a result of these advances, demand response (DR) is being developed and deployed on a more continuous basis, across multiple time frames and multiple markets. Many power systems have established DR integration programs, while others are in the process of designing programs appropriate to their system. The interest in deploying DR programs is evident from the vast amount of work that has been published in recent years [2, 3, 4, 5, 6].

DR is often described as the change in electrical energy usage by end-use customers from their normal consumption patterns in response to changes in the price of electricity over time, to incentive payments [7] or to signals from the system operator. It is widely acknowledged that DR is not a new concept. Historically, demand shedding was used for emergency contingency response and, in more recent times, DR programs have been targeted at large industrial energy users [8]. DR is thus now much broader than traditional emergency load shedding resources as it is in a position to provide a range of system services over varying time frames, and it is also possible to elicit an increase in consumption as well as a reduction. While there is experience with DR programs targeted at large industrial energy users [8], there is also now more of an interest in extending DR to all sectors; industrial, commercial and residential. However, DR is, in general, an underutilized resource [2], but it is touted as having significant potential benefits [3, 7, 9].

Some of these benefits relate to the fact that DR is in a position to make a positive contribution to all three of the energy policy pillars. The inclusion of variable renewable generation, such as wind and solar, in the generation system portfolio can present a number of issues for system operators. Variable renewable generation can increase price volatility and can lead to a greater requirement for ancillary services (AS) [10], which could impose barriers on the desire to strive for greater levels of renewable generation, and thus environmental sustainability. Fortunately, DR can facilitate the integration of renewable generation by the provision of many of the required AS. DR also has the potential to shift system demand to more favorable periods and thus can assist with renewable curtailment reduction thus helping to foster greater exploitation of non-fossil energy sources and improve security of supply,

the first energy policy pillar. Furthermore, DR could assist with meeting occasional peaks in electricity [11] and could thus, potentially, reduce system costs [4] and harmful power plant emissions at times of peak demand [12], thereby tackling the second energy policy pillar, environmental sustainability.

It should be noted that the increase in renewable generation is not necessarily a driver for the investigation and research in the area of DR. The significant and rapid development in telecommunications, control systems and computation has pushed DR to the fore, to such an extent that DR is now considered to be a power system asset. This would have been the case irrespective of the increase in renewable generation penetration. It is merely a coincidence that DR and renewables are developed in parallel. Additionally, regulatory changes and electricity market reform have played a major part in allowing DR to become a more viable resource [12]. Thus, the addition of DR could assist in adding diversity of resources to the market, as well as increasing the number of market players, thereby increasing economic competitiveness, the third energy policy pillar.

However, there are many challenges associated with both the deployment and the evaluation of DR. There is considerable uncertainty surrounding the value of DR from the power system point of view and surrounding the potential revenue for those engaging in DR programs. This impedes wide-spread deployment of demand-side technologies, which in turn impedes the evaluation process of the resource.

Despite this lack of understanding surrounding the value of DR, it is apparent from the literature that there is considerable potential for DR to provide significant value to the power system. Thus, if DR proves to be the neat and sophisticated solution that many claim it to be, failure to exploit the resource is clearly suboptimal. On the other hand, if the potential value of DR is over estimated, considerable resources could be invested in order to exploit a service which ultimately cannot compete effectively. There is evidently a pressing need to quantify the potential system and market value of DR. Indeed, it has been suggested that the limited DR capability at present in the US [7] is set to rise if it is shown that the “*flexibility it offers is valuable and properly valued*” [13]. Evaluation of the DR resource is clearly a vital step in its large scale deployment and could increase understanding of the impacts of DR on the power system and electricity markets.

An understanding of the value of DR would give an indication of the potential opportunities that exist within particular markets [14], could inform which sectors

DR is most amenable to operating in and could provide clarity for potential market participants and investors. Knowledge of the value could also provide guidance on the cost of DR enabling technologies, whether or not market incentives are necessary to exploit DR resources, and what those incentives should be.

One important aspect of value of a DR resource is its contribution to system generation adequacy. Generation adequacy is the existence of sufficient generating capacity to meet system demand and the contribution a resource makes to generation adequacy is usually referred to as its Capacity Value (CV). An appreciation of the CV of DR could enable a comparison of DR with other resources and assist in the analysis of the impacts of DR on long-term power system planning. From a utility's point of view, a resource with a low CV also has reduced economic value [15]. Thus, an appreciation of the potential CV of a DR resource could provide a better understanding of its overall value to the system and its potential economic value. Additionally, as CV metrics are often used in Capacity Payment Mechanisms [16], appreciation of the CV of DR would readily lead to a better understanding of the potential capacity revenues available for DR programs.

It is also very timely to consider the effects of including DR in a Capacity Remuneration Mechanism (CRM), as European electricity markets are currently undergoing significant change due to European Union (EU) legislation driving the integration of electricity markets across Europe in an effort to create a fully liberalized internal electricity market [17]. By the end of 2017, Ireland must have a new wholesale market, which will be known as the Integrated Single Electricity Market (I-SEM), in place. In the current market structure, the Single Electricity Market (SEM), generators are given a payment known as the capacity payment, to incentivize generating firms to be available to generate when and as required, irrespective of the extent to which the generating plant is used [18]. The current CRM in Ireland involves annually distributing a pot of money, known as the capacity pot, among generators based on their availability [19]. The exact format of the new CRM is still undergoing development. Additionally, some countries in Europe, unlike Ireland, do not have a history of CRMs and are currently exploring options that best suit their systems. Thus, it is prudent to consider how DR will be included in and will impact upon a capacity market and CRMs in general.

1.1 Research Hypothesis and Objectives

In this thesis, it is hypothesized that DR resources can provide value to the electrical power system, particularly in relation to generation adequacy, and can have a positive impact on electricity markets. The objectives of this research are as follows:

- Identification of the challenges and barriers associated with evaluation and deployment of DR resources.
- Development of a methodology for assessing the contribution of DR to generation adequacy and determination of the potential CV.
- Appreciation of the factors influencing the CV of DR.
- Consideration of the impact DR resource provision of multiple services can have on the CV of DR resources.
- Analysis of the impact a DR resource can have on electricity markets.
- Determination of whether there is an optimal bidding strategy for a DR aggregator.

1.2 Research Methodology

As will be discussed in Chapter 2, an integration study is ideally suited to the examination of the contribution of DR to the electrical power system. An integration study typically seeks to determine the impact of including a resource in the power system. To that end, an integration study includes production cost simulations, which often involve a Unit Commitment and Economic Dispatch (UCED) methodology [20], which is beneficial in that it allows assessment of both energy production and provision of ancillary services. Additionally, an integration study would involve generation adequacy (or CV) calculations, analysis of the impact of new resource on power system dynamics and power flows [20], among others.

An integration study approach is adopted in this thesis to ascertain the contribution of DR to the electrical power system. Figure 1.1 provides an overview of this methodology. Parallels will be noted between Figure 1.1 and the recommended methodology for wind integration studies (see Figure 2.1 in Chapter 2).

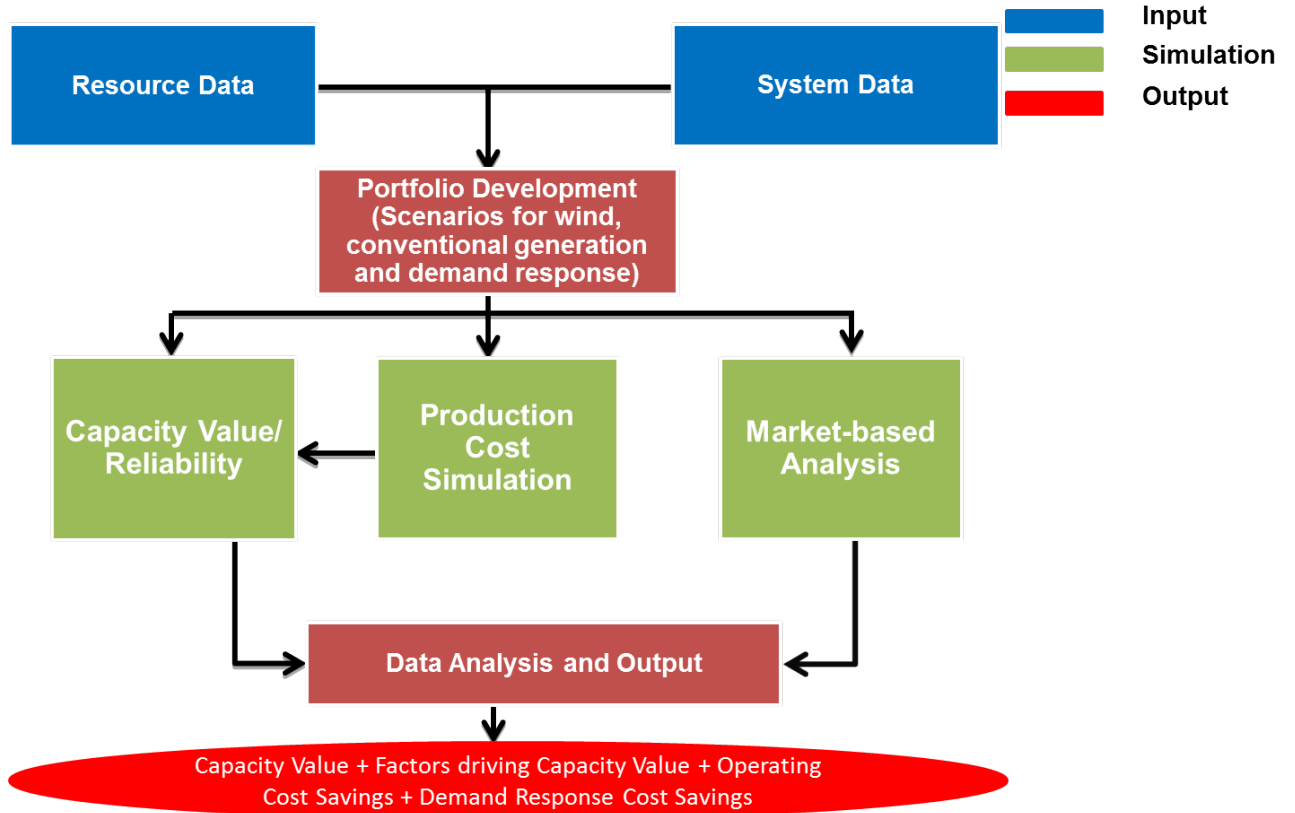


Figure 1.1: Demand response integration study: Thesis methodology overview

The focus is placed on the generation adequacy aspects of an integration study, as this is an area where DR can directly contribute to the power system and participate in electricity markets. These aspects are analyzed through the use of production cost simulations and CV calculation methodologies. Specific focus is also placed on the market implications of DR participation and, accordingly, a market model is developed and employed in this thesis, using a Mixed Complementarity Problem (MCP) framework.

1.3 Thesis Contributions and Structure

The contributions and structure of this thesis are as follows:

- The completion of an extensive review of the broad DR literature and identification of the key challenges facing the deployment and evaluation of large-scale DR. The main opportunities for DR are also discussed, recommendations

provided and areas of important potential research highlighted. This is presented in Chapter 2.

- CV metrics and calculation methodologies are explored in Chapter 3. One of the key contributions of Chapter 3 is the development of an estimation method for calculation of the CV of DR and determination of indicative results and key trends. This is achieved through the adaptation of a methodology typically employed for estimating the CV of wind generation. Through utilization of this method, an understanding of the importance of including DR operational and physical constraints in CV calculations is obtained.
- Chapter 4 extends the work presented in Chapter 3 and involves specific analysis of one specific type of load-shifting DR resource, smart electric thermal energy storage devices for space and water heating, using a coupled Building Energy Model (BEM) and UCED methodology with Monte Carlo (MC). This includes detailed modeling of the physical characteristics of the resource, accounting for consumer behavior and operational decisions on both the supply-side and demand-side of the power system. The key contribution of this chapter is the inclusion of reserve provision from the DR resource within the CV analysis.
- Chapter 5 examines the impact a DR resource providing energy and reserve can have on electricity markets through the inclusion of DR in a Mixed Complementarity Problem (MCP) framework. The chapter also explores the participation of DR in CRM, also using an MCP framework. The key contribution of this chapter is the consideration of the simultaneous participation of DR in all three electricity markets.
- Chapter 6 presents a summary of the main results and conclusions from this thesis. It also indicates some areas of potential future work.

Literature Review

The contents of this chapter are largely reported in a journal paper entitled ‘Challenges and barriers to demand response deployment and evaluation’ which was published in Applied Energy [21]. A full citation can be found in the section titled “Publications Arising”. This chapter provides a broad, high-level review from across the demand response literature. Literature pertaining specifically to the material presented in each individual chapter is reviewed in that chapter.

2.1 Introduction

As was mentioned in Chapter 1, it is widely acknowledged that DR is not a new concept, with demand shedding having historically been used for emergency contingency response. However, there has been much interest recently in more continuous DR, that is, DR on a continuous time-frame, and DR across all sectors. The reason for this burgeoning interest relates to the potential for DR to assist in integrating variable generation [3], the fact that it could be a more cost-effective means to meet occasional peaks in electricity than peaking plants [11] and could thus, potentially, reduce system costs [4] and harmful power plant emissions at times of peak demand [12]. Furthermore, as previously acknowledged, in parallel to developments in variable renewable generation penetration, there have been rapid advances and cost reductions in telecommunications, control systems and computation, which has resulted in greater, potentially cost-effective, controllability and flexibility of demand-side resources. Additionally, regulatory changes and electricity market reform have played a major part in allowing DR to become a more viable resource [12].

There has been significant and beneficial work in the area of DR in recent years, highlighting the potential benefits of DR [7, 9]. Studies have shown that using DR to facilitate the integration of variable renewable generation is technically feasible [3, 9], but there are barriers that limit the use of demand-side strategies for integrating wind and solar, some of which are identified in [22].

There has also been much work reported in the literature focusing heavily on detailed modeling of individual buildings and demand control systems, as well as in the telecommunications and equipment required to implement DR [23, 24, 25, 26]. There has, however, been less work in the evaluation of DR from a power system perspective. Despite significant advances in power system analysis in recent times, many traditional power system models have neglected to incorporate DR. There is now a need for these models to be enhanced in order to better account for the unique characteristics of the demand-side [5]. Additionally, there is a requirement to develop tools which are appropriate for quantifying the impacts of DR resources on market performance, generator dispatch and other system effects [27].

Despite the many barriers associated with DR deployment, some of which are documented and explored in this chapter, it is apparent from the literature that there is considerable interest in DR because of its potential to provide significant value to the power system. There is evidence that DR could deliver some system services more reliably than conventional generation [2]. There is undoubtedly a pressing need to quantify the potential system and market value of DR. Evaluation of the DR resource is clearly a vital step in its large-scale deployment and could increase understanding of the impacts of DR on the system and markets.

DR can potentially enter into the energy, capacity and AS markets and avail of multiple revenue streams. DR is capable of participating in the energy market by providing services such as peak-shaving and load-shifting. Such services could help to reduce system demand at times of typically high prices, potentially reducing output from expensive peaking plants and thereby lowering system costs. A study reported in [28] found that the benefits of DR also lie in the realm of avoided capacity costs, thus DR could be a prominent player in the capacity market. The literature has also shown that one of the benefits of DR lies in its ability to assist with facilitation of variable renewables through reserve provision [9]. It has been illustrated that in some cases demand can provide some responses which are greater than the responses garnered from generators [2, 29]. This illustrates the potential for DR to operate in AS

markets. Indeed, it is indicated in [14] that an understanding of the value of DR gives an indication of the potential or opportunities that exist within particular AS markets.

An understanding of the value of DR should also inform which sectors DR is most suited to operating in. The authors in [30] advocate focusing DR activities on specific large consumers, consumers who are capable of responding appropriately to real-time prices. This is, in effect, exploiting the easiest resources first. Similarly, the work in [31] demonstrates that most of the total benefit of exposing all sectors to real time pricing (RTP) can be achieved by implementing DR through RTP in the industrial and commercial sectors only. The work in [31] is based on one specific power system and consequently, it is important to be aware that the results are likely to be quite system specific.

To-date there has been minimal work on evaluating DR from a system level. As a result, considerable uncertainty exists regarding how DR is going to be deployed and the potential revenue for those engaging in DR programs. This consequently impedes the evaluation process of the resource. Similarly, the lack of understanding and knowledge relating to the value of DR imposes a barrier on widespread deployment. This results in a ‘Chicken and Egg’ type dilemma; DR will be deployed if it is valuable to do so, but an understanding of the methods of deployment and operation of DR is necessary before evaluation studies can produce meaningful results.

This chapter contributes to the literature by identifying the barriers and uncertainties that exist for widespread DR deployment and by then showing how these barriers could impact upon the analysis of the value of the DR resource. A number of recommendations for evaluating DR resources are also provided. Section 2.2 explores opportunities that exist for DR to participate in electricity markets and some associated barriers. Section 2.3 then discusses the barriers to deployment associated with the demand-side of the power system as a result of the uncertainty surrounding consumer behavior and due to the requirement for a greater flow of data and information. Section 2.4 examines potential methodologies for determining the value of DR resources to the power system and discusses how the uncertainty and barriers in DR deployment can impact upon the methodologies. Section 2.5 ties together many of the elements discussed in this chapter and also identifies issues for synthesis and dissemination of evaluation analysis results. Section 2.6 concludes this chapter.

2.2 Demand Response Opportunities and Barriers

DR represents a paradigm shift in how we view electricity systems since electrical load can now appear on both sides of the market, supply and demand. As previously mentioned, an appreciation of the impact of DR implementation on electricity markets would be vital for determination of its value and could indicate which markets would benefit most from DR participation and which markets consumers should consider participating in. The following subsections explore the different services DR can provide and look at some of the barriers associated with deploying DR programs.

2.2.1 Energy Services

DR is capable of participating in the energy market, by offering services such as peak-shaving, load-shifting and energy arbitrage. For more detail regarding such DR services see [32, 33]. Peak-shaving and load-shifting programs can assist in reducing the need for expensive peaking units and in flattening the load profile, by reducing demand at times of high prices and, in the case of load-shifting, by increasing load at times of lower electricity market prices. Such DR programs are usually triggered in response to a price signal from the energy market. Many of the price-based DR programs, such as RTP programs, time-of-use programs and Critical Peak Pricing (CPP) programs are predicated on the exposure of customers to time-varying energy market prices.

RTP allows demand to respond to fluctuations in the price of electricity [34] and exposing customers to real-time energy market prices can mitigate high price spikes. It is shown that RTP is more economically efficient in the short-run than offering customers a flat tariff [35, 36]. RTP is just one of many proposed means for deploying DR programs. For a more in-depth discussion of other DR programs, the reader is directed to [37] and [38] and to a workshop report drafted by the US Department of Energy [33]. For a more detailed discussion and comparison on the welfare impacts of flat tariffs and different off-peak and peak prices, the reader is directed to [35] and to the highly influential work by Fred C. Schweppe in 1988 [39].

Interestingly, many “*energy market meltdowns*” in recent years can be attributed to a lack of price-responsive load [40] and could have been alleviated by DR programs. Additionally, the work in [41] shows that the more price-responsive demand that is available the more competitive the long-term electricity market and the more reliable the system becomes.

There is clearly a benefit in deploying DR programs in the energy market. However, such programs will only be implemented if they are proven to be worthwhile, but in order to assess whether or not it is valuable to society and to the power system requires data and experience from real-life DR programs. Without certainty that it is valuable, the required investment will not be available to implement such programs.

2.2.2 Capacity

As well as providing energy services, the demand-side is also in a position to provide capacity. The purpose of capacity markets is to ensure that there is an incentive for investors to “*build adequate capacity in line with consumer preferences for reliability*” [42] and to ensure that there is enough generating capacity in the long-term. The authors in [43] suggest that a capacity market is needed in most restructured electricity markets, while others argue that there is no need for a capacity market. There are many market failures which induces a need for capacity markets [44]. One such market failure is the absence of an active or responsive demand-side. This absence results in a situation whereby customers cannot signal their desired level of reliability of electricity supply.

Capacity Remuneration Mechanisms (CRM) can be broadly divided into ‘price-based’ mechanisms or ‘quantity-based’ mechanisms. A ‘quantity-based’ mechanism is considered in this thesis as it is expected that the CRM for the I-SEM will take the form of a capacity auction. A ‘quantity-based’ mechanism involves the system operator contracting ahead for a fixed amount of capacity [18]. This fixed amount of capacity is typically equivalent to the expected peak demand [18]. While CRMs have been in existence for some time, there is still considerable research in the area of optimal design of capacity markets [18].

As already alluded to, DR can provide capacity and thus receive capacity payments or operate in capacity markets and ultimately contribute to generation adequacy. This could help to alleviate the need for considerable investment in conventional generation. In [45] it is indicated that DR is ideally suited to participation in capacity markets, because the DR resource would be paid for simply being available to reduce load or provide capacity but may not actually be required to provide a response. Indeed, according to [46], in PJM, over 90% of the revenue earned by DR is from the capacity market, highlighting the importance of the capacity market for attracting DR investors. However, it is important to note that the magnitudes of potential capacity

payments are largely dependent on the requirements for capacity as will be seen in Chapter 5.

One of the difficulties with trying to generalize a discussion on capacity services is that there is a large difference in capacity procurement mechanisms. Some systems simply do not have capacity markets (or payments) and in these situations the method of procuring capacity is entirely through the energy market. In such cases, regulators need to ensure that average electricity prices are sufficiently high in order for generators to recover their costs [47]. In systems where there are CRMs, there is considerable on-going debate as to their effectiveness and there is significant uncertainty regarding the future of capacity markets. This presents a challenge for DR investors as they can not rely upon sure and certain revenue from capacity mechanisms in the coming years. Chapter 5 of this thesis considers DR participation in a CRM.

Considerable differences also exist in ancillary service markets and definitions, presenting further problems for DR deployment.

2.2.3 Ancillary Services

Ancillary services (AS) are services which the system operator employs over various time frames to maintain the supply-demand balance on a continuous basis [48]. The increasing penetration of variable and uncertain renewable generation has led to a greater need for AS and flexibility products [10] and certain types of DR are well-placed to provide some of these services [49, 50]. Coupled with this is the fact that the prices paid for AS provision can be considerable. In particular, it has been illustrated that a number of DR resources can offer spinning reserve and regulation services and thus they could earn considerable revenue.

Some systems, the New York Independent System Operator (NYISO) and PJM for example, permit participation of DR resources in their AS markets [22], however, it is noted in [51] that, in the US, DR resources are minor players in AS markets in most jurisdictions. One of the barriers is that there are many market rules for AS provision and the varying AS rules and requirements can limit the *‘possible scale of DR deployment’* [29]. In [51] it is suggested that the barriers associated with *‘bulk power system service definitions’* should be the first barriers to be dealt with in order for DR to even be in a position to provide bulk system services.

The problem is that definitions of AS have typically been built up heuristically and, traditionally, conventional generating plants were the sole providers of these services. Such definitions, when ‘hard-wired’ into the market design, could be considered to be discriminatory [52] and may preclude DR programs from participating. For example, in the California ISO (CAISO), a resource that is providing an AS must be capable of maintaining its capacity for a specified period of time, otherwise it will not be certified, and thus cannot bid into the AS markets [14], potentially excluding DR. A similar rule is in place in the Midcontinent Independent System Operator (MISO). Other market rules such as those governing minimum resource size [51] and those pertaining to control and telemetry [14] can hinder DR resources from participating. Additionally, rules requiring resources providing regulation to have the same capacity to move in both directions impedes many DR resources from participating, as DR resources typically favor movement in one direction over the other [51].

There may be scope for adaptation or relaxation of AS rules in certain cases. Indeed, according to [51], some systems have recently amended rules governing minimum size of resource. A revision of AS definitions and market rules to accommodate the physical capabilities of resources, generating and demand-side alike, is highly recommended. This could allow for greater participation in DR programs and also permit better exploitation of the available DR resources. It is recommended that focus is placed on the physical characteristics and capabilities of DR when designing DR programs. This recommendation concurs with [2], which stresses that the physical power system reliability requirements are of greater importance than the market rules. Ultimately the physical requirements should inform market design to best exploit the capabilities of the available resources [2].

The stringent requirements for AS resources, may preclude DR from participating in a potentially lucrative market and may therefore act as a disincentive for investment in DR technology. This would impede the deployment and growth of DR in general. A related issue is the fact that increasing the penetration of DR resources can suppress AS market prices [29]. This effect is observed for many resources, however, it is more pronounced in the case of DR since it is a resource that has very low or no opportunity costs. This suppression of prices may cause prices to fall to a level that is no longer attractive to investors. This price suppression, coupled with the stringent AS requirements discussed previously, would limit the amount of DR available for

providing AS, which in turn would restrict the amount of DR available to participate in capacity and energy markets.

Additionally, as alluded to earlier, a similar effect is noticeable in capacity markets —the magnitudes of potential capacity payments are largely dependent on the requirements for capacity. While capacity payments may be enticing for DR participants, it is inevitable that the higher the penetration of DR, the less capacity required. Thus, the capacity payments may fall to a level that may no longer be attractive for all DR resources, depriving other markets of valuable resources. Furthermore, DR can provide services other than those discussed here, such as congestion management or network investment deferral [9]. Therefore, lack of deployment of DR in one market has a knock-on effect to other markets and to other DR services. It is recommended that all revenue streams are considered holistically as a result of this intrinsic interrelationship between multiple DR services.

While AS definitions can pose considerable barriers to the deployment of DR, the greater interaction of the demand-side with the power system also presents its own issues. This is as a result of the need for greater data collection and an increase in information flow between the supply-side and the demand-side of the power system, as well as the limitations on the DR resource as a result of consumer behavior and preferences. These issues are explored in the next section.

2.3 Behavioral and Informational Barriers

Underpinning the entire DR paradigm lies the consumer. The authors in [8] express the view that the greater involvement of consumers in ensuring maintenance of system reliability and uninterrupted electricity flows, a realm that was once solely occupied by conventional generators, is one of the main advantages of DR. If customers can understand the impact of their electricity consumption has on their expenditure, they may be able to adapt their behavior, helping to smooth load profiles and to reduce system costs. However, it is contended here that this greater level of consumer involvement also represents not only one of the greatest barriers to deploying DR but also creates challenges for evaluating DR.

2.3.1 Consumer Behavior

There are numerous benefits associated with DR and many different ways to implement DR, but if the consumer is in anyway inconvenienced they may disengage and possibly withdraw from the program or will demand higher payments or incentives [53]. The authors in [54] explore the motivations and barriers associated with individual consumer decisions on whether to install micro-generation in their home. They indicate that inconvenience can be a dissuading factor in the uptake of micro-generation. While micro-generation is a separate, but related issue, the work in [54] does highlight an important point related to consumers; financial motivations are consistently more important than desires to help improve the environment. The importance of financial motivations is worth highlighting, especially considering there is considerable uncertainty surrounding the potential revenue of engaging in a DR program, as mentioned earlier.

In [35] a related financial issue is raised, namely the fact that savings on customers' electricity bills may not be sufficient enough to warrant investment in equipment and to compensate for the inconvenience of following electricity prices on a continuous basis, when they may only be required to react on rare occasions. Of course, this will be dependent upon the type of program being implemented and the degree of customer participation required. Chapter 5 examines the cost savings attributable to DR aggregators, and thus to the consumer, as a result of DR participation in various electricity markets. It is found that, in many scenarios, there are savings to be made through market participation, but that these savings are typically low. If financial issues are the key motivating factor in adoption of DR programs and it is shown that consumers are not in a position to either recoup their initial investment in DR technologies or make savings on their future electricity bills, there will be minimal engagement in DR. This is evidently a barrier to widespread adoption of DR programs.

It was found in a study in [55] that despite receiving feedback on energy consumption from in-home displays, the majority of study participants continued with their everyday routines and habits [55]. This is a perfect example of unanticipated or somewhat irrational consumer behavior - an issue that needs to be considered in the evaluation of DR deployment. This study also highlights that there is a need to foster greater DR awareness and understanding and to furnish consumers with sufficient information regarding DR programs so that they can make well-informed decisions.

In an effort to avoid this unanticipated consumer behavior, it is assumed for all work in this thesis that there is centralized control of the DR resources, whether by the system operator or by a DR aggregator.

In [53] it is explained that there are two objectives of demand which is involved in a DR program: satisfy customer demand and deliver a reliable resource to the power system when required. These two objectives are usually in competition. This is because the primary role of electricity is consumption. Consequently the length of time the load can offer a response and the frequency of resource deployment is limited since ultimately any program will be disruptive to the consumer. Thus, utilities will be reluctant to dispatch the DR resource regularly [45]. This is an important point to consider when analyzing the value of the resource and is also connected to the physical characteristics and nature of electricity loads and thus is an important consideration when modeling DR resources. In Chapter 3, the responses provided by the DR resources considered are limited to a maximum duration of 2 hours to align with this reluctance of utilities and to accommodate consideration of the energy-limited nature of the resources.

The main challenge is to fully appreciate the limitations of the DR resource as a result of consumers' behavior and preferences and to appropriately account for these in DR implementation. It is crucial to represent these limitations in the evaluation analysis and to understand the motivations consumers have for engaging in or rejecting a DR program. It is important to be mindful of the effect unanticipated consumer behavior can have on the DR characteristics and to deal with it appropriately in the evaluation process.

2.3.2 Baselines

DR performance is typically computed as the difference in the actual demand level and a baseline level so, effectively, consumers are being paid for what they do not use [56]. The North American Energy Standards Board's definition of a demand baseline is an estimate of the electricity that would have been consumed by a customer in the absence of a demand response event [57]. The key point to be taken from this definition is the use of the word 'estimate'. It is naturally challenging to measure or calculate what would have occurred and thus, fundamentally baselines are imperfect [58].

Consequently, there are concerns regarding DR baselines and suggestions that the challenge of establishing a baseline for DR may be a serious impediment to the deployment of DR and to methodologies for determining the true value of certain DR programs in the market. Indeed, the authors in [45] suggest that a DR baseline could play an important role in determining the value DR brings to the electrical system [58]. Thus inaccurate baselines lead to inaccurate DR evaluation.

In [59] it is questioned whether baseline predictions are correct and it is suggested that such calculations may not even be possible for different types of customers. It is understood that it is possible to obtain a reasonable baseline for commercial and industrial loads, where the loads can be directly controlled and closely monitored. However, for smaller devices and for end-uses with irregular or unpredictable power consumptions, establishing a robust and accurate baseline can be more difficult. Ultimately, when a load is dependent upon consumer behavior and where the power consumption cannot be directly controlled, it is typically more difficult to establish a baseline. In Chapter 4, a very specific resource is examined - Electric Thermal Storage Heating for both space and water heating. Due to the aforementioned assumption in this thesis of centralized control of DR, it is reasonable to consider that these devices are both controllable and capable of being closely monitored by an aggregator, thereby avoiding input, and the associated uncertainty, from consumers, thus justifying the direct use of a baseline in the modeling approach.

The implications of inaccurate baselines can be far reaching —the performance of a particular program would not be accurately ascertained and, for certain programs, baselines form the basis for customer remuneration for their participation, which is evidently a serious deployment issue. Without a robust method for determining the baseline, program participants could be under compensated. This could reduce customer willingness to participate. On the other-hand, consumers could be over compensated thereby increasing system operational costs, reducing the potential impact of some of the benefits ascribed to DR programs. Indeed, the authors in [60] analyze a number of different baseline calculations methods for residential consumers. They show that, particularly for incentive-based DR programs, baselines play an crucial part because they determine the incentive prescribed to customers and thereby influence their decisions.

In [61], the relative merits and flaws of a number of baseline calculation methodologies employed in the US are discussed. The author concludes that, in

many cases, an adjustment to the baseline calculation is needed to more accurately estimate customer energy usage. This highlights that baseline estimation is not a straightforward procedure. While the authors in [60] focused on residential consumers, there has been considerable work on analyzing the uncertainty in demand response baseline models used in commercial and industrial facilities [62], which also details the inherent complexities involved in calculating a baseline. According to [62], baseline models for calculating baselines for commercial and industrial load are generated by averaging electricity consumption for a particular load over a number of days and thus can be inaccurate. The authors in [62] thereby advocate the use of regression-based baseline models to overcome this.

All of the methods discussed in [60, 61] require the calculation of a baseline for each individual DR unit, for each individual customer or each individual DR resource. If the residential sector is to fully participate in DR programs and if individual baselines are to be established it is clear to see that there could be a dramatic amount of data required. The quantity of data that would be required for baseline calculations, however, is but one of the data-related issues and such issues will be explored in the next section.

2.3.3 Data Issues

There have to date been limited numbers of large-scale DR deployment projects and consequently, in some circumstances, there is insufficient data available for performing DR analysis. This limitation on data and the lack of previous experience pervades most of the challenges and barriers associated with DR. Furthermore, even if sufficient data were available, there could be implications for its accuracy. According to [22], DR participants have historically overestimated their likely performance during declared curtailment events. Additionally, the problems associated with asymmetric information may need to be accounted for in the deployment and evaluation of DR. This is because individual customers will always know more about their true electricity use than the load serving entity or aggregator and can likely profit from that knowledge [63].

The uncertainty in the deployment of DR, the associated terminology and the lack of sufficient data will inherently impact upon the evaluation methodologies, which will be discussed in section 2.4, as it is necessary to model and represent, insofar as possible, the operation of the real power system and interactions with the consumer.

It is important to understand the DR resource and to appreciate the uncertainty and challenges that exist and then represent DR as best as possible and to account for any uncertainty in the evaluation methodologies applied to DR.

2.4 Evaluation Methodologies

The US Department of Energy [7] concluded that without a robust analytical method to estimate the value of DR it is almost impossible to justify the system reliability improvement attributable to it. A poor methodology leads to a poor understanding of the value of DR and the implications of this for the power system could be considerable.

It was previously acknowledged in section 2.3.3 that limited data is an issue for evaluation of DR. Consequently, it is recommended that analysis is performed using currently available data or performed based on reasonable assumptions pertaining to market size, DR penetration level etc. This should then be followed by a series of updates and iterations on the assumed figures as more data is made available and as greater understanding of the effects of DR on the system and the markets is gained from demonstration projects.

This section explores some of the existing methods which are deemed suitable for estimating the value of DR and identifies some of the associated challenges as a result of the uncertainty in the deployment of DR.

2.4.1 Integration Benefit Concept

Power system level studies have been performed for decades in order to understand the impacts of various resources on the power system. However, the advent of variable renewable sources has spurred greater interest from policy makers, academics and system operators alike. In recent years integration studies of variable renewables have dominated the realm of system level studies and has been an area of considerable activity, as evidenced by the number of studies documented in the literature [20, 64, 65].

An integration cost is usually defined as the extra cost imposed on a system when a resource is added, compared to the situation without that resource [64]. Increasing penetrations of variable renewables have a significant impact on power system operational costs. There is no fuel cost associated with variable renewables

and they can often displace some expensive fossil fuel-based generating units. Thus, there is an operational cost saving. On the other hand, integration of variable renewables necessitate the procurement of greater levels of AS, thus imposing extra operational costs on the system. Consequently, significant focus is placed on this integration cost associated with the introduction of variable renewables but ultimately there is a net operational cost saving.

DR, however, for the most part, has no fuel costs associated with it but is crucially also is in a position to provide AS, among other services, thereby reducing system operational costs. Like variable renewables, the net result of incorporating DR should be an operational cost saving. However, in the case of DR, the saving is potentially two-fold; fuel savings as a result of potential generation displacement and the provision of AS from a cheaper resource. Therefore, the integration cost in the case of DR should be a negative cost, or a benefit, and thus this cost saving could be referred to as an integration benefit.

There is considerable merit in adopting an integration study approach since, as stated earlier, these studies are well documented in the literature and considerable experience and understanding has been gained in recent years [20]. Furthermore, integration studies represent a significant set of methodologies to quantify the impacts of resources on the power system. Applying the concept of an integration benefit to the analysis of the inclusion of DR on the power system could provide a means to ascertain its value. Knowledge of the value could provide an indication as to which markets and sectors are most suited to demand-side participation, a view which is also expressed in [14].

The advantages of using an integration study approach cannot be overlooked and, consequently, such an approach is explored in greater detail here.

2.4.2 Integration Study

An integration study typically includes production cost simulations, which often involve UCEDs, allowing for an assessment of flexibility, cost and emissions impacts [20]. One of the advantages of a production cost simulation is that it allows assessment of both energy production and provision of AS. Additionally, an integration study would involve generation adequacy (or CV) calculations, analysis of the impact of new resource on power system dynamics and power flows [20]. A recommended approach for a wind integration study is shown in Figure 2.1.

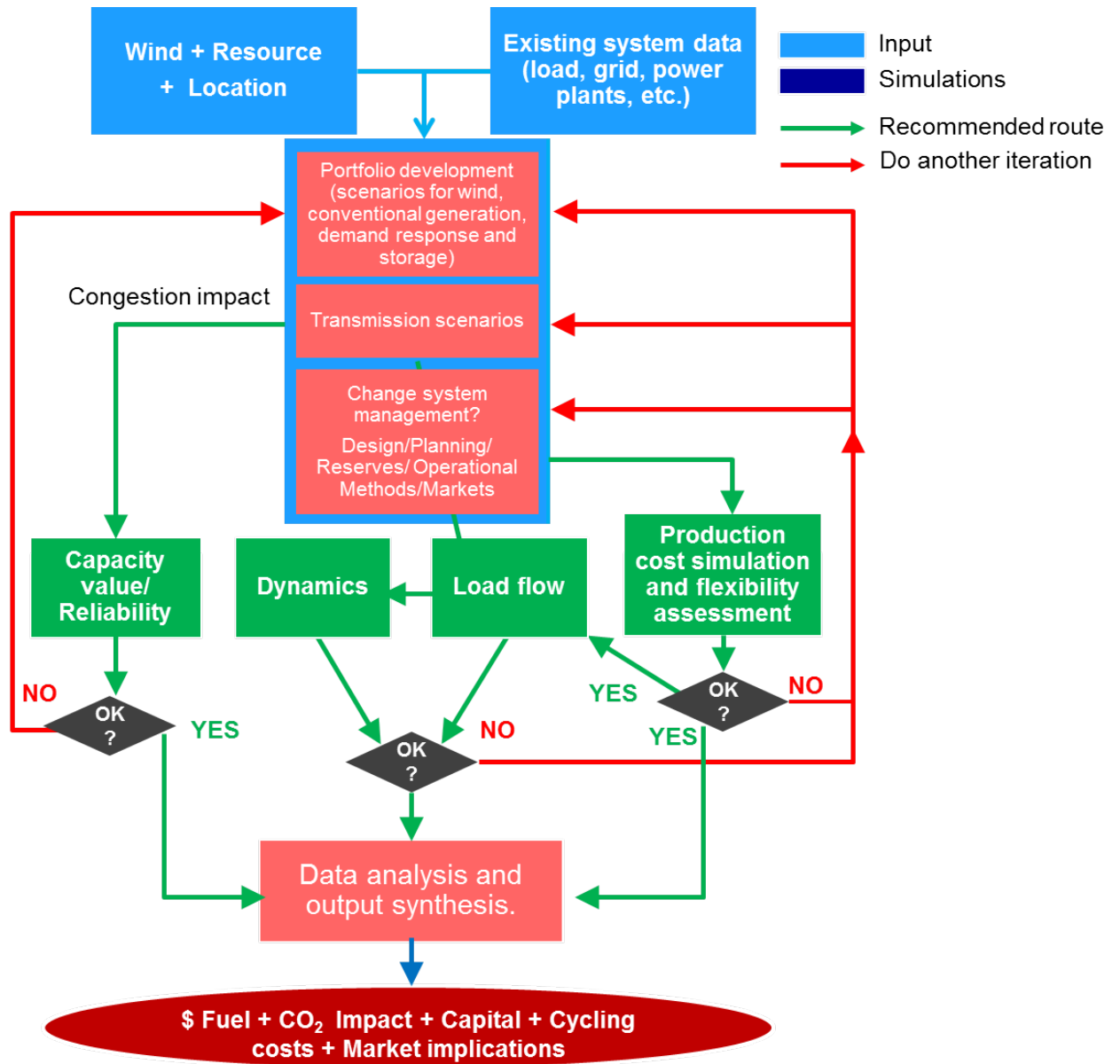


Figure 2.1: Wind Integration Study Components: Flow chart of recommended route
Reproduced with permission [20].

While, all of these assessments are an integral part of an integration study, the focus in this thesis is placed on the operation (production cost simulation) and generation adequacy aspects (CV calculation) of an integration study as these are the areas where DR can contribute to the power system and impact upon electricity markets.

2.4.2.1 Production Cost Simulation

The methodology employed in an integration study is determined by the ultimate aim of the study [64]. There are two common approaches for determining integration costs. One involves simply adding the capacity of the new power source or the DR resource into the production cost simulation and comparing production costs with a base case. The second involves replacing existing capacity with the new power source or the DR resource in the portfolio used in the production cost simulation and comparing production costs with a base case [20]. The difference in costs before and after the inclusion of the technology being assessed is often viewed as being one of the key components of the integration cost. While many approaches can achieve the same end goal—the estimation of an integration cost—the underlying methodologies employed and the assumptions used can be significantly different.

Ascertaining power system operating costs under different scenarios and under a range of conditions, i.e. a production cost simulation, is well understood and follows well-defined methodologies [13]. However, defining the base case for any integration study is often a source of conjecture. The authors in [64, 66] highlight that study results are highly dependent on the reference or base case to which the system costs with the inclusion of the new technology are compared. It is acknowledged in [67] that the definitions of the reference or base cases used in integration studies ‘*is an area of significant disagreement among experts in this field*’. It is recommended in [20] that the results from integration studies should be detailed with regard to the assumptions made to ensure that the weaknesses and caveats are kept in mind.

A recommended approach for an integration study would involve comparing and contrasting a number of different portfolios and a number of scenarios in a manner similar to that employed in [68], rather than simply comparing one scenario without DR and one scenario with DR. It will be noted in this thesis that a considerable number of different scenarios are explored when examining the impact of DR, in an effort to comply with this recommended approach.

It is also highly recommended that, in evaluating DR, the time-varying nature of electricity markets are adequately incorporated into the power system models in order to obtain robust and realistic results from production cost simulations. For example, the volatility of electricity and AS prices over time can have serious implications for the revenue earned by DR resources. It was found in [14] that in the CAISO South region, for the years 2009 to 2011, the winter months had large evening peaks in

regulation up prices while the early morning hours had lower prices. Similarly, in the West Reserve Zone of New York ISO, for all seasons, the spinning reserve prices in the early morning hours are close to zero [14]. The importance of including this daily and seasonal temporal variation in market clearing price when assessing the value of DR participation in AS markets is highlighted in [14]. This would suggest that there are limited time periods over which to earn revenue through participation in an AS market, which may reduce the overall value of DR for the participant. The work in [69] highlights the importance of correctly modeling the variation in price of electricity over time when examining DR. The authors in [69] suggest that in certain cases for certain systems this will necessitate the incorporation of price correlation into the production cost model.

2.4.2.2 Capacity Value Calculations

CV calculations are an important component of integration studies and CV is a measure of the contribution a resource can make to generation system adequacy [70]. The CV of a generator is often defined as *'the amount of additional demand that can be served due to the addition of the generator, while maintaining the existing levels of reliability'* [70].

One of the challenges associated with CV calculations is the fact that the methodology was developed with conventional generators in mind and then adapted and applied to variable renewables [70]. While this methodology has also been adapted to provide a high level estimate the CV of DR [71], which is presented and extended in Chapter 3, the traditional CV methodology does not inherently capture all of the impacts on generation adequacy associated with the operation of DR. This is the motivating factor behind the material presented in Chapter 4.

A feature of DR that is inherently different to the characteristics of a conventional unit pertains to availability rates and the probability of availability. In traditional CV calculations, individual generator availabilities are assumed to be independent of each other. This is not the case for the wind and solar resource and, indeed, the same cannot be said of DR resources.

For CV calculations, another one of the required assumptions is that the failure performance of a unit is independent of the system load and the outage pattern of other units [72]. This is not true of DR resources because they will inherently be dependent on system load. For example, a refrigeration unit will not be available to

respond to a signal to reduce load, if the unit was not on and operating in the first instance.

It can be appreciated that the availability of conventional generation depends upon the mechanics of electricity generation and upon the control systems in place, while wind generation availability is contingent upon natural phenomenon and the physical resource. The wind resource, although variable and uncertain, can be represented to a certain degree of accuracy and there is significant wind data and wind time series available. DR availability, on the other hand, depends upon the mechanics of the devices and upon the control systems, but crucially, also depends upon consumer behavior and user preferences. It is this influence that introduces challenges to the representation of the DR resource availability. It is crucially important, as is recommended in [73], that the wind, load and DR data is time-synchronized in CV calculations and the limited DR data can be a considerable challenge in this regard.

There is significant work to be carried out regarding the CV of DR so that system operators can decide how best to incorporate DR into the daily operating strategies and into long-term planning considerations. If shoe-horning DR analysis into pre-existing methodologies for CV calculation, the breakdown of these assumptions represents a major challenge to modeling and evaluating the DR resource and indicates the potential requirement for a novel alternative approach. Chapters 3 and 4 of this thesis contribute to the development of alternative approaches for estimation of the CV of DR resources.

2.4.3 Modeling Demand Response in Integration Studies

It is recommended that production cost simulations and CV calculations pay close attention to the physical characteristics of the DR resources. These resources have a number of unique characteristics as well as characteristics in common with conventional generation. As is the case with conventional generating units, it is crucial that the DR constraints are modeled appropriately in order to accurately reflect the physical characteristics of the DR unit. DR has constraints equivalent to the response rates, ramp rates, rated capacities and has minimum down time and maximum up time constraints typically associated with conventional generating units. Like conventional units, DR is only possible if the unit is online and available. However, unlike conventional units, which have significant ramp rate limitations, DR units have smaller physical time constants and thus higher ramp rates. Therefore

it is highly improbable that DR units would need to be constrained on. It is the unique characteristics that present challenges for modeling DR across all domains and indicates the need for high resolution models in order to capture all of the physical characteristics of DR. Unlike most conventional generators, a DR resource is energy-limited, similar to hydro generation and pumped storage. After a DR unit has provided a service to the system, the energy which the unit relinquishes may need to be recovered at a later stage. If these recovery periods are uncontrolled, a number of units' recovery periods may coincide and this may inadvertently create an additional, unintentional peak in the system demand. Such unintentional peaks in system demand emerge in Chapter 3 and it is found that this has a considerable impact on the CV of DR. A method is developed in [74] for quantifying the effect increasing penetration of DR resources providing load-shifting can have on electricity markets. They show that load reduction in one period can reduce the market price, but shifting that load to another point in time, can cause a rise in the market price at that time. This effect is also seen in the results presented in Chapter 5.

The energy-limited nature of DR is often viewed as being similar to conventional storage units and thus in many places in the literature, load-shifting units are modeled and considered as storage units, for example in [6] and in [30]. This energy-limited nature constrains the duration of response the DR unit is capable of providing and there may also be a limit on the number of times a DR unit can be called upon to provide a response. The energy storage dimension of the DR resource introduces an element of time-shifting, making the calculation of its availability at a specific point in time complex, since the probability of DR availability in a certain hour is contingent on whether it was available or used in previous hours. It is recommended that these constraints and unique characteristics are incorporated into the model of DR used in the production cost simulation and in the CV calculations to obtain realistic results.

The challenge of the time-varying price of electricity and AS, discussed in the section 2.4.2.1, is compounded by the time-varying availability of DR resources themselves; DR may simply not be in a position to provide a response during periods of high prices, further eroding its potential revenue stream. Some DR resources can respond within minutes or even seconds of a signal or a disturbance and the duration of the response can be on the time scale of minutes or hours. Capturing these inter-temporal characteristics is paramount to adequately evaluating the DR resource and

time-synchronized data is a necessity in this regard. However, as mentioned earlier, there is a dearth of sufficient data of this kind available at present.

There is also a need to represent unanticipated consumer behavior in demand-side models and this represents a major paradigm shift from the traditional power system studies and could present multiple challenges. However, it has been shown that aggregation reduces the irregularity and diversity associated with loads [75]. The authors in [75] even suggest that aggregation of a few thousand households transfers DR from the residential sector into a uniform response '*enough to consider it as a system resource*'. It is based on this that aggregation is highly recommended when modeling and analyzing the DR resource.

One of the benefits of using an integration study methodology is the ability to incorporate stochastic elements in the unit commitment algorithm. In recent times, there has been considerable work in developing stochastic unit commitment tools to account for the uncertainty associated with variable renewable generation, particularly wind [76, 77]. Thus, significant expertise exists to allow the incorporation of the stochastic elements of DR in production cost simulations.

2.5 Summary

The key message of this chapter is that there is considerable uncertainty surrounding the value of DR. Consequently there can be concern regarding the potential revenue for DR program participants and a hesitancy for others to engage in a program. A greater understanding of the value is required for widespread deployment of DR programs. Evaluation of the DR resource is clearly a vital step in its large-scale deployment and could increase understanding of the impacts of DR on the system and markets. One of the challenges faced when evaluating DR is the need for awareness of consumers' preferences and different motivating factors. It is vitally important that the limitations of the DR resource and the effect of unanticipated consumer behavior are borne in mind. This, however, can be hampered by a lack of suitable data and information, as previously discussed.

The terminology used to describe DR programs can also be problematic for the evaluation of DR. It is noted in [56] that despite the fact that some system operators have established DR programs, while many more are developing such schemes, the many ways in which to execute a DR program and the inconsistencies associated with program definitions pose considerable barriers. There are also numerous ways

in which to classify DR programs, for example, in terms of application or in terms of the signaling mechanisms. It is common to classify DR programs based on their incentive structure [78], while programs can also be classified based on whether they are activated by price or by system reliability requirements [79]. Within each of these classifications there are many different DR programs and there is often considerable overlap. This can create difficulties when making comparisons between different programs and it may not be possible to draw conclusions from one system and apply them to another.

From earlier in this thesis, it is clear that DR can provide multiple products and that there is a relationship between them. Thus, failure to develop DR in one market can deprive other markets. Offering multiple services to the power system is not only beneficial from the power system point of view, but also allows DR participants to potentially avail of numerous revenue streams. As a result, it is highly recommended that DR programs are not studied and evaluated separately, but are assessed as part of the larger power system so as to appropriately account for their combined impact. In [59] it is noted, however, that there is a significant challenge in evaluating DR programs holistically. The alternative, on the other hand, to evaluate the programs separately, is unacceptable since, ultimately, any program will be integrated into the larger power system and thus there will be interactions and interdependencies and the programs will form a small part of a larger portfolio of resources.

This need for a comprehensive holistic approach indicates that an integration study is a very suitable, and recommended, approach to adopt for the evaluation of DR. The ideal approach would involve detailed modeling of the physical properties and energy-limited nature of the DR resource in a system operation simulation. This model would also capture the vagaries of consumer behavior insofar as is possible. The system operation simulation would involve co-optimization of energy and ancillary service markets. This is in order to determine the operation of DR resources when in competition with other technologies and strategies [22] and to take account of the intrinsic relationship between multiple DR services. Co-optimization is also noted as being of particular importance in [29]. The simulation would also utilize time-synchronized data to ensure inter-temporal variations are accounted for. A series of different portfolios and scenarios would be examined and compared. The production cost simulation would be combined with a CV assessment of the resource enabling examination of the contribution the DR resource can make to power system reliability.

When an integration study is completed, however, one issue remains - the synthesis and dissemination of experience, understanding and study conclusions. Such tasks are inhibited by the appreciable differences between systems coupled with the disparity in AS terminology and DR program classifications and definitions. It should also be remembered that all power systems are different and the results of system studies are inevitably system-specific. CV results, for example, are dependent on the generation portfolio and thus the requirement for capacity in a particular system.

Additionally, despite the fact that DR can provide multiple products, some systems do not operate markets for capacity, but provide capacity payments, while others do not have AS markets and yet there are AS payments. A related issue is the fact that interpretations of the definition of capacity can be arbitrary, while AS definitions and rules vary considerably from system to system. A revision of AS definitions and market rules to accommodate the physical characteristics of DR resources is highly recommended. This could permit greater participation in DR programs and also allow better exploitation of the available DR resources.

The differences between systems have implications for evaluating DR, preventing transfer and application of the evaluation framework from one system to another. While the methodologies may be applicable, or at least adaptable, to all systems, the results are unlikely to be directly transferable. That being said, however, it is still vital that such studies are performed in order to more fully understand the value of DR and to make informed decisions regarding the future of DR.

2.6 Conclusions

The main contributions of this chapter lie in the identification of key DR deployment barriers, from across the range of the DR literature and from many different aspects of the DR concept, in the discussion of how the uncertainty in DR deployment can impact upon evaluation methodologies and in the introduction of the concept of integration benefit. Additionally, numerous recommendations are provided.

This chapter shows that there is a pressing need to quantify the value of DR. DR could be crucially important to the power system given that it can provide multiple services and it is paramount to be aware of how DR compares with other resources. It is also vital to understand the price that DR could receive in the various electricity

markets and to appreciate the characteristics of DR which are most desirable from a power system point of view.

However, as noted, there is a 'Chicken and Egg' type scenario at present in the DR field; uncertainty regarding the value of the DR resource compromises investment interest and understanding of the ideal manner in which to deploy DR programs, which culminates in limited experience with a consequent shortage of suitable data. This leads to a lack of clarity as regards how best to represent the DR resource and incorporate it in valuation methodologies and results in a minimal understanding of the value of DR. This thesis aims to address some aspects of this lack of understanding regarding the potential value of DR.

It was identified in this chapter, that the capacity market could be a key market for DR resources to participate in, but that there is considerable work required. The determination of the CV of DR resources is noted as being an area requiring significant research and necessitates the development of new calculation methodologies, or at the very least, modification of the conventional approaches. This is the motivation behind the material presented in Chapter 3 which provides a preliminary analysis of the CV of different DR resources.

Estimations of the Capacity Value of Demand Response Resources

The following chapter is largely based on a published paper in the proceedings of the IEEE Power and Energy Society General Meeting 2014 called ‘A Methodology for Estimating the Capacity Value of Demand Response’ [71]. It was a collaboration between the thesis author, Marissa Hummon, formally of the National Renewable Energy Laboratory (NREL), Sila Kiliccote of Lawrence Berkely National Laboratory (LBNL) and Ookie Ma of the US Department of Energy. A full citation can be found in the section titled “Publications Arising”. Additional work considering energy recovery, which was not considered in [71], is also discussed here.

3.1 Capacity Value Metrics and Calculation Methodologies

Capacity value, (CV), by definition, as mentioned in the previous chapter, represents the contribution a generator can make to generation system adequacy [70], which is the existence of sufficient generating resources to meet the system demand. In this thesis, CV represents the contribution any resource (supply-side or demand-side) can make to system adequacy, which allows extension of the concept of CV to DR. CV is often used as a means of comparing different resources and is often employed in capacity payment mechanisms [80] and calculations. It is important to be aware, as noted in [16], that CV is not a quantity which can be calculated directly from observed data. Additionally, there are a wide variety of possible definitions and thus multiple potential values. Thus, CV is not a single definitive value, but is an indication only. Adequacy indices, and by extension the CV estimates, are not an absolute measure of

the system generation adequacy. Nevertheless, they prove to be very useful metrics for the comparison of power system resources and they continue to be used extensively in the literature.

There are numerous metrics for representing the CV of a resource, but the Effective Load Carrying Capability (ELCC) is one of the most commonly used metrics [81]. The concept of the ELCC of a resource was introduced by [82]. The ELCC of a generator is defined as the amount by which a system's load can increase when the generator is added to the system, while maintaining the system's adequacy [83]. Other metrics include the Equivalent Firm Capacity (EFC) or the Equivalent Conventional Capacity (ECC). The EFC is defined as the capacity of a 100% reliable generation that can be replaced by the resource while maintaining the same level of system adequacy [84]. The ECC is similar to the EFC except that the generating unit against which the resource under examination is compared is not a 100% reliable generating resource [84].

The typical approach for determining the generation adequacy contribution of a new resource involves:

1. Modeling the system **without** the new resource and determining the adequacy indices.
2. Modeling the system **with** the new resource and determining the new adequacy indices.
3. Comparing the adequacy indices of each case and increasing the load (or reducing conventional generating capacity) in the case of the system with the new resource until the adequacy indices match that of the 'without' case. The additional load is the ELCC (and the reduction in conventional generating capacity is the EFC or ECC), expressed in Megawatt (MW).

The CV estimate is obtained by expressing the ELCC (or EFC or ECC) as a percentage of the capacity of the new resource.

3.2 Capacity Value of Different Power System Resources

Table 3.1 illustrates the range of CV estimates from the literature for different power system resources. Each resource has different characteristics which ultimately results in the considerable range of CV estimates and necessitates the use of different

methods to obtain CV estimates. The key characteristics and methodologies for determining the CV of a range of resources are presented here.

Table 3.1: Typical capacity value estimates for different power system resources from the literature

Resource	Capacity Value
Wind	13% - 23% [70, 83]
Wave	2%-22% [83]
Tidal	3.3%- 7.6% [81]
Solar PV	40% [85]
Concentrating Solar	60% - 90% [84]
Storage	53% - 95% [86]
Conventional Plants	71% - 91% [87]

3.2.1 Capacity Value of Conventional Generation

Historically, conventional generation was the main resource on the power system. Consequently, much of the literature is devoted to the development of methodologies for estimating the CV of such resources. The most common method employed to determine the CV of conventional generation involves a two-state Markov Chain, with the states representing whether the unit was operating or not. Statistics relating to the repair and failure rates of different generating units are widely available and are employed to determine the generating unit's probability of being available, which are then used in CV calculations. In the early 1970s, this method was extended in order to more appropriately determine the CV of peaking plants, since they are not required to be available at all times throughout the year, and consequently, a four state Markov Chain method was developed [88].

3.2.2 Capacity Value of Renewable Generation

Despite their wide-spread use, it has been suggested that wind power availability cannot be appropriately represented using the Markov Chain methods, due to the loss of information relating to wind and load correlation [70]. Instead a preferred methodology for calculating the CV of wind is proposed in [70] which, for the 'with wind' case, subtracts a time series for the wind power output from the load giving a net load profile. The Loss of Load Expectation (LOLE), which is chosen as the adequacy index, for the 'with wind' case is compared with the reference LOLE from

the 'without wind' case and a constant amount of load is added across all hours until the new LOLE reaches the reference LOLE.

The CV of wave power is determined in [83]. An algorithm was employed to generate an availability matrix for each wave power device in a wave farm, accounting for forced outage rates, mean repair times and maintenance times. This matrix was then applied with wave data and a typical wave device's power matrix to determine the behavior of all the devices in a wave farm. This time series of the behavior of the wave devices is then used to determine a new load profile.

The work in [84] estimates the CV of Concentrating Solar Power (CSP) plants in a similar manner to tidal barrage in [81]. The operation of the CSP plant is optimized to determine the maximum amount of energy that can be generated by the CSP plant in each hour subject to thermal energy storage availability and plant operating constraints [84]. Instead of using the more conventional ELCC metric, the authors chose to use the ECC. The method employed calculates the LOLE when CSP is added to the system and then calculates the LOLE when a benchmark unit only is included in the system. The ECC is found by adjusting the nameplate capacity of the benchmark unit until the LOLE is the same as the LOLE for the system with the CSP plant.

The CV of large tidal barrages is discussed and calculated in [81], by assessing the contribution tidal generation can make during the winter peaks on the United Kingdom (UK) system between 2005 and 2009. In order to incorporate the operational issues associated with tidal barrages, the barrage is modeled in three operational modes. The time series for the operation of the tidal device is taken and included in the ELCC calculation as negative demand, similar to the approach used for wind power in [70].

3.2.3 Capacity Value of Storage

CV calculations for storage devices are inherently complex due to their energy-limited nature and because the operational strategies of storage plants typically involve shifting energy over time, similar to the thermal energy storage component of CSP plants. In [89], it is suggested that in order to gain an appreciation of its CV, storage can be treated, in relation to its ability to meet peak demand, as a conventional unit with an energy limit. The storage device is therefore modeled in [89] in a product cost simulation to obtain a time series of the energy availability, taking into account the

key characteristics of the storage technology. A slightly different approach is taken in [86] where a dynamic program is used to generate operational schedules for a storage device engaged in energy arbitrage. These schedules are used to determine the probability that the device has energy available in each time period and these probabilities are then used as forced outage rates in traditional generation adequacy calculations.

3.2.4 Capacity Value of Demand Response Resources

As stated previously in this thesis, an appreciation of the CV of DR could allow for a comparison of DR with other resources. While there has been significant work recently in developing appropriate methodologies for calculating the CV of conventional generation and of wind generation, and despite the fact that DR, particularly in the US, has historically been used to provide capacity to the power system, there has been very limited work in the area of the CV of DR.

The work in [11] looks at CPP and peak time rebate programs (PTR) and determines the CV of a DR resource. Their work assumes that peak time occurs between noon and 6 PM on twelve summer days. An assumption is made in [11] regarding the amount of load that can be used for DR and there is no consideration of different end-use types, while the work presented here utilizes time series of DR availability for various end-use categories and explicitly analyzes the impact of a response duration constraint.

DR has similarities with many different generating technologies (see Table 3.2): like peaking plants, DR may be in a position to contribute during high demand periods and like renewable resources, DR availability can be variable. DR is also dependent on an underlying resource which is largely driven by consumer behavior. Furthermore, DR has an element of storage associated with it and depending on its operation, DR availability in one time period is contingent on previous and subsequent hours.

As discussed in Chapter 2, it is not possible to utilize traditional ELCC calculations for examining DR resources. An alternative approach is required. A methodology for estimating the CV of DR resources is now introduced.

Table 3.2: Key capacity value related characteristics of different power system resources

Resource	Key Characteristics
Wind & Wave	daily and seasonal variation, intermittent, typically has no storage element
Tidal Barrage	daily and seasonal variation, variable, largely predictable, has a storage element
Concentrating Solar Power	daily and seasonal variation, intermittent, has a storage element
Large Scale Storage	large storage element, usually dependent on operational strategies employed
Demand Response	storage elements possible, daily and seasonal variation, consumers constraints

3.3 Methodology for Estimation of the Capacity Value of Demand Response Resources

To incorporate operational issues associated with DR due to the response duration constraint as a result of the energy-limited nature of the DR resource, the methodology proposed in this section employs availability time series for DR, in a similar manner to the methods proposed in [70, 81, 83], where the resource output is modeled as negative load. It is appreciated that DR can provide many different services to the power system, however, only DR availability and the resultant CV of DR are considered in this chapter. Subsequent chapters will explore the CV of DR when the resource provides multiple services.

It is assumed here that DR is being dispatched in response to high system load levels. It should be highlighted that the capacity represented in the DR availability profiles is the capacity available for dispatch. It does not reflect the total amount of DR capacity that is installed, as there is no way to ascertain such information from the availability profiles. Furthermore, the availability profiles do not reflect capacity that would be needed to guarantee the associated level of availability. While this is an extremely important issue, it is beyond the scope of this chapter. Therefore, here it is being assumed that the necessary technology and market requirements are in place so that load is capable of responding to dispatch instructions and providing 100% of the available capacity when required. The constraints being applied here represent an extreme case and different resources will have their own specific constraints relating to duration of response and frequency of response. Furthermore, optimal dispatches are not involved in this methodology.

3.3.1 Generation of Demand Response Availability Profiles and Data

The work in [1] provides a unique approach to estimate DR availability profiles for thirteen end-use loads within the Western Interconnection in the US for the year 2020. For each end-use, an annual hourly load profile for each balancing authority area was estimated using Itron energy data and California Commercial End-Use Survey, Residential Appliance Saturation Study and census data. These load profiles were further evaluated by [1] (by enablement technologies, DR programs, targets and goals to create value for DR) and filtered by ‘shedddable’, ‘controllable’, and ‘acceptable’ filters by end use as well as aggregation to obtain an estimate of the amount of load available. These profiles were obtained through collaboration with Marissa Hummon, formally of NREL, Sila Kiliccote of LBNL and Ookie Ma of the US Department of Energy and used in this thesis. Henceforth, they are referred to as the ‘availability profiles’. In addition to these DR availability profiles, forecasted system demand for four utilities are obtained from the work in [1], and used with permission in this thesis. These utilities are:

- Avista Corporation (AVA) and Puget Sound Energy (PSE) , two winter peaking systems (labeled AVA and PSEI, respectively, in Figure 3.1 obtained from [90])
- Nevada Power (NEVP) and Los Angeles Department of Water and Power (LDWP), two summer peaking systems (labeled NEVP and LDWP, respectively, in Figure 3.1)

A winter peaking system is a power system which experiences its highest load hour in one of the winter months, while a summer peaking system has its peak load hour during the summer months. The normalized system loads obtained from the data from [1] for each of the four systems examined is illustrated in Figures 3.2, 3.3, 3.4 and 3.5, indicating whether each system is a summer peaking or winter peaking system. These systems were used because their system peak load levels were broadly similar to the Irish power system and due to the fact that typical Irish generator data was employed for the analysis. Thus, no amendments had to be made to the generator data prior to its utilization.

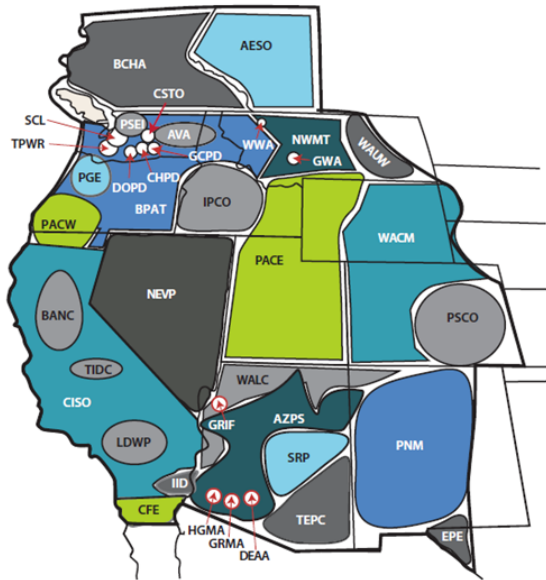


Figure 3.1: Western Interconnection Balancing Authorities [90]

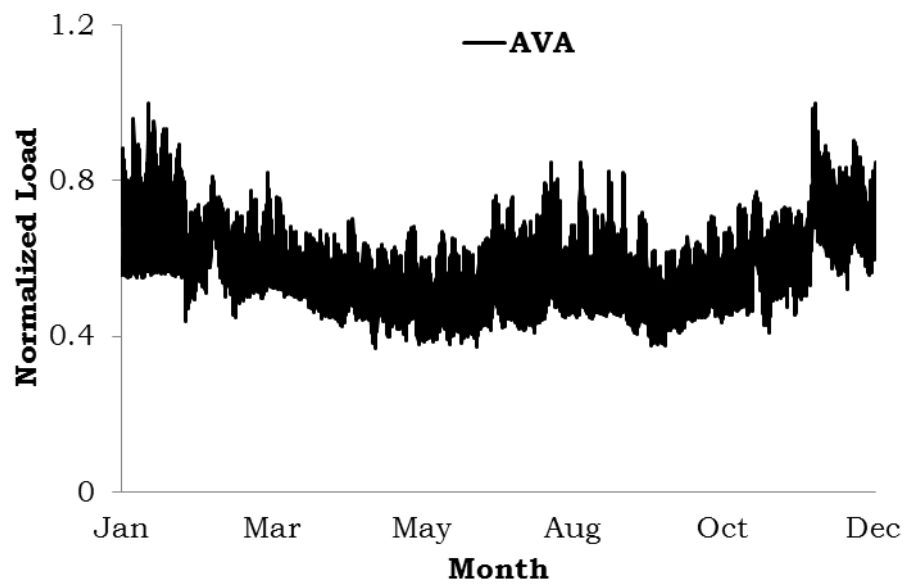


Figure 3.2: Normalized load for AVA

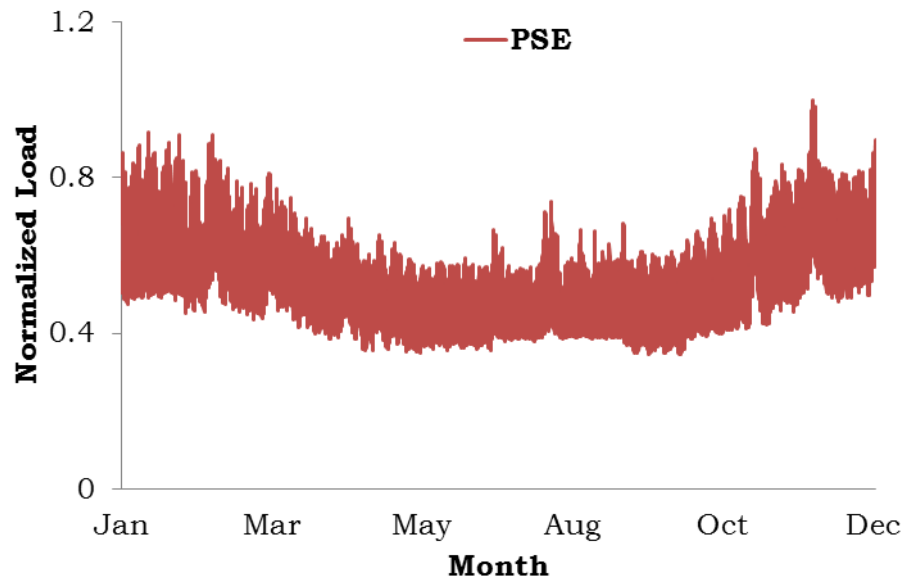


Figure 3.3: Normalized load for PSE

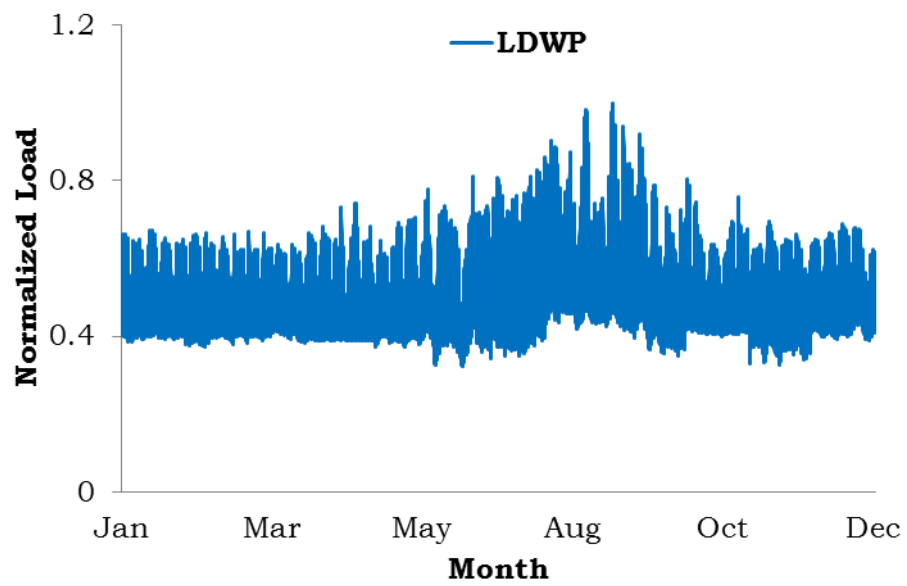


Figure 3.4: Normalized load for LDWP

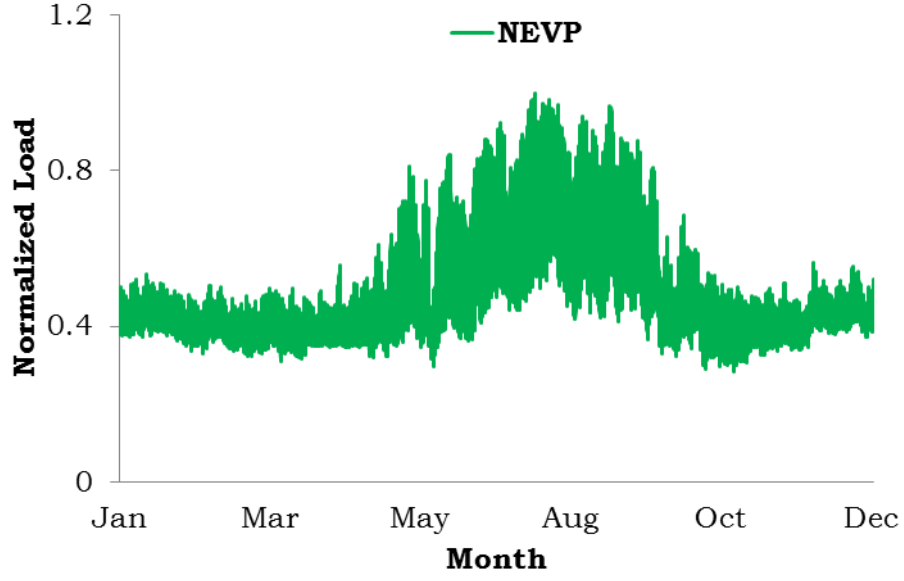


Figure 3.5: Normalized load for NEVP

3.3.2 Effective Load Carrying Capability Calculations

The code adapted and employed in this chapter was developed and discussed in [73]. As the intention of this chapter is mainly illustrative, i.e to estimate the CV and understand the impact of the energy-limited nature, indicative values for conventional plant capacity and availability rates on the Irish system are used. The same plant capacities and availability rates are employed for all four power systems. These values are shown in Table A.1 Appendix A.

A Capacity Outage Probability Table (COPT) is created iteratively using the availability rates and the installed capacities for each unit. A COPT outlines the probability of availability of each possible generation level on the power system [83]. Comparing the COPT with the load time series for the year, the Loss of Load Probability (LOLP) for each hour is calculated based on Equation (3.1) and Figure 3.7. A sum of the LOLPs for the year gives the LOLE in hours per annum, without DR (see Equation (3.2)). This is the reference LOLE ($LOLE_{ref}$).

$$LOLP_{ref}^t = prob (GEN^t < DEM^t) \quad (3.1)$$

$$LOLE_{ref} = \sum_t LOLP_{ref}^t \quad (3.2)$$

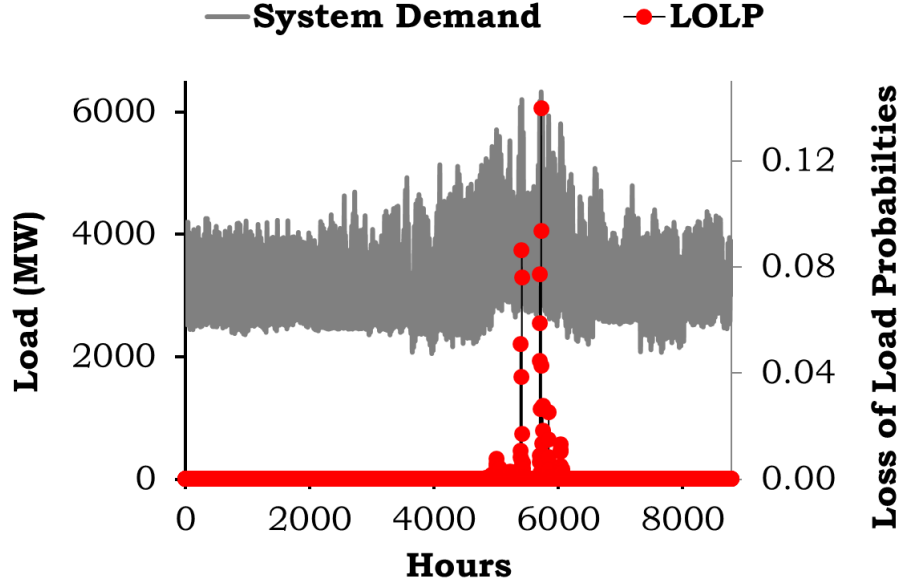


Figure 3.6: System demand and LOLP for each hour of the year for Los Angeles Department of Water and Power

Figure 3.6 shows the result of applying the above methodology to the summer peaking system, LDWP, and illustrates the correlation between the demand and LOLP. As expected, the hours with highest demand, have the highest LOLP and this indicates that these are the hours in which there is most value to be obtained through operating DR resources, from a CV perspective.

Dispatching DR is achieved in this chapter by modeling DR as negative generation, which is similar to the approach taken in [70, 81, 83], and calculating the net load. The new LOLP for each hour is calculated, as is the new LOLE for the year with DR ($LOLE_{DR}$). Constant load that is not dispatchable is then added incrementally to each hour of the year until $LOLE_{DR}$ for the year with DR is the same as the $LOLE_{ref}$. The amount of extra load, ΔL , which is added is the ELCC.

$$LOLP_{DR}^t = prob \ (GEN^t < DEM^t - DR^t) \quad (3.3)$$

$$LOLE_{DR} = \sum_t LOLP_{DR}^t \quad (3.4)$$

$$\sum_t prob (GEN^t \leq DEM^t) = \sum_t prob (GEN^t \leq DEM^t - DR^t + \Delta L) \quad (3.5)$$

While the CV of generating resources is usually expressed as a percentage of the installed capacity or rated capacity, the same convention cannot be applied to the DR resources in this chapter, as the capacity is not known. Since there is no measure of the CV of DR in wide-use, here it is represented as a percentage of its peak availability (see Equation (3.6)) and, in certain places, as a percentage of peak system demand. In Chapter 4, the installed capacity of the resource in question is specified and thus, in Chapter 4 the CV is expressed as a percentage of the installed capacity of all devices considered to be part of the resource.

$$CV = \frac{ELCC}{PEAK_{AVAIL}} \quad (3.6)$$

Different amendments to the DR availability profiles, representing different cases for examination, are now presented. Firstly, the DR resources are dispatched ignoring any energy recovery requirements in Section 3.3.3.1. Following this, an energy recovery period is considered in Section 3.3.3.2. The ELCC method is employed in this thesis as it was the methodology recommended by [20] for use in a wind integration study.

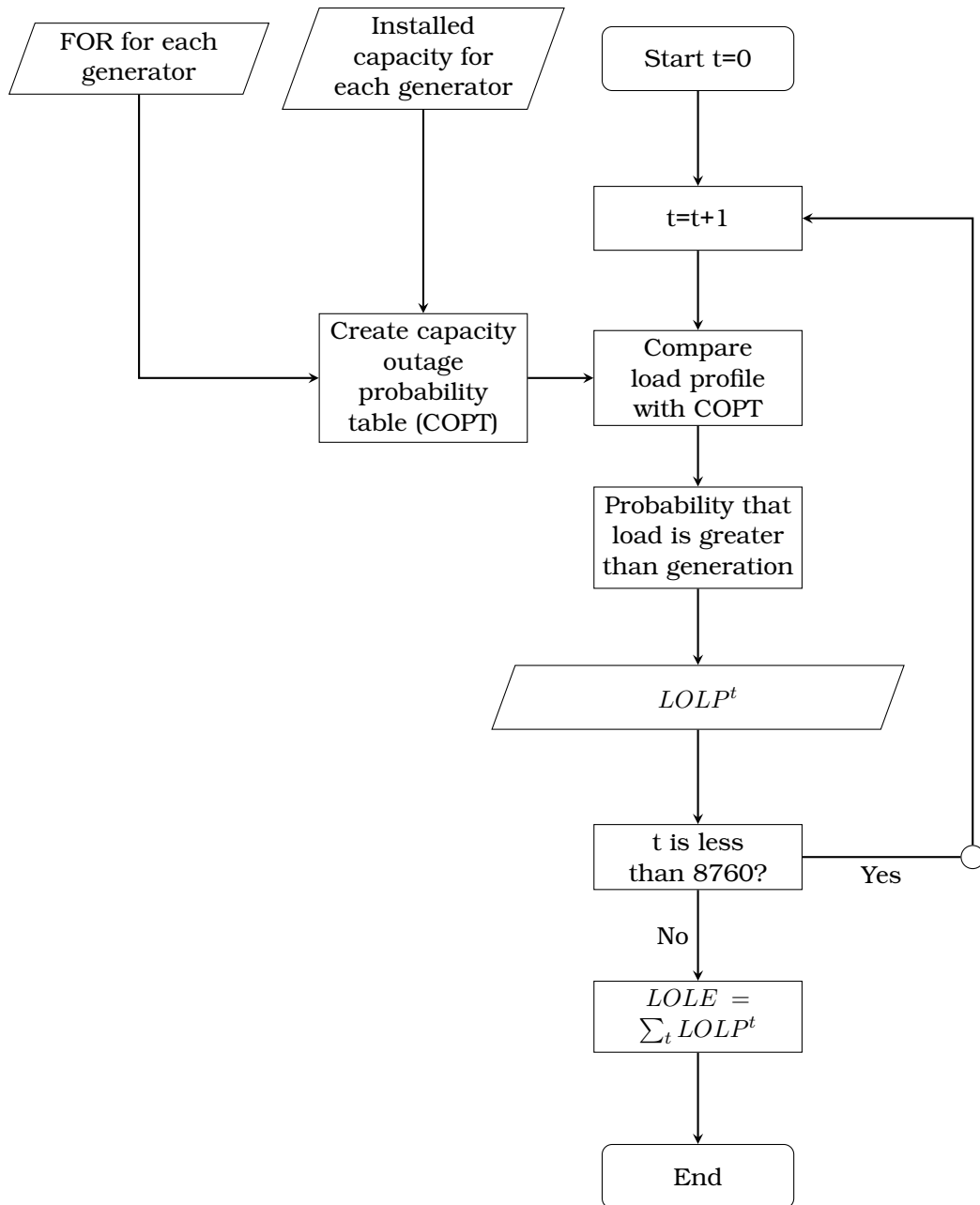


Figure 3.7: Conventional LOLE calculation methodology

3.3.3 Amending Demand Response Availability Profiles

Table 3.3 illustrates the operational constraints associated with the DR resources analyzed in this chapter. As can be seen, each of the resources have similar but resource specific constraints. However, for the sake of simplicity in employing the amendment algorithms, a single version of these individual constraints was applied to all resources, irrespective of the type of resource analyzed. As discussed in Section 2.3.1, and according to [45], utilities will be reluctant to dispatch DR resources regularly since ultimately any program will be disruptive to the consumer. Consequently, in this chapter, a maximum response duration of 2 hours for each event is set for all cases as this is the lower limit on the maximum duration of the resource examined (see table 3.3).

Table 3.3: Product constraints for the DR resources used in this chapter obtained from [1]

DR Resource	Operation Restrictions		
	Min Duration	Max Duration	Energy re-charge
Commercial Heating	5 minutes	N/A	100% within 24 hours
Data Centers	N/A	4 hours	100% within 24 hours
Municipal Pumping	N/A	2 hours	100% within 24 hours
Residential Cooling	5 minutes	N/A	100% within 24 hours
Residential Water Heating	5 minutes	N/A	100% within 24 hours
Waste Water Pumping	N/A	3 hours	100% within 24 hours

3.3.3.1 Cases for Examination - No Energy Recovery

A number of different cases are examined here, assuming that there is no energy recovery requirement imposed on the deployment of the DR resources. A parameter *TRIGGER* is introduced. This is the level of load above which a DR event is required and initiated. For example, in case 2: Top 20 Load Hours and 1 Hour Energy Constraint, described below, the *TRIGGER* value corresponds to hour 20 on the load duration curve. This is illustrated in Figure 3.8. Although DR is only dispatched for less than 100 hours during the year, the LOLE and ELCC calculations takes the whole year into account.

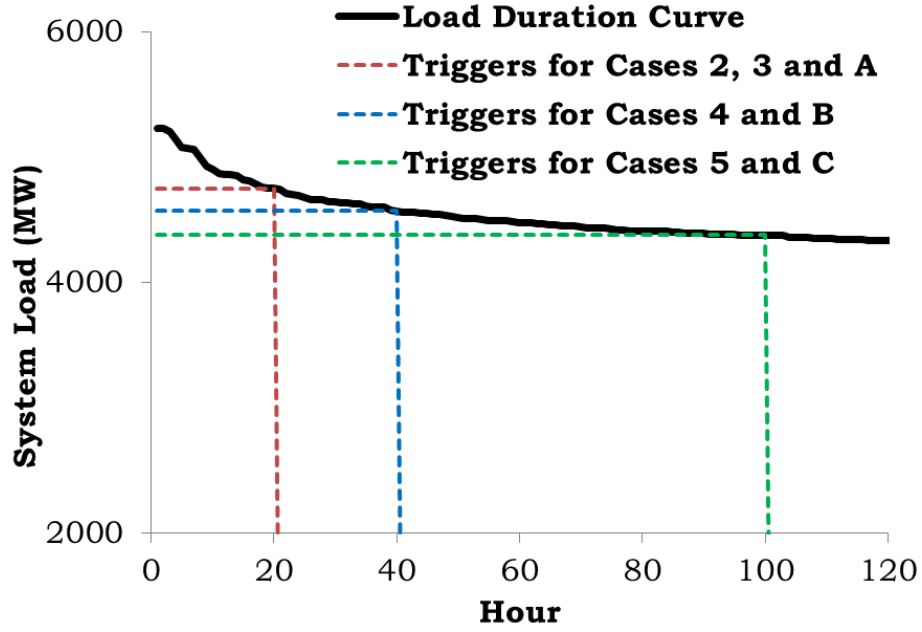


Figure 3.8: Load duration curve for Puget Sound Energy and determination of *TRIGGER* levels for all cases

Once the *TRIGGER* level has been determined, it can then be employed to determine the hours in which DR would be required using:

if $DEM^t \geq TRIGGER$ **then**

$$DR_{req}^t = 1;$$

else

$$DR_{req}^t = 0;$$

end if

The cases examined **without** energy recovery include:

1. **Unconstrained Demand Response Profile** - This case involves estimating the CV for the DR resource without any constraints. This gives the maximum potential ELCC value and thus represents the upper limit on the CV of the DR resource. This is essentially equivalent to completely removing the resource from the power system.
2. **Top 20 Load Hours and 1-Hour Response Duration** - For this case, perfect foresight of when the top 20 load hours will occur and thus when DR will be dispatched is assumed. This implies that DR will not be dispatched at all times of the year, which is reasonable given that electricity is a good and customers

gain utility through its consumption. The DR availability profiles in this case are amended by adding in a response duration constraint, which accounts for the fact that if DR is deployed in one hour it is not available for the next hour. Algorithm B.1.1 is employed to determine the DR availability profiles for this case.

3. **Top 20 Load Hours and 2-Hour Response Duration** - In this case, the DR availability profiles are amended to account for the fact that if DR is dispatched for two hours during the peak demand periods it is not available for the next two hours. Algorithm B.1.2 is employed to determine the DR availability profiles for this case.
4. **Top 40 Load Hours and 2-Hour Response Duration** - For this case, the perfect foresight assumption is relaxed, because the demand level above which DR responds is decreased, therefore load may reach a level which triggers DR but due to the imposed response duration constraints, DR may not be able to operate at the hours of highest LOLP. DR is dispatched here during the top 40 load hours (case 4) to highlight that a reduction in DR *TRIGGER* level increases the time period over which it would be beneficial for DR to offer a sustained response, but that the energy-limited nature of the resource impedes this and severely impacts on the CV.
5. **Top 100 Load Hours and 2-Hour Response Duration** - This case is the same as case 4 except the top 100 load hours are used to trigger DR. Again this was in order to determine the impact of increasing the frequency DR is triggered during the year. Algorithm B.1.2 is employed to determine the DR availability profiles for this case.

Figure 3.9 illustrates how the DR resource is dispatched under cases 3, 4 and 5. At 16:00 and 17:00, for example, the system load exceeds the threshold for dispatching DR, *TRIGGER*, and it is assumed that the DR resource is deployed (red markers in Figure 3.9). Since, the resource has been deployed it is now unavailable for the next two hours and thus its availability is zero in hours 18:00 and 19:00 (orange markers in Figure 3.9).

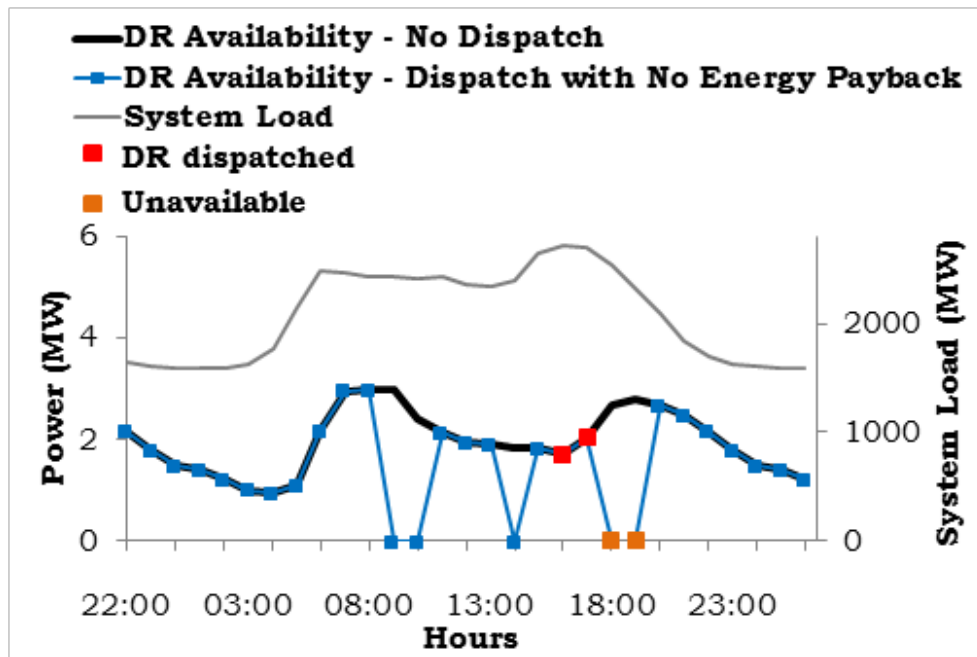


Figure 3.9: Demand response amendments for Avista Corporation residential water heating with no energy recovery constraint

3.3.3.2 Cases for Examination - With Energy Recovery

In the previous cases discussed above, load reduction occurred for a pre-specified period of time and no consideration was given to the fact that following load reduction events, there is often a required period of energy recovery. Incorporating energy recovery * into the analysis requires manipulation of the availability time series' which are used in Equations 3.3 and 3.5 which necessitates the development of a new algorithm to dispatch the DR resources (see Appendix B). The parameter *TRIGGER* is also used here and determines whether DR is required in each hour, as per Figure 3.8. A maximum of 2 hours response duration is set for all cases. The cases to be examined **with** energy recovery include:

- A **Top 20 Load Hours** - DR is dispatched in a particular hour if the system load in that hour exceeds the *TRIGGER* level that corresponds to the system load in hour 20 on the load duration curve. This case is similar to case 3 above, in that *TRIGGER* level is the same and the response duration is the same. Algorithm B.2.1 is employed to determine the DR availability profiles for this case.

*The results pertaining to energy recovery did not appear in [71].

B Top 40 Load Hours - DR is dispatched in a particular hour if the system load in that hour exceeds the *TRIGGER* level that corresponds to the system load in hour 40 on the load duration curve. This case is similar to case 4 above, in that *TRIGGER* level is the same and the response duration is the same. Algorithm B.2.1 is employed to determine the DR availability profiles for this case.

C Top 100 Load Hours - DR is dispatched in a particular hour if the system load in that hour exceeds the *TRIGGER* level that corresponds to the system load in hour 100 on the load duration curve. This case is similar to case 5 above, in that *TRIGGER* level is the same and the response duration is the same. Algorithm B.2.1 is employed to determine the DR availability profiles for this case.

Figure 3.10 illustrates how the DR resource is dispatched under cases A, B and C. As in Figure 3.9 at 16:00 and 17:00, the system load exceeds the threshold for dispatching DR and it is assumed that the DR resource is used. Since, the resource has been used it is now unavailable for the next two hours and the energy used as part of the DR dispatch must be recovered, or paid back. Thus, in hours 18:00 and 19:00 (green markers in Figure 3.10) the power of the DR is negative and is equal to the DR dispatched in hours 18:00 and 19:00, respectively. Since this amended load profile, DR_t , is subtracted from the system load, the negative availability of the DR resource results in an increase in overall system load at those hours (see Equations 3.3 and 3.5).

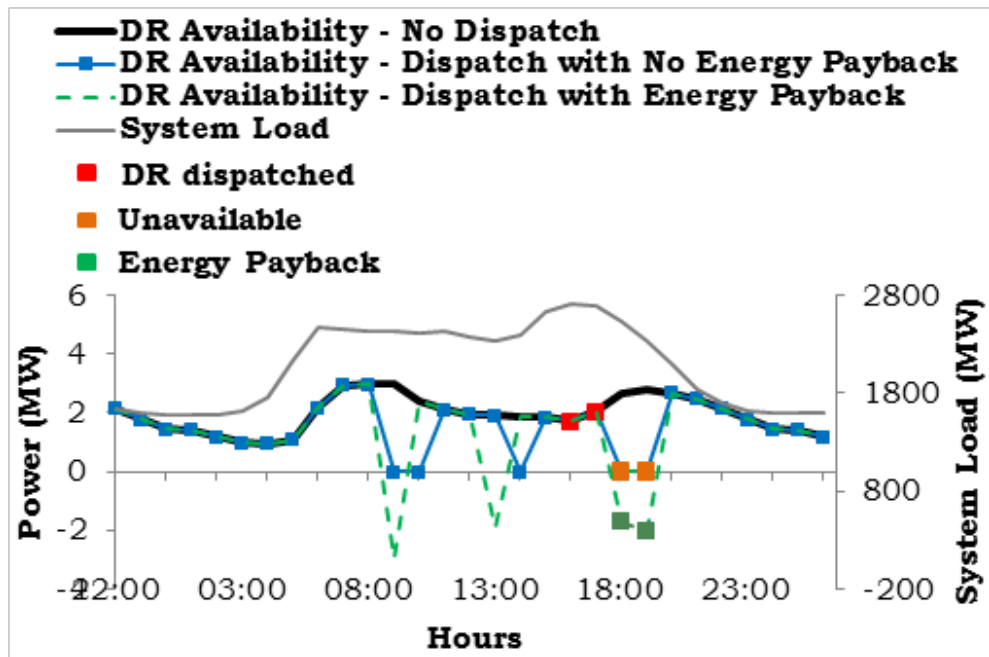


Figure 3.10: Demand response amendments for Avista Corporation residential water heating with an energy recovery constraint

3.4 Results

As expected, the CV of DR is very dependent on whether the resource is available at times of high LOLP. Unless DR is providing capacity when the LOLP is high, its CV is low. For example, unless a DR resource in LDWP provides a response during the hours denoted with red markers in Figure 3.6, that resource will have a low CV. The results also highlight that including consideration of the energy-limited nature of the DR resources in the estimation of the CV has a profound effect on the results. The inclusion of energy recovery in conjunction with the energy-limited nature further impacts upon the CV estimate. Section 3.4.1 presents the results obtained from the analysis, without consideration of the need for energy recovery following dispatch of the DR resources, while Section 3.4.2 provides results produced when energy recovery is incorporated into the analysis. Section 3.4.2 also compares the CV estimates obtained with and without the energy recovery constraint.

3.4.1 Results with No Energy Payback

Tables 3.4, 3.5, 3.6 and 3.7 illustrate that the upper limit on the potential CV of DR (Case 1 from section 3.3.3.1, which does not involve any amendments to the DR availability profiles) can be quite significant. This is particularly true for resources such as data centers and waste water pumping, as these are resources which have constant availability at all times, resulting in CV estimates of 99% and 100%. The other DR resources have more variable availability profiles and thus the CVs associated with them are typically much lower. Figure 3.11 illustrates the contrasting availability profiles of two different DR resources in the same system. The diversity in the availability profiles between resources contributes to the considerable range in CV estimates presented in Tables 3.4, 3.5, 3.6 and 3.7.

It is noted from Tables 3.4, 3.5, 3.6 and 3.7 that the maximum potential CV estimates for similar DR resources in different systems are broadly consistent. For example, commercial heating in the two winter peaking systems, AVA and PSE have estimated values of 59% and 57%, respectively, while the residential water heating resources have respective estimated values of 64% and 66%. Similarly, in the two summer peaking systems, LDWP and NEVP the residential water heating resources have estimated CVs of 31% and 32% respectively.

Table 3.4: Maximum potential ELCC (MW) (Case 1.) and capacity value estimates for Avista Corporation for different end-use resources

Resource	ELCC (MW)	CV
Commercial Heating	3.2	59%
Data Centers	1.5	100%
Municipal Pumping	0.4	51%
Residential Water Heating	2.4	64%
Waste Water Pumping	0.3	100%

Table 3.5: Maximum potential ELCC (MW) (Case 1.) and capacity value estimates for Puget Sound Energy for different end-use resources

Resource	ELCC (MW)	CV
Commercial Heating	4.0	57%
Data Centers	2.8	100%
Municipal Pumping	0.9	46%
Residential Water Heating	5.1	66%
Waste Water Pumping	0.8	99%

Table 3.6: Maximum potential ELCC (MW) (Case 1.) and capacity value estimates for Nevada Power for different end-use resources

Resource	ELCC (MW)	CV
Data Centers	2.8	99%
Municipal Pumping	0.7	62%
Residential Water Heating	1.3	32%
Waste Water Pumping	0.5	99%

Table 3.7: Maximum potential ELCC (MW) (Case 1.) and capacity value estimates for Los Angeles Department of Water and Power for different end-use resources

Resource	ELCC (MW)	CV
Commercial Cooling	35.8	90%
Commercial Heating	0.1	4.5%
Data Centers	3.1	100%
Municipal Pumping	2.4	50%
Residential Cooling	19.4	77%
Residential Water Heating	1.4	31%
Waste Water Pumping	1.8	96%

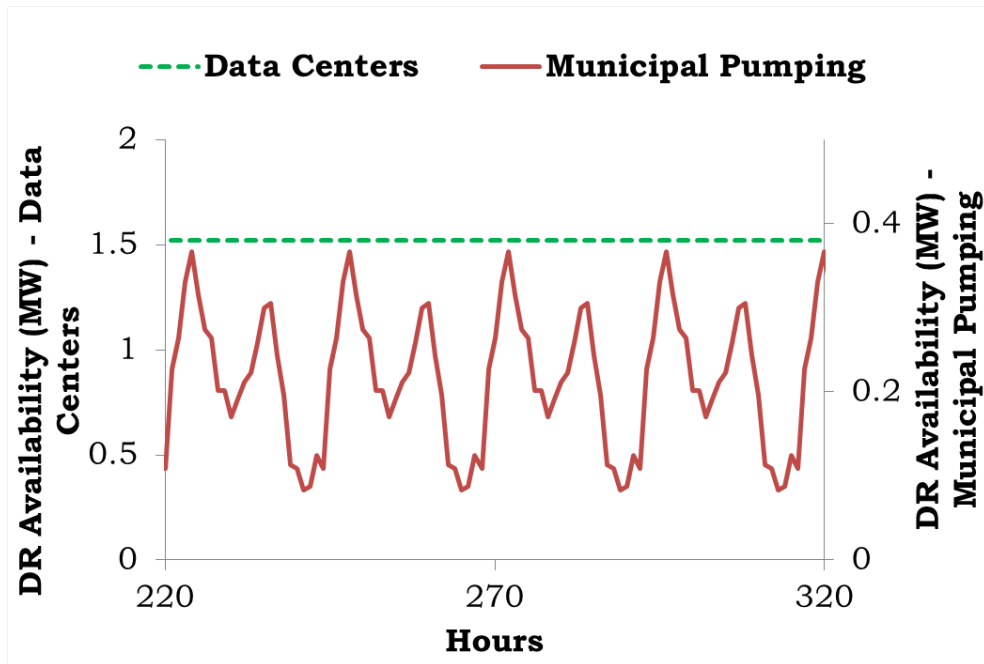


Figure 3.11: Data Center availability -v- municipal pumping availability in Avista Corporation highlighting considerable differences between the two resources

It can also be seen that, even within the same system, different DR resources can receive very different CV estimates, which is as expected. Table 3.7 shows that the CV estimate of commercial heating in LDWP is only 4.5%, while the CV estimate of residential cooling is 77%. Furthermore, the commercial heating CV estimate in the two winter peaking systems (AVA and PSE) is above 57%, far greater than the 4.5% estimate for LDWP.

This considerable difference is a result of the fact that the availability of the commercial heating resource in LDWP is confined to winter periods and LDWP is a summer peaking system. Consequently, the availability of the commercial heating resource in LDWP does not coincide with hours of high LOLP (see Figure 3.12), limiting the ability of the commercial heating resource to shift away from hours of high LOLP and contribute to generating adequacy. This is in great contrast to the residential cooling resource in LDWP, and the commercial heating resource in PSE as the period of high availability of this resource coincides with the hours of high LOLP (see Figures 3.13 and 3.14 respectively).

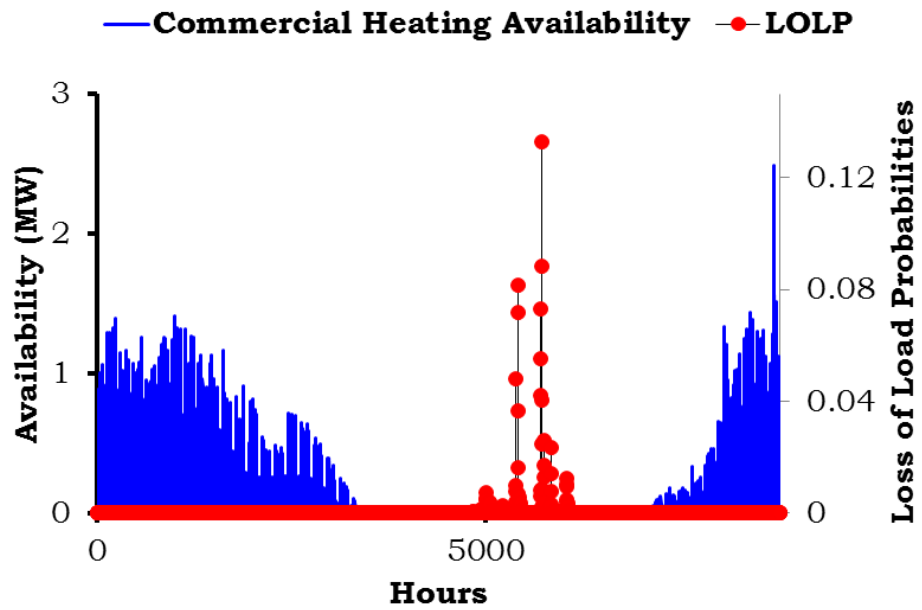


Figure 3.12: Commercial heating availability and LOLP for each hour for Los Angeles Department of Water and Power

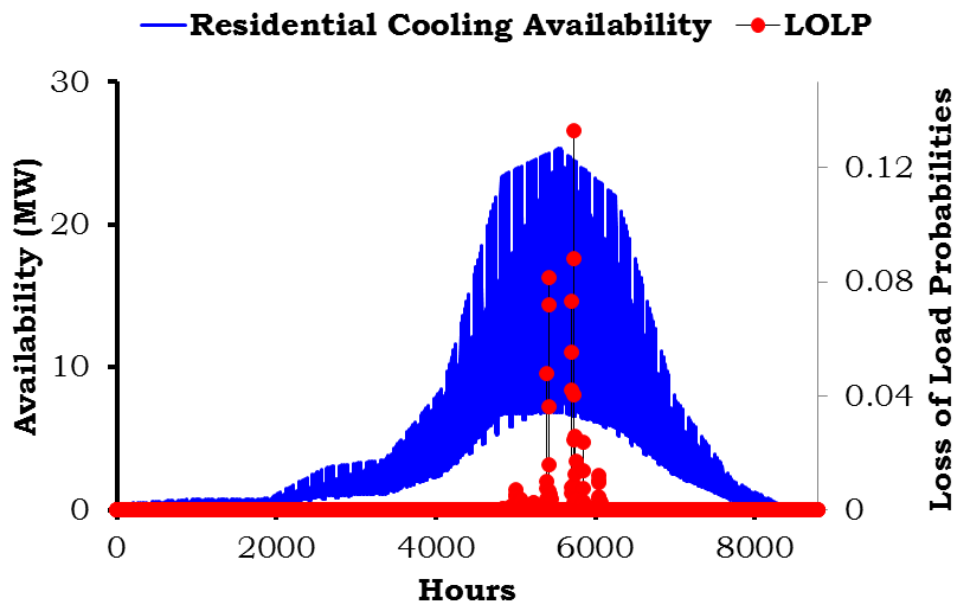


Figure 3.13: Residential cooling availability and LOLP for each hour for Los Angeles Department of Water and Power

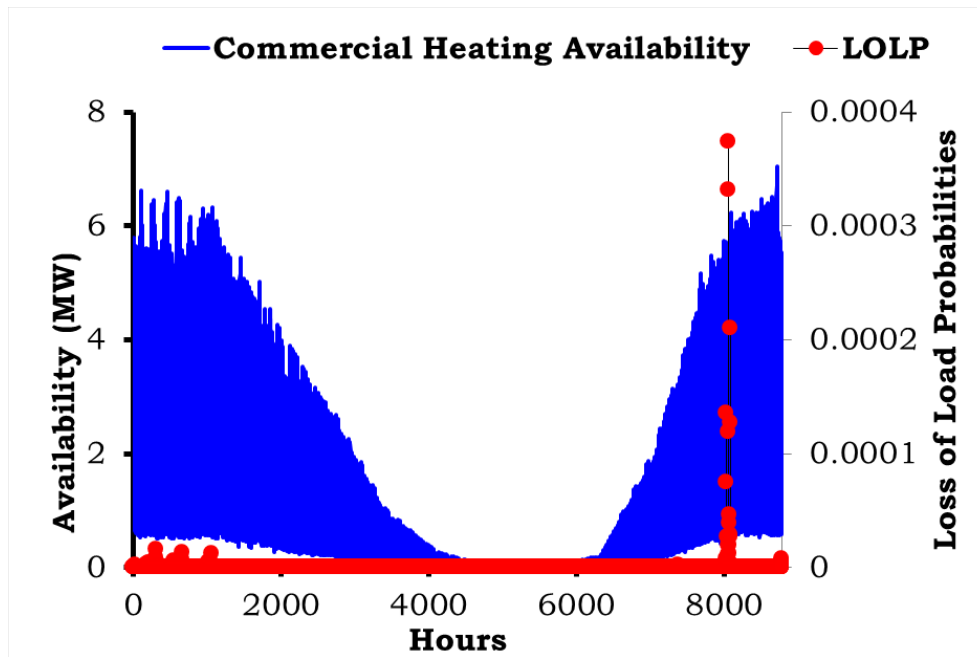


Figure 3.14: Commercial heating availability and LOLP for each hour for Puget Sound Energy

Although the upper limit on the potential CV of DR can be quite significant, Tables 3.8, 3.9, 3.11 and 3.10 illustrate that the CV is very sensitive to the energy-limited nature of the resource (Cases 2 through 5 from Section 3.3.3.1), resulting in lower CV estimates. In many cases, increasing the number of hours in which DR operates (case 3 through 5) has no major benefit for system adequacy and reduces the CV.

This is because, due to its energy limitation, the more often DR operates the greater the likelihood that DR will not be in a position to be dispatched in an hour where the LOLP is high. Furthermore, DR may be triggered due to high system demand levels and sustains a response for two hours, but then is unavailable for the following two hours, despite the fact that the hours in which DR recovers may actually have high LOLPs.

Table 3.8: ELCC (MW) estimates for Avista Corporation for different end-use resources **without** energy recovery

Resource	Case	1.)	2.)	3.)	4.)	5.)
Commercial Heating	ELCC (MW)	3.2	2.9	2.4	2.3	2.0
	CV	57%	52%	43%	41%	36%
Data Centers	ELCC (MW)	1.5	1.43	1.4	1.2	0.9
	CV	99%	94%	92%	79%	59%
Municipal Pumping	ELCC (MW)	0.4	0.3	0.3	0.3	0.3
	CV	59%	45%	45%	45%	45%
Residential Water Heating	ELCC (MW)	2.4	2.0	1.8	1.5	1.4
	CV	64%	54%	48%	40%	38%
Waste Water Pumping	ELCC (MW)	0.3	0.18	0.18	0.18	0.18
	CV	100%	67%	67%	67%	67%

Table 3.9: ELCC (MW) estimates for Puget Sound Energy for different end-use resources **without** energy recovery

Resource	Case	1.)	2.)	3.)	4.)	5.)
Commercial Heating	ELCC (MW)	4.0	3.8	2.4	1.9	2.2
	CV	57%	54%	34%	27%	31%
Data Centers	ELCC (MW)	2.8	2.4	1.6	1.2	1.4
	CV	100%	89%	60%	45%	53%
Municipal Pumping	ELCC (MW)	0.9	0.8	0.5	0.3	0.4
	CV	46%	53%	26%	16%	21%
Residential Water Heating	ELCC (MW)	5.1	4.7	2.9	2.2	2.4
	CV	66%	61%	38%	29%	31%
Waste Water Pumping	ELCC (MW)	0.8	0.7	0.5	0.2	0.4
	CV	99%	89%	65%	26%	52%

Table 3.10: ELCC (MW) estimates for Nevada Power for different end-use resources **without** energy recovery

Resource	Case	1.)	2.)	3.)	4.)	5.)
Data Centers	ELCC (MW)	2.8	2.2	2.2	2.0	1.5
	CV	99%	80%	80%	72%	54%
Municipal Pumping	ELCC (MW)	0.7	0.4	0.4	0.3	0.2
	CV	62%	37%	35%	26%	17%
Residential Water Heating	ELCC (MW)	1.3	1.2	1.1	0.9	0.7
	CV	32%	29%	27%	22%	17%
Waste Water Pumping	ELCC (MW)	0.5	0.2	0.2	0.2	0.2
	CV	99%	46%	43%	43%	43%

Table 3.11: ELCC (MW) estimates for Los Angeles Department of Water and Power for different end-use resources **without** energy recovery

Resource	Case	1.)	2.)	3.)	4.)	5.)
Commercial Cooling	ELCC (MW)	35.9	29.7	23.7	17.2	12.4
	CV	90%	78%	60%	43%	31%
Commercial Heating	ELCC (MW)	0.1	0.1	0.1	0.1	0.1
	CV	4.5%	4.5%	4%	4%	4%
Data Centers	ELCC (MW)	3.1	2.2	1.9	1.2	1.2
	CV	100%	74%	63%	40%	40%
Municipal Pumping	ELCC (MW)	2.4	1.9	1.5	1.212	0.9
	CV	50%	40%	31%	7%	5%
Residential Cooling	ELCC (MW)	19.4	15.5	12.2	8.7	7.7
	CV	77%	61%	48%	34%	30%
Residential Water Heating	ELCC (MW)	1.4	1.0	0.7	0.4	0.4
	CV	31%	21%	15%	9%	9%
Waste Water Pumping	ELCC (MW)	1.8	1.5	1.1	0.8	0.6
	CV	96%	80%	58%	42%	31%

Table 3.9 shows that the CVs of some of the DR resources in PSE increase slightly between Case 4 and Case 5. This system's load profile characteristics results in sustained periods of high LOLPs. Note how the LOLP values in Figure 3.14 are concentrated around hour 8000. Such sustained periods of high LOLPs are better targeted by a pattern of dispatching DR at a lower *TRIGGER* level followed by unavailability for two hours and then dispatching DR for a further two hours at the peak, as opposed to waiting until a higher *TRIGGER* level to dispatch DR. Figure 3.17 shows an example of this: at the *TRIGGER* level corresponding to hour 40 on the load duration curve the ELCC estimate of the data center resource is lower than the corresponding ELCC estimate at the *TRIGGER* level corresponding to hour 100 on the load duration curve. This suggests that careful consideration is needed regarding the operating strategies of DR in different systems, to increase the likelihood that the hours during which the LOLPs are highest are targeted with DR dispatch decisions.

Although the commercial cooling resource in LDWP may have reasonably high CV estimates for all cases, Table 3.12 highlights that the ELCC as a percentage of peak system demand is less than 1% (column 3) for all cases. In AVA, the ELCC of residential water heating as a percentage of peak demand is considerably less (Table 3.13). This result has serious implications for the economic value of the resource since, individually, a DR resource's ability to compete with other resources could be hampered by its small size, relative to the overall system demand.

Table 3.12: Estimated capacity values for commercial cooling in Los Angeles Department of Water and Power (summer peaking system) **without** energy recovery constraint

Case	% of Peak Capacity	% of Peak Demand
1)	90%	0.6%
2)	78%	0.5%
3)	60%	0.4%
4)	43%	0.3%
5)	31%	0.2%

Table 3.13: Estimated capacity values for residential water heating in Avista Corporation (winter peaking system) **without** energy recovery constraint

Case	% of Peak Capacity	% of Peak Demand
1)	64%	0.09%
2)	54%	0.07%
3)	48%	0.06%
4)	40%	0.05%
5)	38%	0.05%

Table 3.14 summarizes the CV estimates from the analysis without energy recovery and indicates the dichotomy in CV between load types in different systems. It is found that loads which are valuable for DR in terms of CV differ from system to system and whether the system peak demand occurs in the winter or summer months. It can be seen that in summer peaking systems commercial heating has a very low CV, compared to a CV above 30% in winter peaking systems.

Commercial heating is typically not available in the summer, precisely when it would be most desired from a reduction of LOLP point of view for a summer peaking system. It can be seen that residential water heating has a higher CV in winter peaking systems compared with summer peaking systems. Residential cooling, on the other hand, has a higher contribution to generation adequacy in summer systems, compared to winter systems, where the residential cooling resource is non-existent.

It is noted that data centers, municipal pumping and waste water pumping have high CV estimates for all systems as a result of the fact that these resources have a more constant availability in all time periods. The results presented in Table 3.14 highlight that the CV of different DR resources is highly system specific and there is a need for careful consideration regarding which DR resources are exploited in each system.

Table 3.14: Comparison of range of CV for winter and summer peaking systems as a percentage of the peak DR capacity **without** energy recovery

System Resource	Winter	Winter	Summer	Summer
	System 1	System 2	System 1	System 2
	(AVA)	(PSE)	(NEVP)	(LDWP)
Commercial Heating	36-41%	27-31%	N/A	4%
Data Centers	59-79%	45-53%	54-72%	40%
Municipal Pumping	45%	16-21%	17-26%	5-7%
Residential Cooling	N/A	N/A	N/A	30-34%
Residential Water Heating	38-40%	29-31%	17-22%	9%
Waste Water Pumping	67%	26-52%	43%	31-42%

3.4.2 Results with Energy recovery

As expected, the inclusion of an additional constraint, namely the need for energy recovery following the dispatch of DR, consistently reduces the CV, when compared with the results where energy recovery is not considered. The reason for this is illustrated in Figure 3.15 which compares the availability profiles of a DR resource with and without energy recovery following a DR dispatch event. In the case with no energy recovery, it is assumed that DR is dispatched at 16:00 and 17:00 (red markers) and is thus unavailable in hours 18:00 and 19:00 (orange markets). However, in the case with energy recovery, if DR is dispatched at 16:00 and 17:00 (red markers), the necessary energy recovery occurs in hours 18:00 and 19:00 (green markers). This results in negative DR availability.

Since the DR availability is subtracted from the system load in Equation 3.3, it is clear to see that this would constitute an increase in overall system demand at hours 18:00 and 19:00. Thus, this would suggest a smaller contribution to system generation adequacy compared with the case where there is no energy recovery. This is because, rather than operating as a load-shedding resource, as is the situation in the cases without energy recovery, the resource is now operated as a load-shifting

resource. This difference has a profound effect on the resultant CV estimates as will now be discussed.

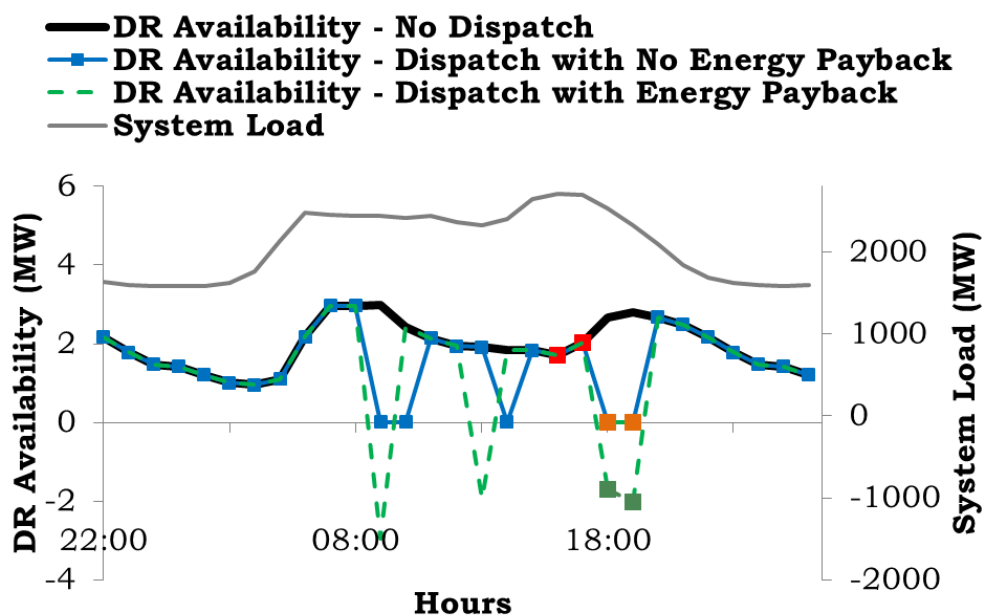


Figure 3.15: Comparison of DR amendments for Avista Corporation residential water heating with and without an energy recovery constraint

Tables 3.15 to 3.17 compare the ELCC estimates obtained for the different DR resources, with and without energy recovery. A graphical comparison is presented in Appendix C, while CVs for the cases with energy recovery are provided in Appendix D. As can be seen, there is a clear reduction in the ELCC estimates when the energy recovery constraint is considered, as is expected. In some cases, most notably in the LDWP system, Table 3.18, over utilization of the DR resources results in negative, or at least extremely low, ELCC estimates, indicating that the operation of the DR resources for more than 20 hours per year can actually disimprove the generation adequacy of the power system.

Figures 3.16 through 3.18 illustrate the clear downward trend in ELCC estimates for the data center resources across all four systems. Similar trends are seen for the other DR resources and such graphs are presented in Appendix C. In some cases, DR operation accounting for energy recovery periods produces negative ELCC estimates.

Table 3.15: Comparison of ELCC (MW) estimates for Avista Corporation for different end-use resources with and without energy recovery

Resource	Energy Constraint	20 Hours	40 Hours	100 Hours
Commercial Heating	without	2.4	2.3	2.0
	with	1.8	1.4	1.1
Data Centers	without	1.4	1.2	0.9
	with	0.9	0.7	0.4
Municipal Pumping	without	0.3	0.3	0.3
	with	0.18	0.18	0.18
Residential Water Heating	without	1.8	1.5	1.4
	with	1.3	1.0	0.8
Waste Water Pumping	without	0.18	0.18	0.18
	with	0.18	0.18	0.18

Table 3.16: Comparison of ELCC (MW) estimates for Puget Sound Energy for different end-use resources with and without energy recovery

Resource	Energy Constraint	20 Hours	40 Hours	100 Hours
Commercial Heating	without	2.4	1.9	2.2
	with	1.1	0.2	0.6
Data Centers	without	1.6	1.2	1.4
	with	0.6	neg	0.1
Municipal Pumping	without	0.5	0.3	0.4
	with	0.2	neg	0.04
Residential Water Heating	without	2.9	2.2	2.4
	with	1.4	0.1	0.4
Waste Water Pumping	without	0.5	0.2	0.4
	with	0.1	neg	neg

Table 3.17: Comparison of ELCC (MW) estimates for Nevada Power for different end-use resources with and without energy recovery

Resource	Energy Constraint	20 Hours	40 Hours	100 Hours
Data Centers	without	2.2	2.0	1.5
	with	1.9	1.0	0.43
Municipal Pumping	without	0.4	0.3	0.2
	with	0.4	0.2	0.01
Residential Water Heating	without	1.1	0.9	0.7
	with	0.8	0.5	0.2
Waste Water Pumping	without	0.2	0.2	0.2
	with	0.2	0.2	neg

Table 3.18: Comparison of ELCC (MW) estimates for Los Angeles Department of Water and Power for different end-use resources with and without energy recovery

Resource	Energy Constraint	20 Hours	40 Hours	100 Hours
Commercial Cooling	without	23.7	17.2	12.4
	with	9.6	neg	neg
Commercial Heating	without	0.1	0.1	0.1
	with	0.1	0.1	0.1
Municipal Pumping	without	1.5	0.34	0.9
	with	0.7	neg	neg
Residential Water Heating	without	0.7	0.4	0.4
	with	0.4	neg	neg
Residential Cooling	without	12.2	8.7	7.7
	with	6.6	1.0	neg
Data Centers	without	1.9	1.2	1.2
	with	1.2	neg	neg
Waste Water Pumping	without	1.1	0.8	0.6
	with	0.6	neg	neg

The fact that some of the estimates are zero, indicating there is no contribution to generation adequacy, is a result of the characteristics of the systems' loads. The PSE system load profile has low statistical variance in comparison with the AVA system load profile, the other winter peaking system examined, while LDWP has much lower statistical variance compared with NEVP, the other summer peaking system. Low statistical variance indicates lower variability in system load, which suggests less variability in load around the system peak load hours, or more sustained peak periods. This was seen earlier in Figure 3.14, which showed that the hours of high LOLP are concentrated in just a few hours of the year.

Consequently, operating DR for the first two hours of a sustained peak period, followed by energy recovery in the next two hours often inadvertently creates an additional, unintentional peak in the system demand. When these additional unintentional peaks are created, the operation of the DR resources may no longer have a net positive contribution to generation adequacy over the year, which can result in negative ELCC estimates.

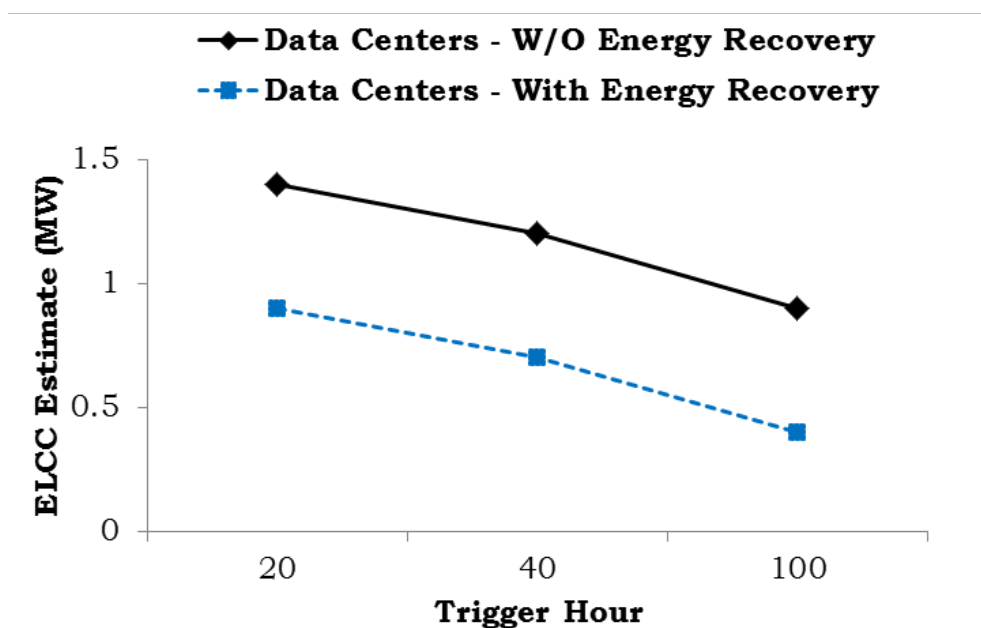


Figure 3.16: Data center ELCC (MW) estimates in Avista Corporation, with and without energy recovery

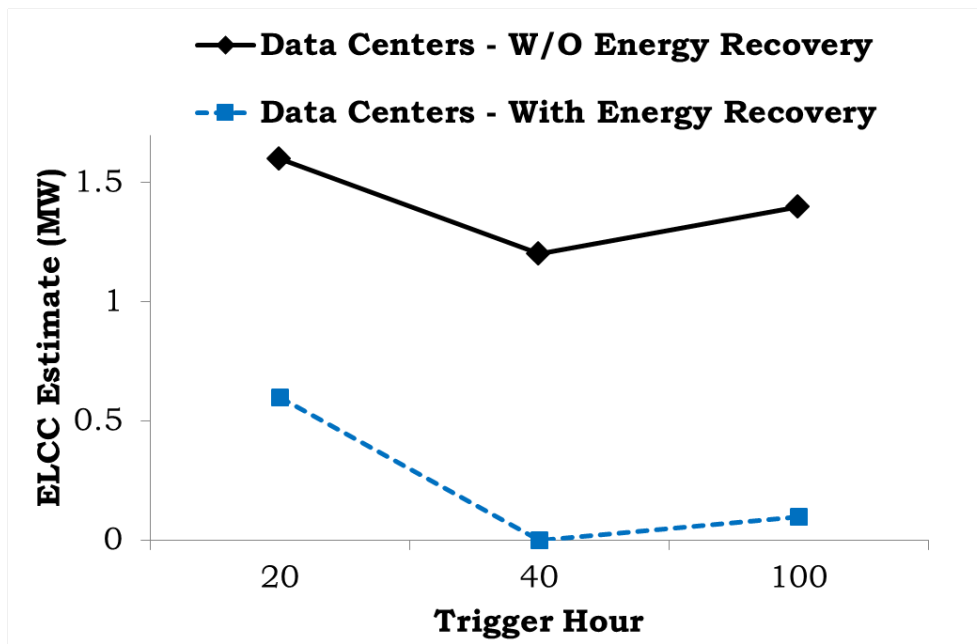


Figure 3.17: Data center ELCC (MW) estimates in Puget Sound Energy, with and without energy recovery

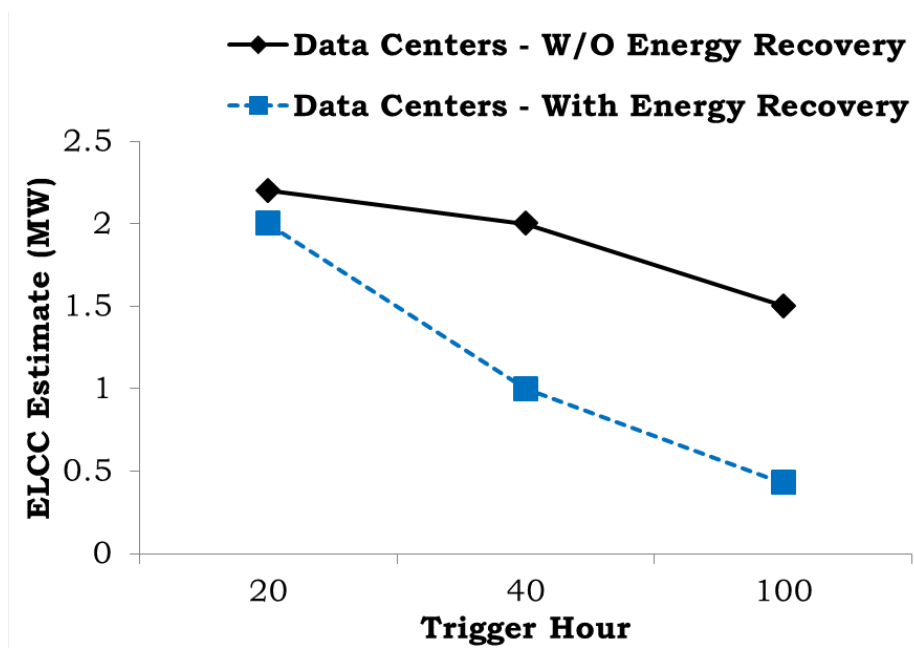


Figure 3.18: Data center ELCC (MW) estimates in Nevada Power, with and without energy recovery

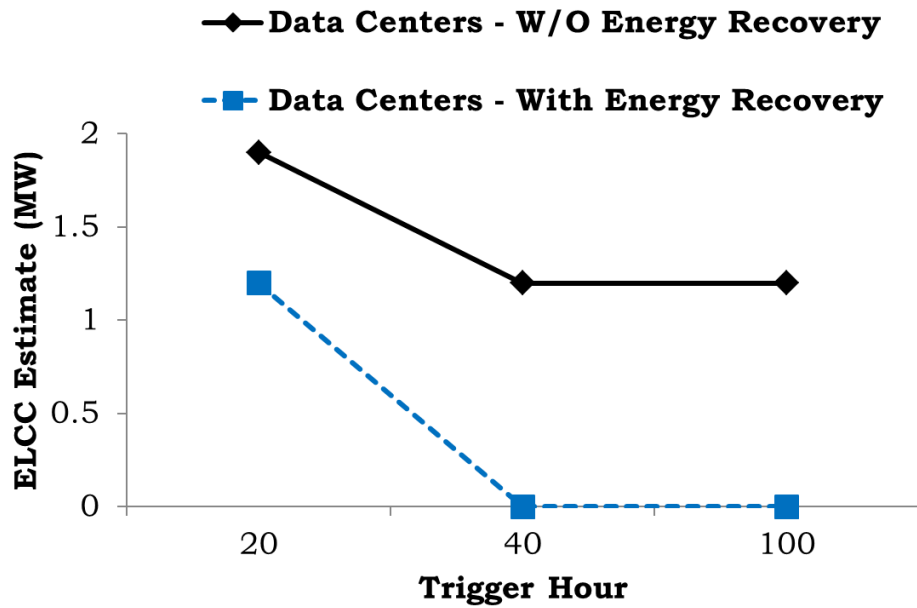


Figure 3.19: Data center ELCC (MW) estimates in Los Angeles Department of Water and Power, with and without energy recovery

Table 3.19 summarizes the CV estimates from the above analysis and allows for direct comparison with the cases without energy recovery in Table 3.14. It is found that, in all cases, there is a considerable reduction in the CV estimates when energy recovery is considered, as is expected.

Table 3.19: Comparison of range of capacity values for winter and summer peaking systems as a percentage of the peak DR capacity **with** energy recovery

	Winter System 1 (AVA)	Winter System 2 (PSE)	Summer System 1 (NEVP)	Summer System 2 (LDWP)
Commercial Heating	20-32%	8-15%	N/A	4.5%
Data Centers	26-60%	2-21%	15-69%	40%
Municipal Pumping	45%	2-10%	1-30%	15%
Residential Cooling	N/A	N/A	N/A	4-26%
Residential Water Heating	22-36%	1-18%	4-20%	10%
Waste Water Pumping	68%	18%	38%	33%

3.5 Conclusion

This chapter presents a preliminary methodology for estimating the CV of different DR resources which are subjected to a response duration constraint which represents the energy-limited nature of the DR resource. The analysis provides an indicative assessment of the CV of several different DR resources and highlights some general trends. The most valuable DR resources in terms of CV differ from system to system and whether the system peak demand occurs in the winter or summer. Perhaps, most importantly, the ELCC of each DR resource is found to represent only a very small portion of the system demand, which may have implications for the economic value of the resource. Chapter 5 of this thesis will explore the value of a DR resource to the power system and the savings apportioned to a DR resource as a result of its participation in electricity markets, and thus its competition with other resources.

This chapter demonstrates that there is a need to be fully aware of the constraints on the resource availability when assessing DR as they can have significant impacts on the CV. The results and analysis also highlight that considerably more detailed work is required to gain a full understanding of the CV of DR. This chapter indicates that the CV of DR varies widely depending on the type of resource considered. Thus, the CV estimate is specific to each DR resource. This suggests there is a need for thorough investigation of specific resources individually.

It is clear from the results presented here that the CV of DR is affected by the manner in which the DR resource is deployed. It is also found that dispatch of the DR resources utilizing the algorithms developed in this chapter can create unintentional, additional system peaks. This is a result of the fact that the algorithms employed here do not incorporate any optimization, resulting in negative ELCC estimates and a negative contribution to generation adequacy. If optimization of the availability profiles are considered more favorable patterns for dispatching the resource during periods of sustained peak load levels could be obtained. This indicates that careful consideration is needed regarding the dispatching and operating strategies of DR, to increase the likelihood that the hours during which the LOLPs are highest are targeted with DR dispatch decisions.

As discussed in Chapter 1, it is again acknowledged here that increasing wind generation penetration on today's power systems, as well as the variable nature of wind, is one of the key drivers of interest in DR. However, wind generation was not considered in this chapter for two reasons. Firstly, the aim of this chapter was

limited to providing an insight into the CV of several different DR resources and to highlighting some general trends. Secondly, supply-side operational decisions, which would include wind generation, were not considered in this chapter. This is a limitation of this chapter and is an area requiring future work, which is acknowledged in Chapter 6.

Similarly, as the deployment of the DR resource is ultimately dependent on the operation of the power system and the consumers, it is prudent to further analyze the CV of DR by incorporating consideration of power system operations and consumers' requirements. Further research is required to refine and improve the proposed methodology and incorporating production cost simulation or UCED with modeling of the DR resource-specific constraints and its energy-limited nature. This would lead to the establishment of a more accurate dispatch schedule for the DR resources, thus providing a more detailed and more realistic understanding of the CV of DR. This, and the need to investigate a specific resource, is the motivation behind the work presented in Chapter 4.

Capacity Value Estimates for a Specific Load-Shifting Resource

The content of this chapter is presented in a journal paper which is currently in preparation entitled ‘Capacity Value Estimation of a Load-Shifting Resource using a Coupled Building and Power System Model’. A full citation can be found in the section titled “Publications Arising”. This paper, and thus this chapter, are the result of collaboration between the thesis author and Olivier Neu, which is the result of wider collaboration in the Electricity Research Centre which culminated in a technical report entitled “Initial Cost Benefit Analysis Comparison for Domestic Smart Electric Thermal Storage (SETS) vs Direct Resistive Heating (DRH) on the Irish All-Island Power System, Electricity Research Centre & GlenDimplex collaboration”. The author also acknowledges Daniel Burke, who provided the Unit Commitment and Economic Dispatch model and the load-shifting resource model used in this chapter.

4.1 Introduction

As mentioned in Chapter 3, methodologies for calculating the CV of some renewable generating technologies have been proposed by [70, 81, 83, 84]. These approaches typically involve subtracting a time series for the resource power output from the system load giving net system load profiles and incorporating the load profiles with the approach in [91]. This was the basis upon which the methodology presented in Chapter 3 was built. However, while the approach was successful in providing indicative, high level CV estimates, a number of shortcomings of that approach were noted, namely the fact that there is no optimization problem incorporated into the

algorithms and thus unintentional, additional system peaks are recorded due to uncontrolled recovery. Furthermore, consideration of the operations of the supply-side of the power system nor the potential provision of reserve from the DR resources were not considered.

It is evident from the discussion in previous chapters that, in order to assess the CV of load-shifting resources, an approach incorporating the operational and time-dependent characteristics of DR resources, as well as the consumers' requirements and the physical characteristics of the resource, should be employed. Additionally, in order to determine the impact the interactions of the DR resource with the power system can have on the CV of such a resource, a model of power system operations is paramount. Additionally, DR is in a position to provide multiple services, the intrinsic relationship between these services necessitates that they are analyzed holistically.

Despite the fact that there have been some contributions to the literature in the area of demand-side contribution to generation adequacy [92, 93, 94], all of these fail to incorporate supply-side inter-temporal operational decisions or consideration of reserve provision from the DR resource. It is important to recognize that a DR resource will alter the overall system demand profile, which in turn will impact upon the operation of the marginal unit, changing generator output and system prices, which may consequently impact the DR operation. As the vast majority of DR programs respond to system prices and other system conditions, this feedback of the DR impact on prices and power system operational decisions requires that the supply-side and the demand-side are co-optimized. This, to the knowledge of the author, has received limited consideration in the literature to-date when assessing the contribution of the demand-side to generation adequacy, with the notable exception of [95].

A fully integrated model of active DR comprising heating systems with thermal energy storage systems and the electric power system is presented in [95]. It is highlighted that only such an integrated model is '*able to simultaneously consider all the technical and comfort constraints*' [95] of the power system and the DR resource, respectively. The heat demand model coupled with the power system model which is employed here is similar to that proposed in [95]. However, the model in this chapter differs to the model in [95] where reserve provided by the DR resource is not considered. While the work in [96] considers DR reserve provision, it is not considered

as part of analysis of the contribution of DR to generation adequacy, as is the case here.

Previous work in the literature used sequential MC simulations, for example in [92, 93] it was employed to determine the contribution of load-shifting to the generation adequacy of the power system. MC methods are regularly employed to assess system generation adequacy and are often used due to their conceptual simplicity: each sampled scenario can be viewed as a possible ‘history’ of system operations [97]. The approach in this chapter also involves sequential MC allowing consideration of chronological or time-dependent issues [98]. This is particularly important when dealing with a load-shifting resource. Both [92] and [93] employ the IEEE Reliability Test System and model a demand-side load-shifting initiative which transfers energy from the peak hours to the off-peak hours [92]. The authors in [92] determine an ELCC for their demand-side program. However, in [92], load-shifting occurs to meet a pre-specified peak and no consumers’ constraints are included. In addition, there is no consideration taken of the operational decisions on the supply-side of the power system. The work in this chapter, however, examines the operation of one specific load-shifting resource, electric thermal storage heating for both space and water heating, allowing explicit modeling of the load resource with energy and power limitations as well as a specific consumer’s constraints. Furthermore, in this thesis, the load-shifting resource operation is co-optimized with the supply-side of the power system.

An assessment framework for determining the contribution load-shifting DR provides to generation adequacy is proposed in [94] and a framework for determining capacity credit (or CV) metrics of DR is presented in [99]. Crucially, the authors of [94] and [99] take account of load restoration, or energy recovery as it was referred to in the previous chapter, which is a key characteristic of many DR resources, in conjunction with consumers’ constraints. However, load-shifting is performed via a load-scheduling algorithm with the objective of minimizing peak load, omitting consideration of the operational decisions from the supply-side of the power system, therefore the supply-side and the demand-side are not co-optimized. The adequacy indices are determined via a sequential Monte Carlo (MC) simulation which alters generation availability. In [94], the MC simulation is used to randomly create a time series of generator availability status which is then compared with the system load

to obtain adequacy metrics. Notably, provision of reserve from the DR resource is excluded.

This chapter contributes to the literature by estimating the CV of a specific load-shifting resource that can participate in the energy and AS markets but is also inherently contributing to generation adequacy. A load-shifting resource has an implicit contribution to generation adequacy, not by design, but simply as a result of the manner in which it is operating. It is this inherent contribution and the impact of DR operational decisions on this contribution which are examined in this chapter. This is achieved through the use of a thermal energy storage load model and a power system model, which permits co-optimization of the supply-side and the demand-side. Such a resource was chosen because it has thermal storage elements, making them very suitable for load-shifting, whilst still maintaining the ability to meet customer heating demands. For the remainder of the chapter, the 'load-shifting resource' will be taken to refer to *both* space and water thermal storage heating devices, with all devices considered to be one resource, centrally controlled.

A MC method is also employed in this chapter, incorporating the combined model, and applied to the estimation of the CV of the load-shifting resource. The heat demand model used in this chapter is derived from a Building Energy Model (BEM), while the power system model is a UCED algorithm. The coupled use of these models permits co-optimization of the supply-side and demand-side, detailed generation adequacy contribution analysis of one specific load-shifting resource, with resource-specific, time-related constraints, which is advocated in [100], and a consumer energy requirement profile for the resource for each point in time, representing an improvement over the work in [92] and [93]. Additionally, the inclusion of a UCED algorithm enables consideration of the interaction of the operation of the DR resource and the operations of the power system, as well as examination of the impact of reserve provision from the DR resource. The incorporation of the UCED algorithm extends the work presented [94].

The layout of this chapter is as follows: Section 4.2 outlines the two CV calculation methodologies employed in this chapter. In Section 4.2.1 the load-shifting resource and power system operations model, incorporating Unit Commitment and Economic Dispatch, used here are described, while the CV calculation approaches are introduced and explained in Section 4.2.2. Section 4.3 details the test system and the specific case study examined in this chapter. Section 4.4 presents the results of the

study, with emphasis on the contribution of the DR resource to generation adequacy and the effect the operation of the DR resource can have on the aforementioned contribution. Section 4.6 summarizes and concludes the chapter.

4.2 Methodology

Figure 4.1 provides an overview of the methodology adopted in this chapter, illustrating the use of a BEM to generate inputs for a load and power system operations model for the application of CV calculations. To reiterate, this chapter contributes to the literature by estimating the CV of a specific load-shifting resource that can participate in both the energy and AS markets but is also inherently contributing to generation adequacy. A description of the BEM which is employed in this chapter, the load-shifting resource and the power system operations model is provided in Section 4.2.1 and the methodology used to estimate the CV is described in Section 4.2.2.

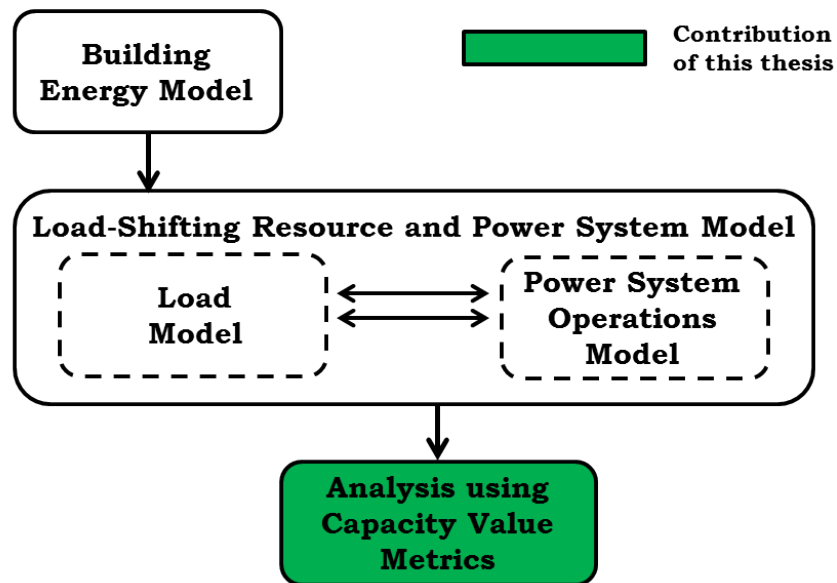


Figure 4.1: Overview of the methodology employed in this chapter

4.2.1 Load-Shifting Resource Model and Power System Operations Model

4.2.1.1 Building Energy Model

The development of ‘archetype models’, which are representative of a group of dwellings and dwelling loads, facilitates modeling and simulation of the performance

of building stock populations as a whole. This is achieved by considering the simulated energy performance from each individual archetype and the maximum share of buildings that could be represented by that archetype within the building stock considered.

A set of reference dwellings is modeled in detail by Olivier Neu (collaborator on the work presented in this chapter), using EnergyPlus, a deterministic building energy analysis and thermal load simulation program [101]. These models are converted into building performance simulation ‘archetypes’ by integrating high space and time resolution operational data. The models are validated by comparing against the Sustainable Energy Authority of Ireland’s Dwellings Energy Assessment Procedure methodology, in terms of annual space and water heating requirements and daily hot water requirements. The authors in [102] indicate that it is crucially important to understand consumers’ behavior prompting the use of time-of-use survey data in order to extract the behavioral patterns of building residents, in terms of occupancy and use of electrical appliances. In [103] the importance of considering weather conditions, building design, occupancy rate and behavior in the determination of heating requirement profiles is noted. The model by Olivier Neu *et al.* [104], which applies the Markov Chain Monte Carlo techniques developed in [105] to time-of-use survey activity data is employed in this chapter. This model, reproduced with permission from Olivier Neu, depicted in Figure 4.2 and discussed in [104], is used to develop activity-specific profiles for occupancy and domestic hot water requirements, thus taking into account consumers’ behavior.

The set of dwelling archetypes is used to generate annual profiles for space and domestic hot water heat demands on a fifteen-minute basis based on maintaining setpoint temperatures of 21 degrees and 65 degrees, respectively. These consumer end-use heating time series are converted to hourly-resolution and scaled for use in the power system operations model. They are represented in the power system operations model as $P_{heat,REQ,arch}^{space,t}$ and $P_{heat,REQ,arch}^{water,t}$, for the space and water heating requirements, respectively.

It should be noted that although only space and domestic hot water heat demand profiles are used in this chapter, a wide range of consumers’ energy requirement profiles can be generated from the BEM and used within a power system operations model. This would require a different DR resource model in the power system model,

but the combined methodology and proposed CV calculation approaches can be considered to be generic.

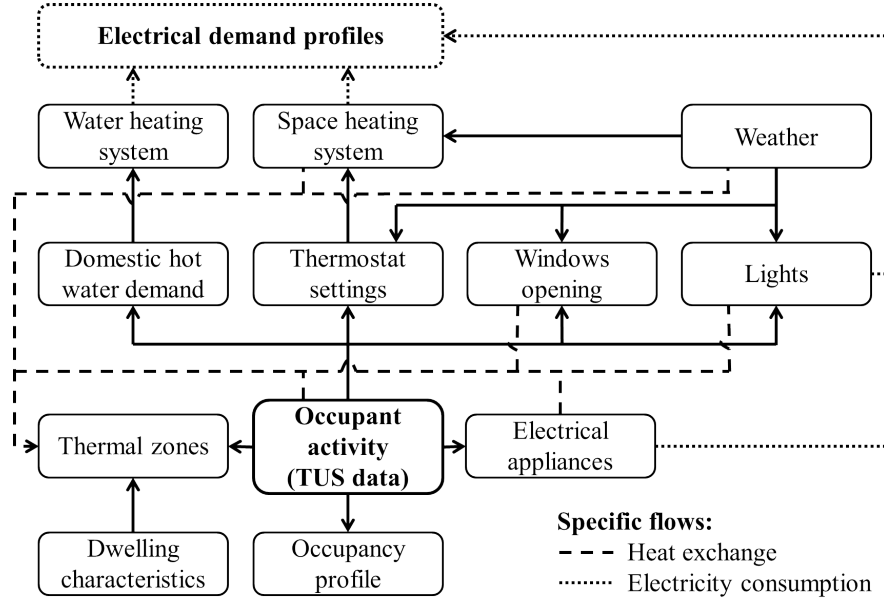


Figure 4.2: Bottom-up structure of the building model. Reproduced with permission from Olivier Neu *et al.* [104]

4.2.1.2 Power System Operations Model

A standard UCED, developed by Daniel Burke, is used with an hourly time resolution, a 12 hour look-ahead and the co-optimization of reserve and energy. A fixed parameter for the Value of Lost Load (VOLL) and a parameter for the Value of Insufficient Reserve (VOIR) for the inability to provide sufficient Primary Operating Reserve (POR) are included. The load-shifting resource is capable of providing both POR and downward reserve. The objective of the UCED is to minimize system operating costs, *Costs*:

$$\min Costs = \sum_t [\sum_g p_g^t \times U_g^t \times COST_g + \sum_h p_h^t \times U_h^t \times COST_h + (VOLL \times dem_{lost}^t) + (VOIR \times por_{unmet}^t)], \quad (4.1)$$

subject to a number of constraints, the most pertinent to the work presented in this chapter are:

$$\sum_g p_g^t + \sum_h p_h^t + dem_{lost}^t = DEM_{BASE}^t + \sum_{arch} (p_{elec,arch}^{space,t}) + \sum_{arch} (P_{elec,arch}^{water,t}), \quad (4.2)$$

$$p_g^t \leq P_g^{MAX}, \quad (4.3)$$

$$p_h^t \leq P_h^{MAX}, \quad (4.4)$$

$$por_{REQ}^t = por_{dr}^t + por_g^t + por_h^t + por_{unmet}^t, \quad (4.5)$$

$$por_{REQ}^t \geq 0.75 \times \max(p_g^t), \quad (4.6)$$

$$dres_{total}^t = dres_{dr}^t + dres_g^t + dres_h^t, \quad (4.7)$$

$$dres_{total}^t \geq 100. \quad (4.8)$$

The index g represents conventional thermal generation, while h represents hydro generation. Equation (4.2) is the supply-demand equation, where DEM_{BASE}^t is the original system load profile less the consumer heating requirements, which are satisfied through the operation of the DR resource. Equations (4.3) and (4.4) ensure that the power output of conventional and hydro generators, respectively, cannot exceed the installed capacity, while Equations (4.5) and (4.7) model the reserve provision from both the supply-side and the load-shifting resource. The value of 0.75 is chosen in Equation (4.6) as 75% of the largest infeed is the POR requirement constraint for the island of Ireland, as specified by the Irish power system operator, Eirgrid [?]]

In addition, the objective function is subject to generator reserve provision constraints and availability constraints due to forced outages. Specific DR resource constraints, which are described next (Equations (4.9) - (4.22)), are also incorporated into the UCED. The power system model employed here permits co-optimization of energy and AS, which was noted in Chapter 2 as being of particular importance when evaluating DR.

4.2.1.3 Load-Shifting Resource Model

The load-shifting resource, the electric thermal storage heating devices, are modeled within the power system operations model and scheduled in conjunction with the conventional and hydro generators to meet the system operational cost minimization objective function, Equation (4.1), whilst subject to a number of resource specific constraints. It is assumed that the system operator has direct control of the heating devices but must ensure that the consumers' requirements are always satisfied.

The equations describing the constraints and the operation of the load-shifting resource include power rating (Equations (4.10) and (4.16) for space and water, respectively), heat output constraints (Equations (4.11) and (4.17) for space and water, respectively) and energy storage constraints. Equations (4.9), (4.12), (4.13), (4.15), (4.18) and (4.19) model the thermal storage components of the devices in a similar manner to conventional storage devices taking account of the energy storage level, the electrical power consumed, the heat output and any associated losses. Equations (4.14) and (4.20) ensure that the consumer heating requirements are always satisfied, taking any losses into account. It should be noted that the losses associated with the space heating devices can contribute to meeting the space heating requirements (see Equation (4.14)).

Space Heating Devices

$$\frac{e_{arch}^{space,t}}{t} = \frac{e_{arch}^{space,t-1}}{t} + p_{elec,arch}^{space,t} - p_{heat,arch}^{space,t} - losses_{arch}^{space,t} \quad (4.9)$$

$$p_{elec,arch}^{space,t} < RATING_{elec,arch}^{space} \quad (4.10)$$

$$p_{heat,arch}^{space,t} < RATING_{heat,arch}^{space} \quad (4.11)$$

$$e_{arch}^{space,t} < E_{MAX,arch}^{space} \quad (4.12)$$

$$losses_{arch}^{space,t} = 1 - \eta_{arch}^{space} \times e_{arch}^{space,t} \quad (4.13)$$

$$p_{heat,arch}^{space,t} + losses_{arch}^{space,t} > P_{heat,REQ,arch}^{space,t} \quad (4.14)$$

Water Heating Devices

$$\frac{e_{arch}^{water,t}}{t} = \frac{e_{arch}^{water,t-1}}{t} + p_{elec,arch}^{water,t} - p_{heat,arch}^{water,t} - losses_{arch}^{water,t} \quad (4.15)$$

$$p_{elec,arch}^{water,t} < RATING_{elec,arch}^{water} \quad (4.16)$$

$$p_{heat,arch}^{water,t} < RATING_{heat,arch}^{water} \quad (4.17)$$

$$e_{arch}^{water,t} < E_{MAX,arch}^{water} \quad (4.18)$$

$$losses_{arch}^{water,t} = 1 - \eta_{arch}^{water} \times e_{arch}^{water,t} \quad (4.19)$$

$$p_{heat,arch}^{water,t} > P_{heat,REQ,arch}^{water,t} \quad (4.20)$$

DR Reserve Provision

$$res_{max}^{t,down} = \sum_{arch} RATING_{elec,arch}^{water} - p_{elec,arch}^{space,t} + \sum_{arch} RATING_{elec,arch}^{space} - p_{elec,arch}^{space,t} \quad (4.21)$$

$$dres_{dr}^{t,down} < res_{max}^{t,down} \quad (4.22)$$

$$por_{dr}^t < por_{dr,max}^t \quad (4.23)$$

$$por_{dr,max}^t = \sum_{arch} p_{elec,arch}^{space,t} + p_{heat,arch}^{space,t} \quad (4.24)$$

The variable $e_{arch}^{space,t}$ is the energy in storage at time t , $p_{elec,arch}^{space,t}$ is the electrical power consumed, $p_{heat,arch}^{space,t}$ is the heat output and $losses_{arch}^{space,t}$ are the losses. The parameters $P_{heat,REQ,arch}^{water,t}$ and $P_{heat,REQ,arch}^{space,t}$ are the consumer heating requirements, for water and space, respectively.

As can be seen in Equations (4.21) and (4.24), POR of the load-shifting resource is modeled as the operating electrical power consumed by the resource, while downward reserve capability is the difference between the installed capacity of the resource and the operating electrical power consumed.

The load-shifting resource operation as modeled here is operated with the objective of providing the least-cost solution for operating the power system. Alternative modes of operating the load-shifting resource may result in different contributions to generation adequacy, but they are not considered here.

4.2.2 Capacity Value Calculation Methodology

Two CV calculation methodologies are employed in this chapter and they are presented in this section.

4.2.2.1 Monte Carlo Method - Simulation Approach

As mentioned earlier, it is vital to consider the operations of the power system. Utilizing MC simulations permits consideration of the changes in power system operation that result from varying generator availability. The MC-based method employed here involves random sampling from an exponential distribution of the status of each of the conventional generators based on their Forced Outage Rate (FOR) and Mean Time to Repair (MTTR) data (see Equations (4.25) - (4.28)), to then determine the hours during which a generating plant outage occurs using Algorithm 1.

Equation (4.25) determines the average number of outages per year for generator, g , which is used to determine the parameter μ (Equation (4.26)) which represents the mean time between outage events (similar equations are employed for hydro generation). Equations (4.27) and (4.28) denote that random samples are taken from an exponential distribution with parameter μ and then rounded to give the hours in which generator outage events occur.

$$OUTAGES_g^{avg} = \frac{FOR_g * 8760}{MTTR_g} \quad (4.25)$$

$$\mu = \frac{1}{OUTAGES_g^{avg}} \quad (4.26)$$

$$\tilde{R}^{event_{num}} \sim exp^{\mu} \quad (4.27)$$

$$eh = round(\tilde{R}^{event_{num}} \times 8760) \quad (4.28)$$

Since the interaction between the operational decisions of the power system and the operation of the DR resource including DR reserve provision is of key interest here, it was necessary to perform a UCED in each MC simulation. The availability of each generator is simulated for every hour of the year. These generator availability profiles are then used in the power system operations model to determine the generation adequacy metrics of the power system (see the algorithm in Figure 4.3).

```

for all gen do
    Calculate average outages (Equation 4.25);
    Initialize outage event number:  $event_{num} = 1$ ;
    Generate random variables (Equation 4.27);
    Determine next event hour ( $eh$ ) (Equation 4.28);
    while  $t < 8760$  do
        if  $t = eh$  then
            Outage occurs;
             $U_g^t = 0$  until  $t + MTTR_g$ ;
             $event_{num} = event_{num} + 1$ ;
             $eh = eh + MTTR_g + round(\tilde{R}^{event_{num}} \times 8760)$ ;
        else
            There is no outage:  $U_g^t = 1$ ;
             $t = t + 1$ 
        end
    end
end

```

Algorithm 1: Creating generator availability profiles

The number of hours during which there is a loss of load event is represented by the Loss of Load Hours (LOLH), $LOLH$, while the sum of the magnitude of the loss of load events over the year is given by the Loss of Load Magnitude (LOLM), $LOLM$.

These metrics are obtained using Algorithm 2. The cumulative average of these two metrics are calculated after each simulation and the simulations are repeated until the values have converged, that is, until the coefficient of variation of the metrics has fallen below the tolerance level. Once the values have converged, the converged values of the LOLM and the LOLH are taken to be the Loss of Energy Expectation (LOEE) and the LOLE, respectively.

```

for each year do
  for each  $t$  do
    if magnitude of loss of load event  $> 1MW$  then
       $LOLM^t =$  magnitude of loss of load event;
       $LOLH^t = 1$ ;
    else
       $LOLM^t = 0$ ;
       $LOLH^t = 0$ ;
    end
  end
   $LOLM^{year} = \sum_t LOLM^t$ ;
   $LOLH^{year} = \sum_t LOLH^t$ ;
end

```

Algorithm 2: Determining adequacy metrics

The ELCC metric in the MC-based approach is determined iteratively using the algorithm in Figure 4.4 until the long-run generation adequacy metrics for the case without load-shifting and for the case with load-shifting are equal. The CV of the load-shifting resource is then estimated by dividing the ELCC by the installed capacity of the load-shifting resource, Equation (4.29), as is the convention for wind, wave and tidal generation.

$$CV = \frac{ELCC}{Cap_{Installed}} \quad (4.29)$$

This differs from Equation (3.6) in Chapter 3 where the peak availability of the DR resource is the denominator. This is because the installed capacity of the devices

examined in this chapter are known, while no such information was available for the DR resources examined in Chapter 3.

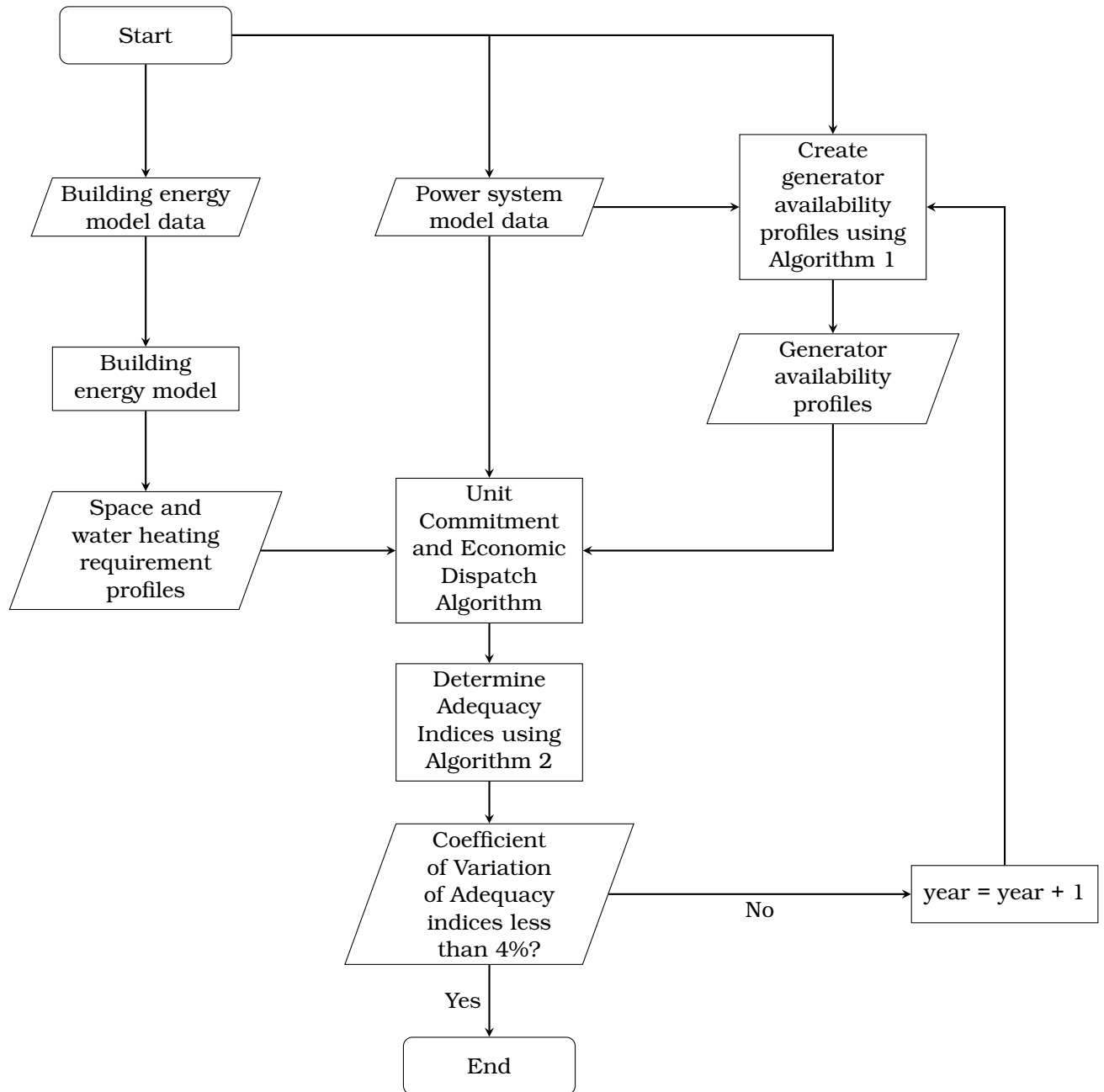


Figure 4.3: Monte Carlo based UCED assessment of power system adequacy indices incorporating outputs from the EnergyPlus building energy model

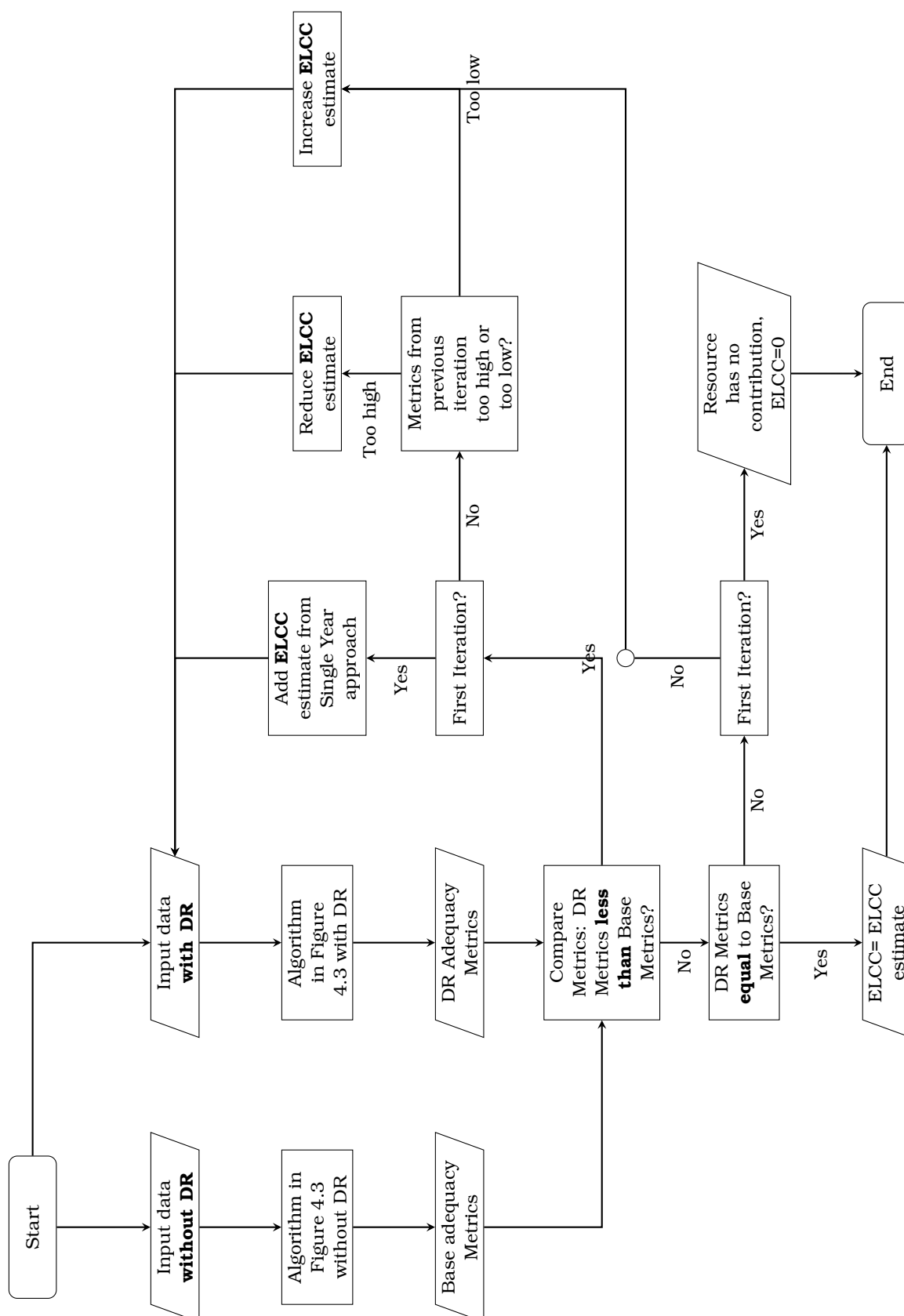


Figure 4.4: Methodology for determining ELCC using iterative approach

4.2.2.2 Single Year Method

This method is a simplification of the full MC-based methodology in an effort to reduce computational time. This approach runs one full year of the UCED algorithm with the load-shifting resource, assuming all the generating plants on the system are fully available. The time series of the new system demand, Dem_{new}^t , is obtained following a single run of the UCED simulation by incorporating the electrical requirement for the load-shifting resource with the baseline load of the power system (Equation 4.30):

$$Dem_{new}^t = DEM_{BASE}^t + \sum_{arch} (p_{elec,arch}^{space,t}) + \sum_{arch} (p_{elec,arch}^{water,t}) \quad (4.30)$$

where DEM_{BASE}^t is the original system load profile less the consumer heating requirements, as before, and where $p_{elec,arch}^{space,t} + p_{elec,arch}^{water,t}$ is the electrical power requirement from the load-shifting resource. This new load profile is used in the conventional approach for determining the ELCC of a resource and hence the CV, and in a similar manner to the approach employed in [70, 81, 83] for wind, wave and tidal barrage generation, respectively. This method creates a COPT and compares this table with the load profile to calculate the generation adequacy metrics. A fixed amount of load (ELCC) is added to each hour of the new load profile, the generation adequacy metrics are recalculated and the process is repeated until the adequacy metrics are equal to the reference metrics, which are the generation adequacy metrics of the case without the load-shifting resource. This method builds upon the methodology proposed and used in Chapter 3 and in [71].

4.3 Test System

Ireland was chosen as a test system because it is a small, island system and because of the ready availability of data pertaining to the generating units. Additionally, capacity payments are currently available for DR aggregators in Ireland based on their availability to reduce their demand, though they may not be called to respond. Although this differs from the resource considered in this chapter, the existence of capacity payments for demand-side resources in Ireland does indicate that there is potential for load-shifting resources to participate in the future. Furthermore, the archetype models and consumer end-use heating time series' developed by Olivier Neu are predicated on the building stock in Ireland and the Irish National time-of-use survey.

4.3.1 Irish power system

The Irish system portfolio comprises 53 conventional thermal plants and 15 hydro plants, with a total installed capacity of 8891 MW (see Table A.2 in Appendix A). The load time series used in the power system operations model and the weather data used in the EnergyPlus model are all time-synchronized and based on 2009 data. This is to ensure that any correlations between weather and system load are incorporated into the analysis, as is recommended in [73]. The same weather data is used for all of the archetypes models.

It should be noted that the Irish power system currently has an over-capacity [106] of over 1900 MW with a maximum all-time system peak load of 6900 MW. Due to this over-capacity, there is little need for additional resources and thus it is likely that the contribution of a DR resource to generation adequacy would be limited. Thus, the peak load is increased to reduce the generation margin and to represent a system which requires additional resources. A peak load of 8490 MW was found to be the peak load at which the generation portfolio used in this chapter results in an LOLE equivalent to the target of 8 hours/year for the power system on the island of Ireland [106]. Consequently, a peak load level of 8500 MW is employed, which reduces the generation capacity margin to 391 MW. The 2009 system load data was normalized and scaled linearly to 8500 MW.

The VOLL for this study was chosen to be €10 million per MW * and the VOIR was chosen to be one quarter of the VOLL. Such a high level of VOLL was selected due to the fact that the optimization gap for the UCED algorithm was relaxed to 2% in an effort to speed up the simulation. In order to minimize the amount of load lost due to the wide optimization gap, rather than purely due to generation adequacy issues as is the intention, the VOLL parameter was increased. With a low level of VOLL, a situation may occur whereby costs are minimized by incurring a loss of load rather than starting an expensive generating unit, as would happen in reality. Thus a high level of VOLL guarantees that load is never shed for economic reasons. The optimization gap employed for the single year UCED used as part of the Single Year Method was 0.5%. The UCED model is solved on an Intel Xeon 2.7 GHz processor with 256 GB of RAM. The coefficient of variation used for the MC-based method was 4%. The UCED algorithm was developed using the Generic Algebraic Modeling System

*Estimating VOLL is a nearly impossible task, according to [43].

(GAMS) and the CPLEX solver. Matlab was used to perform the pre and post-processing and data analysis.

4.3.2 Irish Housing Stock and the Load-Shifting Resources

Five Irish reference dwellings are considered over different construction periods, representative of the majority of the national building stock in Ireland. Surveyed data within both the Irish and the UK housing stocks is used to estimate the maximum number of dwellings with electrical space and domestic hot water heating systems. This is necessary to facilitate the analysis of the scale up of the resource from individual archetype dwellings to a national scale [104, 107]. Apartments are deemed to be the most relevant dwelling types for the installation of thermal storage space heating systems. For this analysis, it is assumed that there are 100,000 apartments in Ireland with electric space heating, both direct resistive heaters (15%) and storage heaters (85%)[†]. These storage heaters operate on a fixed schedule. This stock of resistive and storage heaters is replaced by electric thermal storage heating devices capable of load-shifting and reserve provision and whose operation is optimized on a daily basis within the power system operations model. Similarly, these apartments are assumed to have electric water heaters and these are also displaced by thermal energy storage water heating devices. For the 100,000 apartments, this equates to an installed capacity of thermal storage heating devices, both space and water, of 695 MW.

The five reference dwellings are converted into building performance simulation archetypes by integrating the necessary operational data with a high space and time resolution. Six of these archetypes are utilized in this thesis to represent the 100,000 apartments. These archetypes differ depending on whether they are mid-floor flats or top floor flats and whether they are assumed to be new or existing apartments.[‡]

The operational data subset used is built upon the bottom-up approach proposed by Neu *et al.* [104]. Markov Chain Monte Carlo techniques are applied to the 2005 Irish National time-use survey (TUS) activity data [108] to develop activity-specific profiles for occupancy. From such profiles, domestic hot water requirements and demand for electrical appliances are obtained [104, 107], thus taking into account consumers' behavior. The electrical demand profiles obtained based on the 2005 TUS

[†]This information was obtained through direct correspondence with the manufacturer of these specific thermal storage heating devices.

[‡]For further detail relating to the breakdown of the archetypes see Appendix E

data were found in [104] to compare well with the Smart Metering Trial data which took place in Ireland in 2009-2010, thus justifying the use of the TUS data in the BEM. Additionally, the 2005 TUS activity data is largely representative of the year 2009, ensuring time synchronization with the 2009 demand and weather data, as was recommended by [73] and discussed in Chapter 2.

As previously, stated, the archetypes capture variations in heat requirements for domestic hot water and space heating on a fifteen-minute basis, for different occupancy profiles. The occupancy profiles are based on whether it is assumed the occupants are working full-time, and thus part-time occupants, or whether they are full-time occupants of the dwelling. The heat requirements represent the amount of energy (Kilowatt-Hour (kWh)), or the average power requirements (Kilowatt (kW)), that are required from the heating system(s) in order to meet the set-point temperatures scheduled for space and domestic hot water. These consumer end-use heating time series are converted to hourly-resolution for use as constraints in the power system operations model, ensuring that consumer heating requirements are always met (see Equations (4.14) and (4.20)).

It is important to be aware of how the base case selection and the operations of both the load-shifting resource and the power system affect the CV of the resource. Understanding the effects of residential occupancy profiles and whether or not the resource is capable of providing reserve could enable tailored DR programs to allow for better exploitation of the resource, and indicate what services the resource should provide. A number of different cases are examined in this analysis, including:

1. Full-time occupants

A with DR reserve provision - in this case, the DR is capable of load-shifting and assisting in meeting the system reserve requirements

B without DR reserve provision - in this case, the DR is only capable of load-shifting

2. Part-time occupants

A with DR reserve provision - in this case, the DR is capable of load-shifting and assisting in meeting the system reserve requirements

B without DR reserve provision - in this case, the DR is only capable of load-shifting

4.4 Results

Sections 4.4.1 and 4.4.2 will focus on the results pertaining to generation adequacy and CV estimates, while sections 4.4.3 to 4.4.6 will discuss the results of the different cases examined.

4.4.1 Impact of DR on Generation Adequacy and DR Capacity Value Estimates

Figure 4.5 show the LOLH and LOLM metrics obtained from the MC simulations for one of the cases examined. The markers represent the average value for every thirty MC-UCED simulations. As can be seen, there is a marked improvement in the adequacy metrics following the addition of the load-shifting resource.

It is found that a reduction, and therefore an improvement, in both the LOLH values and the LOLM of between 20% and 30% can be achieved by displacing the current installment of electric water and space heating devices from 100,000 apartments in Ireland with the electrical thermal storage heating devices, the load-shifting resource. The case without load-shifting resource has a higher LOLH value compared to case with the load-shifting resource, illustrating the reduction in LOLH. As can be seen, in both Figure 4.5, when the load -shifting resource is combined with an ELCC of 62 MW, the values converge to approximately the same adequacy metrics obtained without the load-shifting resource. ELCC estimates for the other cases were found in a similar manner and the results are presented in Table 4.1.

The LOLH and LOLM improvement attributable to the resource is as a result of the decoupling of the electrical and thermal requirements, in comparison with the original space and water heating devices in the 100,000 apartments. The decoupling succeeds in shifting a portion of the heating requirement away from peak system load hours, thereby reducing the likelihood of loss of load events. This decoupling is illustrated in Figure 4.7 where it can be seen that the load-shifting resource operation (blue trace) is considerably different from the consumers' heating requirement (orange trace). Additionally, there is an overall load reduction due to the fact that the new devices are more efficient, but this plays a minor role in the generation adequacy improvement.

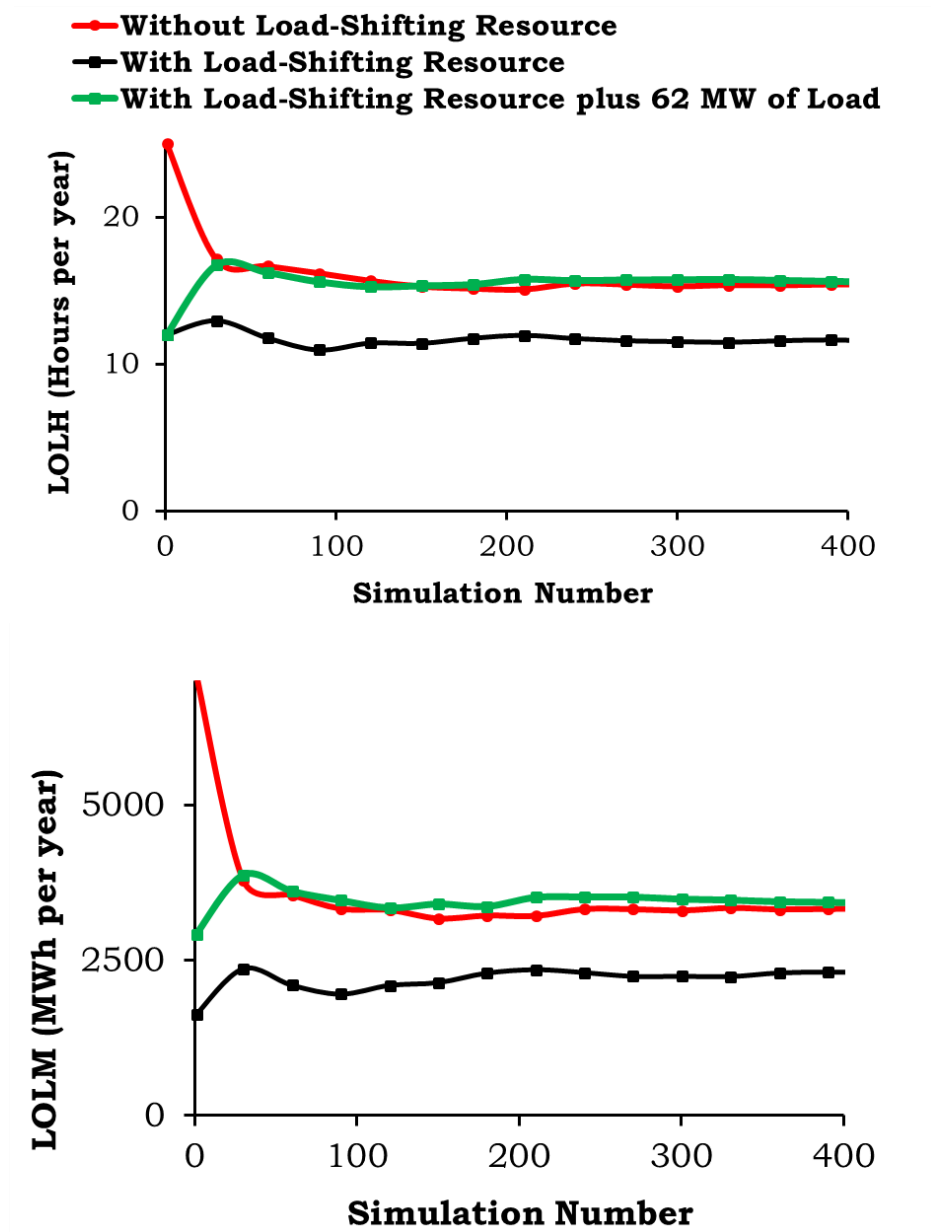


Figure 4.5: LOLH and LOLM convergence using Monte Carlo-based approach for the case with full-time occupants and reserve provision from the demand response resource

Table 4.1: Monte Carlo-based estimates compared with the Single Year method estimates for all cases

Case	Metric	MC	Single-Year
Full-time occupants	ELCC	62 MW	59 MW
with DR Reserve	CV	9%	8.5%
Part-time occupants	ELCC	43 MW	37 MW
with DR Reserve	CV	6%	5.3%
Full-time occupants	ELCC	78 MW	75 MW
without DR Reserve	CV	11%	10.8%
Part-time occupants	ELCC	60 MW	55 MW
without DR Reserve	CV	8.6%	7.9%

Figure 4.6 illustrates the effect load-shifting can have on the load duration curve and thus on generation adequacy; where possible, the addition of the load-shifting resource succeeds in reducing load for the majority of the top 100 load hours. These are the hours when the loss of load probability is higher and thus shifting away from these hours reduces the risk of losing load. Consequently, the load-shifting resource is contributing positively to system generation adequacy. However, crucially, as illustrated by the overlapping heat requirement and heat output in Figure 4.7 and 4.8, the heating requirements of the consumer are always met.

The work in [92] also shows a measurable improvement in system generation adequacy through load-shifting. However, the improvement observed (50%-90%) in [92] is greater than the improvements observed in this chapter. This is for two reasons. Firstly, in [92] load-shifting is applied to the entire load on the system, while here load-shifting applies to one specific type of resource, in 100,000 apartments. Secondly and more importantly, the work presented here captures the physical constraints of the devices and consumers' requirements through the application of the BEM.

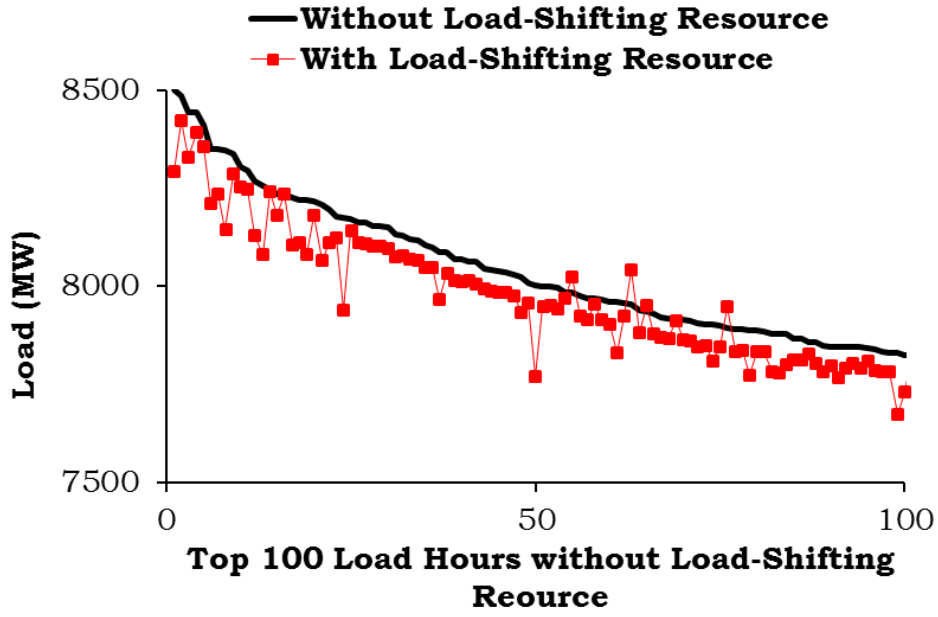


Figure 4.6: Load duration curve for the top 100 load hours with and without the load-shifting resource

4.4.2 ELCC and CV Estimates

It is clear that the resource does provide a reduction, and therefore an improvement, in the generation adequacy metrics of the power system. Consequently, by definition, the resource is contributing to generation adequacy and thus must have an ELCC value and, hence, a CV. Using the MC-based method, an ELCC value was obtained for all different cases examined. These values were found to be between 43 and 78 MW (Table 4.1) and this suggests CVs between 5% and 11%. These values compare well with the ELCC estimates presented in [99] which suggests a CV of about 12.5% for a similarly sized DR resource. While this is undoubtedly a positive contribution to generation adequacy, it is relatively small. However, the magnitude of the CV is comparable to other similar resources: Wind (13% - 23% [70, 83]); Wave (2% - 22% [83]) and Tidal (3.3% - 7.6% [81]). This is unsurprising given the similarities between load-shifting and renewables resources mentioned earlier. In contrast, however, the CV of the load-shifting resource is very low in comparison with conventional generating plants: 71%-97% [87]. See Table 3.1 in Chapter 3.

There are a number of factors driving the CV results for this resource. The load-shifting resource is inherently linked with the consumers' requirements and thus is

not always capable of shifting demand away from the hours when the power system is vulnerable to loss of load events (Figure 4.6). While certain hours of the year may have high probabilities of losing load associated with them, the load-shifting resource may not be in a position to shift demand away from those hours. It is found that in some cases during hours where there is a high LOLP there is still considerable operation from the load-shifting devices as they are constrained to meet the requirements of the consumers. The impact of the consumers' constraints is illustrated in Figure 4.7, which shows that there is operation of the load-shifting resource at hours 9pm and 10pm, despite the fact that the system load level is at or close to 80% of the system peak. Normalized system load highlights when the system peak hours occur.

Figure 4.8 shows that, for those same hours, there is a heating demand that cannot be met by the heat that has been previously stored by the device since the storage element has been depleted. Consequently, in order to satisfy the consumers' requirements, in some instances, the device must operate as a direct resistive heater at certain hours of the day.

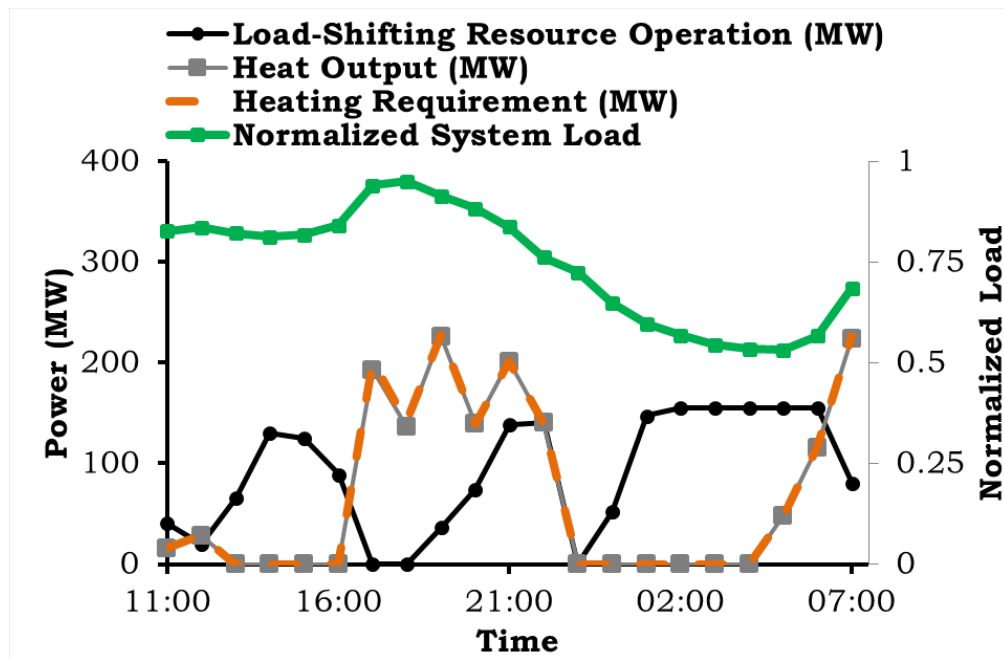


Figure 4.7: Typical operation of the load-shifting resource, heat output and heating requirements.

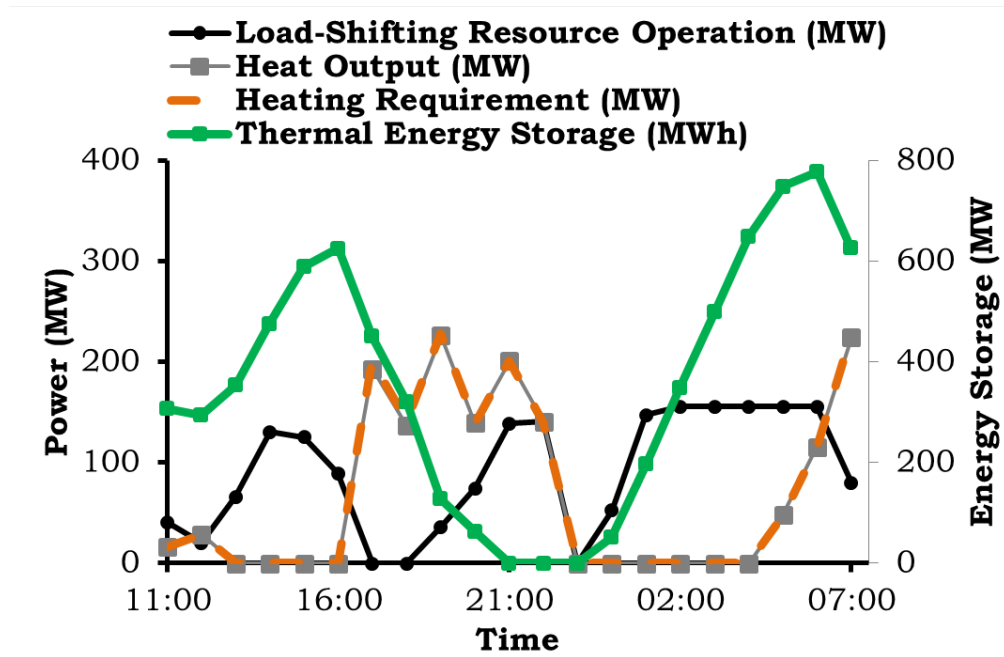


Figure 4.8: Typical operation of the load-shifting resource, heat output, heating requirements and the energy storage component

Power system operations play a major role in influencing the CV results. On many occasions it may not be economic for the resource to shift away from high LOLP hours and it was found that hours of high LOLP are not always coincident with hours of high system marginal price (SMP). Figure 4.9 illustrates this.

As can be seen from Table 4.1, the CV estimates obtained for the electric water and space heating resource considered here are quite low. While these estimates are lower than might be anticipated for a load-shifting resource, comparison with the estimates obtained for similar resources in Chapter 3 illustrates that these results are consistent. In Chapter 3, preliminary estimates for residential water heating resources suggests CVs between 26% and 67% when energy recovery is not included. However, when an energy recovery constraint is incorporated into the analysis, the CV estimates drop to between 1% and 36%, depending on the resource and power system considered. Thus, when a resource operation is constrained to ensure consumer requirements are satisfied and with the inclusion energy-limitations it is unsurprising that the CV of such a DR resource falls to between 5% and 11%, the values obtained in Table 4.1.

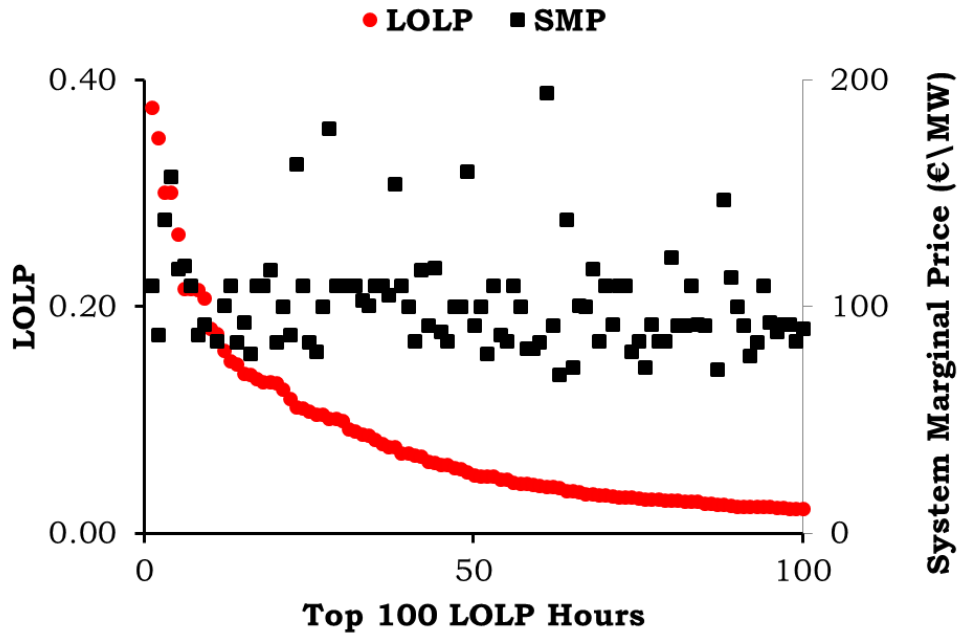


Figure 4.9: LOLP duration curve versus the SMP

4.4.3 Impact of Base Case

Using the MC-based method, two different base cases are examined. The first is where the heating demand in the apartments, both space and water, is met entirely by direct resistive heating. The second is the base case used in Section 4.4.1, where there are both direct resistive heaters (15%) and storage heaters (85%)(which operate on a fixed schedule) meeting the space heating demand in the 100,000 apartments, as is assumed to be the current scenario in Ireland. The water heating demand in this second case continues to be satisfied by direct resistive heating. For the two different base cases, the same model is employed, but the input parameters are changed as necessary.

Table 4.2: Monte Carlo-based results for the case of full-time occupants, with load-shifting resource capable of providing reserve for two different base cases

Metric \ Case	100% Resistive	85 % Storage & 15% Resistive
	Resistive	& 15% Resistive
ELCC	180 MW	62 MW
Capacity Value	26%	9%

Taking the ELCC values calculated, Table 4.2, and dividing by the installed capacity of the load-shifting resource, using Equation (4.29), suggests a CV of up to 26% for the base case with 100% direct resistive heating (DRH). However, for the base case with 85% storage heating devices (which operate on a fixed schedule) and 15% DRH, which reflects the current scenario, the CV is 9%, as seen earlier.

Heating demands often occur at or close to evening peaks, at times when the system has a high LOLP. Consequently, if 100% of the heating demand in the 100,000 apartments is met by DRH, the system peaks will be higher and the system will have a lower generation adequacy. If this demand is satisfied by a combination of 85% storage heating devices and 15% DRH devices, some of the heating demand that coincides with the system peak is already shifted away from high LOLP hours and the system has a higher generation adequacy than in the case with 100% DRH. Thus, using the base case with 100% DRH sets a generation adequacy baseline that is higher than the generation adequacy baseline set when combination of 85% storage heating devices and 15% DRH devices is employed. Consequently, more load can be added to the system until the generation adequacy baseline is met in the former case and consequently it has a higher ELCC and thus CV, as seen in Table 4.2.

4.4.4 Impact of Methodology

Due to the considerable computational effort involved in the MC-based methodology, the Single Year method was also used to assess its feasibility for utilization in this part of the analysis. As illustrated in Table 4.3 the MC-based approach and the Single Year Method produce only slightly different results; the MC method provides estimates which are slightly higher than the Single Year method, as expected. Comparing the methodologies based on computational intensity, however, shows a stark difference between the two approaches; the MC-based approach takes more than 500 hours to produce ELCC results, while the runtime for the Single Year approach, including the UCED, is less than 1 hour. Consequently, the remaining sensitivities were performed using the Single Year method.

Table 4.3: ELCC results and computational times for the base case of space heating demand met by storage (85%) and direct resistive heating (15%) for the Monte Carlo-based (ELCC from Table 4.2) and Single Year methods

Method	ELCC	Time
Monte Carlo	62 MW	500 hours
Single Year	59 MW	1 hour

The Single Year Method results in a slight underestimation of the ELCC of the load-shifting resource with respect to the MC-based method. The reason for this case can be explained by applying the component of the Single Year Method for calculating the ELCC to each MC simulation. In the Single Year Method, all generating units are assumed to be available at all times and the system is marginally more reliable than in the MC-based method where generating units are randomly on forced outage. Thus, when the component of the Single Year Method for calculating the ELCC is applied to every simulation year from the MC simulation, there is a range in the ELCC estimates. This is clearly illustrated in Figure 4.10.

In Figure 4.10, high ELCC estimates coincide with MC simulation years where there are considerable numbers of generating units on forced outage, while lower ELCC estimates correspond to years where fewer, or no, generating units are forced out. As will be noticed in Figure 4.10, the lowest ELCC estimate when the Single Year Method is applied to each MC simulation is equivalent to the Single Year Method, where all generating units are available.

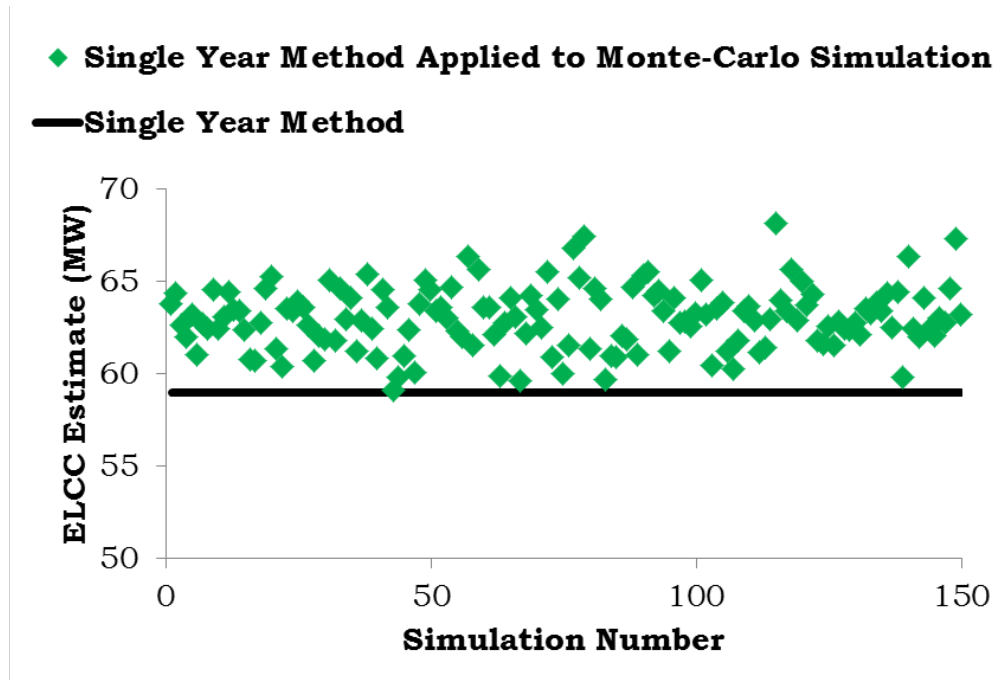


Figure 4.10: ELCC Estimates: Example of Single Year Method applied to each individual Monte Carlo Simulation

4.4.5 Impact of Reserve Capability

Table 4.1 illustrates the larger ELCC values of the load-shifting resource in the case where the resource is not capable of providing reserve. From the system operator point of view, providing reserve may be the best manner in which to operate the load-shifting resource. However, it has been found here that doing so impacts upon the CV of the load-shifting resource (see Table 4.1). This is because a proportion of the resource is held back from providing load-shifting and consequently there is a trade-off between providing reserve and contributing to generation adequacy.

As can be seen in Figure 4.11, while the resource succeeds in load-shifting in both cases, when the load-shifting resource is capable of providing reserve it has less load-shifting potential to shift away from peak system load hours (as illustrated by the normalized system demand) in comparison to the case where reserve is not provided. This is a result of the fact that when the load-shifting resource provides reserve, the optimization is constrained by the upper limit on the installed capacity of the resource which in turn limits the amount of load-shifting. Thus, by not providing reserve, the resource is inherently more flexible.

Additionally, in the case where reserve is provided, there is significantly more electrical demand from the load-shifting resource at or close to the peak demand

hours. This results in a lower ELCC value thereby highlighting the reserve-generation adequacy trade-off. The type of service, or combination of services, required will be system specific and will depend on the energy markets and the consumer preferences. Future work should determine a DR aggregators' optimum services provision portfolio in the case where the resources are capable of playing in multiple markets. Scoping work of this nature is presented in Chapter 5.

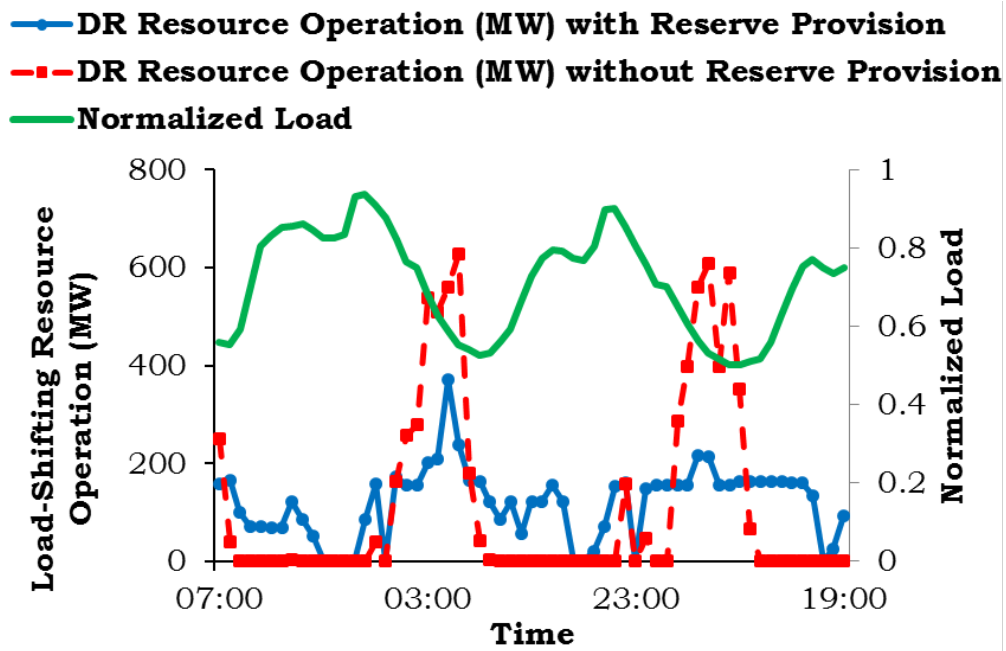


Figure 4.11: Operation of the load-shifting resource, with and without reserve capability, for the case with full-time occupants.

4.4.6 Impact of Occupancy Profile

The type of occupancy is found to have an effect on the estimated CV of the resource (see Table 4.1). In the case of the full-time occupants, the resource has a greater ability to shift away from peak load hours than would be the case for part-time occupants. This is a result of the interplay between the different constraints on the resource and the objective of minimizing system operational costs. From Figure 4.12 it can be seen that the full-time occupants have a relatively continuous space heating demand over most of the day during January, which is similar for most of the year, except the summer months, and thus it is cost-effective to store high levels of energy early in the day, while prices are low, in anticipation of the heating requirements and

energy storage losses during the day, particularly when the losses associated with the space heating devices can contribute to the space heating requirements (see Equation 4.14).

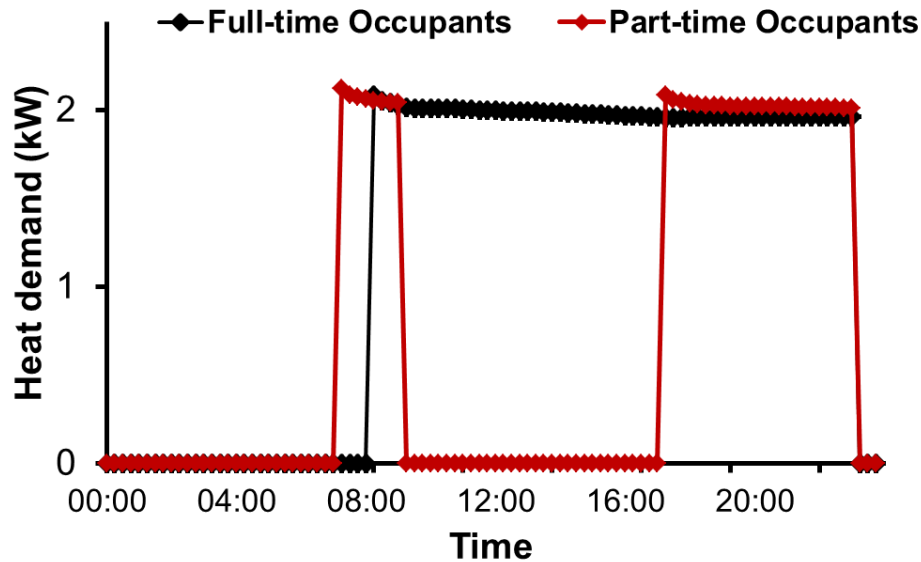


Figure 4.12: Average January weekday space heating requirements for the two types of occupants

In contrast, part-time occupants typically have more discontinuous heating demands during the day. Thus, charging the storage components early in the day would be a more expensive option for the part-time occupancy profiles, as there would be energy losses throughout the day but no significant heating requirements to satisfy. Consequently, there are greater amounts of energy available in the storage components in the case of the full-time occupants, compared to part-time occupants, as the evening peak approaches. Thus, there is greater scope for shifting. This, as would be expected, has an impact on the ELCC estimates and the CV for the resource.

An understanding of the impact differing occupancy profiles can have on the contribution of load-shifting resources to the power system could enable tailored programs and could indicate which customers are most suited to engaging in DR programs.

4.5 Discussion

It should be noted that numerous years worth of data are required in order to accurately determine the CV of a resource. According to [73], data for ten years is required in order to attain results for the CV of wind with minimal over and under estimations. Ideally, multiple years should also be analyzed when determining the CV of DR resources. However, the development of the archetype models and the integration of high space and time resolution operational data to create building performance simulations in EnergyPlus, is a very involved and time-consuming undertaking. Additionally, such an undertaking requires relying on numerous external sources of data. Thus, due to the computational constraints and due to the lack of sufficient data, only one year of data was analyzed here. Thus, the results in this paper are specific to that one study year, 2009. Yet, while it is agreed that the capacity value of resources change depending on the year examined, as discussed in [73], the results for the resource in this paper should not change dramatically, although more work is required. This is a result of the fact that, as discussed in this chapter, the resource is heavily constrained by the need to meet consumers' heating requirements at each point in time. Thus, while only one study year is analyzed here, the results succeed in highlighting the impact of including operational constraints of both the DR resources and the power system. Additionally, the results indicate that there is a requirement for future work on the CV of DR resources with more data as well as work on improving the calculation methodologies that require less data but are as accurate.

Some of the limitations of the BEM used in this chapter are related to the reliance on external data for a single year, in conjunction with the fact that, by definition and due to the complexity of each building model, the individual archetypes are representative building models. Ideally, electricity demand in households would be represented on a one-minute time resolution [104] to capture the necessary details and volatility. However, from a power system perspective and because each of the representative loads are to be scaled and aggregated, a fifteen minute resolution was used. Additionally, the TUS is only based on results from analysis of 1089 adult participants in 567 households [104]. However, as noted in Chapter 2, limited data pervades the area of DR at present and thus, nevertheless the use of this BEM in conjunction with CV analysis represents a significant contribution to the literature.

It is acknowledged that the LOLH and LOLM values obtained in the analysis presented in this chapter, which are estimates for the LOLE and LOEE metrics, are high, perhaps higher than would be acceptable for power system operators [109]. However, as noted in [20], when the adequacy of a power system is low, i.e. when LOLE and LOEE values are high, the higher the value of the added resource, whether it is a supply-side resource or a DR resource. Thus, the case examined in this chapter could be viewed as quite a competitive environment for DR and determining the CV of a DR on a power system with a low level of adequacy will immediately indicate if a DR resource can have a considerable contribution to the power system. Altering the base case used (Section 4.4.3) essentially decreased the adequacy of the system and illustrated that the DR resource examined has a slightly greater contribution to generation adequacy. However, even in this case of very low adequacy, where a resource should have a high CV, the CV estimate for the DR is low, indicating that there is little potential for this load-shifting DR resource to have substantial contributions to generation adequacy.

It is also recognized that while consumers' heating requirements can be guaranteed at all times during normal power system operating conditions, this cannot be assured during periods whereby the reserve committed by the DR resource is actually deployed, i.e. during emergency operation. This is because the amount of reserve committed by the DR resource is now interruptible load, and is shed. Thus, it is acknowledged that it may not be possible to ensure that heating requirements are satisfied at all moments during emergency operation. However, on the Irish system for example, POR is only deployed over a very short time-frame, typically 0-15 seconds post-event and events which trigger the use of POR are very rare. The short-time frame over which POR is deployed prohibits its consideration in a model with a time resolution of one hour, as is the case in this chapter. Additionally, it is assumed here that any inconvenience caused to customers as a result of failure to meet their heating requirements during emergency operation is compensated by the revenue they receive by permitting their devices to be committed to providing POR and downward reserve.

It should be noted that numerous years worth of data are required in order to accurately determine the CV of a resource. According to [73], data for ten years is required in order to attain results for the CV of wind with minimal over and under estimations. Due to the computational constraints and due to the lack of sufficient

data, only one year of data was analyzed here. Thus, the results in this chapter are specific to that one study year, 2009.

It will be noted that there is no wind generation considered in this chapter. Future work on estimating the CV of DR resources should consider the interaction of the DR resource with wind generation. One of the challenges with incorporating wind generation into generation adequacy-related analysis, with the specific intention of determining the CV of a DR resource, is that there is a close interaction between wind generation and DR operation [3, 9, 110]. Consequently, it can be difficult to ascertain whether the reliability improvement is as a result of the wind generation only, the DR resource only or both wind and DR. This complicates the allocation of CV estimates to the DR resource. Thus, perhaps a new methodology is required in order to enable the separation of the different contributions to adequacy of the different resources.

4.6 Conclusion

This chapter provides a greater understanding of the impact of load-shifting resources on generation adequacy by giving an indication of the CV a specific load-shifting resource which can provide multiple services. The CV estimates determined in this chapter indicates that load-shifting can have a positive impact on system generation adequacy. However, the results highlight that the CV estimates are influenced by the operation of the resource, i.e. whether or not it is capable of providing reserve, and by the consumers' constraints. This concurs with the work in [53], where it is stated that DR operation is severely restricted by the energy constraints and the requirement to meet end-consumer's need.

The trade-off between the multiple services provided by this resource suggests that there is a pressing need for market-based analysis to determine the optimal portfolio of services a load-shifting resource should provide and to determine the impact on electricity markets of DR participation. This trade-off and desire to investigate the optimal portfolio for DR is the motivating factor behind the work presented in the next chapter, Chapter 5.

Market Impacts of Demand Response Participating in Multiple Electricity Markets

This chapter is the result of collaboration between the thesis author, Muireann Lynch and Mel Devine (Economic and Social Research Institute). It largely based on a journal paper which is currently in preparation entitled 'Impact of Demand Response Participation in Energy, Reserve and Capacity Markets '. A full citation can be found in the section titled "Publications Arising".

5.1 Introduction

On account of the fact that DR can contribute to generation adequacy and thus has a CV, as shown in [71, 94], as well as in Chapters 3 and 4, albeit low for load-shifting DR resources in comparison to thermal generation, it is necessary to consider how a load-shifting DR resource would impact upon electricity markets. The ability of DR to contribute to three electricity markets (energy, reserve and capacity markets) individually was discussed in Chapter 2. While, there has been considerable work reported in the literature in the area of DR participation in electricity markets, there has been little work exploring the impact of DR participating in all three electricity markets simultaneously. This chapter of the thesis aims to inform the discussion on these topics.

As was discussed in Chapter 2, a holistic approach to analyzing and evaluating DR should be adopted, including co-optimization of energy and ancillary service markets.

This chapter considers an electricity system with energy and reserve markets, as well as a CRM in the form of a capacity market (see Figure 5.1). Multiple firms with different initial endowments of generating capacity participate in the three electricity markets in an effort to maximize their profits. Each firm decides the level of generation, reserve provision, capacity bid, investment and exit, subject to physical constraints, operating, maintenance and investment costs and the market clearing prices. A DR aggregator is also considered, responsible for scheduling the operation of a load-shifting DR resource capable of providing reserve, but also implicitly contributing to generation adequacy as in Chapter 4, whilst ensuring consumers' energy requirements are satisfied. Similar to the generating firms, the DR aggregator seeks to maximize their profit subject to resource constraints and the market clearing prices. The type of DR resource considered is a load-shifting DR resource. Price-responsive load is also included. Wind generation is also incorporated, however wind only contributes to the energy market.

A number of different cases are examined here, considering different combinations of DR market participation. These cases are discussed in detail in section 5.2.4. This chapter contributes to the literature by presenting a novel method for approximating the contribution of a DR resource to generation adequacy and thus it's ability to participate in a capacity market. This approach permits the inclusion of the DR resource in the capacity market model. In order to model these markets Mixed Complementarity Problems (MCPs) are employed. MCPs involve the definition of complementarity pairs for variables and equations [111]. This means that bounded variables must be mapped to complementarity inequality. An MCP with function F seeks to find vectors x and y such that:

$$0 \leq F_x(x, y) \perp x \geq 0, \quad (5.1)$$

and

$$0 = F_y(x, y), \quad y \text{ free}. \quad (5.2)$$

Here x represents the nonnegatively constrained variables with associated non-negative F components denoted F_x while y represents the free variables associated with components F_y that must equal zero exactly [112]. Note: the “perb” notation, $0 \leq a \perp b \geq 0$, is equivalent to $a \geq 0$, $b \geq 0$ and $ab = 0$.

MCPs are typically made up of the combination of Karush-Kuhn-Tucker (KKT) conditions for optimality from multiple interconnected optimization problems in addition to the market clearing conditions that connect the problems. Assuming the individual optimization problems are convex, the Karush Kuhn Tucker (KKT) conditions are both necessary and sufficient for optimality. Consequently, solving the MCP solves the different player's optimization problems simultaneously and in equilibrium. The benefits of the MCP approach include the ability to optimize the problems of multiple players' simultaneously. Furthermore, MCPs allow primal and dual variables to be constrained together. For example, in the formulation presented here, the output of all generators, and the DR resource, and prices are constrained together via market clearing conditions. An MCP is a particularly useful method when computing market equilibria which cannot be represented by an optimization model [111].

Many economic problems can be expressed as MCPs and, consequently, MCPs have been widely deployed in the literature for electricity market analysis. Höschle *et al.* [113] employ an MCP to analyze the impact of a CRM on total system costs. Ventosa *et al.* [114] employ an MCP to represent the participation of hydro and thermal generation in a competitive market. A case study of the Spanish electric power system is used in Liang *et al.* [115] employing an energy and regulation reserve market model. Bushnell [116] presents a framework for studying the competition between multiple firms with hydrothermal generation portfolios. Khalfallah [117] compares three investment incentive mechanisms (1. capacity payment, 2. forward capacity market, 3. reliability options) and finds that market-based mechanisms would be the most cost-efficient way of ensuring long-term system adequacy and encouraging earlier and adequate new investments. Lynch & Devine [18] utilize a stochastic MCP to examine the effects of refurbishment on electricity prices and investment in generating technology. Awad *et al.* [118] utilize an MCP to determine the optimal operating strategy of an energy storage system and the market clearing price simultaneously in perfect competition.

In recent years there has been a move towards incorporating price-responsive demand in MCP. Daoxin *et al.* [119] include both renewable energy generation and price-responsive demand in their MCP. The inclusion of price-responsive demand is achieved through the use of a control parameter which reflects the response of consumers to changes in price. However, constraints on the price-responsive demand

are not taken into account and reserve provision is not considered. While there has been research examining the interaction of DR with high levels of wind penetration, reserve markets with DR participation have only been incorporated through the use of a Unit Commitment and Economic Dispatch (UCED) algorithm [4, 120]. Both [4] and [120] examine the provision of both energy and reserve from DR using a UCED algorithm and assess the impact on operation and cost savings.

Conejo *et al.* [121] propose an hourly real-time DR model. The demand model seeks to minimize the cost of meeting the load minus the utility of the customers [121]. Unlike the model in [119], the model developed in [121] crucially includes physical constraints pertaining to the DR, including a minimum energy consumption constraint, and ramping limits on hourly load levels. However, again no consideration is given to reserve provision by the DR resource.

In the first case study in [120], demand is modeled as a constant value modified by a sinusoid. Additionally, the authors in [120] chose to only include two generation units in the portfolio; a base load unit and a peaking unit. Furthermore, they have assumed that DR is a fixed fraction of the total system demand at each point in time. In the second case study in [120], hourly demand and wind data is used and DR is assumed to be 5% of the system load in each hour. Rather than assume a percentage of the system load which is shiftable, this chapter instead utilizes actual data for the demand-side resource requirements and the power system demand.

Nekouei *et al.* [122] provide a game-theoretic approach for DR. Interplay between aggregators and generators is formulated as a Stackelberg leader game, in which the aggregator influences the spot price of electricity. The consumer seeks to minimize its load curtailment costs, while the aggregator seeks to minimize the aggregate inconvenience of customers. They applied the Stackelberg game to analyze the profitability of DR in South Australia. Crucially, reserve provision by the demand-side is not considered in the model in [122].

The authors in [5] employ an MCP model to minimize costs and incorporate price responsive demand. In their work on determining an optimal generation mix with DR and wind penetration, De Jonghe *et al.* [5] reviewed some of the literature and found that there have been no generation technology mix models that integrate energy efficiency programs, DR to hourly varying prices and dynamic operating constraints simultaneously. In their paper, they propose three methodologies for integrating short-term demand responsiveness into a technology mix optimization model, one of

which is a complementarity programming method [5] and utilize the same DR models in each method. As regards the complementarity programming method employed in [5], the authors formulate a mixed linear complementarity problem, not dissimilar to what is presented in this chapter. The key difference is the manner in which the DR is represented. A reference price and quantity demanded for each hour is considered and elasticity assumptions are used in [5]. The reference price is the quantity weighted average of the hourly energy prices over the time horizon and the short-term demand function that expresses quantity demanded as a function of relative deviations from the reference price [5]. A load-shifting DR resource is modeled here, separate from elastic or price-responsive demand, with the ability to provide reserve and contribute to generation adequacy, complete with an energy limit constraint.

The key contribution of this chapter lies in the development of an MCP framework for examining the impact of a load-shifting DR resource providing multiple services in multiple electricity markets, simultaneously. Analysis of multiple service provision from DR resources within an MCP has not yet been considered in the literature, to the best of the author's knowledge. Additionally, only recently has multiple revenue streams accessible to electricity storage devices received attention, specifically in [123]. However, unlike the work in [123] which only considers energy services provision over different time frames and for different applications, this chapter considers energy, reserve and capacity service provision for a DR resource.

This chapter is organized as follows: Section 5.2 introduces the MCP methodology employed and details the DR aggregator problem. Input data and case study information is discussed in Section 5.3. Sections 5.4 and 5.5 presents the results of the various case studies and sensitivities, while Section 5.6 discussions the key findings.

5.2 Methodology

This section details the generators' and DR aggregator problems as well as the market clearing conditions, under competitive market conditions. The corresponding KKT conditions are also derived and presented here. Throughout this section parameters are denoted with capitals, primal variables are denoted with lower case lettering. Variables in parentheses, beside constraints, are the Lagrange multipliers associated with the constraints and are denoted with lower-case Greek letters.

5.2.1 Generator Problem

Each firm chooses the amount of generation ($gen^{t,i,j}$), reserve provision ($reserve_{Gen}^{t,i,j}$) and capacity bid ($cap_{bid}^{i,j}$) (all in MW), as well as investment ($invest^{i,j}$) and exit ($exit^{i,j}$), for all of their generating units in order to maximize their profits, $\Pi_{Gen}^{i,j}$, which consists of profits from the energy, reserve and capacity markets, Π_{Energy}^i , $\Pi_{Reserve}^i$ and $\Pi_{Capacity}^i$, respectively. Exit, in this context, is the decommissioning of a unit, or part of a unit. Exit and investment decisions (MW) are not constrained to be integer variables. The equations defining each generators problem are given below, where there are i firms and j technologies. Generator i 's problem is:

$$\max_{\substack{gen \\ exit \\ invest \\ cap}} \Pi_{Gen}^i = \sum_j \Pi_{Energy}^{i,j} + \sum_j \Pi_{Reserve}^{i,j} + \sum_j \Pi_{Capacity}^{i,j}, \quad (5.3)$$

where

$$\Pi_{Energy}^{i,j} = \sum_t (gen^{t,i,j}) \times (\lambda^t - MC^{i,j}), \quad (5.4)$$

$$\Pi_{Reserve}^{i,j} = \sum_t (reserve_{Gen}^{t,i,j}) \times \mu^t, \quad (5.5)$$

$$\Pi_{Capacity}^{i,j} = (cap_{bid}^{i,j}) \times (\kappa) - (invest^{i,j}) \times ICOST^j - (CAP^{i,j} - exit^{i,j}) \times MCOST^j, \quad (5.6)$$

subject to:

$$gen^{t,i,j} + reserve_{Gen}^{t,i,j} \leq CAP^{i,j} - exit^{i,j} + invest^{i,j}, \quad (\theta_1^{t,j}), \quad \forall t, i, \quad (5.7)$$

$$cap_{bid}^{i,j} \leq CAP^{i,j} - exit^{i,j} + invest^{i,j}, \quad (\theta_2^{i,j}), \quad \forall t, j, \quad (5.8)$$

where i is an index representing each different generator, j represents the generating technology and t is the time index. The variables λ_t , μ_t and κ (€) represent the prices associated with the energy, reserve and capacity markets, respectively. Each are exogenous to the generators' problems but are variables of the overall model determined via the market clearing conditions. All of the generating firm's primal variables are constrained to be non-negative.

The parameter $MC^{i,j}$ (€/MW) denotes the marginal cost of generating firm i and each technology j , $ICOST^j$ (€/MW) represents the investment cost of generating technology j , while $MCOST^j$ (€/MW) is the maintenance cost associated with technology j . The parameter $CAP^{i,j}$ (MW) represents the initial endowment of generating capacity for each firm i and for each technology j .

Equation (5.3) is the objective function of the generating firm. Each generating firms choses how to participate in each market in order to maximize their profit. Equation (5.4) represents the energy component profit of the generator and consists of the revenue obtained from the energy market less the marginal cost $MC^{i,j}$ of producing energy. Equation (5.5) denotes the reserve component of the generator's profit. As can be seen, there is no cost component associated with providing reserve as it is assumed that the cost of providing reserve is the opportunity cost of not providing energy. Equation (5.6) represents the revenue from the capacity market less investment costs and maintenance costs associated with providing capacity. Equation (5.7) constrains the power and reserve provided by a generating unit to be strictly less than or equal to the installed capacity of the unit, taking any exit and investment decisions into account. Equation (5.8) ensures the capacity bid of each generator does no exceed the installed capacity.

5.2.1.1 KKT conditions for generator providing energy, reserve and capacity

The KKT optimality conditions for all firms are given below.

$$0 \leq gen^{t,i,j} \perp -\lambda^t + \theta_1^{t,i,j} + MC^{i,j} \geq 0, \forall t, i, j, \quad (5.9)$$

$$0 \leq reserve_{Gen}^{t,i,j} \perp -\mu^t + \theta_1^{t,i,j} \geq 0, \forall t, i, j, \quad (5.10)$$

$$0 \leq cap_{bid}^{i,j} \perp -\kappa + \theta_2^{i,j} \geq 0, \forall t, i, j, \quad (5.11)$$

$$0 \leq invest^{i,j} \perp ICOST^j - \sum_t \theta_1^{t,i,j} - \theta_2^{i,j} \geq 0, \forall t, i, j, \quad (5.12)$$

$$0 \leq exit^{i,j} \perp -MCOST^j + \sum_t \theta_1^{t,i,j} + \theta_2^{i,j} \geq 0, \forall t, i, j, \quad (5.13)$$

$$0 \leq \theta_1^{t,i,j} \perp CAP_{i,j} - exit^{i,j} + invest^{i,j} - gen^{t,i,j} - reserve_{Gen}^{t,i,j} \geq 0, \forall t, i, j, \quad (5.14)$$

$$0 \leq \theta_2^{i,j} \perp CAP^{i,j} - exit^{i,j} + invest^{i,j} - cap_{bid}^{i,j} \geq 0, \forall t, i, j. \quad (5.15)$$

As generator i 's problem is convex, the KKT conditions are both necessary and sufficient for optimality [112].

5.2.2 Demand Response Aggregator Problem

The DR aggregator's problem is to choose DR in both the downward and upward direction, dr_{down}^t and dr_{up}^t , respectively, and reserve provision, $reserve_{dr}^t$, so as to maximize revenue, from the reserve market, $\Pi_{Reserve}$, and to minimize costs from the energy market, Π_{Energy} . The total load-shifting performed by the DR resource is the net result of a combination of dr_{down}^t and dr_{up}^t , the upwards and downwards change in demand at each time, t . The energy limitation on the load-shifting modeled here constrains the operation of the resource so that the sum of dr_{down}^t and the sum of dr_{up}^t over 24 hours are equal, in a manner similar to conventional energy storage devices. In the models developed here, DR can only provided reserve in the downward direction (from the DR point of view). This DR reserve is assumed to be analogous to a generator holding upward reserve, permitting the formulation of Equation (5.34) to represent a reserve market. The DR reserve will henceforth be referred to as upward reserve.

Load-shifting DR is chosen here and is modeled in a generic way in this study. It is intended that the model of the DR can be made resource-specific in future work by varying the values of the parameters relating the maximum installed capacity, $DMAX$, the reference demand, $DREF^t$, and whether or not the resource can provide reserve.

As shown in [6, 94, 99], DR is capable of reducing peak demand thereby displacing a certain amount of peaking generating capacity. This capability is referred to as the contribution to generation adequacy of the resource in previous chapters. In [99] a new metric called the Equivalent Generation Capacity Substituted is proposed. The authors in [99] suggest that the Equivalent Generation Capacity Substituted metric indicates the amount of conventional generation capacity that can be displaced by DR

without impacting upon the original level of generation adequacy. In Chapters 3 and 4 the Effective Load Carrying Capability (ELCC) is the metric used, which is the amount by which a system's load can increase when the generator is added to the system, while maintaining the system's adequacy [83]. It should be noted, as mentioned earlier and as highlighted in [81], that there are a variety of possible definitions and calculation methods for capacity value metrics. Thus, there is not one single definite value for the CV of a resource.

Throughout an entire year, the period during which a lack of generation adequacy would be most apparent is the peak demand period. In the MCP models employed here, firms make investment and exit decisions based on the peak system demand and their profitability over the study period. Firms decide to invest in generation if there is a deficit during peak periods and there is scope for them to recoup their investment costs. On the other-hand, firms will opt to exit the market if there is excess generating capacity, displacing their operation at the peak, thus impacting upon their profits. Consequently, it is assumed here that the change in generator investment due to the addition of the DR resource in an MCP model is representative of the contribution of the DR resource to generation adequacy and the DR resource is thus in a position to participate in the capacity market.

It is assumed that, in future electricity markets, reference or baseline demands for DR resources will be knowable and obtainable by DR aggregators, and that reserve markets are non-discriminatory, permitting the participation of DR. It is also assumed that DR aggregators are capable of responding to wholesale electricity market prices. Assuming the DR resource is capable of providing a response (dr_{down}^t and dr_{up}^t) and providing reserve in the same period as well as the ability to participate in the capacity market, the DR aggregators problem is:

$$\max_{\substack{dr_{down} \\ dr_{up} \\ reserve_{dr} \\ cap_{dr}}} \Pi_{dr} = \Pi_{energy} + \Pi_{reserve} + \Pi_{capacity}, \quad (5.16)$$

where

$$\Pi_{energy} = \sum_t (dr_{down}^t - dr_{up}^t - DREF^t) \times \lambda^t, \quad (5.17)$$

$$\Pi_{reserve} = \sum_t (reserve_{dr}^t) \times \mu^t, \quad (5.18)$$

$$\Pi_{cap} = cap_{dr} \times \kappa - MC^{slack} \times slack, \quad (5.19)$$

subject to:

$$dr_{down}^t + reserve_{DR}^t \leq DREF^t, \quad \gamma_1^t, \quad \forall t, \quad (5.20)$$

$$dr_{up}^t + DREF^t \leq DMAX, \quad \gamma_2^t, \quad \forall t, \quad (5.21)$$

$$\sum_{t=t'}^{t'+23} (dr_{down}^t) = \sum_{t=t'}^{t'+23} (dr_{up}^t), \quad \gamma_3^{t'}, \quad \forall t' \in H = \{1, 25, 49, \dots\}, \quad (5.22)$$

$$cap_{dr} \leq \sum_{i,j} INVEST_{NoDR}^{i,j} - \sum_{i,j} invest_{DR}^{i,j} + slack, \quad \gamma_4. \quad (5.23)$$

The parameter $DREF^t$ (MW) represents reference or baseline demand and thus the amount of load consumers would wish to consume at a specific point in time, t . This parameter, in effect, represents the resource demand in the absence of load-shifting. The parameter $DMAX$ (MW) represents the total installed capacity of the DR resource. Equation (5.16) is the objective function of the DR aggregator. The DR aggregator chooses how to participate in each market in order to maximize their profit. Equation (5.17) represents the energy component of the DR aggregator's profit and consists of the revenue obtained from the energy market due to load-shifting as well as the cost of meeting the consumers' reference demand, $DREF^t$. Equation (5.18) denotes the reserve component of the DR aggregator's profit. As can be seen, there is no cost component associated with providing reserve as it is assumed that the cost of providing reserve is the opportunity cost of not participating in the energy market. Constraint (5.20) ensures that, in each time-step, t , the DR aggregator can only shift downwards and can only provide upward reserve (from the point of view of the power system) by an amount less than or equal to the reference demand. That is, there can only be downwards shifting load and reserve if the consumers' appliances are on and available. Equation (5.21) constraints the upward shifting to be less than the installed capacity of the resource, $DMAX$. Constraint (5.22) represents the energy limited nature of the DR resource and ensures that any shifting downwards is balanced by shifting upwards over a 24 hour period, where H is the set containing the first hour

of each day. As is the case for the generator problem, λ^t is determined via the energy market clearing condition, μ^t is determined via the reserve market clearing condition and κ is obtained from the capacity market. All of the DR aggregator's primal variables are constrained to be non-negative.

Equations (5.19) and (5.23) represent the manner in which the DR resource is able to participate in the capacity market. The capacity bid, cap_{dr} , is equal to the change in investment from the 'no DR' case, the parameter $\sum_{i,j} INVEST_{noDR}^{i,j}$, see section 5.2.4, to the case 'with DR' case, the variable $\sum_{i,j} invest_{DR}^{i,j}$ plus an additional slack variable, $slack$ (see Equation (5.23)). The change in investment is an approximation for the generation adequacy contribution of the DR resource.

The slack variable is included in order to ensure that there is no opportunity for the DR aggregator to over-estimate the generation adequacy contribution of the resource. This variable represents generation from an expensive generating unit, MC^{slack} , which would be required to make up any difference between the capacity bid of the DR and the actual, realized generation adequacy contribution of the resource. If the change in investment between the 'no DR' case and the 'with DR' case is zero, the high cost associated with the slack variable forces the variable cap_{dr} to be zero also. Thus, while the slack variable represents generation, its sole function is to ensure that the DR aggregator problem is feasible; there is no participation of this generator in any of the electricity markets.

5.2.2.1 KKT conditions for DR providing energy, reserve and capacity

The KKT optimality conditions for the DR aggregator are given below.

$$0 \leq dr_{down}^t \perp -\lambda^t + \gamma_1^t + \gamma_3^{t'} \geq 0, \quad \forall t, t' \in H, \quad (5.24)$$

$$0 \leq dr_{up}^t \perp \lambda^t + \gamma_2^t - \gamma_3^{t'} \geq 0, \quad \forall t, t' \in H, \quad (5.25)$$

$$0 \leq reserve_{DR}^t \perp -\mu^t + \gamma_1 \geq 0, \quad \forall t, \quad (5.26)$$

$$0 \leq cap_{dr} \perp -\kappa + \gamma_4 \geq 0, \quad \forall t, \quad (5.27)$$

$$0 \leq \gamma_1^t \perp DREF^t - dr_{down}^t - reserve_{DR}^t \geq 0, \forall t, \quad (5.28)$$

$$0 \leq \gamma_2^t \perp D_{MAX} - dr_{up}^t - DREF^t \geq 0, \forall t, \quad (5.29)$$

$$0 \leq \gamma_3^{t'} \perp \sum_h dr_{up}^t - \sum_h dr_{down}^t \geq 0, \quad (5.30)$$

$$0 \leq \gamma_4 \perp \sum_i INVEST_{NoDR}^i - \sum_i invest_{DR}^i + slack \geq 0, \forall t. \quad (5.31)$$

As was the case for the generator's problem presented above, since the DR aggregator's problem is convex, the KKT conditions are both necessary and sufficient for optimality [112].

5.2.3 Market Clearing Conditions

Perfect competition in the electricity markets is assumed, in which all participants are price-takers. The market clearing conditions combine each of the generators problems and the DR aggregator problem. These conditions are:

$$\sum_i Gen^{t,i} = DEM^t + E \times \lambda^t, \forall t, \quad (\lambda^t), \quad (5.32)$$

where the parameter DEM^t denotes the system demand in hour t and the parameter E represents the elasticity of load, which henceforth refers to elasticity associated with demand or price-responsive load. This price-responsive load is distinct from the DR resource's ability to load-shift, which involves energy payback. When DR is included Equation (5.32) becomes:

$$\sum_i gen^{t,i} = DEM^t - DREF^t + dr_{up}^t - dr_{down}^t + E \times \lambda^t \quad \forall t \quad (\lambda^t). \quad (5.33)$$

The reserve and capacity markets, with and without DR participation are shown below:

$$\sum_i reserve_{Gen}^{t,i} = RESERVE_{REQ}, \quad \forall t, \quad (\mu^t), \quad (5.34)$$

$$\sum_i reserve_{Gen}^{t,i} + reserve_{DR}^t = RESERVE_{REQ}, \quad \forall t, \quad (\mu^t), \quad (5.35)$$

$$\sum_i cap_{bid}^i = TARGET, \quad (\kappa), \quad (5.36)$$

$$\sum_i cap_{bid}^i + cap_{DR} = TARGET, \quad (\kappa). \quad (5.37)$$

Equations (5.35) and (5.34) represent the reserve market with and without DR participation, while including a reserve requirement, the parameter $RESERVE_{REQ}$. Similarly, the capacity market with and without DR is represented by Equations (5.37) and (5.36), respectively, where the parameter $TARGET$ is the amount of capacity deemed to be required by the system.

5.2.4 MCP Cases and Sensitivities

The authors in [5] suggest that since their MCP model minimizes costs and incorporates price-responsive demand, it can be viewed as a optimal generation technology mix model. Similarly, the cases employed in this chapter (see Figure 5.1) minimize costs, incorporate both elastic demand and DR, allow for investment and withdrawal of generation and include DR. Thus, the models employed may also be considered optimal generation technology mix models.

The MCP models are developed in GAMS and solved using the PATH solver. Due to the considerable computation time, the MCP analysis is performed for the first 100 days of the year. All of the cases presented are examined by varying the input parameters for the peak load level as illustrated in Table 5.1.

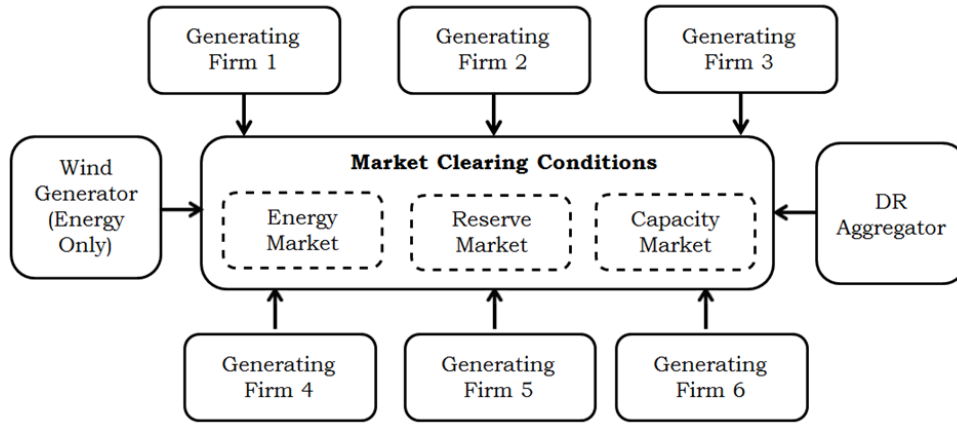


Figure 5.1: Overview of the methodology employed in this chapter

Table 5.1: Wind and load level sensitivities

Peak Load	Wind Level			
2500 MW	0 MW	500 MW	1000 MW	1500 MW
5000 MW	0 MW	500 MW	1000 MW	1500 MW
7500 MW	0 MW	500 MW	1000 MW	1500 MW

The market clearing conditions presented in the previous section are used in conjunction with the generator problems and the DR aggregator problems in different combinations in order to produce a number of different MCP cases (see Table 5.2), an explanation of which is now provided:

A **Without a DR resource** - for the following three cases, the DR aggregator problem is not included, but there can be price-responsive demand.

- 1 **Energy Market Only** - in this model, the generators' problems are solved with just the energy market clearing condition. The MCP consists of the generators' problems, KKT conditions (5.9) through (5.14), and market clearing condition (5.32).
- 2 **Energy & Capacity Market Only** - in this model, the MCP consists of the generators' problems, KKT conditions (5.9) through (5.15), and market clearing conditions (5.32) and (5.36).

3 **All Markets**- in this model, the generators' problems are solved with all three market clearing conditions. There is no load-shifting DR but there is price-responsive demand. The MCP consists of the generators' problems, KKT conditions (5.9) through (5.15), and market clearing conditions (5.32), (5.34), (5.36). In order to solve model 2. and 4. below, it is necessary to first solve this model in order to determine the variable $\sum_{i,j} invest^{i,j}$, which becomes the parameter $\sum_{i,j} INVEST_{noDR}^{i,j}$ in Constraint (5.23) in cases 2. and 4.

B **With a DR resource** - for all of the following cases, the MCP consists of the generators' problems, KKT conditions (5.9) through (5.15). However, the market clearing condition representations differ depending on what markets the DR resource is participating in.

- 1 **DR in Energy Market Only** - in this model all three market clearing conditions are considered, however DR only participates in the energy market. The market clearing conditions in this model comprise Equations (5.33), (5.34) and (5.36). The KKT conditions for the DR resource in this model are KKT conditions (5.24), (5.25), (5.27), (5.28), (5.29) and (5.30) with the variable $reserve_{DR}^t$ fixed to zero.
- 2 **DR in Energy & Capacity Market Only** - in this model all three market clearing conditions are again considered, however DR only participates in the energy market and capacity market. The market clearing conditions in this model comprise Equations (5.33), (5.34) and (5.37). The KKT conditions for the DR resource in this model are KKT conditions (5.24), (5.25), (5.27), (5.28), (5.29), (5.30) and (5.31), with the variable $reserve_{DR}^t$ fixed to zero.
- 3 **DR in Energy & Reserve Market Only** - in this model all three market clearing conditions are considered, however DR only participates in the energy market and reserve market. The market clearing conditions in this model comprise Equations (5.33), (5.35) and (5.36). The KKT conditions for the DR resource in this model are KKT conditions (5.24), (5.25), (5.26), (5.28), (5.29) and (5.30).
- 4 **DR in All Markets** - in this model all three market clearing conditions are considered, with DR participating in all available markets. The market

clearing conditions in this model comprise Equations (5.33), (5.35) and (5.37). The KKT conditions for the DR resource in this model are KKT conditions (5.24) through (5.31).

Table 5.2: DR market participation sensitivities for each load and wind level

	Available Markets		
DR Market Participation	All markets	Energy & Only	Energy & Capacity
No DR (E=0.0)	Y	Y	Y
No DR (E=-0.11)	Y	Y	Y
Energy Only (E=0.0)	Y	—	—
Energy Only (E=-0.11)	Y	—	—
Energy & Reserve (E=0.0)	Y	—	—
Energy & Reserve (E=-0.11)	Y	—	—
Energy & Capacity (E=0.0)	Y	—	—
Energy & Capacity (E=-0.11)	Y	—	—
All Markets(E=0.0)	Y	—	—
All Markets (E=-0.11)	Y	—	—

5.3 Input Data

The input parameters employed here are chosen from a variety of sources. Three generating technologies are considered: baseload, mid-merit and peaking capacity.

5.3.1 System Data

The initial endowment of generating capacity for each firm, $CAP^{i,j}$ is shown in Table 5.3 and the corresponding cost characteristics are presented in Tables 5.4 and 5.5. It can be seen from Table 5.3 that examining a peak load of 7500 MW represents a case where there is considerable under-capacity, while examining a peak load of 2500 MW represents a case with over-capacity.

Firm 1 has baseload capacity only, firm 2 has baseload and mid-merit capacity only, firm 3 and 5 are integrated firms, with capacity in all three generation

technologies. Firm 4 has baseload and peaking capacity, while firm 6 has peaking capacity only. The total initial endowment of generation capacity is 5400 MW which falls short of the peak load of 7500 MW, but exceeds the peak loads of 2500 MW and 5000 MW.

Table 5.3: Initial endowment of capacity $CAP^{i,j}$ for each firm (MW)

Technology	f1	f2	f3	f4	f5	f6	Technology Total
Baseload	1000	800	500	500	400	—	3200
Mid-Merit	—	500	400	—	400	—	1300
Peaking	—	—	200	300	200	200	900
Firm Total	1000	1300	1100	800	1000	200	5400

The marginal costs, maintenance costs and investment costs are all based on the values employed in [18]. The marginal costs in [18] were fixed for each technology. Here some variation between generating units marginal costs is included in order to ensure that there is intra-hour variation in SMP as the operation of load-shifting resources such as that examined here are predicated on varying energy market prices.

Table 5.4: Marginal cost $MC^{i,j}$ of each firm (€/MW)

Technology	f1	f2	f3	f4	f5	f6
Baseload	30	45	55	55	65	—
Mid Merit	—	50	35	—	35	—
Peaking	—	—	93	83	93	93

Table 5.5: Generator cost characteristics (€/MW)

Technology	Maintenance Cost	Investment Cost
	$MCOST^j$	$ICOST^j$
Baseload	25	100000
Mid Merit	12	65000
Peaking	7	45000

The reserve requirement, $RESERVE_{REQ}$, is 500 MW for all cases, unless otherwise stated. This level of 500 MW was chosen as it broadly complies with the reserve requirements for different levels of installed wind for the Irish power system as

reported in [124]. The capacity target, $TARGET$, is 1.2 times the system peak load for all cases. In all cases examined, all firms are assumed to be price-takers, unless otherwise stated.

5.3.2 Demand Data

The reference DR data, denoted as $DREF^t$, used in this chapter is the space and water heating demand profile for 100,000 apartments on the Irish system as determined by [107] and as employed in Chapter 4. To reiterate, this data was obtained through the development of ‘archetype models’ by [107], which are representative of a group of dwellings and dwelling loads. A set of reference dwellings is modeled in detail by [107], using EnergyPlus, a deterministic building energy analysis and thermal load simulation program [101]. These models are converted into building performance simulation ‘archetypes’ by integrating high space and time resolution operational data. The set of dwelling archetypes is used to generate annual profiles for space and domestic hot water heat demands on a fifteen-minute basis. These consumer end-use heating time series are converted to hourly-resolution and used in this chapter as $DREF^t$. The installed capacity of the resource, $DMAX$, is 556 MW, while the marginal cost associated with the slack variable, MC^{slack} is €10,000 /MWh.

An annual system demand profile from Ireland for the year 2009 [125] is examined, and scaled linearly as appropriate to produce the parameter DEM^t , with different peak load levels. For example, when peak load in the following sections is stated to be 7500 MW then, for each hourly timestep, DEM^t is 1.5 times that when peak load is stated to be 5000 MW. Actual wind data from Ireland, also for the year 2009, is employed. Elasticity of demand (E) is chosen to be -0.11 as this corresponds to the elasticity of demand on the Irish system as determined by Di Cosmo & Hyland [126].

5.4 Results: Without Demand Response Participation

This section considers a situation with an energy market and a situation with an energy and a capacity market, without DR participation in order to highlight the importance of a CRM.

5.4.1 Energy Only Markets -v- Energy and Capacity Markets

Figure 5.2 shows the initial endowment of generating capacity for each firm as well as the resultant installed capacity for each firm for two models, energy market only model and energy and capacity markets model, respectively, after the models have been solved. Figure 5.2 highlights the considerable change in market exit decisions that occurs at a peak load level of 2500 MW in the absence of a capacity market. It is noted that there is no change in the investment decisions made, irrespective of whether a capacity market exists or not. While, firm 1 is the only firm to invest, all other firms exit but to different extents depending on the available markets. It is clearly seen that considerably more capacity is removed from the market when there is no capacity market. The resultant installed capacity for all firms for the two other peak load levels are shown in Appendix F.

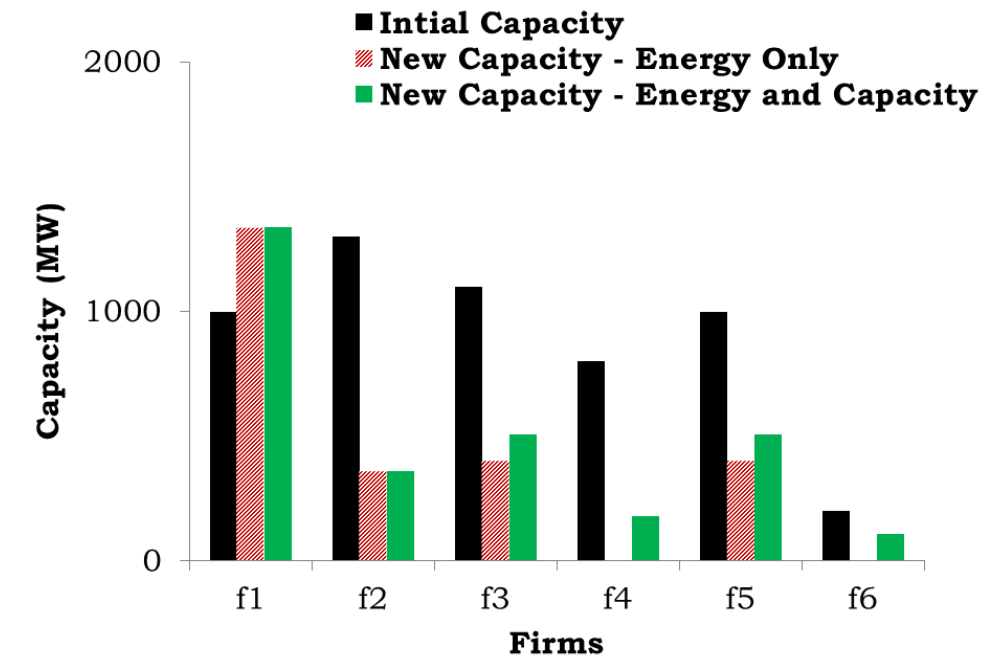


Figure 5.2: Installed generating capacity for each firm with energy only (no *TARGET*) and energy & capacity markets (with *TARGET*) for a peak load of **2500 MW**

Figure 5.3 illustrates the impact the existence of a capacity market can have on the type of technology in the generation portfolio for a peak load of 2500 MW. For all peak load levels examined (figures for the two other peak load levels are shown in

Appendix F), the existence of a capacity market ensures that a certain level of peaking plant is maintained.

Peaking generators, and generators high on the merit order list such as the mid-merit plants, typically have low capacity factors associated with them and thus do not receive substantial payments from the energy market, relying on capacity payments to cover their maintenance costs. In the absence of capacity payments, such generators are less profitable and it is optimal for those firms to remove some of their capacity from the market.

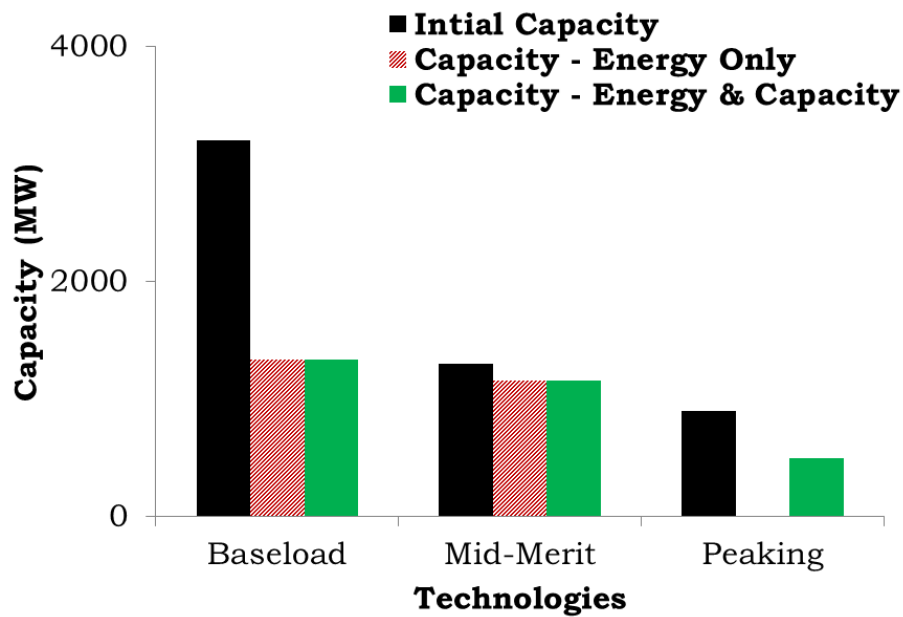


Figure 5.3: Installed generating capacity for each technology with energy only (no *TARGET*) and energy & capacity markets (with *TARGET*) for a peak load of **2500 MW**

The results here show that an energy only market leads to lower levels of installed capacity which concurs with [127]. In the absence of a capacity market the capacity target does not need to be satisfied and thus generators are not incentivized to maintain their units and they chose to exit the market. Thus, in an energy only market, enough capacity is maintained to meet the peak load only, while with both energy and capacity markets, the installed capacity level exceeds the peak load level (see Table 5.6), which is desirable from a generation adequacy and reserve provision point of view. Indeed, using the traditional LOLE calculation methodology employed in the previous two chapters, comparison of the generation adequacy metrics associated

with energy only markets and energy and capacity markets shows that, for peak loads of 2500 MW, 5000 MW and 7500 MW, there is a 55%, 38% and 30% decrease, respectively, in the LOLE metric when moving from an energy only market, to a situation with energy and capacity markets.

This concurs with the work in [109], in which future scenarios were examined to investigate what would happen in a market devoid of capacity payments. It was found that some generators would become financially unviable and would be removed from the portfolio, which would lead to severe adequacy difficulties, with significant deficits and risk to security of supply [109]. Similarly, the work in [113] noted that insufficient remuneration for conventional thermal generators in an energy-only market may result in inadequate generation mixes to meet peak demand.

Table 5.6: Total installed capacity for an energy only market (no *TARGET*) and for energy & capacity markets (with *TARGET*)

Peak Load Level	Energy Only	Energy & Capacity
2500 MW	2500 MW	3000 MW
5000 MW	5000 MW	6000 MW
7500 MW	7500 MW	9000 MW

It is noted that, for the majority of the hours examined, there is little difference in the SMPs with and without the presence of a capacity market (Figure 5.4). This is as expected as the marginal units at most hours remain the same irrespective of whether a capacity market exists or not. However, at the peak load hour, in the absence of a capacity market, there is no additional installed capacity and investment in generation would be required. Thus, the system marginal cost, the cost of increasing the demand by 1 MW, would involve investment thereby pushing the SMP up, since the SMP has capital and operational cost components (see KKT conditions (5.9), (5.12) and (5.13)).

The existence of a capacity market, on the other-hand, incentivizes investment in generating technology. Thus, there is sufficient capacity at the peak hour to meet demand and, as there is no requirement to invest in technology just to meet the demand at that one hour, the marginal cost is simply the marginal cost of providing 1 extra MW of generation. This results in a reduction in the energy prices at the peak

hour. Consequently, there is a marked reduction in the SMP at the peak load hour between the energy only case and the energy and capacity case. Additionally, the magnitude of this reduction in SMP at the peak increases with increasing peak load level.

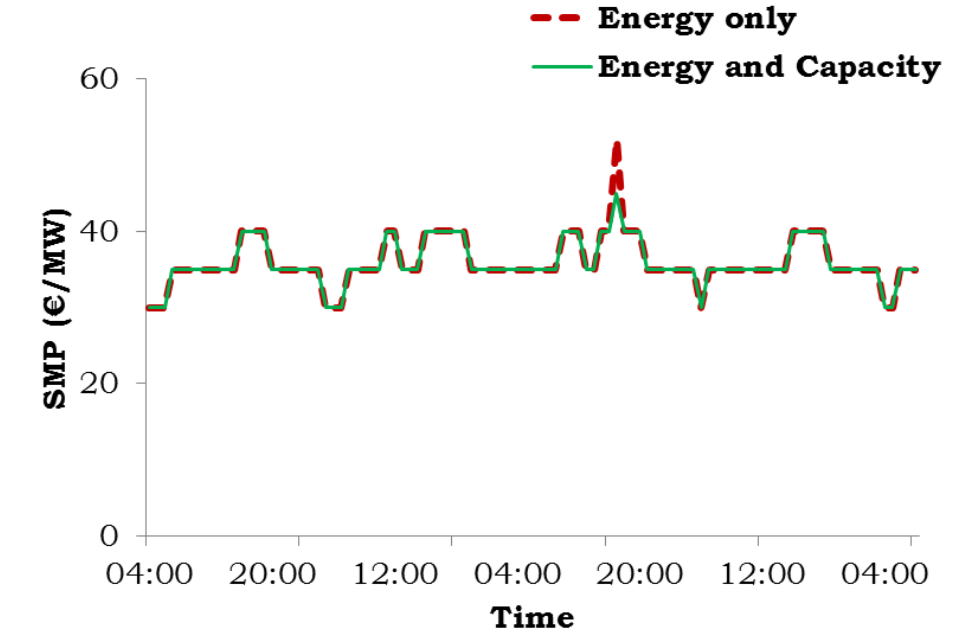


Figure 5.4: System marginal price without demand response for the for four days around the system peak load hour: comparison of energy & capacity markets and an energy only market for a peak load of 2500 MW and no wind generation

A comparison of the total system costs (costs paid by consumers) shows that there is an increase following the addition of a capacity market, which concurs with the work in [113]. This increase is a result of capacity payments (compare columns 2 and 3 in Table 5.7). However, the overall change in system costs is less than the increase as a result of capacity payments. This is because, as previously mentioned, there is a saving in energy costs for the consumer at the peak load hour, when there is a marked reduction in the SMP, which was shown earlier in Figure 5.4.

Table 5.8 displays similar increases in system costs following the introduction of a capacity market, however in this case elastic demand is considered. The inclusion of elastic demand results in slightly lower energy costs, and thus energy savings, compared to Table 5.7 where there is no elastic demand. The capacity payments remain unchanged.

The analysis presented in this section provides evidence for the need for a capacity market and, as the results concur with previous work in the literature, validates the MCP models developed in this chapter. Given the importance of capacity markets, as shown here, for the remainder of this chapter, generating firms will participate in all electricity markets (energy, reserve and capacity markets), unless stated otherwise.

Table 5.7: System costs for an energy only market and for energy & capacity markets **without** elastic demand ($E = 0$), without demand response

Peak Load	Energy Only	Energy & Capacity	Energy Savings	Capacity Payments	Net Change
2500 MW	€138,292,260	€138,295,760	- €17,500	€21,000	€3,500
5000 MW	€279,412,054	€279,437,054	- €125,000	€150,000	€25,000
7500 MW	€420,902,244	€420,939,744	- €187,500	€225,000	€37,500

Table 5.8: System costs for an energy only market and for energy & capacity markets **with** elastic demand ($E = -0.11$), without demand response

Peak Load	Energy Only	Energy & Capacity	Energy Savings	Capacity Payments	Net Change
2500 MW	€137,979,145	€137,982,720	- €17,425	€21,000	€3,575
5000 MW	€279,095,415	€279,120,730	- €124,685	€150,000	€25,315
7500 MW	€420,586,451	€420,624,316	- €187,135	€225,000	€37,865

5.4.2 Impact of Increasing Wind Generation Penetration

Prior to investigating the impact of DR on the various electricity markets, to further explore the validity of the MCP developed in this chapter, the impact of increasing wind generation is first examined. As would be expected, in the absence of the DR resource, increasing the penetration of wind generation, at each peak load level, results in decreasing investment in generating capacity (Figure 5.5). This is because the addition of wind generation alters the net demand profile, lowering the requirement for additional generation. There is also a notable difference in the levels of investment and exit with increasing peak load level, since the same initial endowment of generating

capacity is used in each case, with high peak load levels requiring much greater levels of investment but less capacity leaving the market. Similar graphs for the other two peak load levels are displayed in Appendix F.

In general, it is found that Firm 1 is the only firm to invest in generation despite the fact that baseload generation has the highest maintenance and investment costs and the lowest marginal costs. Firms 3, 4 and 5 all choose to remove some baseload generation from the generating portfolio. This is because the baseload units owned by these firms have extremely low capacity factors. Capacity factor is the ratio of the actual power output of a generator over a specific period to its potential output if the generator were to operate at full capacity continuously over that period. In this case the capacity factor for all baseload units, except the unit operated by Firm 1, is less than 3%. The high maintenance costs associated with baseload units, coupled with low capacity factors, prohibits high returns.

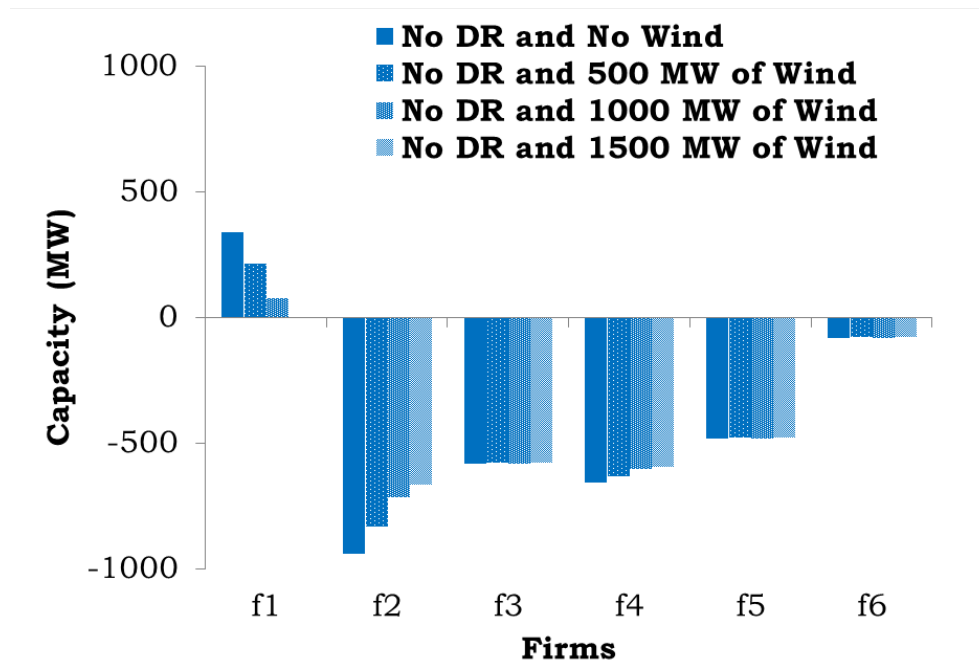


Figure 5.5: Generator net investment (Investment - Exit) under different wind penetrations at a peak load of **2500 MW**, without demand response. Positive capacity represents investment while negative capacity represents exit.

As mentioned in section 5.2.4, a peak load of 2500 MW is less than the initial endowment of installed capacity. In such a case, there is no capacity requirement to invest in generation, which is evidenced by the considerable amount of capacity which

is removed from the generating portfolio in comparison with the level of investment. At this load level, it is more a case of rearranging the generating portfolio rather than investing in new generation.

Interestingly, at this load level of 2500 MW, the level of market exit for Firms 3, and 5 remain relatively constant, irrespective of the amount of wind generation on the system. This relates to the fact that these firms' portfolios include mid-merit peaking generating units. Such units have low marginal costs and very low maintenance costs. Thus, with high levels of wind and low system demand, the net demand is highly variable and mid-merit units are well placed to profit.

5.5 Results: With Demand Response Participation

The previous section examined energy and capacity markets without DR participation. In this section, all markets are considered and DR participation is included. Section 5.5.1 examines the energy market, while Section 5.5.2 discusses the reserve and capacity markets. Sections 5.5.3 and 5.5.4 present the key findings relating to the impact of DR participation on system operating costs and generator profits, respectively. DR aggregator costs and DR participation is subsequently discussed.

5.5.1 Impact Demand Response on Energy Market

The inclusion of DR in the energy market has considerable influence on the SMPs. As can be seen in Figure 5.6 (and Figures G.1 and G.2 in Appendix G), the introduction of the DR resource into the energy market results in an alteration to the SMP *. The DR resource also succeeds in reducing system demand peaks and increasing system demand at the troughs. This is a result of the fact the DR is capable of shifting some of its demand away from peak load hours to off-peak hours. Consequently, the addition of the DR resource results in slight SMP reductions during peak hours, but increased prices during off-peak hours. This concurs with [74], where it is shown that shifting load from one period to another can cause a rise in market price at that time. Figure 5.6 clearly illustrates this leveling of the SMP during off peak hours. Similarly, the work presented in [118] on the impact of energy storage operation also shows that electricity prices reduce at peak periods, but increase at off-peak periods. This is

*Graphs shown in this section are for a peak load of 2500 MW as the variation between 'no DR' case and the 'with DR' case are seen more clearly than for other peak load levels, which are shown in the Appendix.

unsurprising given the similarities in how load-shifting resources and energy storage devices operate.

Comparing Figures 5.6 with G.1 and G.2 in the Appendix, shows that the impact on the SMP profile of adding in the DR resource diminishes with increasing peak load, with the largest impact on SMP occurring in the case with a peak load of 2500 MW. Correspondingly, the system demand profile is altered with the addition of the DR resource, though this effect also reduces with increasing peak load, which is illustrated in Figure 5.7, as well as in Figures G.3 and G.4 in the Appendix.

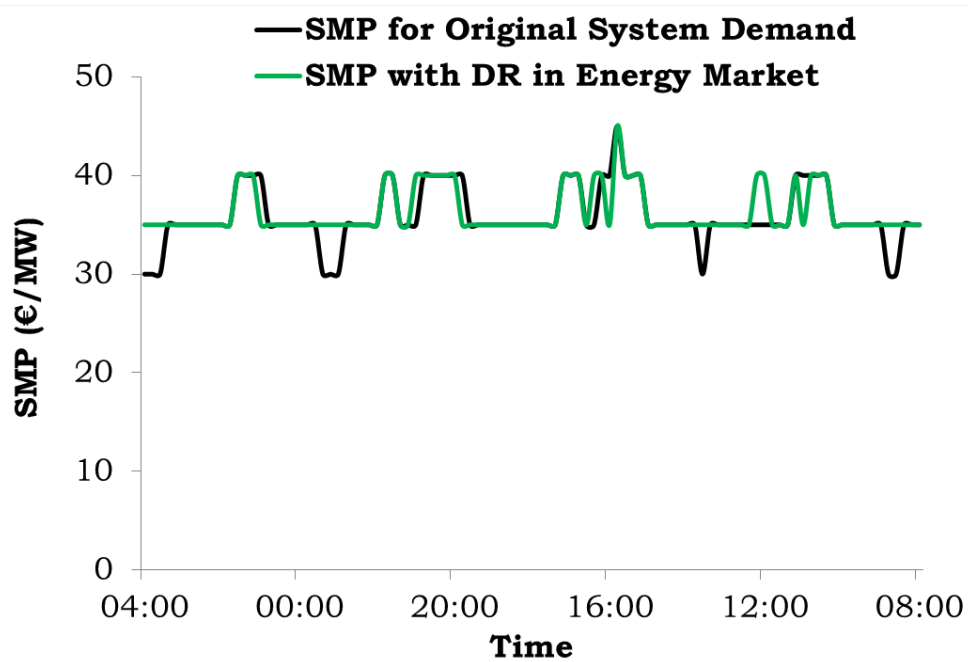


Figure 5.6: SMP with and without demand response over a four day period for a peak load of **2500 MW** and no wind generation

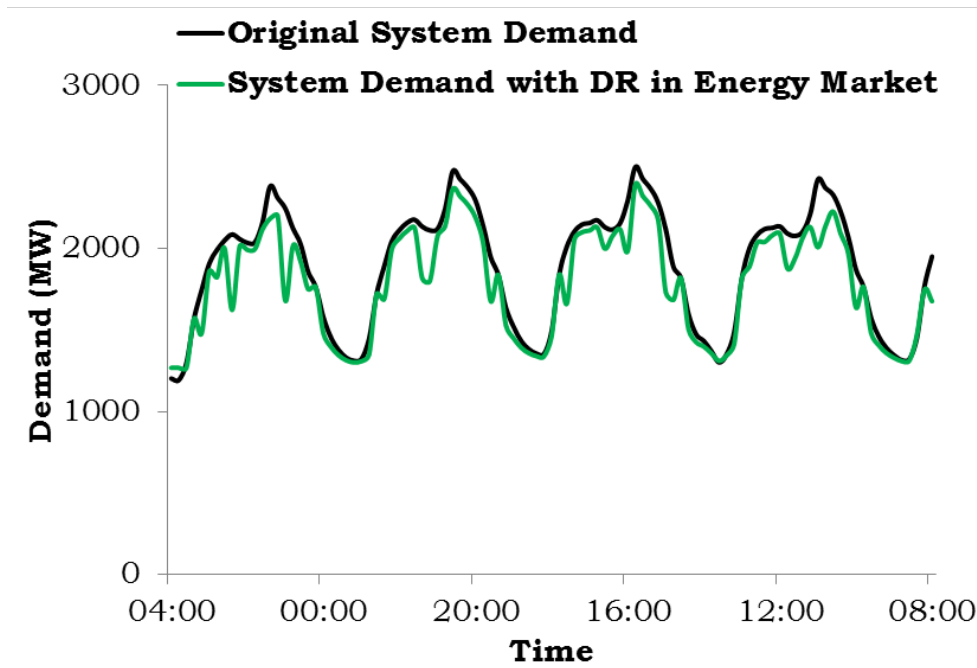


Figure 5.7: System demand with the addition of demand response over a four day period for a peak load of **2500 MW** with no wind generation

The reduction in impact on the demand profiles, and by extension, on the SMP profile, is as expected given that the size of the DR resource relative to the system demand is decreasing with increasing peak load. The DR resource represents 22%, 11% and 7.4% for peak load levels of 2500 MW, 5000 MW and 7500 MW, respectively. Thus, with increasing load, the impact the DR resource has on the system load profile decreases.

The addition of DR in markets other than the energy market has very little impact on the SMP, as seen in Figure 5.8 for a peak load of 2500 MW. A similar negligible change in SMP following inclusion of DR in multiple markets is also noted at the other peak load levels. This minor impact on SMP is because the price variations in the energy market is the key driver of the operational decisions of the load-shifting DR resource and because the dispatch of reserve is not considered.

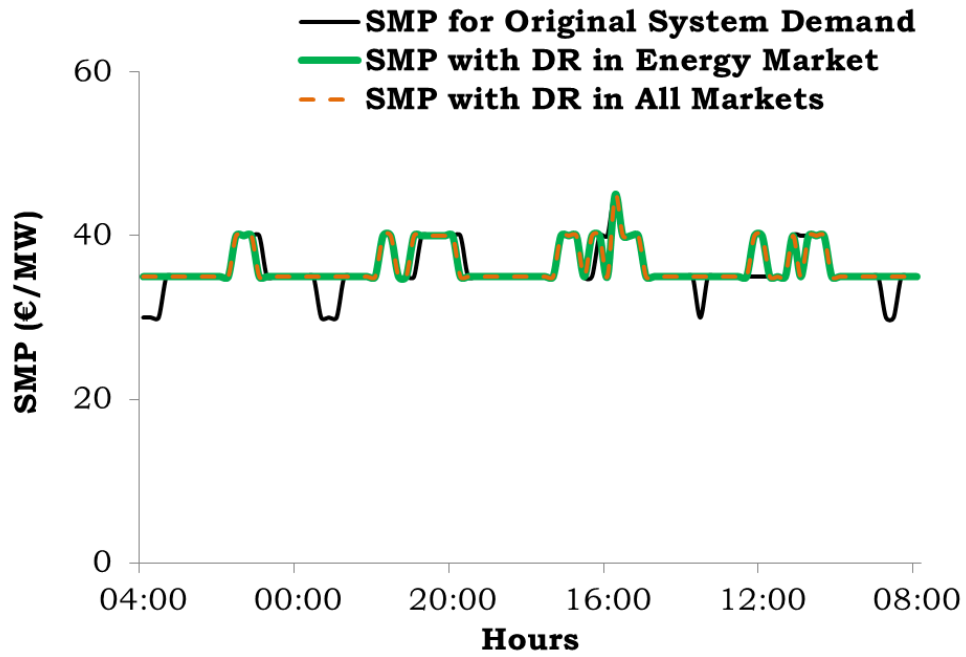


Figure 5.8: System marginal price with the addition of demand response in the energy market only and in all three markets over a four day period for a peak load of **2500 MW** with no wind generation

Furthermore, as will be discussed in the following section, for the majority of cases examined the reserve price is zero at all times and thus there is little incentive for the DR to drastically change its operational decisions in order to provide reserve as there is little revenue to be gained. Additionally, permitting a load-shifting DR resource to participate in more than one electricity market simultaneously is shown to not significantly impact upon the energy market. This is as a result of the energy limitation constraint placed on the DR resource, in order to ensure consumers' needs are satisfied, which restricts how the DR resource can operate.

When the reserve requirement is low, in this case 500 MW, it is found that the participation of DR in both the energy and reserve markets has a negligible impact on the SMP profile. The following section includes an examination of the impact an increase in the system reserve requirement can have on the capacity and reserve markets.

5.5.2 Impact of Demand Response on Reserve Market and Capacity Market

When there are low reserve requirements ($RESERVE_{REQ} = 500$ MW), the reserve prices (μ^t) are always €0 with and without DR inclusion in the reserve market. This is because, at such a low reserve requirement level, the generating firms are investing to meet the capacity target, which far exceeds the reserve requirement, and thus the firms are investing to receive capacity payments rather than to receive revenue from the reserve markets.

As mentioned in Chapter 2, the magnitudes of potential capacity payments are largely dependent on the requirements for capacity. This is indeed found to be the case here. As the peak load level increases, the requirement for generating capacity increases, which in turn results in higher capacity prices and consequently higher capacity payments to those participating in the capacity market. The capacity price is linked to the Lagrange multiplier of Equation (5.8). When $cap_{bid}^{i,j}$ is positive, that is when the firm chooses to participate in the capacity market, $\theta_2^{i,j}$ for firm i and each technology j is equal to the price at which the capacity market clears, κ .

Examining the KKT conditions for the generator problem shows that $\theta_2^{i,j}$ is related to both the cost of investment, $ICOST^j$, and the cost of maintenance, $MCOST^j$ for each firm, as well as to $\theta_1^{t,i,j}$ through both Equation (5.12) and (5.13). In turn, $\theta_1^{t,i,j}$ is dependent on the marginal cost and the SMP, λ , through Equation (5.9), as well as the reserve price μ through Equation (5.10). Essentially the decision for each firm i and each technology j , i.e each unit, to participate in the capacity market is dependent upon the investment, maintenance and marginal costs and the revenue earned from the electricity markets. Any firm and technology whose revenue does not cover costs does not participate in the capacity market. Thus, the price at which the capacity market clears, κ , is the value of the Lagrange multiplier $\theta_2^{i,j}$ for the firm and technology whose revenue exactly equals their costs. These marginal units often do not participate in the energy or reserve markets and so the only costs they incur are maintenance costs. Thus it will be seen that the capacity prices, κ , are regularly equivalent to the maintenance costs of the different technologies (see Table 5.5).

Table 5.9 displays that the capacity price, κ , associated with a peak load of 2500 MW, with and without elastic demand. It is seen that with 0 MW of wind and with an elasticity of $E = 0.0$, κ changes from €3.35 per MW to €7 per MW following the addition of DR. Figure 5.9 shows that following the addition of DR, there is a slight

increase in the installed capacity of peaking plants in the system generating portfolio. Thus, it is not surprising that the capacity price is €7 per MW, as this equates to the maintenance cost of the peaking units. However, with the participation of DR in all markets, there is a reduction in peaking plant capacity. This concurs with the results in [6], which finds that subject to correct pricing, DR resources can ‘*substitute for the energy function of peaking units*’.

Table 5.9: Capacity prices, κ , for **2500 MW** of peak load and with a reserve requirement of **500 MW**

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = 0$	€3.35	€7	€7	€7	€7
	$E = -0.11$	€7	€7	€7	€7	€7
500 MW	$E = 0$	€7	€7	€7	€7	€7
	$E = -0.11$	€7	€7	€7	€7	€7
1000 MW	$E = 0$	€7	€7	€7	€7	€7
	$E = -0.11$	€7	€7	€7	€7	€7
1500 MW	$E = 0$	€7	€7	€7	€7	€7
	$E = -0.11$	€7	€7	€7	€7	€7

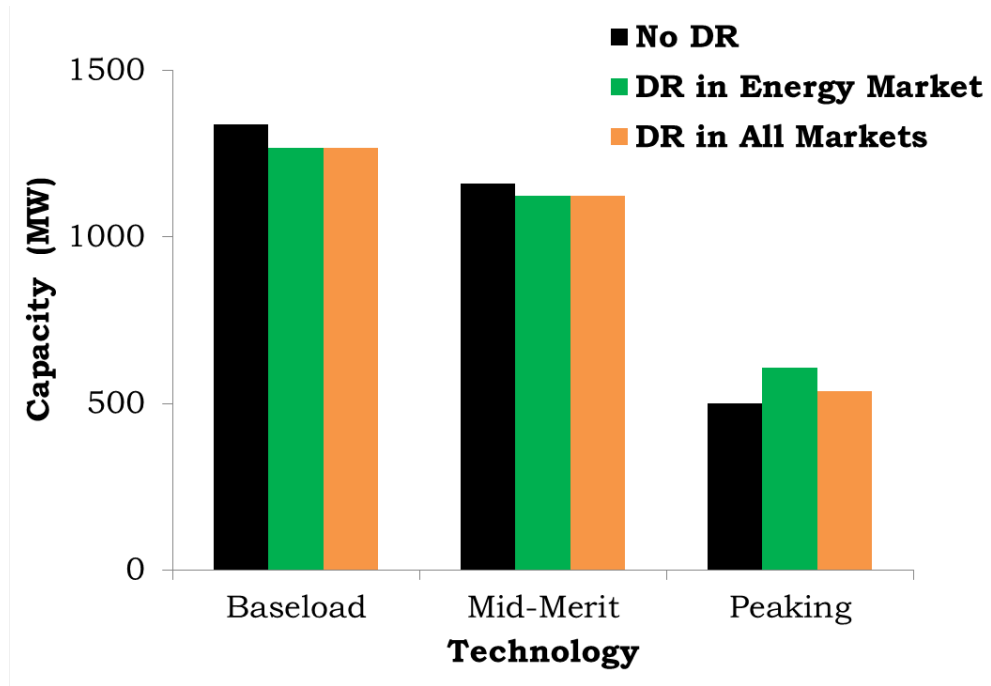


Figure 5.9: Installed generating capacity at a peak load of **2500 MW** with no wind generation

Table 5.10: Capacity prices, κ , for **5000 MW** of peak load and with a reserve requirement of **500 MW**

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
500 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
1000 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
1500 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25

As illustrated in Table 5.10, at a peak load level of 5000 MW, the capacity price increases to €25 per MW for all scenarios and all wind levels examined, with and without DR. Such a capacity price is to be expected given that €25 is the maintenance cost of baseload units and, as shown in Figure 5.10, baseload generation dominates the generating portfolio and there is no change in mid-merit or peaking capacity following the addition of DR.

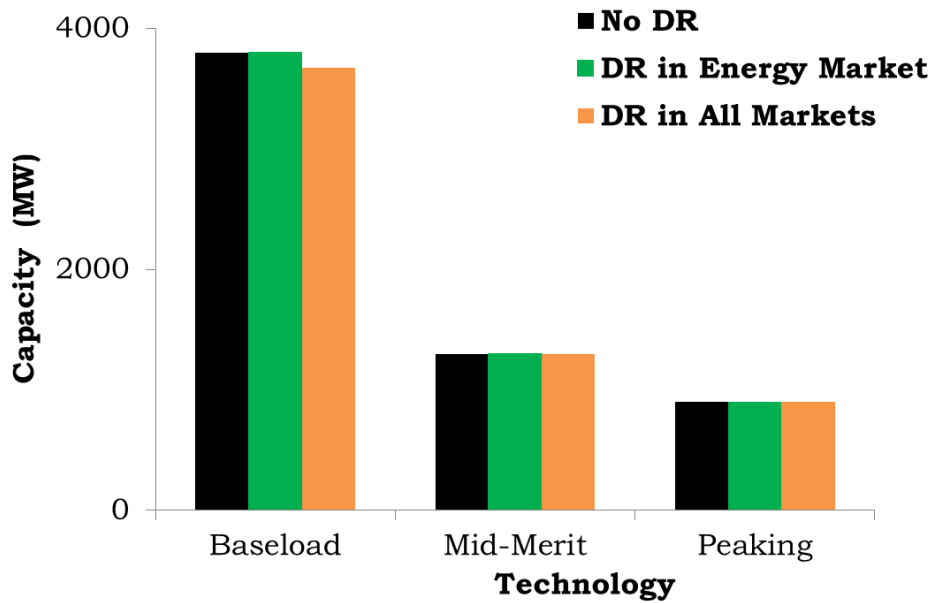


Figure 5.10: Installed generating capacity at a peak load of **5000 MW** and no wind generation

At a peak load level of 7500 MW, the capacity price remains at €25 per MW for all scenarios, except at a wind penetration of 1500 MW, when the capacity prices increase dramatically, as can be seen in Table 5.11. For the case of 1500 MW of wind, the capacity price is €65, which is the marginal cost of the most expensive baseload unit on the system.

In general, it is also noted that accounting for elastic load has little effect on the reserve prices or the capacity market prices, κ . This is as expected as the elastic demand cannot have direct influence on the capacity target, *TARGET*, for the capacity market as it is a fixed value and therefore elastic demand cannot impact upon the capacity bids and the capacity price. Furthermore, as will be seen in the next section 5.5.2.2 in Table 5.14, the capacity bids of the DR resources for all cases

is low relative to the capacity target, *TARGET*, thus DR only forms a small part of the capacity market. Consequently, DR participation in the capacity market results in only very marginal changes to the generating firms' investment and exit decisions, and accordingly only minor changes to the capacity bids of the different firms. As the marginal unit in the capacity market does not change following the addition of DR in general, the capacity price also does not change, which is what is seen for the vast majority of cases examined (Tables 5.9, 5.10 and 5.11).

Table 5.11: Capacity prices, κ , for **7500 MW** of peak load and with a reserve requirement of **500 MW**

Wind Level	Elastic Demand	Demand Response Market Participation				
		No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
500 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
1000 MW	$E = 0$	€25	€25	€25	€25	€25
	$E = -0.11$	€25	€25	€25	€25	€25
1500 MW	$E = 0$	€65	€1370	€1370	€233	€255
	$E = -0.11$	€110	€1402	€1402	€272	€272

However, at a peak load of 7500 MW and with an installed wind capacity of 1500 MW there are considerable changes in capacity price. The addition of DR in the energy market and in both the energy and reserve markets results in dramatic increases in the capacity price. As might be expected, DR participation in the capacity market reduces these capacity prices by a factor of 5 or more. This large increase in capacity price at both high peak load levels and high wind penetration is driven by the suppression in SMPs, which can be seen in Figure 5.11, as a result of high wind generation and DR participation. This suppression in SMPs would result in a reduction in generator profit. However, the generators' problem seeks to maximize profit. Consequently, higher prices are needed to clear the capacity market in order

to cover the costs associated with the high investment at high peak load levels and to maximize their profits. Crucially, system operating costs do not increase drastically for this particular scenario, as will be shown in Table I.3 in Appendix I.

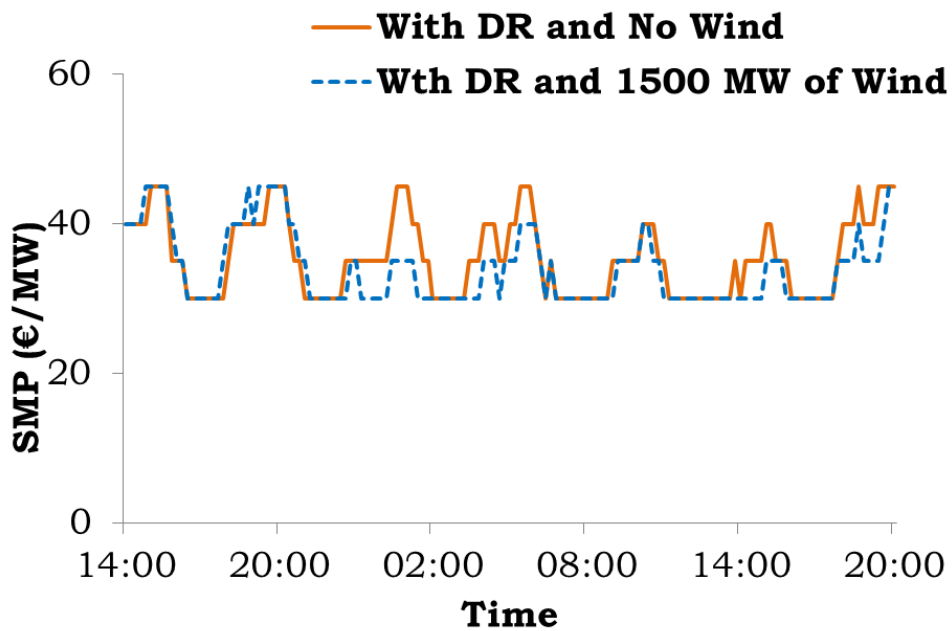


Figure 5.11: System marginal price at a peak load of **7500 MW** with wind generation 20% of peak load

5.5.2.1 Increasing the Reserve Requirement

At the highest load level, 7500 MW, the initial endowment of capacity is insufficient to meet the capacity target, *TARGET*. Thus, increasing the reserve requirement to 1500 MW has no impact on the reserve price, which remains at €0, as the generating firms are continuing to invest in order to meet the capacity target. As sufficient capacity will be installed as a result of meeting this target, there is no cost associated with providing reserve.

At lower peak load levels of 2500 MW and 5000 MW, however, increasing the reserve requirement to 1500 MW impacts upon both the reserve price (at the peak hour only) and on the capacity price. At these lower peak load levels, the necessity to meet the more stringent reserve requirement dominates investment decisions, that is the reserve market constraint becomes binding, and thus firms invest, or stay in the market, in order to meet the reserve requirement, not the capacity target. This results

in capacity prices of €0 for all cases, while the reserve price is extremely low at all hours, except at the peak hour where the reserve price is €25. Figure 5.12 highlights that the increased reserve requirement at a peak load of 2500 MW incentives greater levels of investment in mid-merit and peaking generation.

While the same change in technology is not noted at peak load levels of 5000 MW or 7500 MW, there is a change in the installed capacity of each firm, with a tendency towards greater levels of installed capacity for firms with more expensive baseload units, particularly firm 5. However, as will be shown in section 5.5.3, particularly in Table 5.16, this increase in the capacity of the most expensive baseload unit results in only a very marginal increase in the system operating costs (an increase less than 0.053%) which will be seen later.

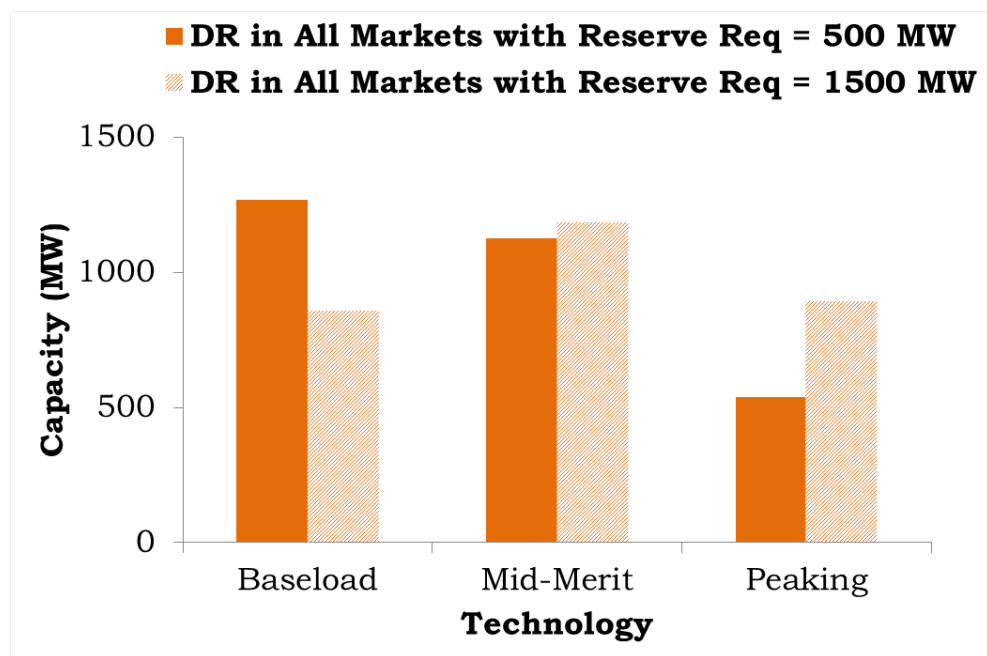


Figure 5.12: Installed generating capacity with increasing reserve requirement with no wind generation at a peak load of **2500 MW**

At a peak load of 7500 MW however, there are capacity prices greater than €0, but it is noted that there is a dependency on whether DR is participating in the capacity market. In the cases without DR participation in the capacity market, the capacity price is €25, while it is €0 when DR does provide capacity. Thus, it can be seen that the inclusion of DR in the capacity market at this high peak load level and high reserve requirement suppresses the capacity price. Here it can be seen that the inclusion of DR in the capacity market results in the erosion of the market it is well

suitable to participation in, as mentioned in Chapter 2. This is a result of the fact there is effectively no cost associated with the DR resources participation in the capacity market.

Table 5.12: Capacity prices at a peak load of **7500 MW** and with a reserve requirement of **1500 MW**

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = 0.0$	€25	€25	€25	€0	€0
	$E = -0.11$	€25	€25	€25	€0	€0

5.5.2.2 Capacity Bids for the Demand Response Resource

Table 5.13 compares the capacity bids of the DR resources obtained from the MCP with the ELCC estimations obtained from the methodology developed and presented in section 3.3.2 in Chapter 3. Recall how constant non-dispatchable load added incrementally to each hour of the year (ΔL) represented the ELCC in Chapter 3 (see Section 3.3.2).

As can be seen, there is only a slight difference between the two approaches. This validates, to a limited extent, the approach for estimating the DR resource's contribution to generation employed in here. Furthermore, a comparison of the capacity bids, cap_{dr} (MW), of the DR resources obtained from the MCP with the estimations obtained from the methodology developed and presented in section 3.4.2 in Chapter 3 reveals that there is consistency in the estimates of the contribution to generation adequacy for similar resources.

Table 5.13: Capacity bid and ELCC of demand response for a peak load level of **7500 MW** with a reserve provision of **500 MW** and without wind generation

Peak Load	Metric	cap_{dr}	ΔL
7500 MW	ELCC	126 MW	132 MW
	CV	23%	24%

The *TARGET* parameter for the capacity market is set to be 1.2 times the peak system load which, in the cases of peaks loads of 5000 MW and 7500 MW, results in a target for capacity exceeding the initial endowment. In order to reach this target there is a requirement for additional generating capacity, and as wind does not participate in the capacity market in these case studies, although it is acknowledged that wind does have a CV, this need for additional capacity results in investment in generation. When DR does participate in the capacity market, the net change in generating capacity is less than the case without DR, as would be expected.

In previous work presented in Chapter 4, it was found that there is a trade-off between the CV estimates of a DR resource and whether that DR resource commits to providing reserve. In such a case, the trade off resulted in a higher CV estimate for a DR resource when it did not commit to providing reserve since the resource had a greater ability to shift demand away from the peak load hours. However, it is found here that there is no change in the CV estimates of the DR resource, irrespective of whether or not reserve is provided. This is a result of the fact that exit and investment decisions are considered here, whereas previously, in Chapter 4, exit and investment decisions were not accounted for.

A comparison of the 'DR in the Energy and Capacity Markets' case and the 'DR in All Markets' case in Figure J.1 in Appendix J illustrates that there is a very slight restructuring of the generating portfolio to account for the lack of reserve provision from the DR resource. Furthermore, as the reserve price is found to be €0 there is little incentive for the DR resource to provide reserve.

It is also noted that the CV estimates here are higher than those determined in Chapter 4. These higher estimates are a result of the fact that there are less DR resource constraints considered within the MCP in comparison to those constraints considered in the UCED in Chapter 4. In particular, the constraint which ensures that consumer's requirements are met at each point in time is omitted here. Instead, a 24-hour energy balance is included.

Consequently, as seen in Figure 5.13, there is a marked difference in the manner in which the same resource is operated. The DR resource operation is contingent on the inclusion, in the case of the UCED dispatch of the resource, or the omission, in the case of the MCP-based dispatch, of the consumer requirement constraint and constraints pertaining to the physical characteristics of the resource.

As can be clearly seen, in the case of the MCP-based dispatch there is no operation of the DR resource around the peak load hours, the hours with the highest LOLP. Consequently, it is unsurprising that such operation is awarded a higher CV. Although this is a limitation of the DR model employed here, this comparison clearly reiterates the importance of modeling the resource as realistically as is possible, which was recommended in Chapter 2 and demonstrated in Chapter 3.

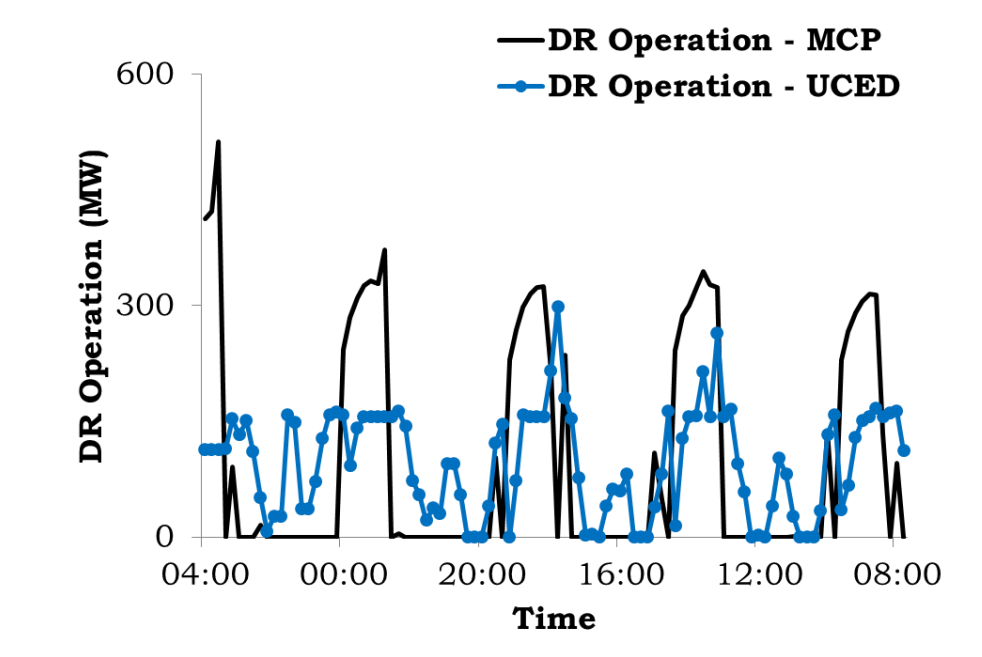


Figure 5.13: Comparison of the demand response resource operation from the Mixed Complementarity Problem employed here with the demand response resource operation from Chapter 4. without wind generation

Table 5.14 displays the capacity bid estimates for the DR resource. From Table 5.14 it can be seen that for a peak load of 2500 MW, the DR resource has the highest capacity bids at a installed wind capacity of 500 MW. However, at peak loads of 5000 MW and 7500 MW, the highest capacity bids are found at installed wind capacities of 1000 MW and 0 MW, respectively. It can be seen that there is no discernible trend.

Changing the marginal cost parameters (see Table K.1 in Appendix K) results in a change in the generating portfolio (Figure K.3), which in turn produces different CV estimates, as shown in Table K.5. However, similarly, there is no clear trend as regards the peak load and installed wind levels for optimal DR contribution to the capacity market. This indicates that perhaps the decision as to whether DR should participate in the capacity market is a case-by-case, or system specific, consideration.

Table 5.14: Capacity bids of demand response, cap_{dr} , with and without reserve provision, with a system reserve requirement of **500 MW**, with elastic demand

Peak Load	Wind Level	cap_{dr} with reserve	cap_{dr} without reserve
2500 MW	0 MW	71 MW	71 MW
	500 MW	82 MW	82 MW
	1000 MW	69 MW	69 MW
	1500 MW	0 MW	0 MW
5000 MW	0 MW	123 MW	123 MW
	500 MW	122 MW	122 MW
	1000 MW	129 MW	129 MW
	1500 MW	110 MW	110 MW
7500 MW	0 MW	126 MW	126 MW
	500 MW	119 MW	119 MW
	1000 MW	122 MW	122 MW
	1500 MW	114 MW	118 MW

5.5.3 Impact of Demand Response on System Costs

In order to determine the system operating costs, Equation (5.38) is used for the model without DR, while Equation (5.39) is employed for all MCP models with DR. These equations represent the cost incurred by the system in meeting demand, reserve and capacity requirements. The energy cost associated with wind participation in the energy market is also considered. These system costs are the total costs paid by consumers, rather than fuel, carbon and other costs incurred by the generating firms.

$$\begin{aligned}
 Cost_{System} = \sum_t \sum_i \sum_j (Gen^{t,i,j} \times \lambda^t + Reserve_{Gen}^{t,i,j} \times \mu^t) + \sum_i \sum_j (Cap_{Bid}^{i,j}) \times \kappa \\
 + WIND^t \times \lambda^t \quad (5.38)
 \end{aligned}$$

$$\begin{aligned}
Cost_{System} = & \sum_t \sum_i \sum_j (Gen^{t,i,j} \times \lambda^t + Reserve_{Gen}^{t,i,j} \times \mu^t) + \sum_i \sum_j (Cap_{Bid}^{i,j}) \times \kappa \\
& + \sum_t (Reserve_{DR}^t \times \mu^t) + Cap_{DR} \times \kappa + WIND^t \times \lambda^t \quad (5.39)
\end{aligned}$$

Table 5.15 displays the system cost savings attainable for the different peak load levels and different wind penetrations, following the introduction of DR. As would be expected, and as a result of the positive impact of DR participation in the energy market, there are considerable system operating cost savings to be obtained following the introduction of DR into electricity markets. This concurs with [74].

The analysis here shows that the increase in system costs with increasing peak load is roughly commensurate with the magnitude of the increase in peak load. For example, the system costs at 5000 MW are effectively two times the system costs at 2500 MW. Similarly, the system costs at 7500 MW are a factor of 1.5 greater than the corresponding costs at a peak load of 5000 MW. The change in system costs, however, with the addition of DR is roughly 6%, 3% and 2%, for peak load levels of 2500 MW, 5000 MW and 7500 MW, respectively. The incorporation of elastic load, as well as DR, produces only small changes in system costs, with a maximum decrease of 0.23% in system costs with the addition of elastic demand compared to the system costs without elastic demand.

The only scenario examined in which the system operating costs change measurably with varying DR market participation is the case with a peak load level of 7500 MW and an installed wind capacity of 1500 MW. This results in two values (final row of Table 5.15). The first, lower value, corresponds to the percentage savings as a result of DR participating the energy market only or in just the energy and reserve market, as it is found there is a negligible difference. For this case, when DR participates in the capacity market there are greater cost savings, resulting in the second, higher value, in the final row of Table 5.15. This is a result of the differences in operating cost depending on the DR market participation combination driven by the significant change in capacity prices noted at this peak load and installed wind level, which are not noted for any other case (see Table 5.11).

Table 5.15: Percentage reduction in system operating costs, including wind costs, following inclusion of demand response

Peak Load	Wind Level	<u>Without Elastic Demand</u>	<u>With Elastic Demand</u>
2500 MW	0 MW	6%	6%
	500 MW	6%	6%
	1000 MW	6%	6%
	1500 MW	7%	7%
5000 MW	0 MW	3%	3%
	500 MW	3%	3%
	1000 MW	3%	3%
	1500 MW	3%	3%
7500 MW	0 MW	2%	1%
	500 MW	2%	2%
	1000 MW	2%	2%
	1500 MW	1% or 2%	1% or 2%

The system operating costs for a range of different scenarios, as well as the percentage change values, are presented in Tables I.1, I.3, I.2 and I.4 in Appendix I. It is found that, for the majority of cases, there is no change between the cases with DR participation in different combinations of energy markets. Once DR is added, the only additional change to the system is participation of DR in the reserve and capacity markets, which, as seen earlier, does not result in changes to either the reserve price or the capacity price. As noted in the previous section, at low levels of reserve requirement (in this case, 500 MW) the reserve price is zero at all times, and participation of DR in the capacity market, which as was noted earlier, for the most part, does not change with the addition of DR.

These cost savings are between 1% and 7% depending on the peak load level and the installed wind capacity, as shown in Table 5.15. The level of these savings decreases with increasing peak load level, as would be expected given that the ratio of the capacity of the DR resource to system load is decreasing pro rata with increasing peak load level. It was also found that, in general, with increasing wind generation levels, the operating cost savings attributed to DR participation also increases. This

concur with the work of [34]. The operating cost savings clearly indicate that inclusion of the DR resource has a positive impact on the electrical power system. This confirms that there is likely a negative cost to integrating DR into the power system and thus there is a likely integration benefit associated with DR. These results also concur with the work in [118], where it is found that an energy storage device can reduce generation costs at peak hours by about 10%.

As mentioned earlier, increasing the reserve requirement has an impact on both the reserve price and the capacity price. Thus it is conceivable that there would be a similar impact on the system operating costs. Table 5.16 compares the system operating costs with two different levels of reserve requirement and illustrates that, as expected, the higher reserve requirement results in slightly increased system operating costs.

Table 5.16: System costs with demand response participating in all markets with different reserve requirements without wind generation

Peak Load	Elasticity	Reserve 500 MW	Reserve 1500 MW
2500 MW	$E = 0$	€130,559,381	€130,607,182
	$E = -0.11$	€130,249,339	€130,296,961
5000 MW	$E = 0$	€271,725,303	€271,734,977
	$E = -0.11$	€271,405,604	€271,415,101
7500 MW	$E = 0$	€413,225,784	€413,444,320
	$E = -0.11$	€412,907,825	€413,126,622

5.5.4 Impact of Demand Response on Generator Profit

Unlike the case with system operating costs, generator profit is not dramatically impacted by the participation of DR in the various electricity markets. In fact, in some cases, generator profit actually increases slightly with the introduction of the DR resource. Crucially, generator profit is more heavily affected by the increasing penetration of wind generation than it is by the participation of DR.

These results differ from the work presented in [128], where it was found that the introduction of more flexible demand generally reduces the profits of generators.

However, in this section, the profits of the generator includes consideration of the cost of investing in new generator technology as per Equation 5.3, while in the work in [128], generator investment decisions, and associated costs, were not included. Consequently, here it is seen that the profit of generators increases slightly due to the fact that the introduction of DR results in less generator investment. Generator profit, not accounting for investment costs, however, reveals that there is indeed a reduction following the introduction of DR, thus confirming the results in [128], albeit for a different system using a different method. The impact of demand elasticity alone, not including DR, on the profit of the generating firms is negligible. This is due to the fact that demand elasticity has only very marginal impacts on the investment decisions of the generating firms and because the high maintenance and investment costs dampen any changes in energy market revenue as a result of elastic demand.

At a low load level of 2500 MW, where the capacity of the DR resource represents 22% of the system peak demand, the inclusion of DR in the energy market and in the energy and reserve markets has an impact on the profit of Firms 3 and 5 (firms with both mid-merit and peaking technologies)) (Figure 5.14), with these firms experiencing less profit compared to the case without DR participation. This reduction in profit is more easily seen by expressing generator profit as a percentage of the 'No DR' case (Figure 5.15). At higher load levels, the impact of DR on the profit of these firms becomes less pronounced. Furthermore, examination of Figures 5.15 to 5.17 illustrates that, once DR is included in the energy market, further participation in the other electricity markets has negligible impact on generator profits.

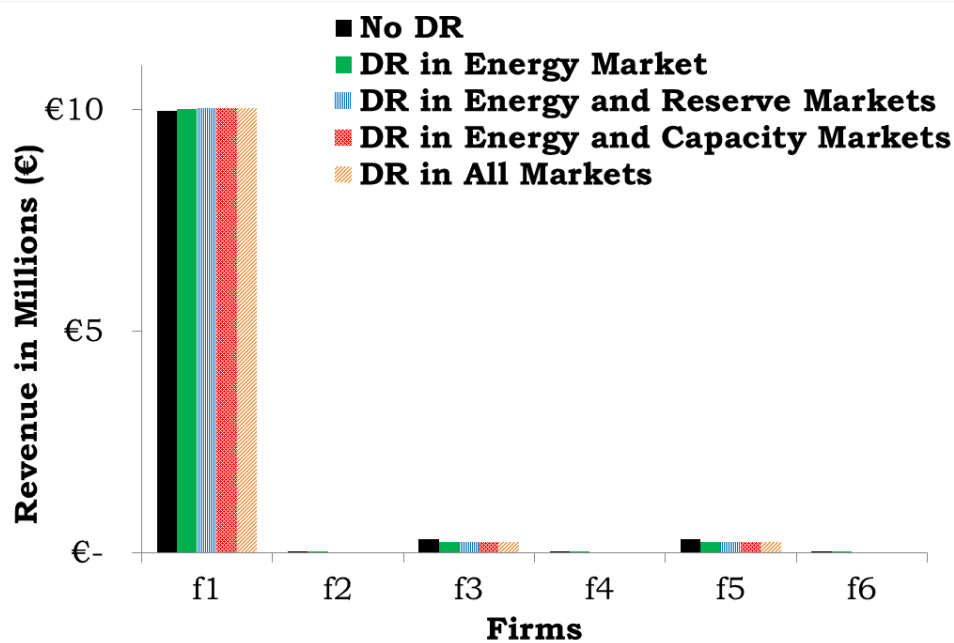


Figure 5.14: Generator profit under different demand response market participation frameworks with 0 MW of wind and with a peak load of **2500 MW**

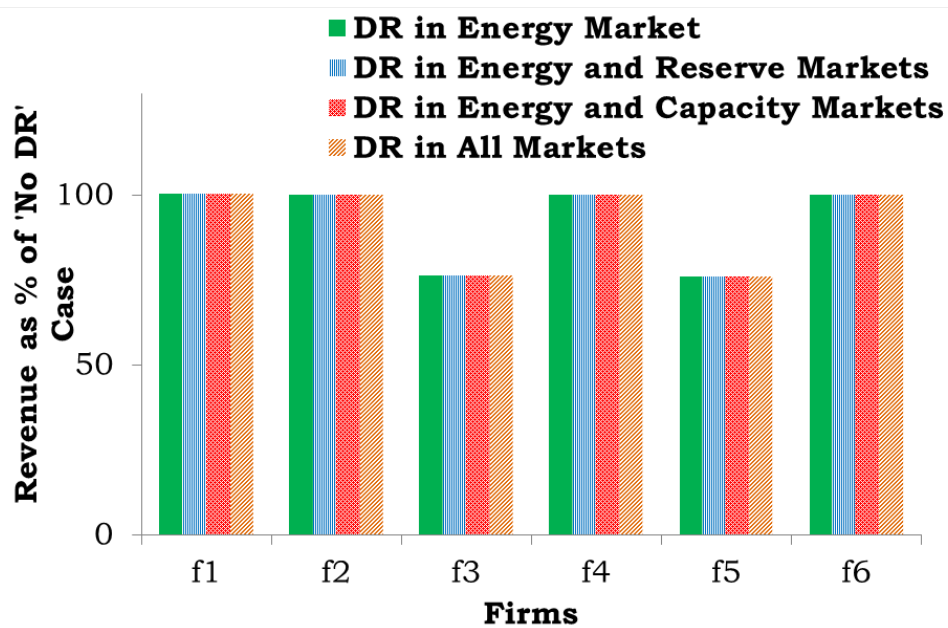


Figure 5.15: Generator profit as a percentage of the 'No DR' case for different demand response market participation frameworks with 0 MW of wind and with a peak load of **2500 MW**

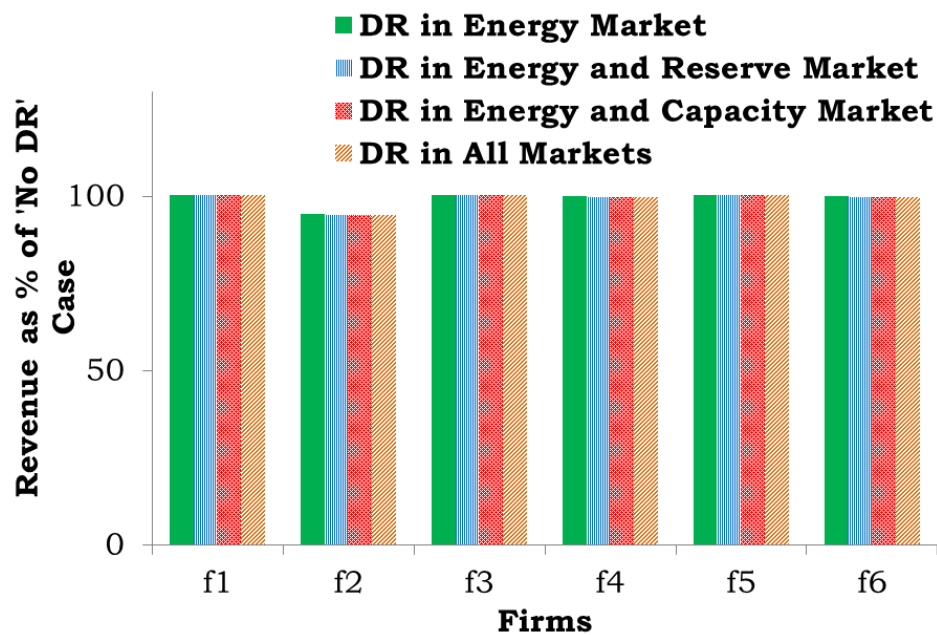


Figure 5.16: Generator profit as a percentage of the 'No DR' case for different demand response market participation frameworks with a peak load of **5000 MW**

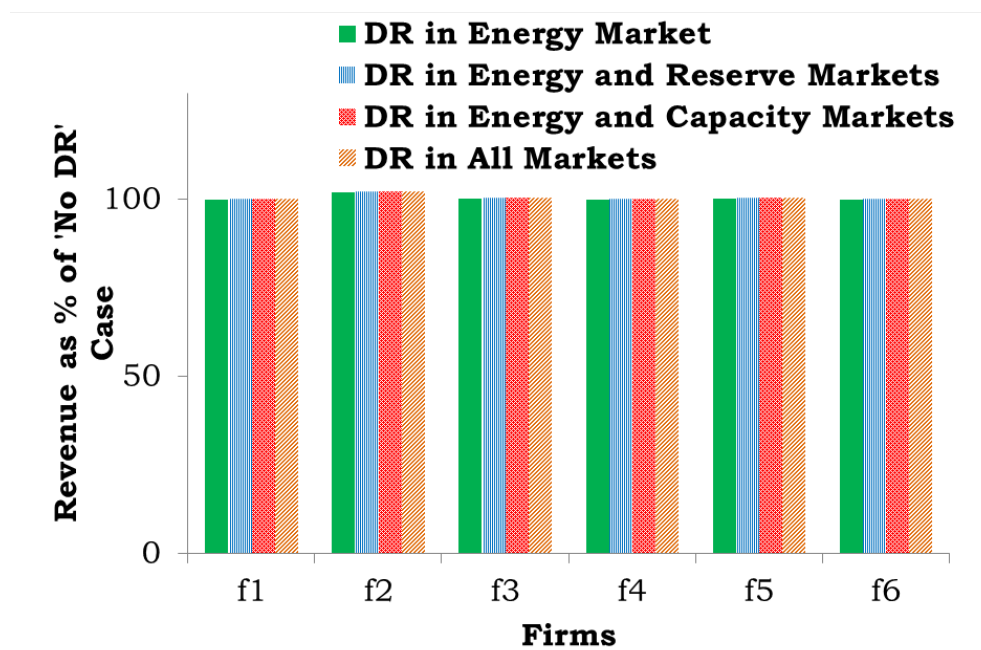


Figure 5.17: Generator profit as a percentage of 'No DR' case for different demand response market participation frameworks with a peak load of **7500 MW**

5.5.5 Demand Response Aggregator Costs

This section examines the impact DR market participation can have on the costs imposed on the DR aggregator. The DR aggregator is responsible for ensuring consumers' loads are satisfied and this requires the purchase of energy from the market. As previously discussed, the aggregator seeks to maximize revenue from the reserve and capacity markets, whilst minimizing energy costs.

Table 5.17 through Table 5.23 indicate that the costs incurred by the DR aggregator decrease following the inclusion of the DR resource in electricity markets, as expected. The aggregator cost savings are between 4% and 12%, depending on the different cases examined. It is found that, in general, with increasing wind generation levels, the operating cost savings attributed to DR participation also increase slightly. This concurs with the work of [34]. However, it is found at a peak load of 5000 MW, that higher wind generation on the system does not necessarily result in higher cost savings for DR aggregators, illustrating the subtlety of the interaction of the DR resource with the power system. Therefore, in general, with wind generation there is price suppression but because of the constraints on the operation of the DR resource it cannot shift appropriately.

Table 5.17: Demand response aggregator costs at a peak load of **2500 MW** and **without** elastic demand included

DR Case	Wind Level	Total Energy Costs Only	Savings	With Capacity Payments	
				Total Costs	Savings
No DR	0 MW	€7,268,572	—	—	—
Energy Only		€6,992,651	4%	—	—
Energy & Reserve		€6,992,620	4%	—	—
Energy & Capacity		€6,992,919	4%	€6,993,413	4%
All Markets		€6,992,919	4%	€6,993,413	4%
No DR	500 MW	€7,273,270	—	—	—
Energy Only		€6,982,895	4%	—	—
Energy & Reserve		€6,982,777	4%	—	—
Energy & Capacity		€6,992,650	4%	€6,992,156	3.9%
All Markets		€6,992,919	4%	€6,992,425	3.9%
No DR	1000 MW	€7,294,062	—	—	—
Energy Only		€6,971,008	4%	—	—
Energy & Reserve		€6,964,518	5%	—	—
Energy & Capacity		€6,964,519	5%	€6,964,007	5%
All Markets		€6,964,518	5%	€6,964,006	5%
No DR	1500 MW	€7,118,854	—	—	—
Energy Only		€6,538,129	8%	—	—
Energy & Reserve		€6,538,128	8%	—	—
Energy & Capacity		€6,538,129	8%	€6,538,129	8%
All Markets		€6,538,128	8%	€6,538,128	8%

Table 5.18: Demand response aggregator costs at a peak load of **5000 MW** and **without** elastic demand included

DR Case	Wind Level	Total Energy Costs Only	Savings	With Capacity Payments	
				Total Costs	Savings
No DR	0 MW	€7,174,428	—	—	—
Energy Only		€6,530,551	9%	—	—
Energy & Reserve		€6,529,881	9%	—	—
Energy & Capacity		€6,530,845	9%	€6,527,760	9%
All Markets		€6,530,073	9%	€6,526,988	9%
No DR	500 MW	€7,181,910	—	—	—
Energy Only		€6,541,809	9%	—	—
Energy & Reserve		€6,541,604	9%	—	—
Energy & Capacity		€6,540,726	9%	€6,537,671	9%
All Markets		€6,541,739	9%	€6,538,684	9%
No DR	1000 MW	€7,111,938	—	—	—
Energy Only		€6,565,703	8%	—	—
Energy & Reserve		€6,565,703	8%	—	—
Energy & Capacity		€6,565,934	8%	€6,562,707	8%
All Markets		€6,565,272	8%	€6,562,045	8%
No DR	1500 MW	€7,192,125	—	—	—
Energy Only		€6,643,309	8%	—	—
Energy & Reserve		€6,643,097	8%	—	—
Energy & Capacity		€6,643,306	8%	€6,640,561	8%
All Markets		€6,643,096	8%	€6,640,351	8%

Table 5.19: Demand response aggregator costs at a peak load of **7500 MW** and **without** elastic demand included

DR Case	Wind Level	Total Energy Costs Only	Savings	With Capacity Payments	
				Total Costs	Savings
No DR	0 MW	€7,193,387	—	—	—
Energy Only		€6,529,452	9%	—	—
Energy & Reserve		€6,528,007	9%	—	—
Energy & Capacity		€6,527,756	9%	€6,524,599	9%
All Markets		€6,527,860	9%	€6,524,703	9%
No DR	500 MW	€7,197,031	—	—	—
Energy Only		€6,529,880	9%	—	—
Energy & Reserve		€6,528,129	9%	—	—
Energy & Capacity		€6,529,062	9%	€6,526,080	9%
All Markets		€6,527,511	9%	€6,524,766	9%
No DR	1000 MW	€7,196,959	—	—	—
Energy Only		€6,528,555	9%	—	—
Energy & Reserve		€6,529,891	9%	—	—
Energy & Capacity		€6,527,786	9%	€6,524,734	9%
All Markets		€6,529,070	9%	€6,526,018	9%
No DR	1500 MW	€7,193,545	—	—	—
Energy Only		€6,528,872	9%	—	—
Energy & Reserve		€6,529,746	9%	—	—
Energy & Capacity		€6,528,855	9%	€6,495,331	10%
All Markets		€6,528,787	9%	€6,500,227	10%

Tables 5.17, 5.18 and 5.19 illustrate the DR aggregator cost savings that can be attained for each of the different scenarios examined, without price-responsive load, i.e. with $E = 0$. Comparing with Tables 5.20, 5.21 and 5.22 shows that there are only very marginal changes in the DR aggregator cost savings with price-responsive load included, i.e. with $E = -0.11$. Table 5.23 displays the DR aggregator cost savings with the higher reserve requirement of 1500 MW for each of the peak load levels. It can be seen that there are only very marginal differences in savings between the higher and lower reserve requirements. This indicates, for the scenarios examined and for the type of DR considered in this chapter, that increased system reserve requirements do not result in greater revenues, and thus cost savings, for DR aggregators.

It is found from Tables 5.17, 5.18 and 5.19 that the DR aggregator savings increase with increasing peak load level. However, for each peak load level and for each wind level, regardless of the markets DR participates in, the percentage cost savings are similar. Neither the addition of elastic demand (Tables 5.20 to 5.22) nor the increase in reserve requirement (Table 5.23) significantly changes the DR aggregator's cost savings.

Table 5.20: Demand response aggregator costs at a peak load of **2500 MW** and **with** elastic demand included

DR Case	Wind Level	Energy Costs	Savings
No DR	0 MW	€7,268,961	—
Energy Only		€6,992,656	4%
Energy & Reserve		€6,992,614	4%
Energy & Capacity		€6,992,332	4%
All Markets		€6,992,118	4%
No DR	1500 MW	€7,113,773	—
Energy Only		€6,538,263	8%
Energy & Reserve		€6,538,263	8%
Energy & Capacity		€6,538,263	8%
All Markets		€6,538,263	8%

Table 5.21: Demand response aggregator costs at a peak load of **5000 MW** and **with** elastic demand included

DR Case	Wind Level	Energy Costs	Savings
No DR	0 MW	€7,174,783	—
Energy Only		€6,530,838	9%
Energy & Reserve		€6,530,013	9%
Energy & Capacity		€6,527,760	9%
All Markets		€6,526,908	9%
No DR	1500 MW	€7,192,125	—
Energy Only		€6,643,309	8%
Energy & Reserve		€6,643,563	8%
Energy & Capacity		€6,640,156	8%
All Markets		€6,640,284	8%

Table 5.22: Demand response aggregator costs at a peak load of **7500 MW** and **with** elastic demand included

DR Case	Wind Level	Energy Costs	Savings
No DR	0 MW	€7,193,387	—
Energy Only		€6,526,963	9%
Energy & Reserve		€6,528,960	9%
Energy & Capacity		€6,523,612	9%
All Markets		€6,525,829	9%
No DR	1500 MW	€7,191,498	—
Energy Only		€6,528,872	9%
Energy & Reserve		€6,529,387	9%
Energy & Capacity		€6,497,859	10%
All Markets		€6,361,990	12%

Table 5.23: Demand response aggregator costs with a reserve requirement of **1500 MW**, without wind generation and **without** elastic demand

DR Case	Load Level	Total Energy Costs Only	Savings	With Capacity Payments	
				Total Costs	Savings
No DR	2500 MW	€7,268,435	—	—	—
All Markets		€6,992,926	4%	€6,992,926	4%
No DR	5000 MW	€7,175,771	—	—	—
All Markets		€6,529,741	9%	€6,529,741	9%
No DR	7500 MW	€7,193,387	—	—	—
All Markets		€6,528,929	9%	€6,528,929	9%

Varying the marginal cost inputs, and thus the generating portfolio, however, it is found that the DR aggregator savings remain relatively constant with increasing peak load level. Again, this hints at the optimal portfolio for DR being highly system dependent, as was the case with the CV estimates presented in Section 5.5.2.2. The DR aggregator costs with these changed marginal cost inputs are displayed in Appendix K.

As discussed earlier in Chapter 2, savings on customers' electricity bills may not be sufficient enough to warrant investment in equipment and to compensate for the inconvenience [35] associated with engaging in a DR program. This does indeed seem to be the case here. Examination of the results suggests that considerable additional savings, over and above the savings as a result of participating in the energy only market, are not awarded to the DR aggregator following the participation in the reserve market. This is because, as discussed previously, the reserve prices for these particular cases were effectively zero at all times. Thus, choosing to participate in the reserve market may not result in considerable reward and may in fact put consumers at a risk of inconvenience, in the form of their load requirements not being met during emergency operating periods, that is periods when the DR resource is called upon to provide reserve. This risk stems from the assumptions employed that any inconvenience placed on customers as a result of failure to meet their heating requirements during emergency operation is compensated by the revenue they receive by permitting their devices to be committed to providing reserve. It is shown here

that the reserve market may not in fact be particularly lucrative for load-shifting DR resources, which concurs with the concerns stated in [35] that the cost savings associated with DR may not justify the inconvenience placed on consumers.

It is acknowledged that some systems do not have reserve markets, but they do have reserve payments. In such a system it may still be worthwhile for a DR resource, such as the type of resource described here, to provide reserve. While there is little benefit for the DR aggregator here to participation in the reserve market, there is a slight increase in cost savings as a result of participation in the capacity market. The capacity payments earned by the DR resource varies depending on the peak load level, since, as discussed earlier, the capacity price varies considerably with peak load level. In general, however, the capacity payment acquired by the DR resource is less than €3,000, except at a peak load level of 7500 MW and with a wind penetration of 1500 MW. The increase in capacity payments in this scenario is not driven by a change in operating decisions on the part of the DR aggregator. Rather it is a result of significantly higher capacity prices (see final two rows in Table 5.11). However, it is found that at lower wind levels and at lower peak load levels, there are no further cost savings available through participation in markets other than the energy market.

Nevertheless, as previously alluded to in Chapter 4, load-shifting DR resources have an inherent contribution to generation adequacy as a result of their operation. Thus, given that the ability of the DR resource to participate in the capacity market is, in effect, a consequence of the operation of the resource in the energy market, there does not appear to be any indication that participation in both the energy and capacity markets results in a trade-off. Consequently, it could be concluded that the optimal portfolio for the load-shifting resource examined here is participation in the energy and capacity markets only.

5.6 Conclusion

The results in this chapter indicate that, in general, the DR resource can have a positive impact on electricity markets. However, this impact is largely limited to the energy market. The participation of DR in the energy market does succeed in reducing variability in SMP, whilst increasing prices at off-peak hours and decreasing peak prices. This concurs with the literature, as was discussed in Chapter 2. However, it was found that including DR in reserve markets had only a minor effect. This is a result of the fact that the reserve prices, μ^t , in the system examined, were zero for the

vast majority of cases and hours, even when the reserve requirement was increased. As discussed in section 5.5.2.1, investment decisions at high peak load levels are dominated by the requirement to meet the capacity target. Thus, as sufficient capacity will be installed, apart from the opportunity cost of not participating in the energy market, there is no cost associated with providing reserve.

As discussed in section 5.5.3 there are significant system operating cost savings to be obtained following the introduction of DR into electricity markets, mainly driven by the impact of the DR resource on the energy market. These cost savings are between 1% and 7%. While these are substantial cost savings, the results presented in this chapter indicate that there is minimal impact on generator revenue following the addition of DR in the various electricity markets as a result of the fact that the generators can strategically amend their decisions as regards how much capacity to invest in and remove from the market as a result of the inclusion of the DR resource.

Using the MCP framework developed in this chapter, future work should consider performing more sensitivities. In particular, different types of DR resources could be examined and varying the parameters, as well as the reference demand profile. Different systems should be analysed by changing the marginal costs and initial endowment parameters used, as well as different load profiles. Furthermore, DR resource other than load-shifting resources should be considered in order to determine the best combination of markets to participate in, as the conclusions provided here are specific to the type of resource examined in this chapter.

Conclusions

6.1 General Conclusions

This thesis identified the key challenges and barriers that exist to evaluating DR. Different approaches for calculating the capacity value of various DR resources were developed enabling a greater appreciation of the contribution of DR to the power system. It is confirmed that the contribution of DR resources to the electrical power system is positive. It has been shown in Chapters 3 and 4 that DR resources do indeed provide a positive contribution to system generation adequacy and thus have a CV associated with them. An understanding of the CV of a DR resource and the factors influencing it, such as is provided in this thesis, can assist in the operation and planning of future power systems with more involvement from the demand-side. Furthermore, it has been illustrated in Chapter 5 that there are significant power system operational cost savings to be attained through the participation of DR in electricity markets. Thus, in general, as discussed in Chapter 2, DR resources have an integration benefit associated with them. This thesis has shown that this benefit is influenced by a number of factors.

6.1.1 Physical Constraints of the Demand Response Resource

The need to be fully aware of the constraints on the resource availability when assessing DR was highlighted in Chapter 3. The CV estimates of DR with and without an energy recovery constraint were compared. The key results show that when energy recovery following DR deployment is considered the CV estimates fall significantly, resulting in very low, or in some cases negative, CV estimates. It is clear from the results presented in Chapter 3 that the CV of DR is affected by the manner in which

the DR resource is deployed, i.e. with or without an energy recovery and the frequency of deployment. Dispatch of the DR resources utilizing the algorithms developed in Chapter 3 can create unintentional, additional system peaks, which results in negative ELCC estimates and a negative contribution to generation adequacy.

The impact of the need for realistic modeling of the DR resources on their contribution to generation adequacy was further illustrated in Chapter 4. It is found that inclusion of a specific constraint to meet consumers' requirements at each point in time results in low CV estimates of the associated DR resource, in comparison with similar resources analyzed in Chapter 3. Furthermore, as is noted in Chapter 5, failure to model details of the physical characteristics of the resource or failure to correctly represent the need to satisfy consumer requirements results in, effectively, over-estimates of the contribution to generation adequacy.

6.1.2 Interaction with the Power System and between Multiple Services

The intrinsic interrelationship between electricity markets is discussed in Chapter 5. It was shown that the inclusion of a capacity market incentivizes investment in generating technology, which also results in a reduction in the energy costs of the power system. This interrelationship is also seen following the introduction of DR.

Chapter 4 provided a greater understanding of the impact of load-shifting resources on generation adequacy by giving an indication of the CV of a load-shifting resource which can provide multiple services. It is found that the provision of reserve services reduces the CV estimates of the load-shifting resource. The trade-off between providing reserve and contributing to generation adequacy is evident from the CV estimates obtained for the different cases examined and illustrates the need to be cognizant of the interactions between services.

In Chapter 5 the load-shifting resource's ability to participate in the energy market was the main driver of the operation of the resource. Thus, changes on the supply-side of the power system, for example the inclusion of greater levels of wind, can alter the system marginal price profile, which in turn can change how a load-shifting resource operates. For example, it was shown in Chapter 5 that for certain peak load levels, higher wind penetration levels on the system do not necessarily result in higher cost savings for DR aggregators. This illustrates that the interactions between power system resources is not always consistent and indicates the system specific nature of power system and DR evaluation analysis.

The results suggest that there maybe no major potential opportunity for a load-shifting resource, such as that considered in Chapter 5, to participate in a reserve market. This is an important result for DR aggregators aiming to determine how best to operate the resources in their portfolios.

6.1.3 Underlying Power System Characteristics

Chapter 3 shows that even completely removing a demand-side resource from the power system does not necessary result in a very high CV. The correlation between the DR resource availability and the hours when the probability of losing load is highest for a specific system, dictates the resultant CV. Furthermore, the most valuable DR resources in terms of CV differ from system to system, depending upon the power system characteristics, whether the system peak demand occurs in the winter or summer. Chapter 5 also showed that the CV estimates were dependent upon the peak load level and the installed wind generation capacity.

The analysis presented in Chapter 4 showed that the generation adequacy of the underlying power system, or the base case, can have a considerable impact upon the CV estimate of the DR resource. In essence, a system with high generation adequacy metrics is a system which is in need of further capacity. Such a system could be considered a suitable environment for DR to make a significant contribution to generation adequacy. However, as shown in Chapter 3 and to a greater extent in 4, DR resources do not typically have significant contributions to generation adequacy.

6.2 Suggestions for Future Work

Throughout this thesis the importance of accurate modeling of DR resources was noted and highlighted. Future work on DR should carefully consider the various physical constraints on the specific DR resources. There is also a need for more accurate modeling of reserve services from DR resources. For example, future modeling of reserve services from DR should include a constraint restricting a load-shifting DR resource from providing reserve when it is in a period of energy recovery, similar to the constraint imposed in [6]. Future work should explore the impact of updated modeling of load-shifting DR resources as more data becomes available from DR programs and trials.

As was mentioned in Chapter 4, no wind generation was considered as part of the generation adequacy analysis. Future work could consider the interaction of the DR resource with wind generation and their combined contribution to generation adequacy, to complement the work in [31].

Using the CV calculation approaches presented in this thesis and the MCP framework developed in Chapter 5 of this thesis, future work could include more sensitivities. In particular, different types of DR resources could be examined by using the generic model of DR developed in Chapter 5 and varying the parameters, as well as the reference demand profile. Different systems could be examined by changing the marginal costs and initial endowment parameters used, as well as different load profiles.

Future work on Capacity Remuneration Mechanisms is also advised. Large increases in capacity price were noted in Chapter 5 at both high peak load levels and high wind penetration levels. These large increases were driven by the suppression of SMPs as a result of high wind generation coupled with DR participation. Consequently, in order to cover investment costs and maintenance costs and to maximize their profits, the capacity market cleared at much higher prices in order to cover the costs associated with the high investment at high peak load levels and to maximize their revenues. This result opens up a number of questions regarding the best manner in which to procure capacity in future electricity markets with high levels of DR participation.

Taking a broader perspective, an interesting area of potential future research could be based around designing the optimal DR resource, at the planning stage, similar to the approach taken for generation planning, but on the demand-side of the power system. This could involve determining appropriate DR resource and device sizing in terms of rated capacity and the size of any storage components. It may consider the necessary amount of redundancy required in order to provide reserve services whilst continuing to meet consumers' requirements even during emergency operation and during deployment of reserve services. This could permit determination of the capital costs, volumes, footprints and other device characteristics. This could then be compared against potential market revenue, allowing a greater understanding of the costs required to develop and implement DR programs.

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Appendix

Plant Data

A.1 Plant Data Utilized in Chapter 3

Table A.1: Sample Power System Plant Data

Plant	Capacity (MW)	1 - FOR	Plant	Capacity (MW)	1 - FOR
1	83	0.97	30	280	0.938562
2	83	0.97	31	112	0.952329
3	117.6	0.96	32	85	0.952329
4	21	0.9756164	33	163	0.832795
5	22	0.9756164	34	104	0.956164
6	19	0.9756164	35	109.5	0.867288
7	24	0.9765753	36	109.5	0.867288
8	258	0.9369041	37	242	0.867288
9	52	0.95	38	480	0.941479
10	88	0.9508219	39	52	0.95
11	90	0.9508219	40	52	0.95
12	90	0.9517808	41	54	0.8677
13	10	0.9760959	42	54	0.868
14	10	0.9760959	43	240	0.86
15	22.5	0.9760959	44	240	0.86
16	22.5	0.9909589	45	73	0.935753
17	54	0.6	46	73	0.978712
18	49	0.6	47	73	0.935753
19	109	0.6	48	73	0.935753
20	15	0.9765753	49	52	0.95
21	4	0.9770548	50	137	0.929315
22	8	0.9765753	51	415	0.98
23	15	0.975137	52	379	0.964
24	15	0.9765753	53	404	0.93
25	4	0.975137	54	343	0.93
26	4	0.975137	55	420.8	0.970986
27	91	0.9500685	56	445	0.98
28	280	0.9366027	57	12.812	1
29	280	0.9410548			

A.2 Plant Data for the Irish System Utilized in Chapter 4

Table A.2: Irish System Plant Data

Plant	Capacity (MW)	1 - FOR	Plant	Capacity (MW)	1 - FOR
1	258	0.964	35	52	0.9966
2	435	0.973	36	101	0.95
3	90	0.981	37	247	0.97
4	90	0.981	38	247	0.97
5	90	0.981	39	170	0.89
6	415	0.98	40	170	0.97
7	404	0.93	41	170	0.95
8	343	0.93	42	404	0.975
9	88	0.981	43	238	0.968
10	163	0.861	44	238	0.968
11	104	0.981	45	58	0.98
12	480	0.96	46	58	0.98
13	83	0.985	47	58	0.989
14	83	0.985	48	29	0.9918
15	388.48	0.964	49	29	0.9918
16	445	0.9665	50	42	0.98
17	54	0.998	51	42	0.98
18	49	0.914	52	17	0.981
19	113	0.983	53	62	0.981
20	54	0.9969	54	21	0.985
21	54	0.9894	55	22	0.985
22	240	0.997	56	19	0.985
23	240	0.95	57	24	0.985
24	285	0.953	58	10	0.985
25	285	0.953	59	10	0.985
26	285	0.953	60	22.5	0.985
27	117.6	0.95	61	22.5	0.985
28	98	0.953	62	15	0.986
29	137	0.952	63	4	0.986
30	58	0.98	64	8	0.986
31	58	0.98	65	15	0.985
32	52	1	66	15	0.985
33	52	0.998	67	4	0.985
34	52	0.987	68	4	0.985

Demand Response Availability

Algorithms used in Chapter 3

B.1 No Energy Recovery

B.1.1 DR Availability Algorithm without Energy Recovery and Maximum 1-Hour Response Duration (Case 2)

```

if  $DR_{req}^t = 1$  and  $DR_{req}^{t+1} = 0$  then
    DR is required for 1 hour;
     $DR^t = DR_{avail}^t$ ;
     $DR^{t+1} = 0$ ;
     $t = t + 2$ ;
else if  $DR_{req}^t = 1$  and  $DR_{req}^{t+1} = 1$  and  $DR_{req}^{t+2} = 0$  then
    DR is required for 2 hours;
     $DR^t = DR_{avail}^t$ ;
     $DR^{t+1} = 0$ ;
     $t = t + 2$ ;
else
    DR is not required
     $t = t + 1$ ;
end if

```

B.1.2 DR Availability Algorithm without Energy Recovery and Maximum 2-Hour Response Duration (Cases 3,4, and 5)

if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 0$ **then**

DR is required for 1 hour;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = 0;$$

$$t = t + 2;$$

else if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 1$ and $DR_{req}^{t+2} = 0$ **then**

DR is required for 2 hours;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = DR_{avail}^{t+1};$$

$$DR^{t+2} = 0;$$

$$DR^{t+3} = 0;$$

$$t = t + 3;$$

else if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 1$ and $DR_{req}^{t+2} = 1$ **then**

DR is required for 3 hours;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = DR_{avail}^{t+1};$$

$$DR^{t+2} = 0;$$

$$DR^{t+3} = 0;$$

$$t = t + 4;$$

else

DR is not required

$$t = t + 1;$$

end if

B.2 With Energy Recovery

B.2.1 DR Availability Algorithm with Energy Recovery and Maximum 2-Hour Response Duration (Cases A, B, C)

if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 0$ **then**

DR is required for 1 hour;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = -DR_{avail}^t;$$

$$t = t + 2;$$

else if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 1$ and $DR_{req}^{t+2} = 0$ **then**

DR is required for 2 hours;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = DR_{avail}^{t+1};$$

$$DR^{t+2} = -DR_{avail}^t;$$

$$DR^{t+3} = -DR_{avail}^{t+1};$$

$$t = t + 3;$$

else if $DR_{req}^t = 1$ and $DR_{req}^{t+1} = 1$ and $DR_{req}^{t+2} = 1$ **then**

DR is required for 3 hours;

$$DR^t = DR_{avail}^t;$$

$$DR^{t+1} = DR_{avail}^{t+1};$$

$$DR^{t+2} = -DR_{avail}^t;$$

$$DR^{t+3} = -DR_{avail}^{t+1};$$

$$t = t + 4;$$

else

DR is not required

$$t = t + 1;$$

end if

Graphical Comparison of ELCC Estimates in Chapter 3

Graphs relating to the results presented in Tables 3.15, 3.16, 3.18 and 3.17 in Chapter 3 are shown here.

C.1 ELCC Comparisons for Avista Corporation

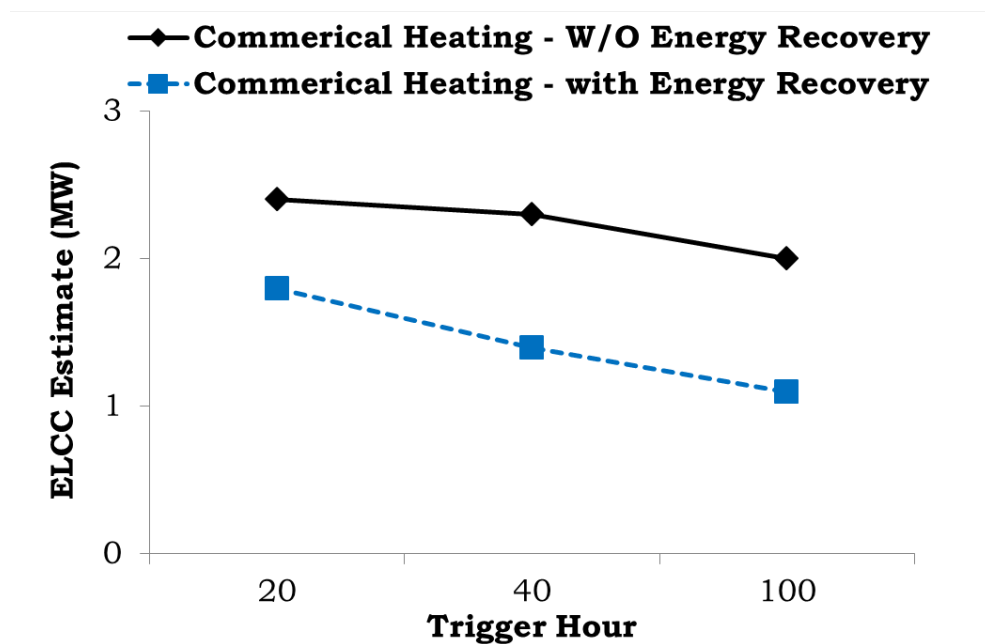


Figure C.1: Avista Corporation commercial heating

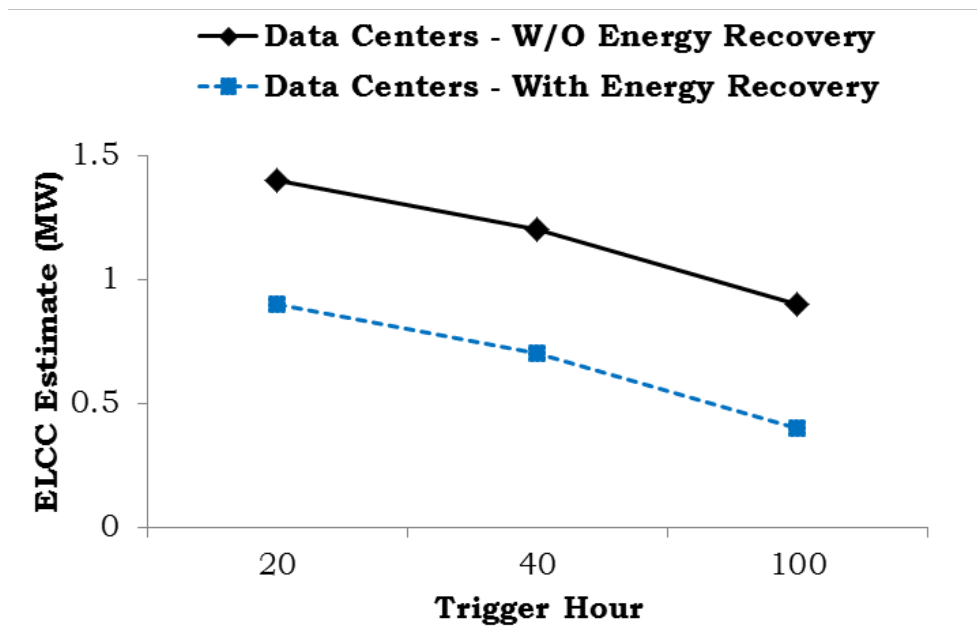


Figure C.2: Avista Corporation data centers

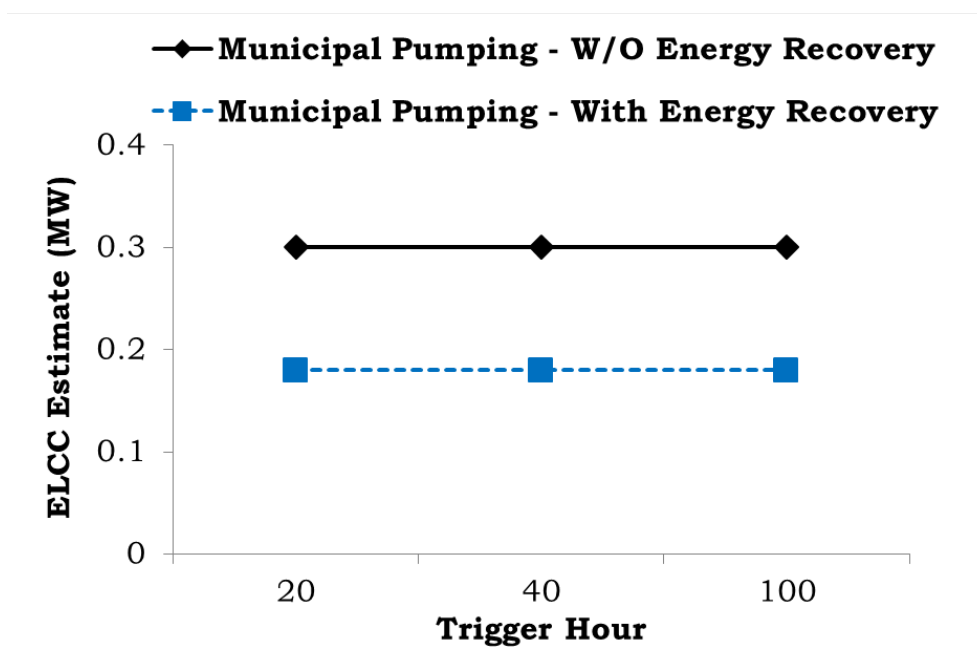


Figure C.3: Avista Corporation municipal pumping

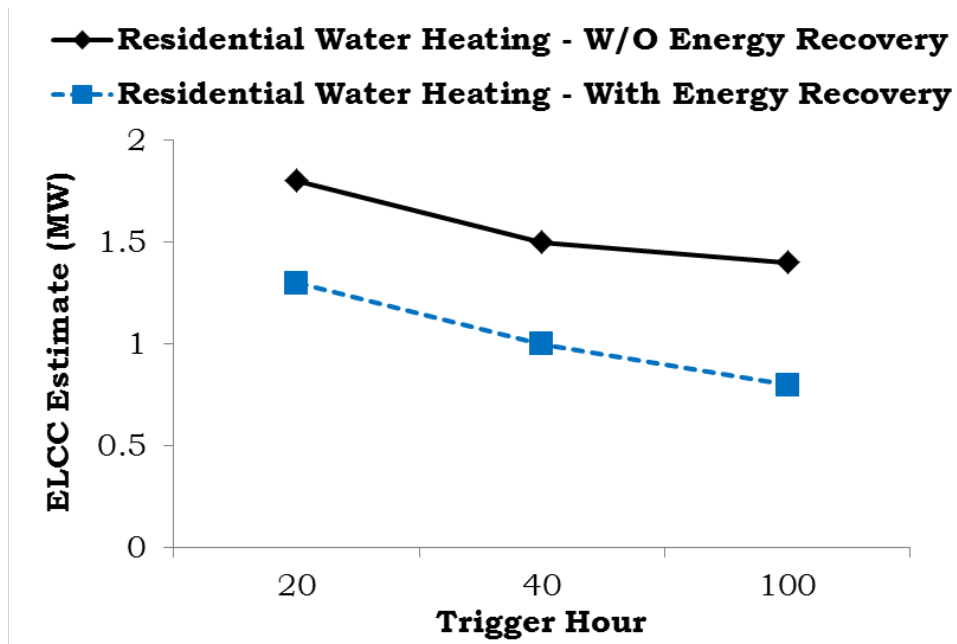


Figure C.4: Avista Corporation residential water heating

C.2 ELCC Comparisons for Puget Sound Energy

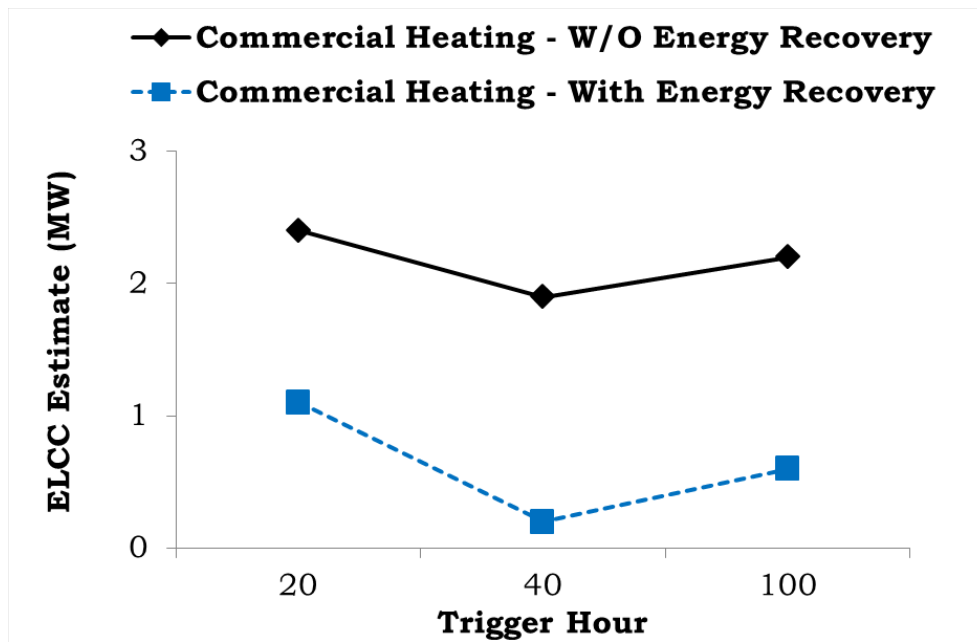


Figure C.5: Puget Sound Energy commercial heating

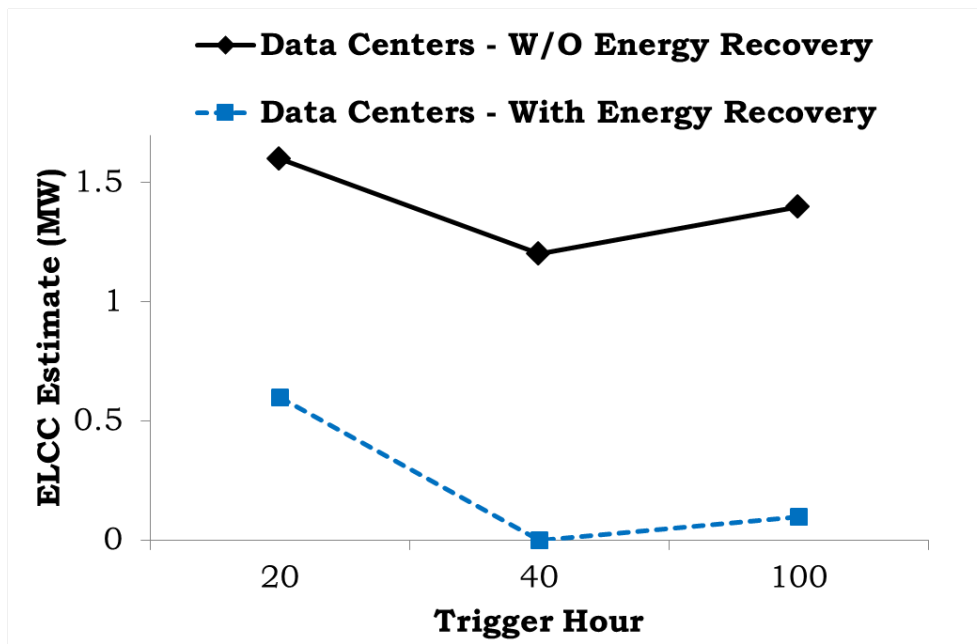


Figure C.6: Puget Sound Energy data centers

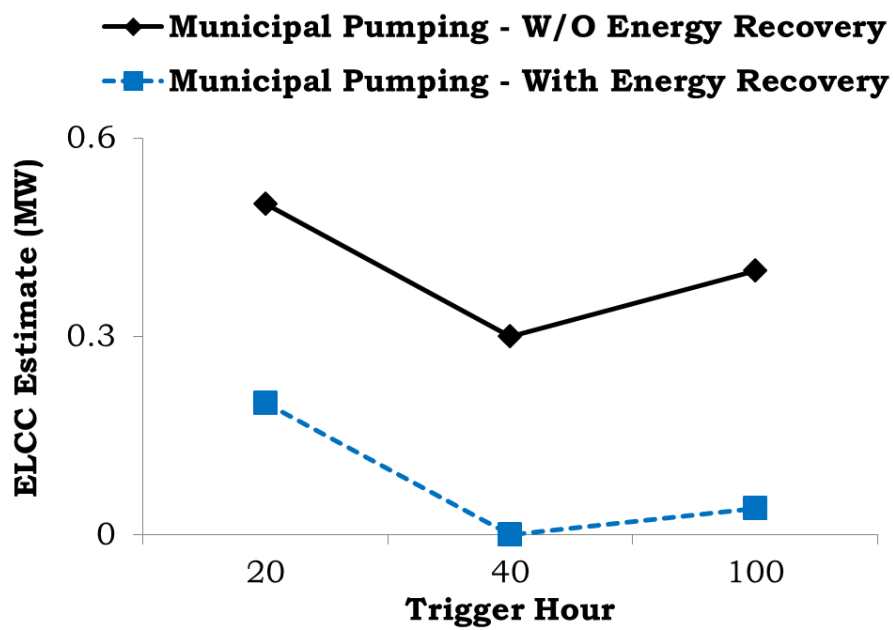


Figure C.7: Puget Sound Energy municipal pumping

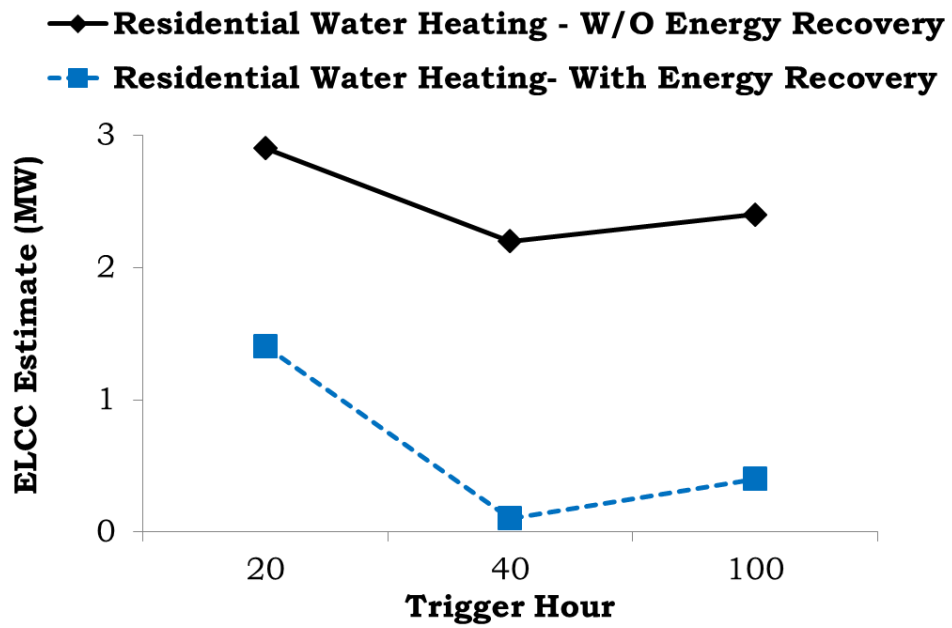


Figure C.8: Puget Sound Energy residential water heating

C.3 ELCC Comparisons for Nevada Power

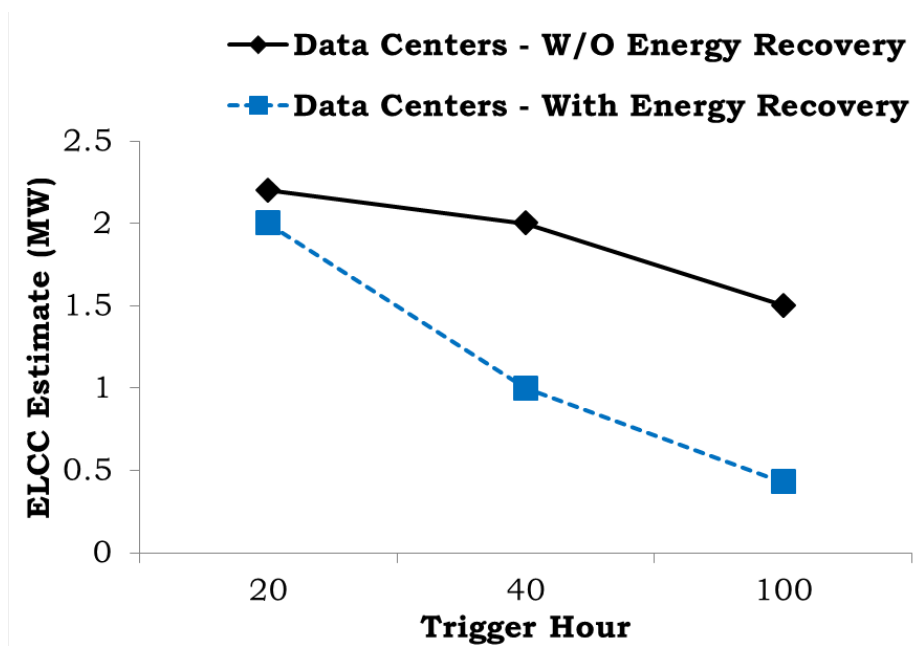


Figure C.9: Nevada Power data centers

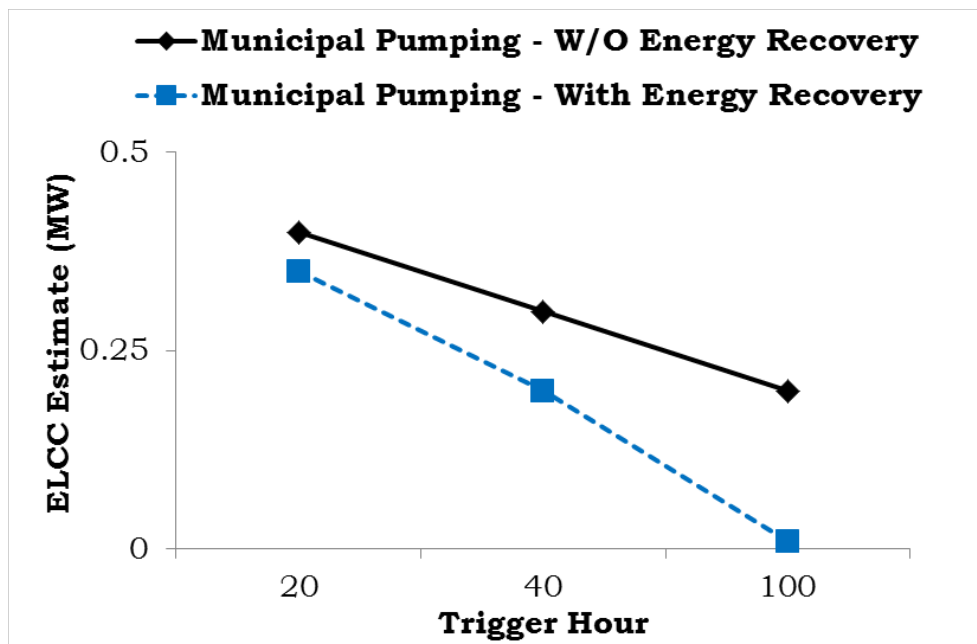


Figure C.10: Nevada Power municipal pumping

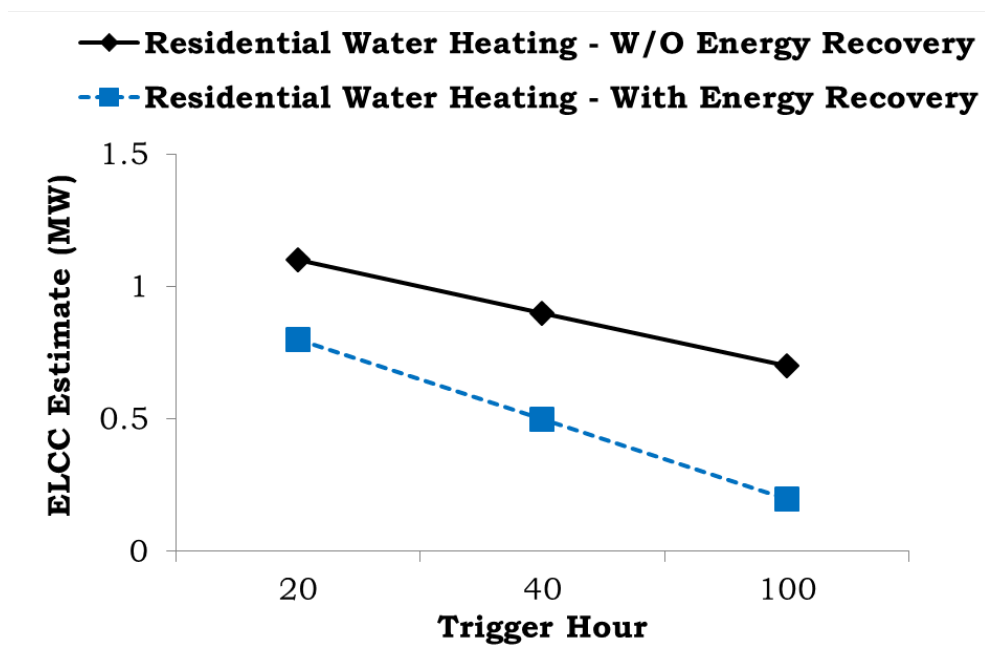


Figure C.11: Nevada Power residential water heating

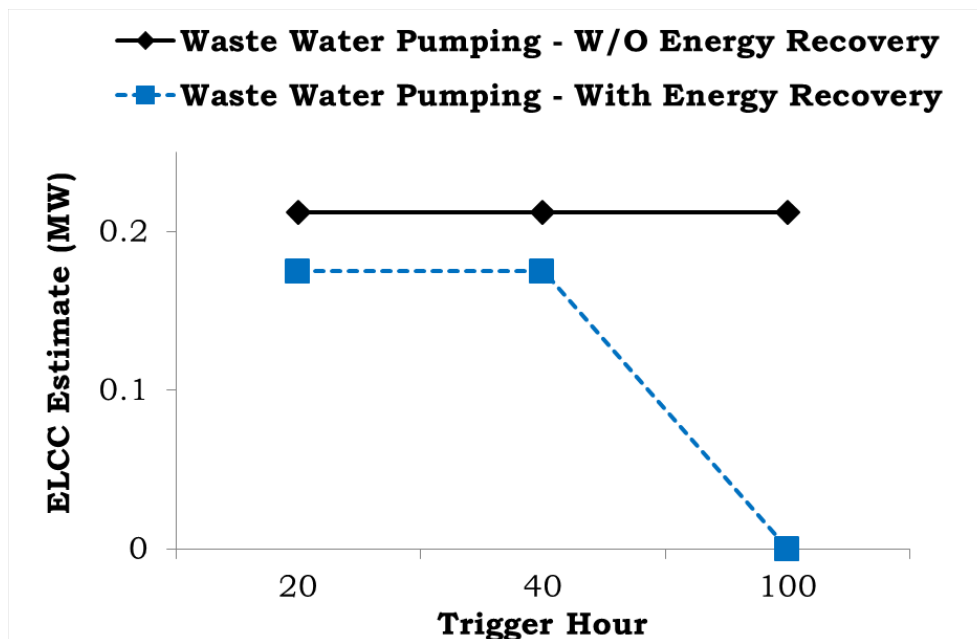


Figure C.12: Nevada Power waste water pumping

C.4 ELCC Comparisons for Los Angeles Department of Water and Power

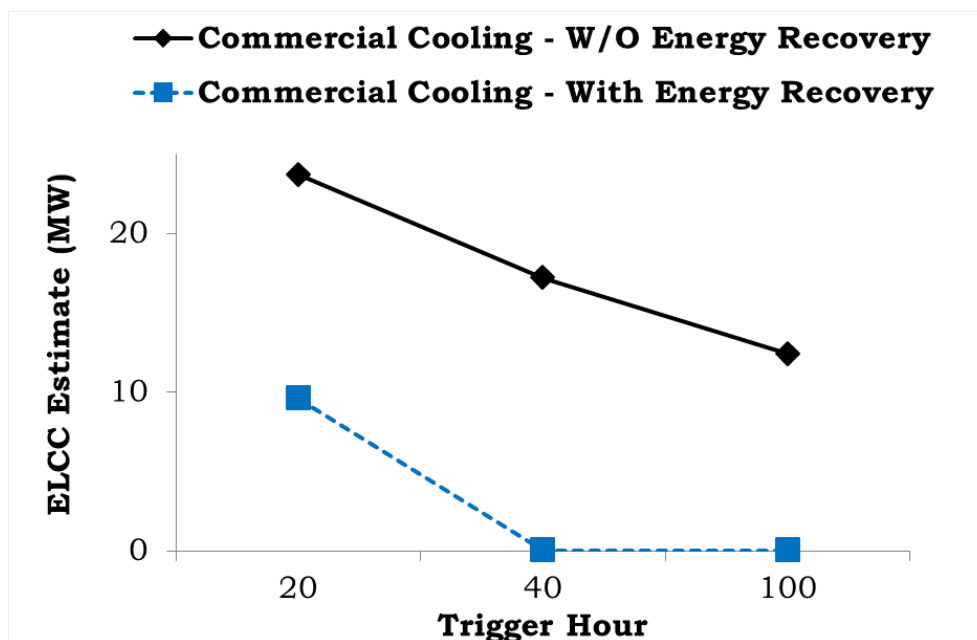


Figure C.13: Los Angeles Department of Water and Power commercial cooling

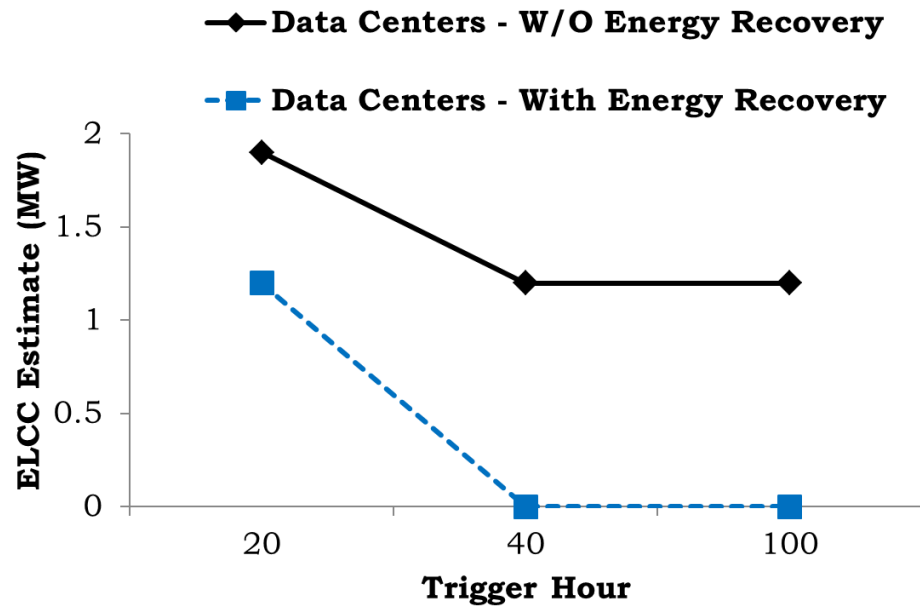


Figure C.14: Los Angeles Department of Water and Power data centers

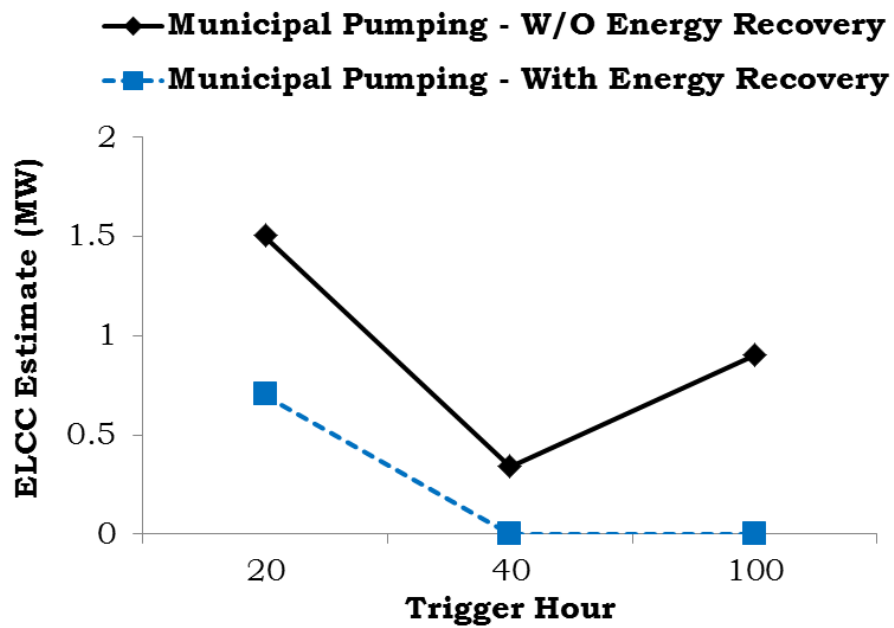


Figure C.15: Los Angeles Department of Water and Power municipal pumping

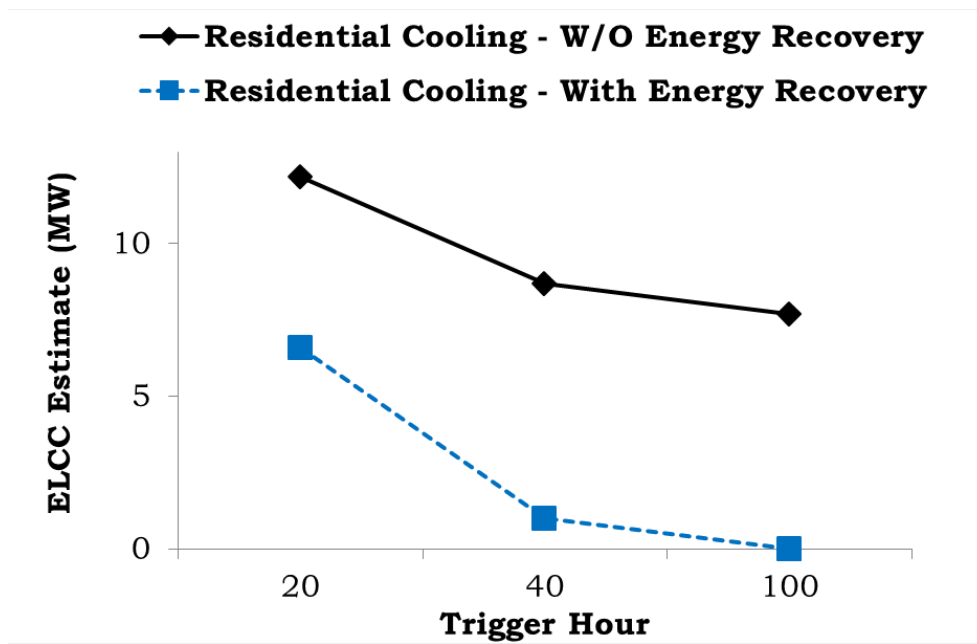


Figure C.16: Los Angeles Department of Water and Power residential cooling

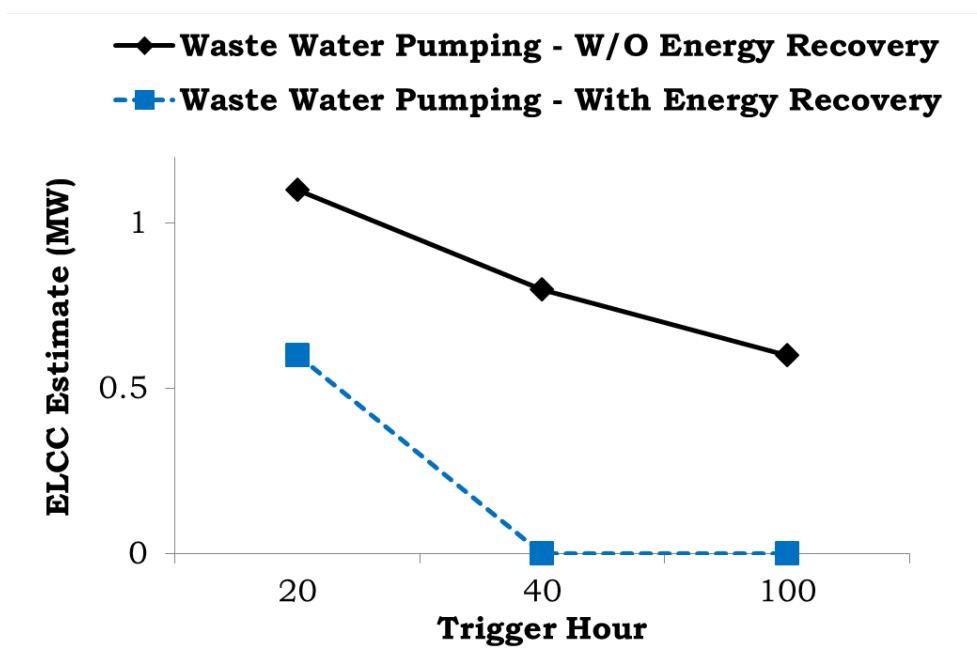


Figure C.17: Los Angeles Department of Water and Power waste water pumping

Capacity Value Estimates for Chapter 3

D.1 Capacity Value Estimates for Avista Corporation

Table D.1: Capacity value estimates for Avista Corporation for different end-use resources **with** energy recovery

Resource	Metric	A	B	C
Commercial Heating	ELCC	1.8 MW	1.4 MW	1.1 MW
	CV	32%	25%	20%
Data Centers	ELCC	0.9 MW	0.7 MW	0.4 MW
	CV	60%	46%	26%
Municipal Pumping	ELCC	0.18 MW	0.18 MW	0.18 MW
	CV	45%	45%	45%
Residential Water Heating	ELCC	1.3 MW	1.0 MW	0.8 MW
	CV	36 %	26 %	22 %
Waste Water Pumping	ELCC	0.18 MW	0.18 MW	0.18 MW
	CV	68%	68%	68%

D.2 Capacity Value Estimates for Puget Sound Energy

Table D.2: CV estimates for Puget Sound Energy for different end-use resources **with** energy recovery

Resource	Metric	A	B	C
Commercial Heating	ELCC CV	1.1 MW 15%	0.2 MW 3%	0.6 MW 8%
Data Centers	ELCC CV	0.6 MW 21%	neg N/A	0.1 MW 2.2%
Municipal Pumping	ELCC CV	0.2 MW 9.9%	neg N/A	0.04 MW 2%
Residential Water Heating	ELCC CV	1.4 MW 18 %	0.1 MW 0.74 %	0.4 MW 5.5%
Waste Water Pumping	ELCC CV	0.1 MW 18%	neg N/A	neg N/A

D.3 Capacity Value Estimates for Nevada Power

Table D.3: CV estimates for Nevada Power for different end-use resources **with** energy recovery

Resource	Metric	A	B	C
Municipal Pumping	ELCC CV	0.4 MW 30%	0.2 MW 20%	0.02 MW 1%
Residential Water Heating	ELCC CV	0.8 MW 20%	0.5 MW 13%	0.2 MW 4%
Data Centers	ELCC CV	1.9 MW 69%	1.0 MW 36%	0.4 MW 15%
Waste Water Pumping	ELCC CV	0.2 MW 38%	0.2 MW 38%	neg N/A

D.4 Capacity Value Estimates for Los Angeles Department of Water & Power

Table D.4: CV estimates for Los Angeles Department of Water and Power for different end-use resources **with** energy recovery

Resource	Metric	A	B	C
Commercial Cooling	ELCC CV	9.6 MW 24%	neg N/A	neg N/A
Commercial Heating	ELCC CV	0.1 MW 4.5%	0.1 MW 4.5%	0.1 MW 4.5%
Data Centers	ELCC CV	1.2 MW 40%	neg N/A	neg N/A
Municipal Pumping	ELCC CV	0.7 MW 15%	neg N/A	neg N/A
Residential Cooling	ELCC CV	6.6 MW 26%	0.9 MW 4%	neg N/A
Residential Water Heating	ELCC CV	0.4 MW 10%	neg N/A	neg N/A
Waste Water Pumping	ELCC CV	0.6 MW 33%	neg N/A	neg N/A

Storage Heating Device

Characteristics in Chapter 4

E.1 Space Heating Devices

Table E.1: Space heating device characteristics

Mid Flat			
	New	Existing cavity	Existing hollow
Number of Homes (in 1000's)	34.883	14.403	25.731
Number of SETS Units per home	2	2	2
Elec Power Rating Per SETS Unit (kWh)	1.56	2.2	2.2
Energy Storage Capability Per SETS Unit (kWh)	10.9	15.4	15.4
Heat Output Capability Per SETS Unit (kWh)	0.7	1.1	1.1
Hourly Efficiency Parameter	0.972	0.975	0.975
Top Flat			
	New	Existing cavity	Existing hollow
Number of Homes (in 1000's)	11.628	4.801	8.577
Number of SETS Units per home	2	2	2
Elec Power Rating Per SETS Unit (kWh)	1.56	2.76	2.76
Energy Storage Capability Per SETS Unit (kWh)	10.9	19.3	19.3
Heat Output Capability Per SETS Unit (kWh)	0.7	1.6	1.6
Hourly Efficiency Parameter	0.972	0.975	0.975

E.2 Water Heating Devices

Table E.2: Water heating device characteristics

Mid Flat			
	New	Existing cavity	Existing hollow
Number of Homes (in 1000's)	34.883	14.403	25.731
Number of SETS Units per home	1	1	1
Elec Power Rating Per SETS Unit (kWh)	3	3	3
Energy Storage Capability Per SETS Unit (kWh)	15.3	15.3	15.3
Heat Output Capability Per SETS Unit (kWh)	0	0	0
Hourly Efficiency Parameter	0.99	0.99	0.99
Top Flat			
	New	Existing cavity	Existing hollow
Number of Homes (in 1000's)	11.628	4.801	8.577
Number of SETS Units per home	1	1	1
Elec Power Rating Per SETS Unit (kWh)	3	3	3
Energy Storage Capability Per SETS Unit (kWh)	15.3	15.3	15.3
Heat Output Capability Per SETS Unit (kWh)	0	0	0
Hourly Efficiency Parameter	0.99	0.99	0.99

Additional Results without Demand Response for Chapter 5

F.1 Variation in Installed Generating Capacity - Firms

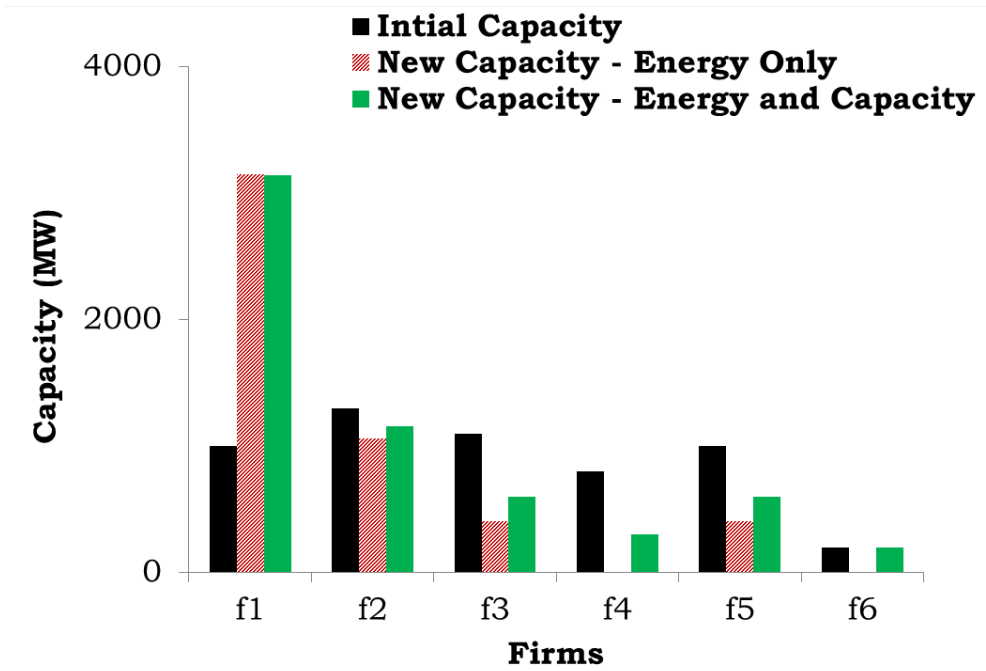


Figure F.1: Installed generating capacity with energy only and energy & capacity markets for a peak load of **5000 MW**

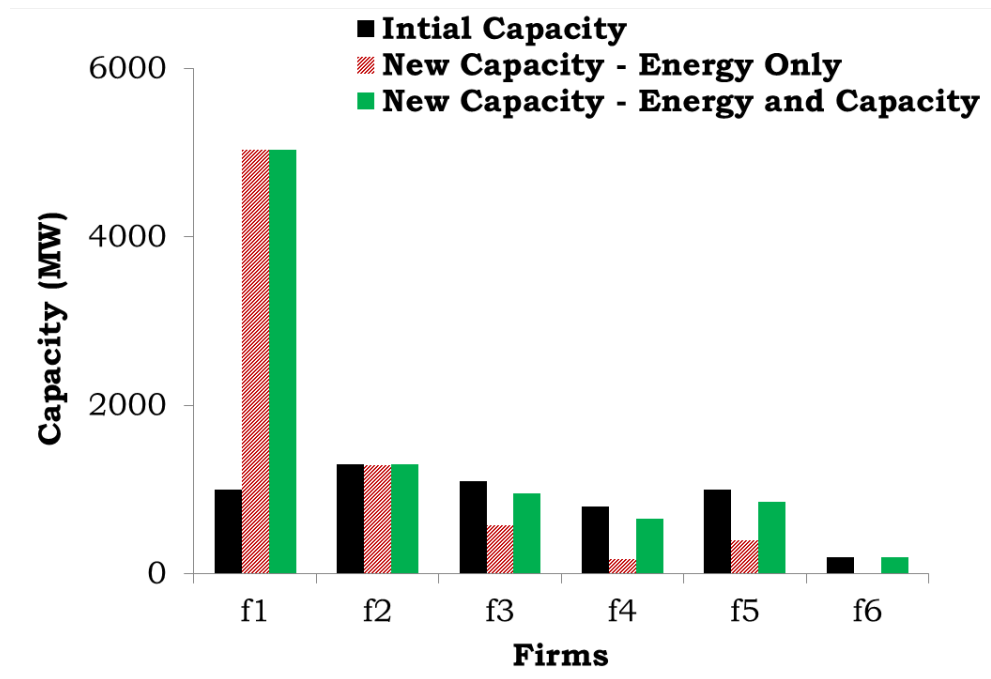


Figure F.2: Installed generating capacity with energy only and energy & capacity markets for a peak load of **7500 MW**

F.2 Variation in Installed Generating Capacity - Technologies

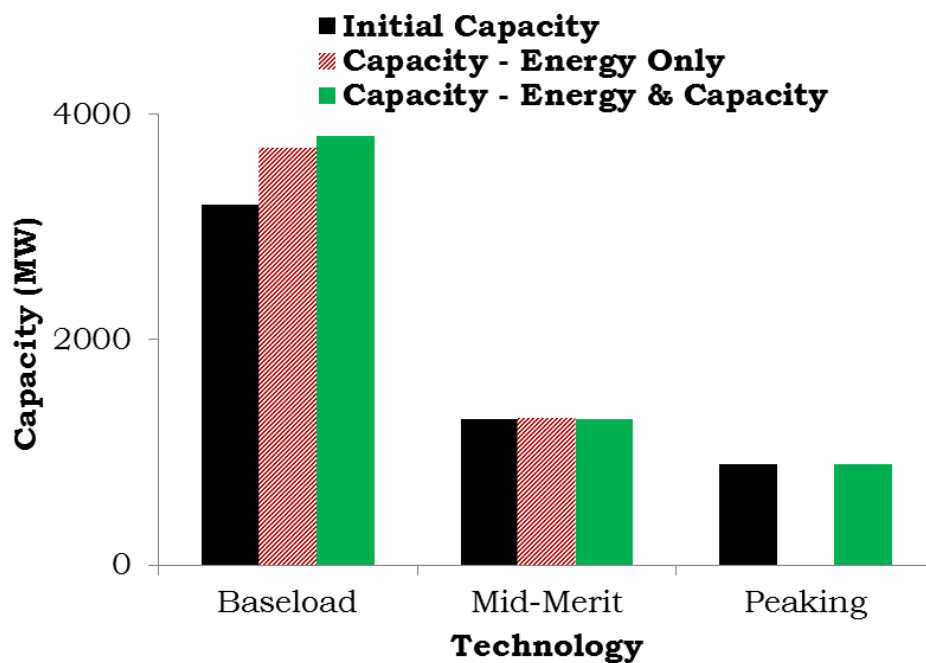


Figure F.3: Installed generating capacity with energy only and energy & capacity markets for a peak load of **5000 MW**

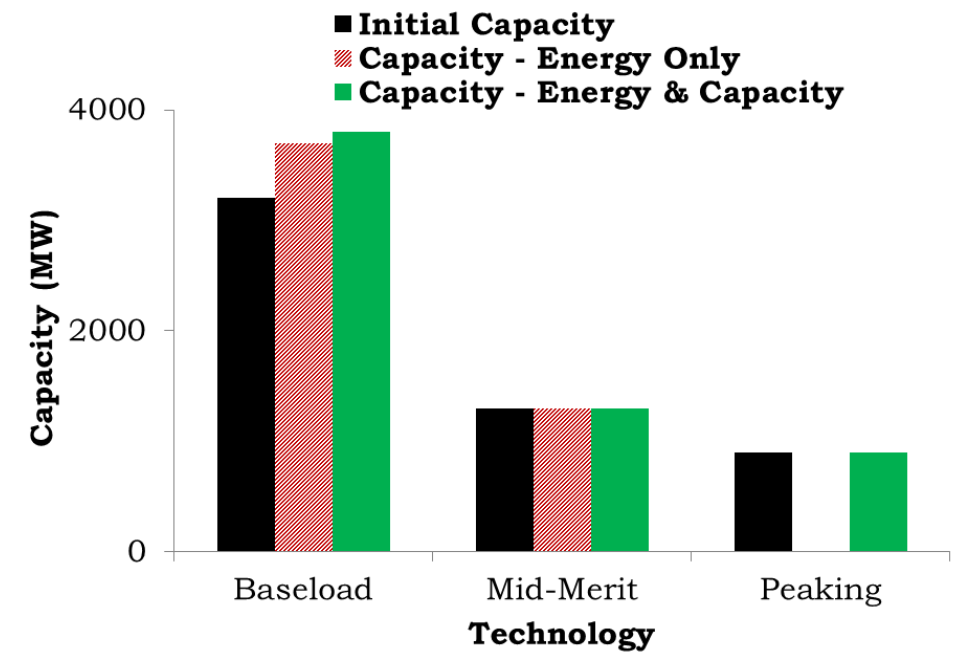


Figure F.4: Installed generating capacity with energy only and energy & capacity markets for a peak load of **7500 MW**

F.3 Variation in System Marginal Price and System Demand Profile

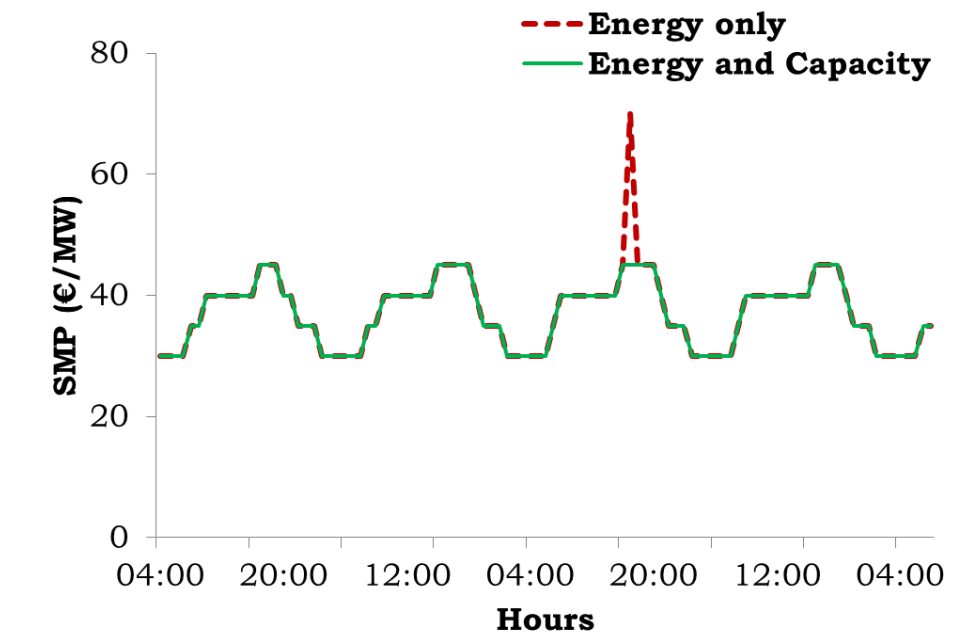


Figure F.5: SMP without DR: Comparison of energy & capacity markets and an energy only market for a peak load of **5000 MW** and no wind generation

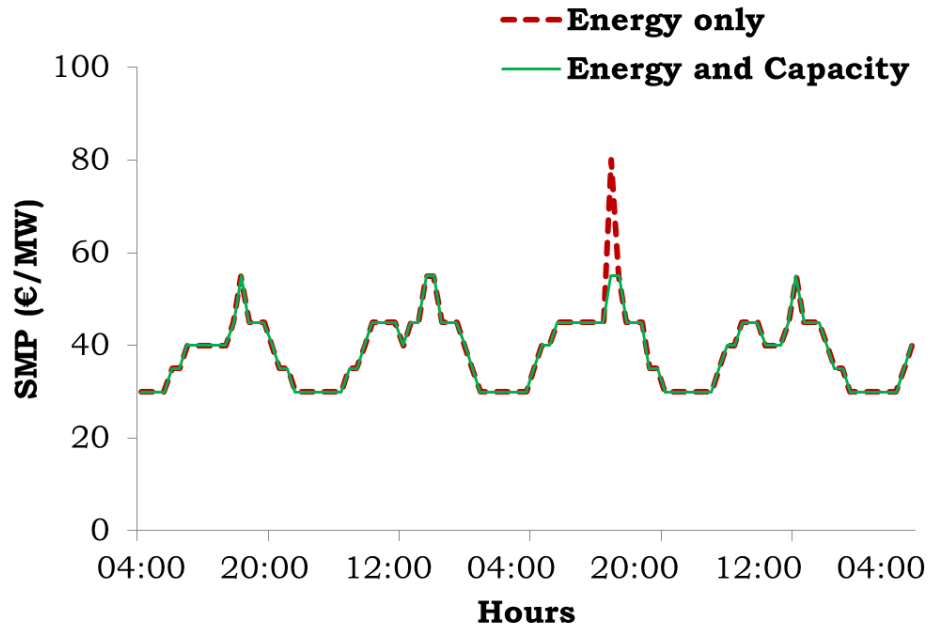


Figure F.6: SMP without DR: Comparison of energy & capacity markets and an energy only market for a peak load of **7500 MW** and no wind generation

F.4 Change in Generator Investment with Different Wind Generation Levels

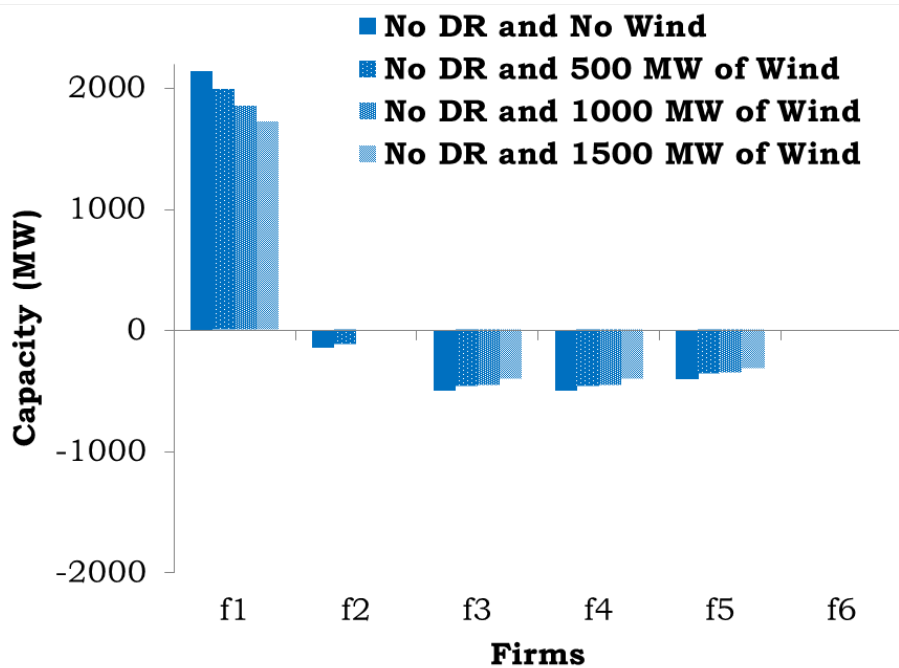


Figure F.7: Generator investment and exit decisions under different wind penetrations at a peak load of **5000 MW**

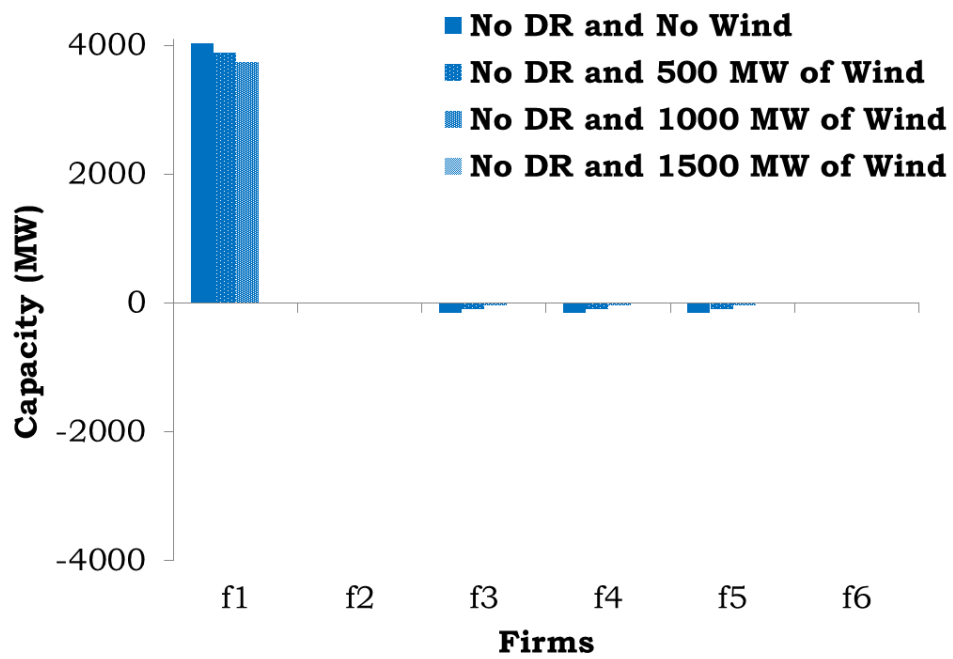


Figure F.8: Generator investment and exit decisions under different wind penetrations at a peak load of **7500 MW**

Change in System Price and Demand Profile with Addition of Demand Response in Chapter 5

G.1 System Marginal Price

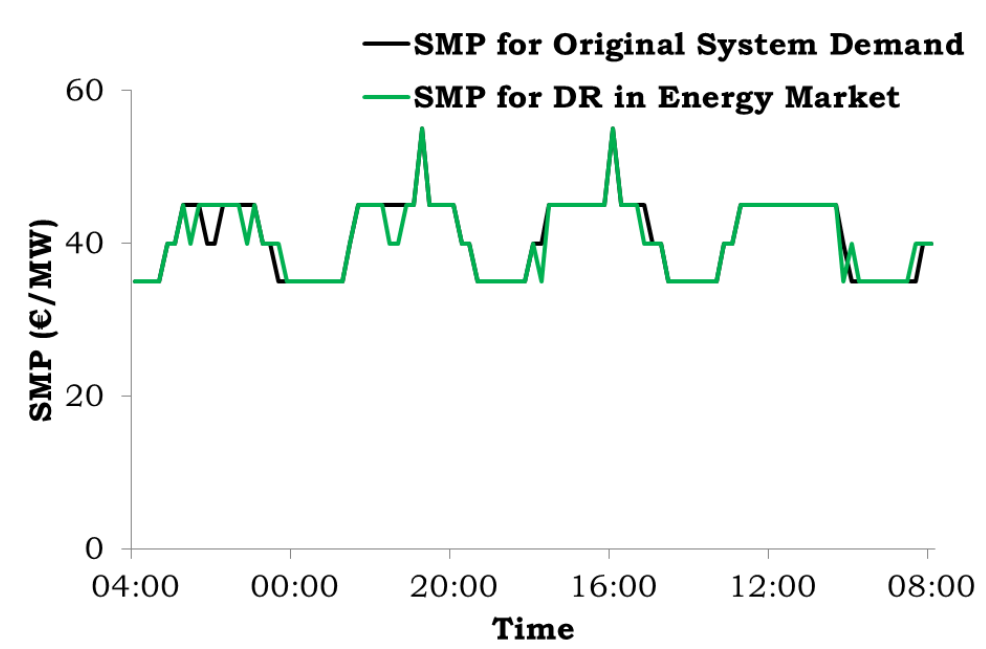


Figure G.1: SMP with and without demand response over a four day period for a peak load of **5000 MW** and no wind generation

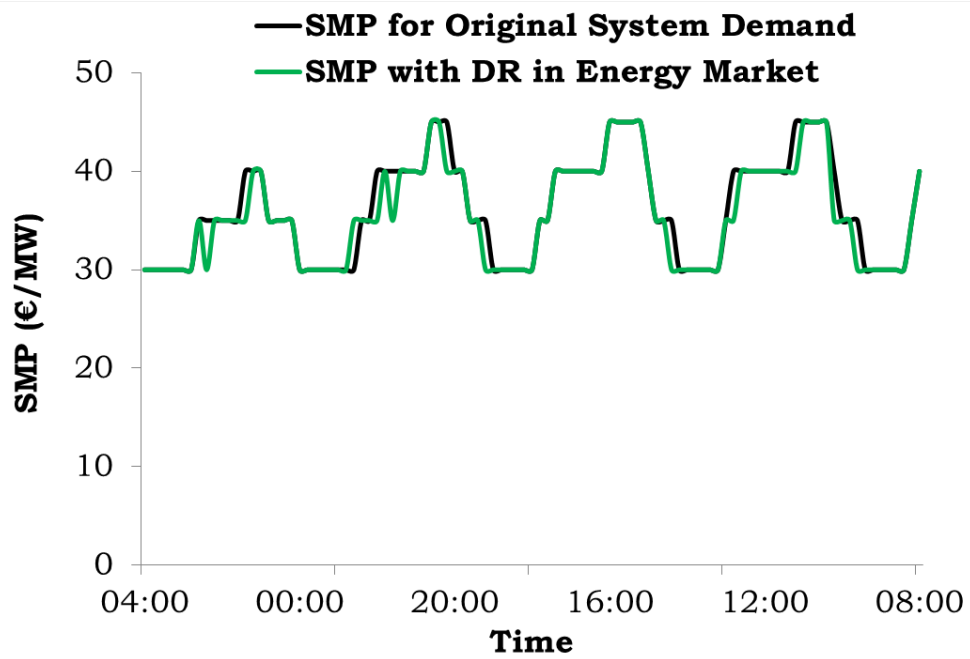


Figure G.2: SMP with and without demand response over a four day period for a peak load of **7500 MW** and no wind generation

G.2 System Demand Profile

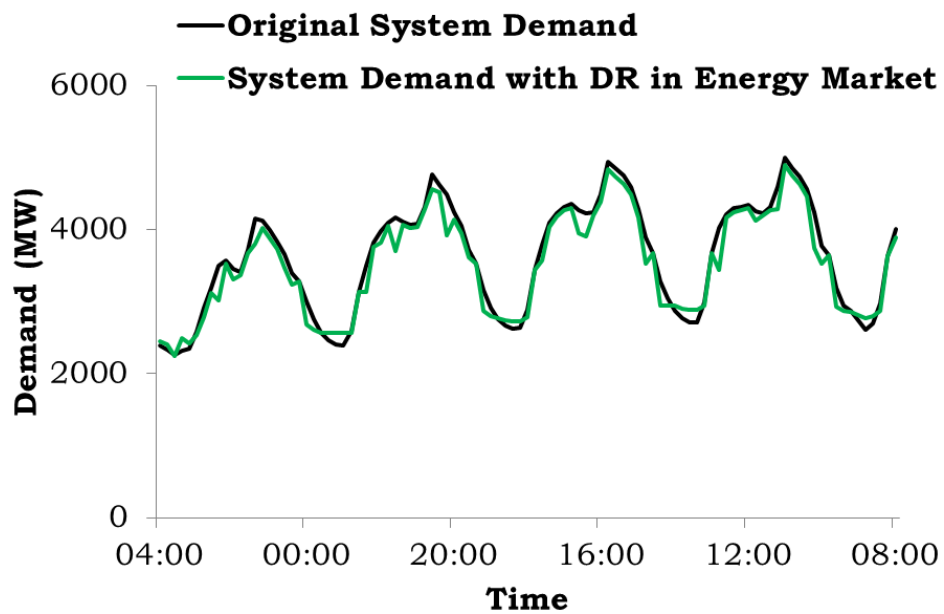


Figure G.3: System demand with the addition of demand response over a four day period for a peak load of **5000 MW** with no wind generation

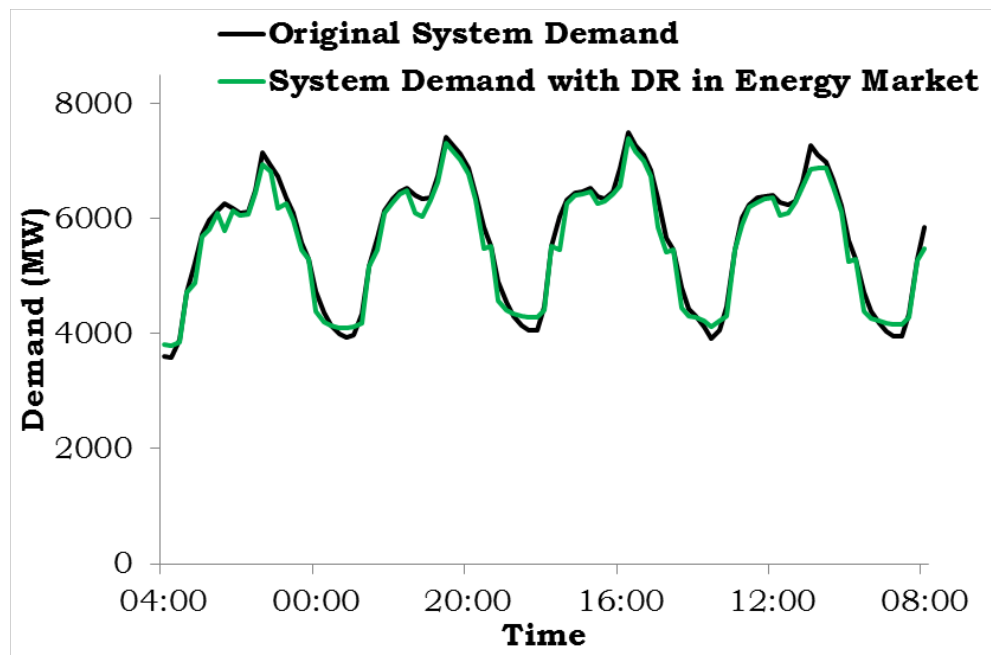


Figure G.4: System demand with the addition of demand response over a four day period for a peak load of **7500 MW** with no wind generation

Capacity Bids of Demand Response without Elasticity in Chapter 5

Table H.1: Capacity bids of demand response, cap_{dr} , with and without reserve provision, with a system reserve requirement of **500 MW**, without elastic demand

Peak Load	Wind Level	CV with reserve	CV without reserve
2500 MW	0 MW	71 MW	71 MW
	500 MW	82 MW	82 MW
	1000 MW	73 MW	73 MW
	1500 MW	0 MW	0 MW
5000 MW	0 MW	123 MW	123 MW
	500 MW	122 MW	122 MW
	1000 MW	129 MW	129 MW
	1500 MW	110 MW	110 MW
7500 MW	0 MW	126 MW	126 MW
	500 MW	119 MW	119 MW
	1000 MW	122 MW	122 MW
	1500 MW	112 MW	115 MW

System Operating Costs in Chapter 5

Tables I.1 and I.3 display the system operating costs for various different demand response market participation combinations for different peak load levels and different installed wind capacities. The tables are discussed in Section 5.5.3.

Table I.1: System operating costs, including wind costs, with different Demand Response market participation combinations **without** elastic demand included

		Demand Response Market Participation				
Peak Load	Wind Level	No DR	Energy Only	Energy & Capacity	Energy & Reserve	All Markets
2500 MW	0 MW	€138,338,447	€130,559,381	€130,559,381	€130,559,381	€130,559,381
	500 MW	€138,343,775	€130,548,482	€130,548,482	€130,548,482	€130,548,482
	1000 MW	€138,308,,949	€130,179,399	€130,179,399	€130,179,399	€130,179,399
	1500 MW	€135,964,589	€125,910,712	€125,910,712	€125,910,712	€125,910,712
5000 MW	0 MW	€279,449,414	€271,725,303	€271,725,303	€271,725,303	€271,725,303
	500 MW	€279,474,534	€271,736,923	€271,736,923	€271,736,923	€271,736,923
	1000 MW	€279,370,359	€271,732,677	€271,732,677	€271,732,677	€271,732,677
	1500 MW	€279,293,061	€271,461,175	€271,461,175	€271,461,175	€271,461,175
7500 MW	0 MW	€420,393,744	€413,225,784	€413,225,784	€413,225,784	€413,225,784
	500 MW	€421,031,410	€413,305,838	€413,305,838	€413,305,838	€413,305,838
	1000 MW	€421,039,668	€413,240,785	€413,240,785	€413,240,785	€413,240,785
	1500 MW	€421,084,116	€417,717,937	€412,442,544	€417,717,937	€412,328,448

Table I.2: Percentage system operating cost savings, including wind costs, with different Demand Response market participation combinations **without** elastic demand included

		Demand Response Market Participation				
Peak Load	Wind Level	No DR	Energy Only	Energy & Capacity	Energy & Reserve	All Markets
2500 MW	0 MW	-	6%	6%	6%	6%
	500 MW	-	6%	6%	6%	6%
	1000 MW	-	6%	6%	6%	6%
	1500 MW	-	7%	7%	7%	7%
5000 MW	0 MW	-	3%	3%	3%	3%
	500 MW	-	3%	3%	3%	3%
	1000 MW	-	3%	3%	3%	3%
	1500 MW	-	3%	3%	3%	3%
7500 MW	0 MW	-	2%	2%	2%	2%
	500 MW	-	2%	2%	2%	2%
	1000 MW	-	2%	2%	2%	2%
	1500 MW	-	1%	2%	1%	2%

Table I.3: System operating costs, including wind costs, with different DR market participation combinations **with** elastic demand included

		Demand Response Market Participation				
Peak Load	Wind Level	No DR	Energy Only	Energy & Capacity	Energy & Reserve	All Markets
2500 MW	0 MW	€138,338,447	€130,249,339	€130,249,340	€130,249,340	€130,249,339
	500 MW	€138,030,978	€130,317,420	€130,317,420	€130,317,420	€130,317,420
	1000 MW	€137,999,006	€129,832,278	€129,832,278	€129,832,278	€129,832,278
	1500 MW	€135,567,215	€125,590,205	€125,590,205	€125,590,205	€125,590,205
5000 MW	0 MW	€279,132,776	€271,405,604	€271,405,605	€271,405,605	€271,405,604
	500 MW	€279,474,534	€271,422,067	€271,422,067	€271,422,067	€271,422,067
	1000 MW	€279,371,979	€271,418,128	€271,418,127	€271,418,127	€271,418,127
	1500 MW	€278,979,306	€271,142,327	€271,142,327	€271,142,327	€271,142,327
7500 MW	0 MW	€420,624,322	€412,907,825	€412,907,825	€412,907,825	€412,907,825
	500 MW	€420,715,722	€412,984,932	€412,984,932	€412,984,932	€412,984,932
	1000 MW	€421,039,667	€412,924,775	€412,924,775	€412,924,775	€412,924,775
	1500 MW	€421,068,019	€417,519,853	€412,535,101	€417,519,853	€412,328,448

Table I.4: Percentage system operating cost savings, including wind costs, with different Demand Response market participation combinations **without** elastic demand included

		Demand Response Market Participation				
Peak Load	Wind Level	No DR	Energy Only	Energy & Capacity	Energy & Reserve	All Markets
2500 MW	0 MW	-	6%	6%	6%	6%
	500 MW	-	6%	6%	6%	6%
	1000 MW	-	6%	6%	6%	6%
	1500 MW	-	7%	7%	7%	7%
5000 MW	0 MW	-	3%	3%	3%	3%
	500 MW	-	3%	3%	3%	3%
	1000 MW	-	3%	3%	3%	3%
	1500 MW	-	3%	3%	3%	3%
7500 MW	0 MW	-	1%	1%	1%	1%
	500 MW	-	2%	2%	2%	2%
	1000 MW	-	2%	2%	2%	2%
	1500 MW	-	1%	2%	1%	2%

Generator Investment and Exit with Different Demand Response Market Participation in Chapter 5

J.1 Investment and Exit at Peak Load of 2500 MW

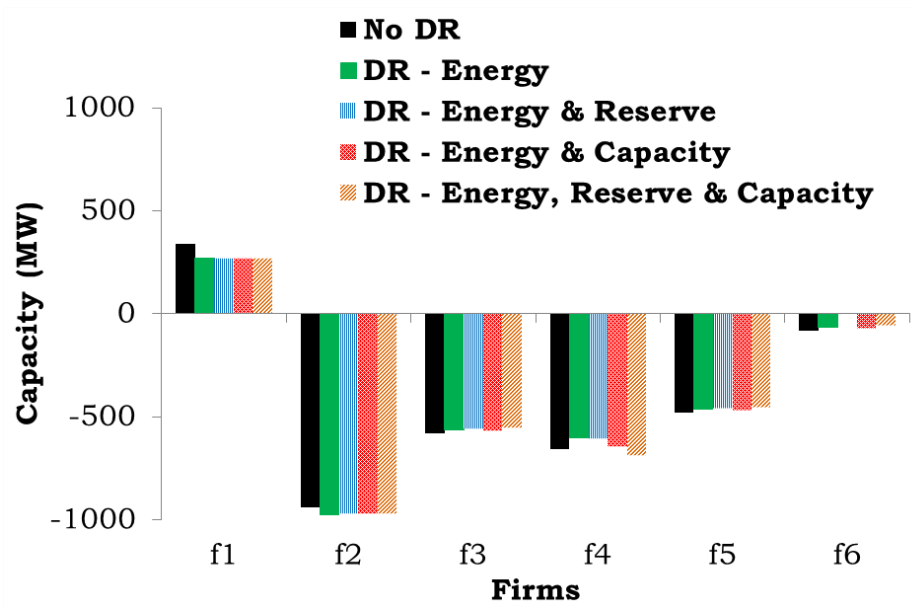


Figure J.1: Generator investment and exit decisions under different DR market participation frameworks with **0 MW** of wind generation at peak load of **2500 MW**

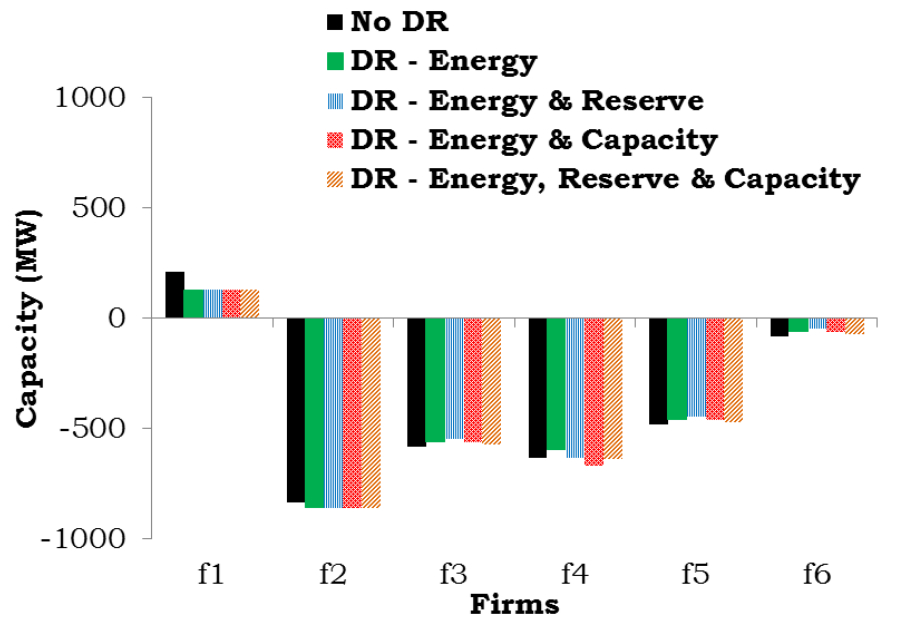


Figure J.2: Generator investment and exit decisions under different DR market participation frameworks with **500 MW** of wind generation at peak load of **2500 MW**

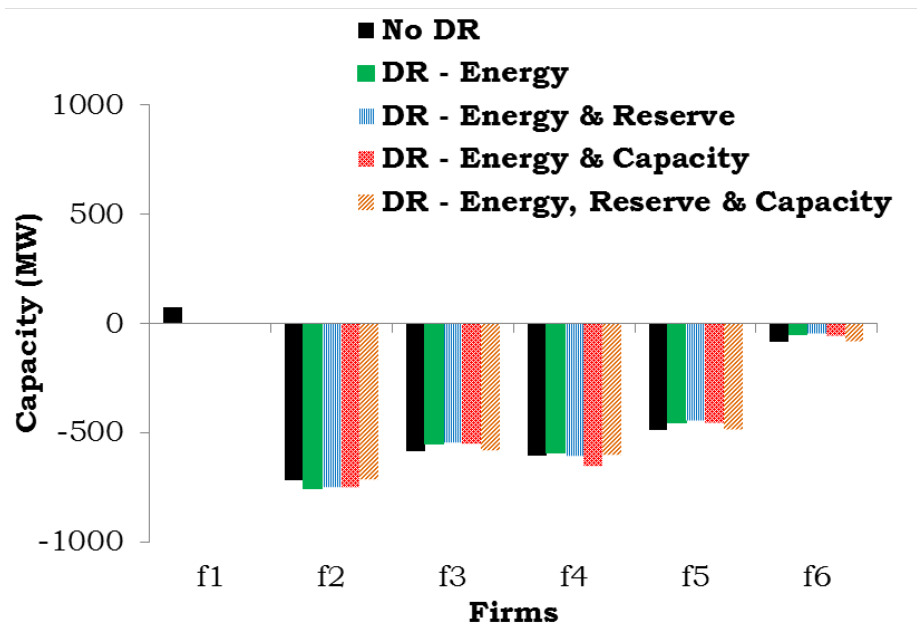


Figure J.3: Generator investment and exit decisions under different DR market participation frameworks with **1000 MW** of wind generation at peak load of **2500 MW**

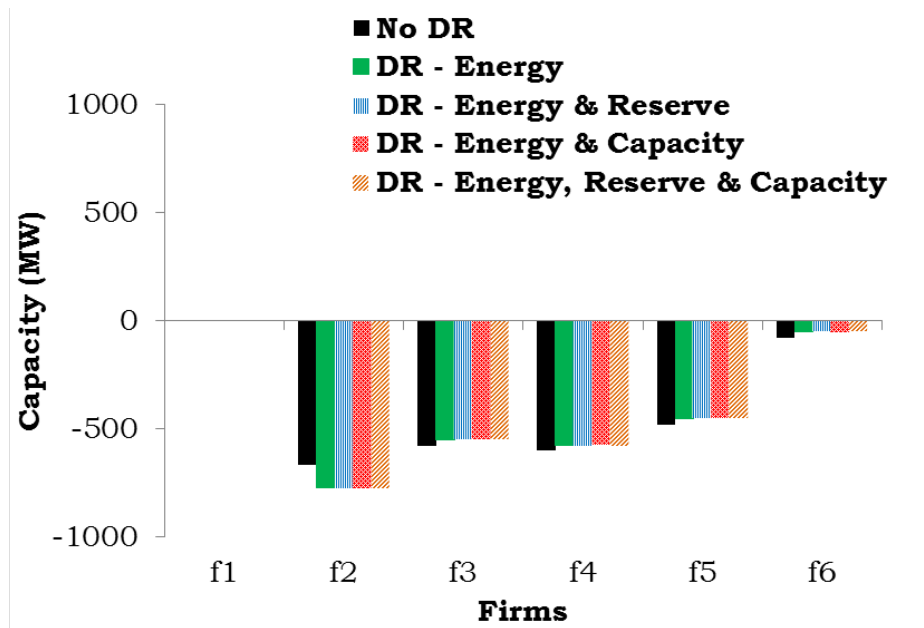


Figure J.4: Generator investment and exit decisions under different DR market participation frameworks with **1500 MW** of wind generation at peak load of **2500 MW**

J.2 Investment and Exit at Peak Load of 5000 MW

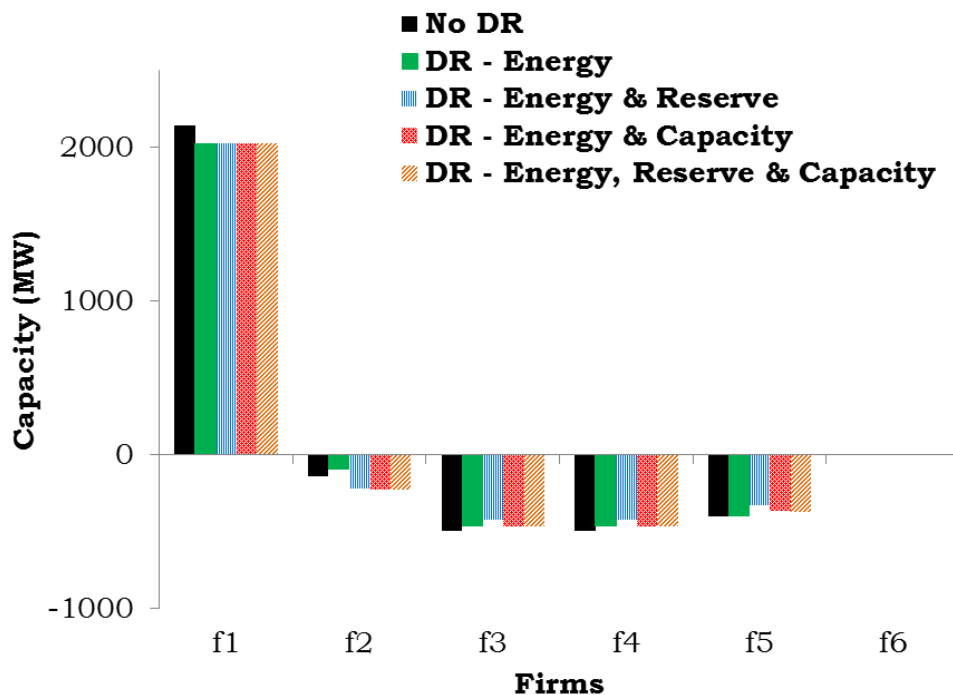


Figure J.5: Generator investment and exit decisions under different DR market participation frameworks with **0 MW** of wind generation at peak load of **5000 MW**

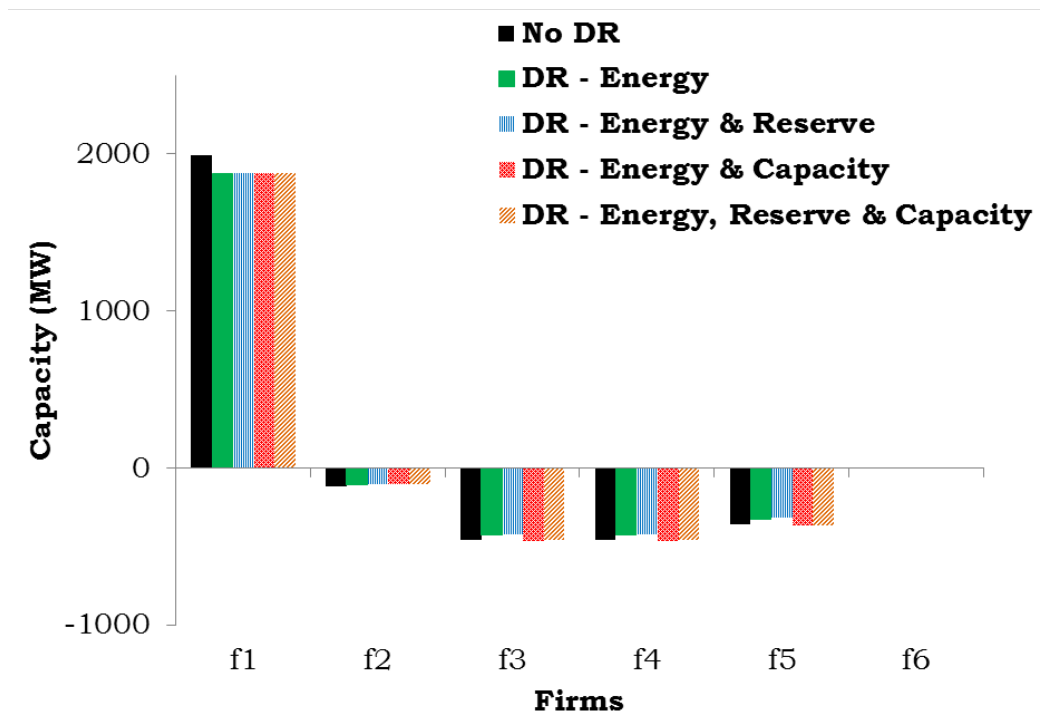


Figure J.6: Generator investment and exit decisions under different DR market participation frameworks with **500 MW** of wind generation at peak load of **5000 MW**

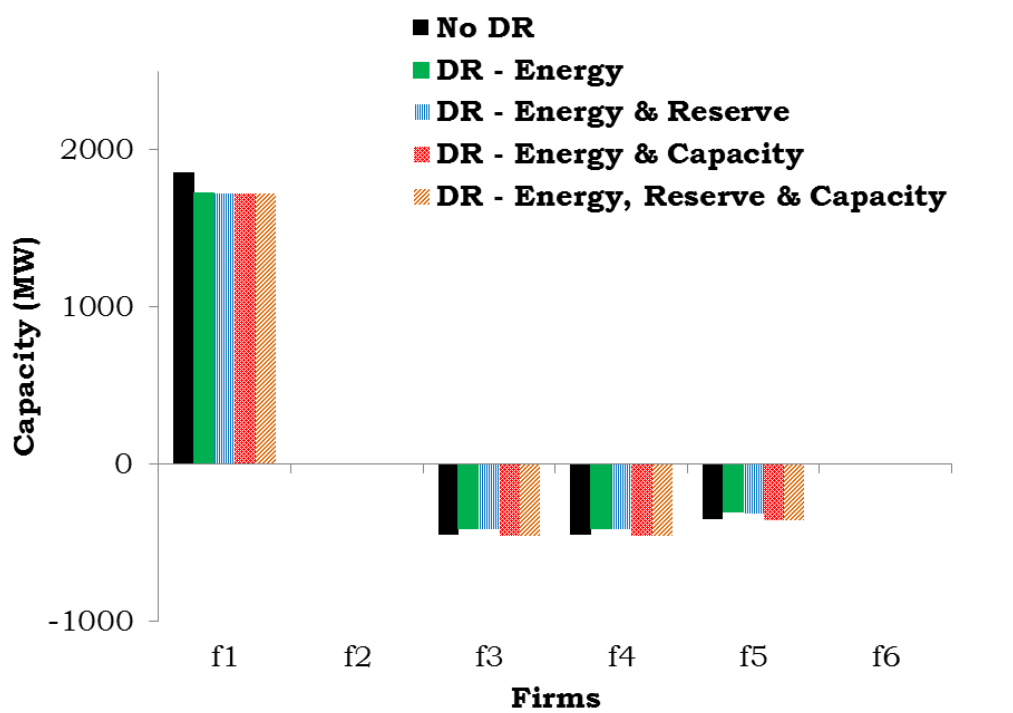


Figure J.7: Generator investment and exit decisions under different DR market participation frameworks with **1000 MW** of wind generation at peak load of **5000 MW**

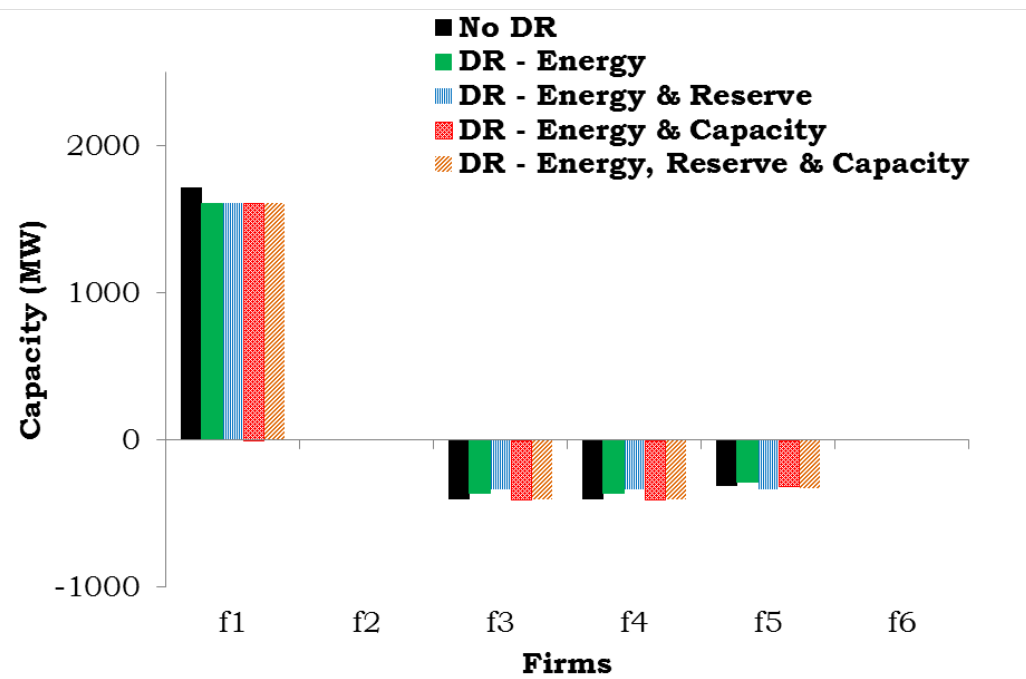


Figure J.8: Generator investment and exit decisions under different DR market participation frameworks with **1500 MW** of wind generation at peak load of **5000 MW**

J.3 Investment and Exit at Peak Load of 7500 MW

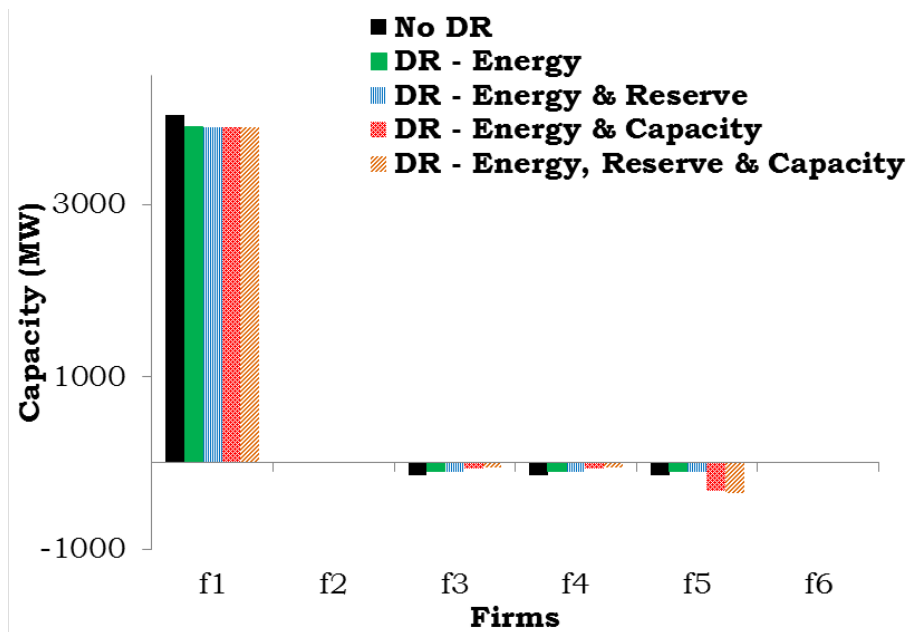


Figure J.9: Generator investment and exit decisions under different DR market participation frameworks with **0 MW** of wind generation at peak load of **7500 MW**

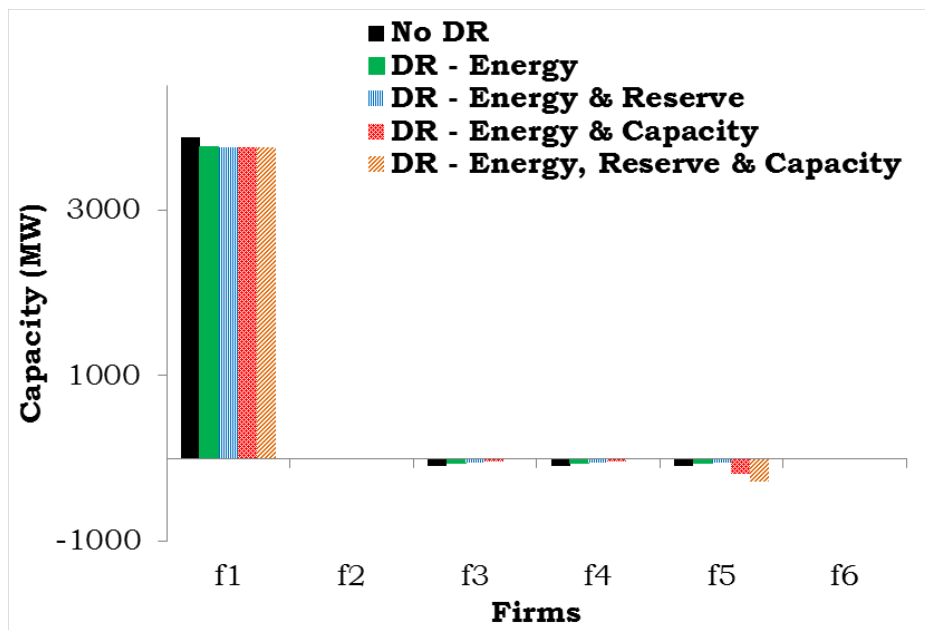


Figure J.10: Generator investment and exit decisions under different DR market participation frameworks with **500 MW** of wind generation at peak load of **7500 MW**

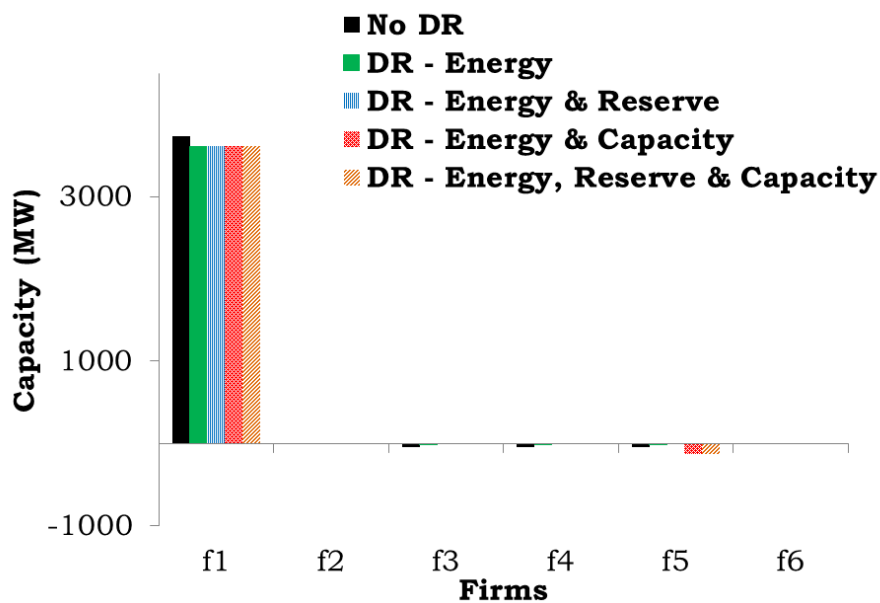


Figure J.11: Generator investment and exit decisions under different DR market participation frameworks with **1000 MW** of wind generation at peak load of **7500 MW**

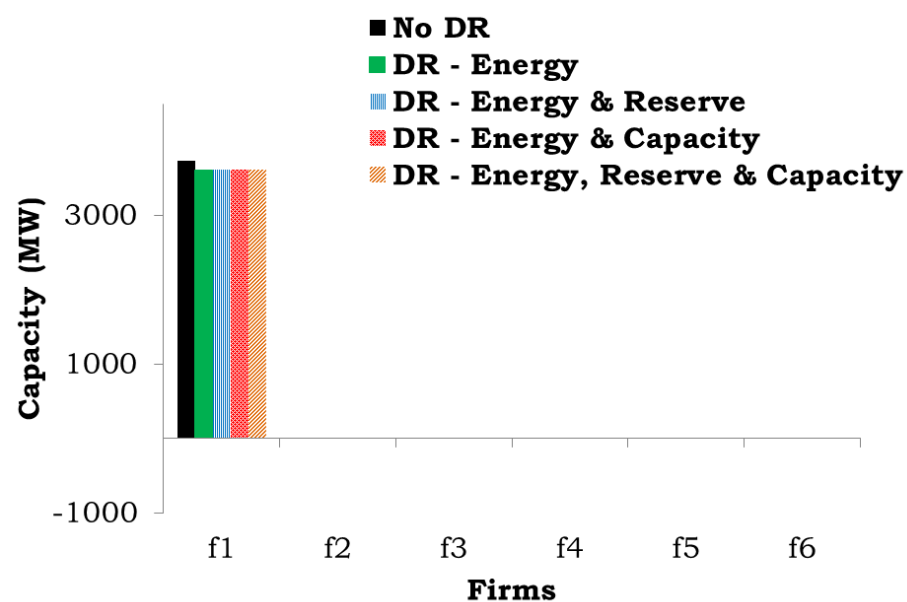


Figure J.12: Generator investment and exit decisions under different DR market participation frameworks with **1500 MW** of wind generation at peak load of **7500 MW**

Marginal Cost Set 2 in Chapter 5

Table K.1: Set 2: Marginal Cost $MC^{i,j}$ of Each Firm (€/MW)

Technology	f1	f2	f3	f4	f5	f6
Baseload	55	45	55	55	65	—
Mid Merit	—	50	35	—	35	—
Peaking	—	—	93	83	93	93

K.1 Change in Generating Portfolios

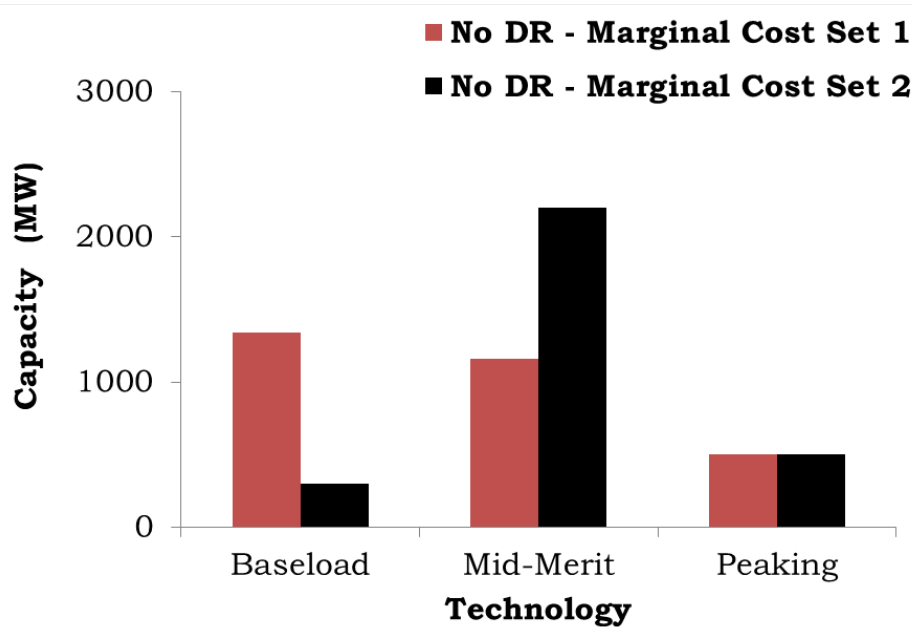


Figure K.1: Generating portfolios with different marginal cost inputs, no demand response and no wind generation at a peak load of **2500 MW**

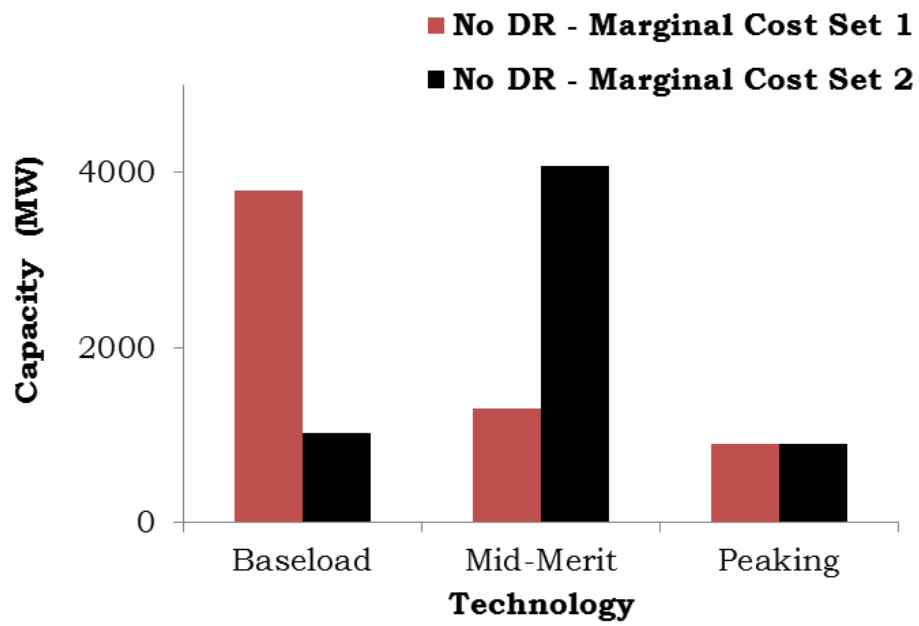


Figure K.2: Generating portfolios with different marginal cost inputs, no demand response and no wind generation at a peak load of **5000 MW**

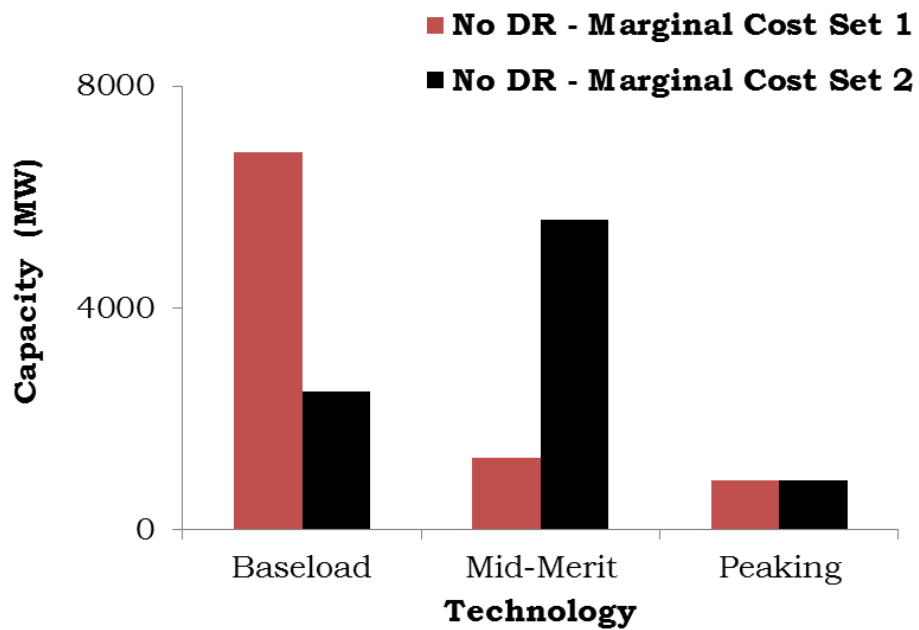


Figure K.3: Generating portfolios with different marginal cost inputs, no demand response and no wind generation at a peak load of **7500 MW**

K.2 Capacity Prices with Marginal Cost Set 2

Table K.2: Capacity Prices for **2500 MW** of peak load and with a reserve requirement of **500 MW** with Marginal cost Set 2

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = -0.11$	€7	€7	€7	€0	€0
500 MW	$E = -0.11$	€7	€7	€7	€7	€7
1000 MW	$E = -0.11$	€7	€7	€7	€7	€7
1500 MW	$E = -0.11$	€7	€7	€7	€7	€7

Table K.3: Capacity prices for **5000 MW** of peak load and with a reserve requirement of **500 MW** with Marginal Cost Set 2

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = -0.11$	€25	€25	€25	€25	€25
500 MW	$E = -0.11$	€25	€25	€25	€25	€25
1000 MW	$E = -0.11$	€25	€25	€25	€25	€25
1500 MW	$E = -0.11$	€25	€25	€25	€25	€25

Table K.4: Capacity prices for **7500 MW** of peak load and with a reserve requirement of **500 MW** with Marginal Cost Set 2

		Demand Response Market Participation				
Wind Level	Elastic Demand	No DR	Energy Only	Energy & Reserve	Energy & Capacity	All Markets
0 MW	$E = -0.11$	€25	€25	€25	€25	€25
500 MW	$E = -0.11$	€25	€25	€25	€25	€25
1000 MW	$E = -0.11$	€25	€25	€25	€25	€25
1500 MW	$E = -0.11$	€25	€25	€25	€25	€25

K.3 Capacity Bids with Marginal Cost Set 2

Table K.5: Capacity bids of demand response with Marginal Cost Set 2

Peak Load	Wind Level	Elastic Demand?	CV with reserve	CV without reserve
2500 MW	0 MW	$E = -0.11$	126 MW	126 MW
	500 MW	$E = -0.11$	120 MW	120 MW
	1000 MW	$E = -0.11$	109 MW	109 MW
	1500 MW	$E = -0.11$	96 MW	96 MW
5000 MW	0 MW	$E = -0.11$	113 MW	113 MW
	500 MW	$E = -0.11$	117 MW	117 MW
	1000 MW	$E = -0.11$	118 MW	118 MW
	1500 MW	$E = -0.11$	112 MW	112 MW
7500 MW	0 MW	$E = -0.11$	126 MW	126 MW
	500 MW	$E = -0.11$	130 MW	130 MW
	1000 MW	$E = -0.11$	113 MW	113 MW
	1500 MW	$E = -0.11$	128 MW	128 MW

K.4 System Operating Costs with Demand Response

Table K.6: Percentage reduction in system operating costs following inclusion of demand response

Peak Load	Wind Level	% Reduction in Costs
2500 MW	0 MW	5.4%
2500 MW	500 MW	5.2%
2500 MW	1000 MW	5.5%
2500 MW	1500 MW	5.7%
5000 MW	0 MW	2.8%
5000 MW	500 MW	2.8%
5000 MW	1000 MW	2.7%
5000 MW	1500 MW	2.8%
7500 MW	0 MW	1.9%
7500 MW	500 MW	1.9%
7500 MW	1000 MW	1.8%
7500 MW	1500 MW	1.8%

Table K.7: System operating costs, including wind costs, with different demand response market participation combinations **with** elasticity included with Marginal Cost Set 2

		Demand Response Market Participation				
Peak Load	Wind Level	No DR	Energy Only	Energy & Capacity	Energy & Reserve	All Markets
2500 MW	0 MW	€152,499,423	€144,207,807	€144,207,807	€144,207,807	€144,207,807
2500 MW	500 MW	€152,505,464	€144,185,683	€144,185,683	€144,185,683	€144,185,683
2500 MW	1000 MW	€152,411,984	€144,065,551	€144,065,551	€144,065,551	€144,065,551
2500 MW	1500 MW	€151,277,075	€142,622,562	€142,622,562	€142,622,562	€142,622,562
5000 MW	0 MW	€306,357,036	€297,931,179	€297,931,179	€297,931,179	€297,931,179
5000 MW	500 MW	€306,386,599	€297,943,444	€297,943,444	€297,943,444	€297,943,444
5000 MW	1000 MW	€306,312,712	€297,913,399	€297,913,399	€297,913,399	€297,913,399
5000 MW	1500 MW	€306,250,677	€297,779,090	€297,779,090	€297,779,090	€297,779,090
7500 MW	0 MW	€460,605,232	€452,110,884	€452,110,884	€452,110,884	€452,110,884
7500 MW	500 MW	€460,599,186	€452,091,533	€452,091,533	€452,091,533	€452,091,533
7500 MW	1000 MW	€460,543,913	€452,041,694	€452,041,694	€452,041,694	€452,041,694
7500 MW	1500 MW	€460,455,996	€452,017,053	€452,017,053	€452,017,053	€452,017,053

Table K.8: Demand response aggregator costs at a peak load of **2500 MW** with Marginal Cost Set 2

DR Case	Wind Level	Energy Costs Only	Savings	With Capacity Payments	
				Energy Costs	Savings
No DR	0 MW	€8,044,875	—	—	—
Energy Only	0 MW	€7,618,437	5.3%	—	—
Energy & Reserve	0 MW	€7,618,031	5.3%	—	—
Energy & Capacity	0 MW	€7,618,064	5.3%	€7,618,064	5.3%
All Markets	0 MW	€7,618,064	5.3%	€7,618,064	5.3%
No DR	500 MW	€8,045,572	—	—	—
Energy Only	500 MW	€7,618,669	5.3%	—	—
Energy & Reserve	500 MW	€7,618,447	5.3 %	—	—
Energy & Capacity	500 MW	€7,618,669	5.3%	€7,617,852	5.3%
All Markets	500 MW	€7,618,453	5.3%	€7,617,616	5.3%
No DR	1000 MW	€8,046,742	—	—	—
Energy Only	1000 MW	€7,676,288	4.6%	—	—
Energy & Reserve	1000 MW	€7,675,710	4.6%	—	—
Energy & Capacity	1000 MW	€7,676,245	4.6%	€7,675,485	4.6%
All Markets	1000 MW	€7,676,245	4.6%	€7,675,485	4.6%
No DR	1500 MW	€7,940,229	—	—	—
Energy Only	1500 MW	€7,404,415	6.7%	—	—
Energy & Reserve	1500 MW	€7,404,415	6.7%	—	—
Energy & Capacity	1500 MW	€7,404,415	6.7%	€7,403,741	6.8%
All Markets	1500 MW	€7,404,415	6.7%	€7,403,741	6.8%

Table K.9: Demand response aggregator costs at a peak load of **5000 MW** with Marginal Cost Set 2

DR Case	Wind Level	Energy Costs Only	Savings	With Capacity Payments	
				Energy Costs	Savings
No DR	0 MW	€8,052,692	—	—	—
Energy Only	0 MW	€7,619,311	5.4%	—	—
Energy & Reserve	0 MW	€7,619,277	5.4%	—	—
Energy & Capacity	0 MW	€7,618,816	5.4%	€7,615,982	5.4%
All Markets	0 MW	€7,618,816	5.4%	€7,615,982	5.4%
No DR	500 MW	€8,050,779	—	—	—
Energy Only	500 MW	€7,619,528	5.4%	—	—
Energy & Reserve	500 MW	€7,619,397	5.4%	—	—
Energy & Capacity	500 MW	€7,619,531	5.3%	€7,616,614	5.4%
All Markets	500 MW	€7,619,100	5.4%	€7,616,183	5.4%
No DR	1000 MW	€8,041,184	—	—	—
Energy Only	1000 MW	€7,619,748	5.2%	—	—
Energy & Reserve	1000 MW	€7,619,274	5.3%	—	—
Energy & Capacity	1000 MW	€7,619,755	5.3%	€7,616,825	5.3%
All Markets	1000 MW	€7,619,364	5.3%	€7,616,404	5.3%
No DR	1500 MW	€8,049,650	—	—	—
Energy Only	1500 MW	€7,619,965	5.3%	—	—
Energy & Reserve	1500 MW	€7,619,463	5.3%	—	—
Energy & Capacity	1500 MW	€7,620,224	5.3%	€7,617,437	5.3%
All Markets	1500 MW	€7,619,495	5.3%	€7,616,707	5.3%

Table K.10: Demand response aggregator costs at a peak load of **7500 MW** with Marginal Cost Set 2

DR Case	Wind Level	Energy Costs Only	Savings	With Capacity Payments	
				Energy Costs	Savings
No DR	0 MW	€8,060,201	—	—	—
Energy Only	0 MW	€7,619,272	5.5%	—	—
Energy & Reserve	0 MW	€7,619,398	5.5%	—	—
Energy & Capacity	0 MW	€7,619,272	5.5%	€7,616,115	5.5%
All Markets	0 MW	€7,576,478	6%	€7,573,321	6%
No DR	500 MW	€8,057,430	—	—	—
Energy Only	500 MW	€7,619,499	5.4%	—	—
Energy & Reserve	500 MW	€7,618,885	5.4%	—	—
Energy & Capacity	500 MW	€7,619,498	5.4%	€7,616,246	5.5%
All Markets	500 MW	€7,619,158	5.4%	€7,615,906	5.5 %
No DR	1000 MW	€8,056,147	—	—	—
Energy Only	1000 MW	€7,619,725	5.4%	—	—
Energy & Reserve	1000 MW	€7,619,039	5.4%	—	—
Energy & Capacity	1000 MW	€7,619,039	5.4%	€7,616,208	5.5%
All Markets	1000 MW	€7,618,889	5.4%	€7,616,058	5.5%
No DR	1500 MW	€8,053,171	—	—	—
Energy Only	1500 MW	€7,620,335	5.4%	—	—
Energy & Reserve	1500 MW	€7,619,249	5.4%	—	—
Energy & Capacity	1500 MW	€7,619,947	5.4%	€7,616,747	5.4 %
All Markets	1500 MW	€7,619,193	5.4%	€7,615,993	5.4%

Vita

Sheila Patricia Nolan was born in Ireland. After completing her Leaving Certificate Examinations at St. Mary's College, Arklow, Co. Wicklow, Ireland in 2007, she studied Mechanical Engineering in University College Dublin (UCD), Ireland. Upon graduating from UCD with a First Class Honours Degree in 2011, she completed her masters in Sustainable Energy Systems at the University of Edinburgh, Scotland, graduating summa cum laude with First Class Honours. In October 2012, she commenced her PhD studies at the Electricity Research Centre, in UCD, Ireland.